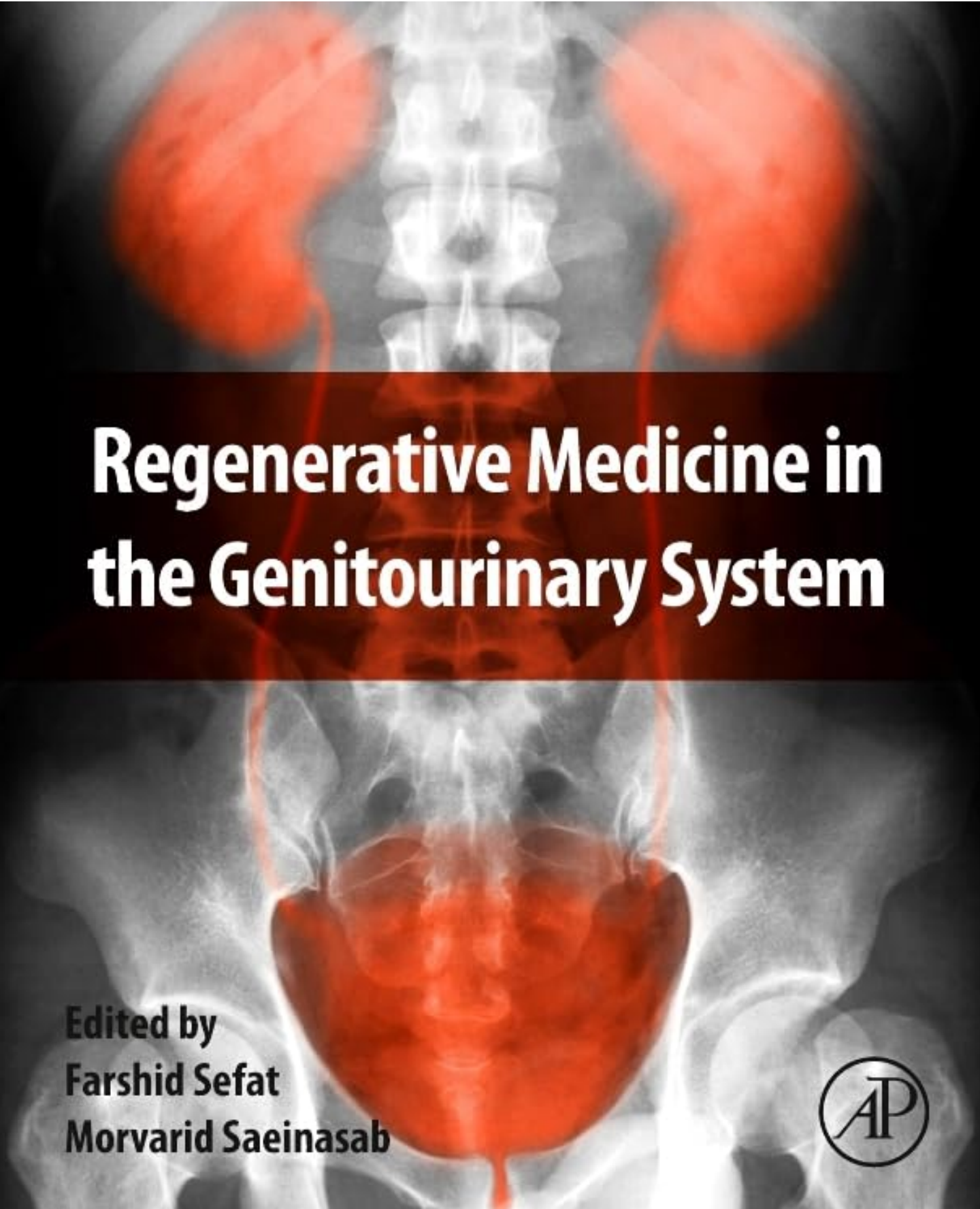




Regenerative Medicine in the Genitourinary System

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Contents

Contributors.....	xv
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SECTION 1 Introduction

CHAPTER 1 Genitourinary tissue engineering: Promises, advances, and challenges.....	3
<i>Haalah Islam, Morvarid Saeinasab and Farshid Sefat</i>	
1.1 Introduction.....	3
1.2 Regenerative medicine	4
1.3 Principles of tissue engineering.....	4
1.3.1 Source of cells.....	5
1.3.2 Biomaterials	6
1.3.3 Fabrication.....	7
1.3.4 Structure and material properties.....	8
1.3.5 Requirements of scaffold design	8
1.3.6 Growth factors.....	9
1.3.7 Decellularization	9
1.4 Genitourinary system anatomy and physiology	9
1.4.1 Urinary structure and function	10
1.4.2 Genital structure and function.....	10
1.5 Genitourinary conditions.....	11
1.5.1 Affecting the ureter	11
1.5.2 Diseases that lead to damage or loss of function of the kidneys.....	11
1.5.3 Conditions that specifically affect females	12
1.5.4 Conditions that specifically affect males.....	12
1.6 Current treatments and potential drawbacks	13
1.7 Genitourinary tissue engineering.....	14
1.7.1 Urethra.....	14
1.7.2 Bladder.....	18
1.7.3 Genital tissues	18
1.7.4 Kidney	19
1.8 Summary	20
References.....	20
Further reading.....	24

SECTION 2 Urinary system

CHAPTER 2 The progress in tissue engineering of kidney.....27

*Sufyan Dalal, Aziz Maleki, Masoud Mozafari,
Morvarid Saeinasab and Farshid Sefat*

2.1	Introduction.....	27
2.2	Anatomy and physiology	28
2.3	History of kidney transplants and treatments.....	30
2.3.1	Biomaterials and scaffolds	31
2.3.2	Fabrication techniques.....	37
2.3.3	Cell culture	41
2.3.4	Growth factor	42
2.4	Tissue engineering of a kidney	44
2.5	Discussion and conclusion	44
	References.....	46

CHAPTER 3 The progress in tissue engineering of bladder.....51

*Mohammad Tayyab, Adnan Haroon, Kamiyar Mohajerani,
Chin Phong Soon, Syafigah Saidin, Morvarid Saeinasab
and Farshid Sefat*

3.1	Introduction.....	51
3.1.1	Bladder physiology and structure	52
3.1.2	Bladder function	54
3.2	Bladder disease.....	54
3.2.1	Urinary incontinence.....	54
3.2.2	Interstitial cystitis.....	55
3.2.3	Bladder cancer.....	55
3.3	Biomaterials for bladder regenerative medicine	57
3.3.1	Scaffold.....	58
3.3.2	Cells	60
3.4	Tissue engineering for bladder.....	61
3.4.1	Engineering for stress incontinence	63
3.4.2	Tissue-engineered cystoplasty treatment for neurogenic bladder	63
3.4.3	Urinary bladder bioprinting.....	64
3.5	Conclusion.....	65
	References.....	66

CHAPTER 4 The progress in tissue engineering of urethra71

*Kamiyar Mohajerani, Mustafa Mansour,
Cristina Tuinea-Bobe, Morvarid Saeinasab and
Farshid Sefat*

4.1	Introduction.....	71
-----	-------------------	----

4.2	Etiology and anatomy	72
4.3	Tissue engineering of urethra.....	72
4.3.1	Cell sources.....	73
4.3.2	Biomaterial scaffolds for urethral regeneration	76
4.3.3	Growth factors.....	82
4.4	Clinical issues and discussion.....	84
	References.....	85
CHAPTER 5	The progress in ureter tissue engineering.....	89
	<i>Hamza Pervez Khan, Ahmed Anwar Ali Bhagat,</i>	
	<i>Mohammad Ali Shahbazi, Morvarid Saeinasab and</i>	
	<i>Farshid Sefat</i>	
5.1	Introduction.....	89
5.2	Ureters anatomy.....	90
5.2.1	General anatomy.....	90
5.2.2	Microscopic anatomy.....	90
5.3	Physiology	90
5.4	Diseases of ureter.....	91
5.4.1	Congenital anomalies.....	91
5.4.2	Tumors and tumor-like lesions.....	91
5.4.3	Ureter stones	92
5.5	Ureter tissue engineering	92
5.5.1	Repair of a ureteral dysfunction.....	94
5.5.2	Different biomaterials used	95
5.5.3	Sources of regenerative cells	96
5.5.4	Preimplantation seeding.....	97
5.5.5	Decellularized tissue and synthetic polymers in scaffolding	98
5.5.6	Repair of full-circumference ureteral defects.....	100
5.5.7	Models using animals	101
5.6	Discussion and challenges.....	102
	References.....	103
SECTION 3	Male reproductive system	
CHAPTER 6	The progress in tissue engineering of penile corporal tissue.....	111
	<i>Jevin Karia, Afzhan Ali, Morvarid Saeinasab and</i>	
	<i>Farshid Sefat</i>	
6.1	Introduction.....	111
6.2	Disease/disorders related to penile corporal tissue.....	111

6.2.1	Balanitis	112
6.2.2	Epispadias.....	112
6.2.3	Hypospadias.....	112
6.2.4	Penile cancer.....	112
6.2.5	Peyronie's disease	113
6.2.6	Phimosis and paraphimosis	113
6.2.7	Priapism	113
6.3	Materials used for transplantation for penile corporal tissue	113
6.4	Physical bodies	114
6.5	Albuginea tunica.....	114
6.6	Latest and previous tissue engineering approaches for penile corporal tissue	115
6.7	Cell/stem cell used in penile corporal tissue.....	116
6.8	Conclusion.....	118
	References.....	118
	Further reading.....	119
CHAPTER 7	The progress in tissue engineering of prostate.....	121
	<i>Olekanye Seema, Wei Zhang, Ximu Zhang, Phil Coates, Afzhan Ali, Morvarid Saeinasab and Farshid Sefat</i>	
7.1	Introduction.....	121
7.2	Anatomy and histology of the prostate.....	122
7.3	Conditions of the prostate.....	122
7.3.1	Prostate cancer.....	122
7.3.2	Prostatitis.....	124
7.4	Need for prostate tissue engineering	125
7.5	Tissue engineering approach.....	126
7.5.1	Injection stem cell therapy	126
7.5.2	Tissue generation	127
7.5.3	Biomaterials.....	128
7.6	Conclusion.....	129
	References.....	129
CHAPTER 8	The progress in tissue engineering of testicle.....	133
	<i>Dacosta Owusu, Kamiyar Mohajerani, Wei Zhang, Ximu Zhang, Phil Coates, Morvarid Saeinasab and Farshid Sefat</i>	
8.1	Introduction.....	133
8.2	Anatomy and physiology of the testes.....	134

8.3	Pathophysiology of the testicles	135
8.3.1	Cryptorchidism	135
8.3.2	Testicular microlithiasis.....	135
8.3.3	Hydrocele	136
8.3.4	Epididymitis.....	136
8.3.5	Varicocele.....	136
8.3.6	Spermatocele.....	136
8.3.7	Testicular torsion.....	137
8.3.8	Hypogonadism.....	137
8.3.9	Testicular cancer	137
8.4	Biomaterials used for testicular tissue engineering	137
8.4.1	Extracellular testicular matrix.....	138
8.4.2	Decellularized testicular matrix.....	141
8.4.3	Natural biomaterials.....	141
8.4.4	Synthetic biomaterials for a testicular implant.....	142
8.5	History of biomaterials used for testicular prostheses.....	143
8.6	Tissue engineering approach.....	144
8.6.1	Cell-based tissue engineering	144
8.6.2	Scaffold-based tissue engineering.....	145
8.6.3	Stem cell used in tissue engineering approaches for testes.....	148
8.6.4	Fibroblast growth factors	149
8.7	Conclusion.....	150
	References.....	150

CHAPTER 9 Tissue engineering applications for erectile dysfunction..... 157

*Asma Poudel Chhetri, Kamiyar Mohajerani,
Chin Fhong Soon, Syafigah Saidin, Morvarid Saeinasab
and Farshid Sefat*

9.1	Introduction.....	157
9.2	Anatomy and physiology of penis	158
9.2.1	Anatomy of corpus cavernosum.....	158
9.2.2	Mechanism of erection.....	159
9.3	Risk factors and diseases associated with ED	159
9.4	Current treatments	160
9.4.1	Phosphodiesterase type-5 inhibitors (PDE5i).....	160
9.4.2	Intracavernosal injection.....	160
9.4.3	Testosterone therapy	160
9.4.4	Intraurethral drugs.....	160

9.4.5 Penile implants	161
9.4.6 Shockwave therapy.....	161
9.4.7 Penile revascularization	161
9.5 Biomaterials used for CC scaffolds.....	161
9.6 Desired properties of scaffolds.....	163
9.7 Cells used	164
9.8 Tissue engineering approach used.....	165
9.9 Engineered prosthesis.....	169
9.10 Engineered corpus cavernosum	170
9.11 Challenges	173
9.12 Conclusions.....	173
References.....	174

CHAPTER 10 Tissue engineering strategies to treat male infertility 179

*Abraham Vudoyra Gracian, Angelos Stavropoulos,
Jacob Elies, Mustafa Mansour, Kamiyar Mohajerani,
Neda Ghaslani, Morvarid Saeinasab and Farshid Sefat*

10.1 Introduction.....	179
10.2 Male reproductive system physiology.....	180
10.2.1 Testicles.....	180
10.2.2 Penis.....	180
10.2.3 Urethra	181
10.2.4 Prostate.....	181
10.2.5 Vas deferens	181
10.3 Male infertility.....	181
10.3.1 Causes of male infertility	182
10.3.2 Evaluation of the infertile male.....	182
10.4 Current treatments	186
10.4.1 Surgical treatment for male infertility.....	186
10.4.2 Hormonal treatment	187
10.4.3 Gamete (sperm) donation.....	188
10.5 Tissue engineering approaches for treating male infertility.....	188
10.5.1 Stem cells	191
10.5.2 Scaffolds.....	192
10.5.3 Decellularized extracellular matrix	193
10.5.4 3D bioprinting.....	194
10.6 Conclusion	194
References.....	195

SECTION 4 Female reproductive system

CHAPTER 11 The progress in tissue engineering of uterus..... 203

*Fereshteh Asgharzadeh, Sima Etebari,
Maryam Moradi Binabaj, Farhat Naz,
Morvarid Saeinasab and Farshid Sefat*

11.1	Introduction.....	203
11.2	Uterus anatomy and physiology	204
11.3	Uterus physiology.....	205
11.4	Common uterine diseases and disorders.....	206
11.4.1	Genetic disorders.....	206
11.4.2	Advanced age	206
11.4.3	Diseases and illnesses.....	206
11.4.4	Accidents and traumas.....	206
11.4.5	Absolute uterine infertility	207
11.5	Uterine histoarchitecture	207
11.6	UTx animal research	208
11.7	UTx human research	210
11.8	Tissue engineering	211
11.9	2D models and 3D models	211
11.10	Cell source	212
11.10.1	Adult stem cells.....	212
11.10.2	Embryonic stem cells	213
11.10.3	Induced pluripotent stem cells.....	213
11.11	Scaffolds.....	214
11.12	Natural and synthetic materials	214
11.13	Decellularized matrices.....	215
11.14	Uterine tissue engineering	216
11.14.1	Cell source for bioengineering of the uterus	216
11.15	Decellularization and recellularization strategy	218
11.16	Clinical research for tissue engineering of the uterus	219
11.17	Summary.....	219
	References.....	228

CHAPTER 12 The progress in tissue engineering of vagina..... 235

*Salamatu Abubakar Ibrahim, Elzarie De Wet,
Mojgan Najafzadeh, Saeed Heidari-Keshel,
Morvarid Saeinasab and Farshid Sefat*

12.1	Introduction.....	235
12.2	Vagina anatomy.....	236
12.3	Causes of vaginal disorders	237

12.3.1	Vaginal cancer	237
12.3.2	Vaginal agenesis.....	238
12.3.3	Posttraumatic stress disorder.....	238
12.3.4	Vaginal candidiasis	238
12.3.5	Vaginal infections.....	238
12.4	Diagnosis of vagina disorder	239
12.5	Current treatment	239
12.5.1	Nonsurgical treatment.....	239
12.5.2	Surgical treatment.....	239
12.6	Tissue engineering.....	240
12.6.1	Synthetic biomaterials	240
12.6.2	Natural biomaterials.....	242
12.7	Stem cells and vaginal reconstruction.....	246
12.7.1	Stem cells	246
12.7.2	Amniotic fluid stem cells	246
12.7.3	Bone marrow-derived stem cells.....	247
12.7.4	Adult stem cell and native progenitor.....	247
12.8	Conclusions.....	248
	References.....	248
	Further reading.....	252

CHAPTER 13 The progress in tissue engineering of cervix..... 253

*Theresa Ofure Okonofua, Elzarie De Wet,
Parichart Naruphontjirakul, Mansour Youseffi,
Morvarid Saeinasab and Farshid Sefat*

13.1	Introduction.....	253
13.1.1	The cervix, its anatomy and physiology	254
13.1.2	Disorders/diseases related to the cervix.....	254
13.1.3	Risk factors of cervical conditions	254
13.1.4	Medication.....	257
13.1.5	Organ tissue transplantation.....	257
13.1.6	Scaffold based approach for tissue engineering	258
13.1.7	Fabrication techniques	259
13.1.8	Scaffold selection for cervical tissue engineering....	259
13.1.9	Cell/stem cell-based tissue engineering approach....	260
13.1.10	Possible growth factor used in tissue engineering of cervix.....	261
13.2	Conclusion	261
	References.....	262

15.5.1 Engineering the endometrium through stem cell sheets	288
15.5.2 Regeneration of endometrium based on cell-scaffold interfaces.....	289
15.5.3 Renewal of endometrium based on a surface-functionalized scaffold.....	291
15.6 Summary	293
References.....	293
CHAPTER 16 Tissue engineering strategies to treat female infertility	299
<i>Ebenezer Adjei, Morvarid Saeinasab and Farshid Sefat</i>	
16.1 Introduction.....	299
16.2 Female reproductive system	301
16.2.1 The ovary.....	301
16.2.2 The fallopian tube.....	301
16.2.3 The uterus	302
16.2.4 The cervix.....	302
16.3 Cause of female infertility.....	303
16.4 Other risk factors of female infertility	304
16.5 Current treatments for female infertility	304
16.5.1 Medication	307
16.5.2 Surgical procedures.....	307
16.5.3 Assisted reproductive technologies	309
16.6 Tissue engineering-based approaches in treating female infertility.....	309
16.6.1 Scaffold-based tissue engineering approach.....	309
16.6.2 Cell/stem cell-based tissue engineering approach	313
16.6.3 Growth factor-based tissue engineering approach	316
16.7 Conclusion	316
References.....	318
Index.....	325

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Genitourinary tissue engineering: Promises, advances, and challenges

1

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1.1 Introduction

The Genitourinary system is large and is made up of many organs, these have vital roles in fulfilling normal and healthy functions of the human body. Therefore, the problems and treatments for these conditions are wide and vast. Over 400 million people worldwide are affected by urinary tract abnormalities (Davis, Cunnane, O'Brien, et al., 2018). The most prevalent urethral abnormalities, urethral strictures, affect 1 in 1000 males over the age of 65. Congenital urethral birth defects such as hypospadias, impact 1 in every 300 births (Pastorek et al., 2021). Subsequently, in the first 5 years following surgery, 20% of urethra reconstruction issues occur (Wang et al., 2021).

Numerous illnesses can affect urological tissues. Most of these diseases are acquired or congenitally present defects. Patients may need surgical reconstruction to restore the genitourinary tract to its normal state, depending on how severe the aberration is and whether the condition is reoccurring. In the present era, it is still challenging to repair and replace these damaged or aberrant tissues (Caneparo, Sorroza-Martinez, et al., 2021).

When organs are damaged or organ failure has occurred, the available surgical procedures include organ transplantation. Surgical interventions are accompanied by a variety of issues, including disease transmission, rejection, and excessive demand, which outweigh the supply. Hence, the need for transplantable tissues and organs is growing, leading to the development and improvement of procedures in regenerative medicine (Guruswamy Damodaran & Vermette, 2018).

High hopes have been placed in regenerative medicine using tissue engineering technology because of its progress and promise for ground-breaking therapeutic approaches. Functional outcomes are still unsatisfactory because of poor healing, extensive scarring, and recurrent fibrosis. A unique tissue engineering technology-based approach is being gradually researched for better, more fruitful results. The potential to create organ-specific grafts makes tissue engineering the most promising

method for enhancing the outcomes of reconstructive urological treatments. In addition, stem cells and growth factors have a unique opportunity to modify the healing environment specifically, which could advance reconstructive urology to the molecular level ([Adamowicz et al., 2019](#); [Chua et al., 2020](#)).

1.2 Regenerative medicine

Regenerative medicine is the process in which the body employs its own mechanisms with assistance from foreign biological material to reproduce cells and reconstruct tissues and organs ([NIH, 2022](#)).

For damaged tissues to be permanently repaired there are certain conditions that must be satisfied. The developed cells must be sufficient to fill the gap in the damaged tissue, they must be able to differentiate into the desired phenotypes, have the appropriate 3D scaffold structure and produce an extracellular matrix (ECM). The engineered cells must be structurally and mechanically biocompatible with the native cell and have a low risk of rejection and minimal associated biological risks.

The ECM of a cell depends on the native tissue type. In addition to cell adhesion proteins like fibronectin and laminin and structural proteins such as collagen and elastin, ECMs also contain proteoglycans, which are protein-polysaccharide complexes where sugars are linked to core proteins; these are commonly glycosaminoglycans (GAGS).

1.3 Principles of tissue engineering

The goal of tissue engineering is to restore physiological function and fulfill the biochemical purpose. The developments in prostheses, reconstructive surgery, transplantation, cell biology, biochemistry, molecular biology, and genetics are all strongly related to tissue engineering. Tissue engineering concepts can be used in a wide variety of applications in order to restore many organs and functions of the human body as demonstrated in [Fig. 1.1](#).

To provide innovative methods and treatments for tissue and organ regeneration, the interdisciplinary area of tissue engineering incorporates components from biology, material science, medicine, and engineering. It describes the use of scaffolds or synthetic material-derived building blocks made of cells for tissue repair. Scaffolds are materials that have been designed to result in favorable biological interaction. They provide a biochemical and physical environment that is similar to that of natural tissue in addition to acting as a support for cells ([Pastorek et al., 2021](#)).

Tissue engineering primarily uses three components: cells, biomaterials, and growth factors (GFs).

1.3.1 Source of cells

Cells are the living part of the tissue, they are responsible for the production of the ECM, cytokines, and proteins; they provide cells with their function and repair properties of the tissue.

- Allograft

An allograft is when tissue is taken from a genetically unrelated donor and used on a patient.

- Xenograft

Xenografts are cells taken from another species and used on a patient; this could be from rabbits, mice, cows, and pigs.

- Isograft

In an isograft, the tissue is taken from a genetically identical donor and used on the patient. Using a twin donor is highly successful due to its low rejection rate.

1.3.1.1 Stem cells

To restore biological functions, several genitourinary tract problems can be treated or replaced. Growing tissue shortages pose a threat to current therapeutic strategies. Therefore, the utilization of stem cells shows promise ([Caneparo, Sorroza-Martinez, et al., 2021](#)). Given that they mimic crucial host-derived biological processes like mitosis, proliferation, differentiation, and death, stem cells could be a useful alternative ([Davis, Cunnane, Mulvihill, et al., 2018](#)).

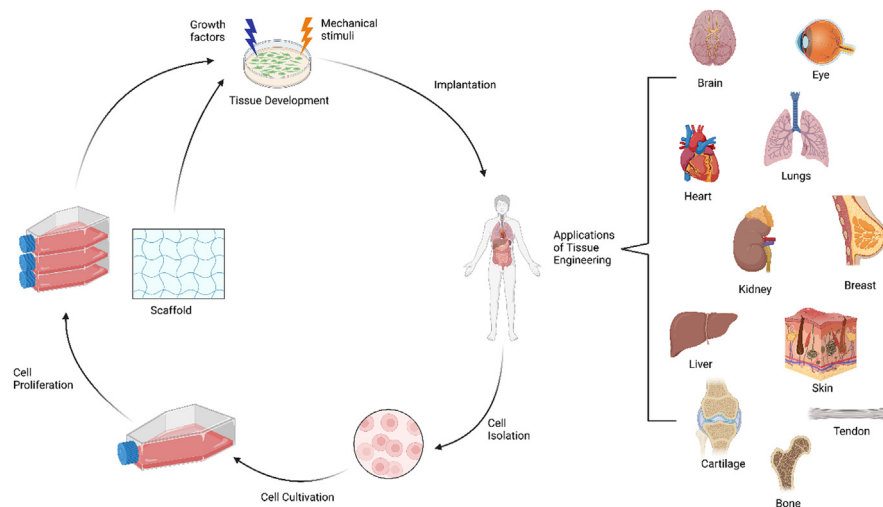


FIGURE 1.1

Overview of tissue engineering.

Source: Created by Biorender.

Due to their stability and capacity for self-renewal, stem cells are helpful in reconstructive urology. Excellent urological regenerating capabilities can be seen in mesenchymal stem cells (MSCs), embryonic stem cells (ESCs), and adult stem cells (ASCs). Numerous stem cell therapies have been used to repair damaged lower urinary tract components. In vitro and in vivo pre-clinical investigations have produced promising outcomes (Davis, Cunnane, Mulvihill, et al., 2018). Adipose-derived stromal stem cells are potential substitutes for the treatments for genitourinary diseases now provided to patients because of their accessibility, high yield when harvested, anti-fibrotic, immunomodulatory, and proangiogenic capabilities (Caneparo, Sorroza-Martinez, et al., 2021; Davis, Cunnane, Mulvihill, et al., 2018).

1.3.2 Biomaterials

Biomaterials come in the form of scaffolds. They provide structural support, and are a framework which allows cell adhesion, and promotes cell proliferation. Scaffolds must be biocompatible and biodegradable into non-toxic components. Biomaterials can be metal polymers or from a natural source.

Bioscaffolding is a biocompatible scaffold that contains and delivers biologics, which are cells, genes, and proteins. If the bioscaffold interacts with the host cells it indicates successful reconstruction or regeneration. Successful bioscaffolding stimulates the cell's native ECM's protein synthesis.

The biomaterial used as the scaffold matrix must be dynamic, while still allowing for cell growth, and therefore must be permeable. The permeability of the scaffold must be in line with that of the native tissue in order for the proper functioning of the organ can continue, where the permeability of the scaffold is not appropriate problems occur. In the case of the bladder, fibrosis results from competition between renewing cells from the native tissue and urine seeping through the pores of the scaffold or biomaterial (Chua et al., 2020).

1.3.2.1 Extracellular matrix

ECM scaffolds are processed mechanically, chemically, and enzymatically to produce tissue that preserves the essential structural components. The goal of properly made ECMs is to offer a biologically active tissue or organ substitute that may successfully integrate into the host tissue and replace the functionality of the damaged tissue. The ECM scaffold should be biocompatible to reduce inflammatory and rejection reactions (Davis, Cunnane, O'Brien, et al., 2018). The decellularized organ-specific ECM scaffolds can either be used as bioinks for 3D bioprinting applications to produce organs, or they can be repopulated and recellularized with the appropriate cells and growth factors. Decellularized ECMs are also used to create hydrogels or to enhance the biological characteristics of synthetic materials (Hasanpour et al., 2018).

To repair a damaged urinary tract segment ECMs are often obtained from swine organs and are decellularized to produce a biocompatible and biodegradable

biomaterial. An integrated tissue-engineered ECM should maintain or restore the urinary tract structure or replace normal functionality (Davis, Cunnane, O'Brien, et al., 2018). Another area of research is a bioengineered ovary. Research suggests when a decellularized ovarian ECM is used as a native scaffold, it can offer an excellent 3D microstructure for regenerative efforts in order to support cell activity (Hasanpour et al., 2018).

1.3.2.2 Scaffolds

A suitable scaffold should have a 3D structure, be non-toxic, and be biodegradable. The scaffold can also be thought of as a cell transport system that promotes cell motility, division, and proliferation (Wang et al., 2020).

There are several types of scaffolds some include hydrogels, woven silk, nano fibers, microfibers, decellularized organs, and composites.

The organic polymer collagen makes up a significant portion of the ECM. It is an excellent substrate for cell attachment and proliferation because it has ligands that act as surface binding sites. When collagen is cross-linked with acetic acid or used in conjunction with other synthetic polymers, its low Young's Modulus (E) can be raised. It is possible to create a porous collagen scaffold using freeze-drying procedures.

Water-soluble polymer chains are cross-linked to form insoluble networks, forming hydrogels. Hydrogels mimic soft tissue's high-water content, they are used to regenerate soft tissue.

Although synthetic polymers can be processed using a variety of methods to produce scaffolds, they lack binding sites and must have adhesive proteins applied to their surface in order to function effectively. The degradation products of the synthetic polymer could be cytotoxic and result in an inflammatory response even if the polymer itself is not.

1.3.3 Fabrication

- Foaming

Hydrogels can be foamed by bubbling CO₂; a cell culture strainer is used to act as a filter and control pore size.

- Electrospinning

Through a network of connected micron-scale fibers, a high-voltage electric field is applied to spin fibers. Through a small nozzle, fibers made from a polymer solution are extruded.

- Rapid prototyping

Complex geometries are created using computer control. To build up successive layers and form a solid scaffold shape. Stereolithography, 3D printing, and selective laser sintering are utilized in rapid prototyping.

- 3D bio-printing

3D bioprinting techniques include thermal inkjet, micro-extrusion, and laser assistance.

- Thermal inkjet printers use electrical heating of the printhead to create air-pressure pulses that drive droplets from the nozzle, whereas acoustic printers use pulses produced by piezoelectric or ultrasound pressure.
- Micro-extrusion printers use pneumatic or mechanical dispensing apparatus to extrude continuous beads of material or cells.
- By focusing lasers on substrate absorption, laser-assisted printers create pressures that push materials containing cells onto a collection substrate.
- Freeze-drying

Freeze-drying of collagen scaffold gives a porous scaffold. Directional cooling can be used to create elongated pores, used for nerve regeneration.

- Decellularization

When all cellular components have been eliminated from a tissue or organ, the remaining structural characteristics of the ECM act as a scaffold.

1.3.4 Structure and material properties

The structural properties of a scaffold are determined by native tissue force and deformation, or stress and strain characteristics. The behavior of the material under various loading conditions, such as tension, compression, torsion, or bending, will differ, as will the associated modulus of elasticity or stiffness; this must be tested to determine the scaffold's suitability.

The scaffold must have sufficient mechanical integrity for handling in surgery, cell differentiation and proliferation. The degradation must degrade at a controlled rate in order for the tissue to become fully formed, via cell deposition of native ECM; the scaffold must be completely reabsorbed at a controlled rate. The chemical makeup of the scaffold will allow for the provision and binding of ligands that affect cell activity as well as the synchronization of the degradation rate with the rate of cell formation. The scaffold must be rigid, have a good elastic and stiffness modulus, and be strong enough mechanically to sustain wear.

1.3.5 Requirements of scaffold design

For cell migration, nutrition transport, and the movement of regulatory elements, scaffolds need a large number of pores. The density of binding sites available for cell attachment determines the top boundary of the pore size, while cell size determines the lower boundary. The geometry of the pore must be compatible with the morphology of the native cells.

1.3.6 Growth factors

GFs, which play a crucial role in tissue regeneration, are biomechanical signals that instruct cells on how to operate via protein or mechanical stimulation. GF treatment for tissue regeneration is constrained by the requirement for large numbers. As a result, high volumes become necessary, which are expensive and increase the risk of side effects. Therefore, new methods are being developed to deliver bioactive chemicals locally and precisely. The required dose of GFs could be decreased as a result of these scaffolds' ability to keep the biomolecules at the fracture site in vivo. These GFs then work to draw endogenous stem cells from nearby tissues and drive their development into the porous scaffold's tissue (De Witte et al., 2018).

Several scaffold-based techniques have been investigated in order to meet the requirements for GF delivery. The utilization of covalent or non-covalent protein binding to the scaffold, micro- or nanoparticles as protein reservoirs or physical protein trapping within the scaffold are all possible GF delivery techniques (De Witte et al., 2018). Table 1.1 shows research that has been conducted for tissue engineering and regenerative medicine of the genitourinary system, and any growth factors that have been used.

1.3.7 Decellularization

Decellularized tissues and organs can serve as scaffolds on which transplanted cells can be placed later. These tissues and organs are devoid of cells and genetic material but retain the detailed structure of the ECM. They are able to mimic the physiological conditions that exist naturally (Guruswamy Damodaran & Vermette, 2018).

Organ and tissue decellularization can result in the creation of bioscaffolds made of naturally occurring matrices that are comparable to native tissue or organs in terms of composition, microstructure, and biomechanical properties while retaining the majority of the signaling cues required for organ development, repair, and physiological regeneration. For different tissues and organs, numerous decellularization techniques have been devised in an effort to strike a compromise between cell loss and ECM maintenance. This is brought on by the pro-inflammatory reactions brought on by the retention of cellular debris from insufficient decellularization. On the other side, a thorough and severe decellularization procedure will denature proteins and bleach growth hormones (Simões et al., 2017). Some decellularization techniques involve the removal of all genetic, protein and cellular components only leaving behind the structural components of the ECM, this is then recellularized using the appropriate cells and GFs.

1.4 Genitourinary system anatomy and physiology

The genitourinary system includes organs from the urinary and reproductive systems, it is a complex system that is composed of the kidney, bladder, urethra, ureter, renal, reproductive organs, and genital structures.

The main aim of the urinary system is the filtration of the blood, which creates the waste product of urine. The organs that compose this system are kidneys, renal pelvis, ureters, bladder, and urethra. The renal and urinary systems aid in the body's elimination of urea, a toxic waste liquid, and maintain a balance of potassium, sodium, and water. When foods high in protein, such as meat and poultry, are digested by the body, urea is formed. Urea is transported by the bloodstream to the kidneys, where it is eliminated in the form of urine in combination with water and other waste products ([Medicine, 2022](#)).

1.4.1 Urinary structure and function

The human body consists of two kidneys and two ureters, these organs have a range of medical challenges. The function of the kidneys is to remove waste products from the body and in turn balance fluid levels within the body, they are also responsible for the production of red blood cells and to regulate blood pressure via hormones. The ureter carries the urine from the kidneys to the bladder. The bladder is the organ responsible for storing urine in the body until it is excreted. The bladder of a healthy individual is able to expand to store urine and contract for the release of the waste product. The urethra is connected to the bladder and is the external organ of the urinary system that empties the bladder. The bladder is located in the lower abdomen, whereas the kidneys are positioned below the ribs in the middle of the back ([Medicine, 2022](#)).

1.4.2 Genital structure and function

The genital structure of the human body consists of both the internal and external reproductive organs. The differences between male and female reproductive organs results in their own unique problems and require medical treatments that are suited to them.

1.4.2.1 Female genitalia

The internal structure of female genitalia includes the vagina, cervix, uterus, ovaries, and fallopian tubes. The external structures consist of the labia majora, labia minora, clitoris, hymen, and vaginal opening ([Clinic, 2022a](#)).

There are many uses for the female reproductive system. Along with permitting sexual activity, it also promotes sexual reproduction. The egg gamete is created in the ovaries. These eggs are then transferred to your fallopian tube during ovulation, where sperm fertilization could occur. The fertilized egg is subsequently placed inside the uterus, where the lining thickens as a result of the normal hormone production associated with menstruation. The fertilized egg grows there after implanting into the denser uterine lining. If egg implantation has not taken place, the uterine lining will deteriorate during menstruation. Sex hormones are also produced by the female reproductive system to maintain a regular menstrual cycle. The number of eggs that a woman will have for the duration of her life is determined during fetal

development, with each menstrual cycle the number of eggs declines which results in decreased fertility in women as they age ([Clinic, 2022a](#)).

1.4.2.2 Male genitalia

The male genitalia is also made up of internal and external components. The internal organs include the vas deferens, prostate and urethra; the penis, scrotum and testicles are the external organs. Similar to the female reproductive system the male reproductive system is responsible for sexual function, additionally it is also responsible for urination ([Clinic, 2022b](#)).

The majority of the male reproductive system is external to the pelvic or abdominal cavity. The urinary portion male reproductive system is responsible for removing waste products from the body via urination. The reproductive function is carried out by producing, storing, and transporting the sperm gamete and semen, this is done by producing the male sex hormones in order to carry out this function ([Clinic, 2022b](#)).

1.5 Genitourinary conditions

1.5.1 Affecting the ureter

- Megaureter

Megaureter is an abnormality that can affect the ureters. The affected ureter is enlarged and is dysfunctional. A megaureter often measures more than 7 mm, while a normal ureter is ~ 4 mm ([Nakamura et al., 2020](#)).

- Neurogenic bladder

In individuals with neurogenic bladder leading to spina bifida, lower urinary tract dysfunction and hypertrophy of the bladder wall with a loss of the viscoelastic characteristics due to extracellular matrix components infiltration are common findings ([Hwang et al., 2023](#)).

- Urinary tract infections

A bacterial infection of the urinary tract that affects predominantly women ([Pereira et al., 2020](#)).

1.5.2 Diseases that lead to damage or loss of function of the kidneys

- Hemolytic-uremic syndrome

Acute renal failure, microangiopathic hemolytic anemia, and sometimes severe thrombocytopenia characterize the uncommon but devastating consequence of bacterial and viral infections ([Birlutiu & Birlutiu, 2018](#)).

- Polycystic kidney disease

Polycystic kidney disease is a main factor in end-stage renal disease, which calls for long-term dialysis or kidney transplantation ([Mohamed et al., 2022](#)).

1.5.3 Conditions that specifically affect females

- Urinary incontinence

Over 200 million people worldwide experience urinary incontinence, and it is becoming more prevalent as the population ages. The condition affects more women due to muscle, tissue, and nerve damage during birth, it can also be because of obesity.

- Premature ovarian failure

Premature ovarian failure, which causes infertility in women, is on the rise as a result of a number of reasons, most notably the widespread use of chemotherapy. Premature ovarian failure has a variety of causes, including autoimmune, iatrogenic, hereditary, and idiopathic ([He et al., 2018](#)).

- Cervical

Women who suffer from ovarian dysfunction face a variety of medical and psychosocial difficulties, including infertility issues and symptoms similar to menopause. Ovarian hormones not only affect the reproductive system but also the breast, blood arteries, and bones ([Hassanpour et al., 2018](#)).

1.5.4 Conditions that specifically affect males

- Hypospadias

A condition where abnormalities in the urinary meatus lead to an abnormal urethral development; there is no urethral opening. The condition affects ~1 in 300 new-born boys ([Abbas et al., 2018](#); [Casarin et al., 2022](#)).

- Urethral strictures

The majority of urethral strictures in adults happen as a result of scarring, which replaces the corpus spongiosum's vascular tissue and results in ischemic spongiofibrosis. Scar tissue replaces the injured urethra, decreasing its lumen and eventually causing lower urinary tract blockage ([Pastorek et al., 2021](#)).

- Testicular torsion

When the tissues around the testicle, commonly known as the “testis,” are not securely attached, it is known as testicular torsion. The testes may then twist around the spermatic cord as a result of this. When this occurs, the testicle's blood supply is cut off ([Care, 2022](#)).

- Cancer

Prostate cancer

The prostate is a small gland located in front of the rectum and immediately below the bladder (back passage). Each year, approximately 48,600 males in the UK are given the diagnosis ([MacMillan, 2022](#)).

1.6 Current treatments and potential drawbacks

Treatments and therapies currently available for genitourinary conditions range from drug therapies to surgical intervention. The severity of the treatment is dependent on the damage to the tissue, the loss of normal bodily function, and the pain to the patient ([Sharma & Basu, 2022](#)). Therefore, each treatment is prescribed on a case-by-case basis.

Over 400 million people worldwide are affected by urinary tract abnormalities ([Davis, Cunnane, O'Brien, et al., 2018](#)). Acquired urethral abnormalities, such as urethral strictures are the most common and affect 1 in 1000 males over 65 years old, and congenital birth disorders of the urethra, such as hypospadias affect 1 in every 300 births ([Pastorek et al., 2021](#)).

The “gold-standard” course of treatment for people with genitourinary disorders is typically bladder augmentation using intestine segments ([Sharma & Basu, 2022](#)). It is argued that this approach cannot provide the requisite impermeability function since the ureter’s major job is to act as a barrier to protect other organs from urine, but the gastrointestinal bowel’s primary role is to absorb nutrients. Therefore, people who have had this operation are typically affected by malignancies, metabolic problems such as acid reabsorption, electrolyte imbalance, upper tract degeneration, and persistent infections ([Caneparo, Sorroza-Martinez, et al., 2021](#)).

The primary benefit of intestinal wall is its propensity for tabularization and capacity to create variously shaped hollow reservoirs that, physically, can make up for damaged or surgically removed portions of urinary tracts. Additionally, intestinal wall is a highly vascularized autologous material with little danger of severe fibrosis or negative biomaterial responses ([Adamowicz et al., 2019](#)).

Reconstructive urological operations using autologous tissue sources are required as a result of pathological illnesses that result in problems with neurochemical and metabolic systems ([Davis, Cunnane, Mulvihill, et al., 2018](#)). End-stage upper and lower urinary tract abnormalities frequently require the use of autologous, vascularized mucus secreting gastrointestinal tissue during surgical repair. This can be to replace the organ or supplement insufficient tissue ([Davis, Cunnane, O'Brien, et al., 2018](#)). Allogeneic transplantation is constrained by the risk of tissue rejection and the availability of these tissues. As the population ages and certain abnormalities become more common, there is a growing need for transplantable tissues but a scarcity of them ([Caneparo, Sorroza-Martinez, et al., 2021](#)).

Surgery, such as urethral dilation (UD) or direct vision internal urethrotomy, is typically used to treat strictures. However, disadvantages of this method have resulted in the European Association of Urology (EAU) opposing this as the only form of treatment. On the other side, open reconstructive procedures like urethroplasty have higher long-term success rates. These treatments frequently involve multiple stages and the use of autologous buccal or skin grafts or flaps (Pastorek et al., 2021).

Some research suggests there is currently no reliable and efficient tissue engineering treatment for urethral strictures, and the cost to use autologous tissue in tissue engineering for urethroplasty is too high. This research also suggests that further research is required even though tissue-engineered oral mucosa has been effectively employed (Chapple, 2020).

1.7 Genitourinary tissue engineering

Methods used in the 1950s included synthetic materials that were not biodegradable, and over time would cease to function as intended. Urologists now have a more effective option thanks to the development of successful biocompatible tissue-engineered biomaterials, namely scaffolds constructed of ECMs. By triggering a host-derived tissue remodeling response after implantation, decellularized, biodegradable ECMs have shown promise for mending urinary tract segments. Both swine urinary bladder matrix (UBM) and porcine small intestinal submucosa (SIS) contain various types of collagens and are often employed as ECMs for urological tissue regeneration. Both ECMs promote the development of natural tissue and offer mechanical and structural support to prevent the collapse of the urinary tract (Davis, Cunnane, O'Brien, et al., 2018). Table 1.1 shows review and research papers related to tissue engineering of the genitourinary system in the last few years.

1.7.1 Urethra

Urethral restoration is typically required to treat congenital urethral deformity brought on by injury or infection and lessen morbidity or mortality. Artificial replacements are required to build a urethral repair scaffold, which aid in wound healing and restore the urethra's functionality after damage (Wang et al., 2020).

A study developed Nano-Laponite/poly(lactic acid-glycolic acid) copolymer (PLGA) fiber scaffolds to replicate the true human urethra tissue microenvironment. The primary materials used were laponite and PLGA, which were synthesized utilizing an electrospinning technique in a 50:50 M ratio (PLGA), has great biodegradability and biocompatibility, good cell adhesion, and rapid in vitro disintegration rate it has been used considerably in tissue engineering and medicinal applications. Modification of the molecular weight and ratio can be used to affect the rate of degradation. Some reports suggest the porosity of PLGA can reach up to 95%. The inclusion of Laponite particles in the scaffold increases the porosity for cell adhesion (Wang et al., 2020).

Table 1.1 Review and research papers related to the genitourinary system.

Paper type			Growth factor	Detail	References
Review	Reconstruction and research	Genitourinary	FGF2HB-EGF	Self-assembly without the use of decellularized biomaterial scaffold and successful genitourinary studies on animals.	Caneparo, Brownell, et al. (2021)
Review	Stem cells	Genitourinary		A summary of the currently available stem cells for genitourinary tissue regeneration.	Caneparo, Sorroza-Martinez, et al. (2021)
Review	Reconstructive urology	Genitourinary		Explores the limitation of urology reconstruction currently face and uses research studies to investigate the applications of tissue engineering.	Adamowicz et al. (2019)
Research	Urine-derived stem cells	Lower urinary tract	VEGF	Using the rabbit urine derived stem cells for a tissue-engineered urethra in a biodegradable scaffold.	Yang et al. (2018)
Research	3D bioprinted urinary patch	Urinary tract (patch)		The use of 3D bioprinting to create a urinary tract patch, to be tested on beagles.	Jin et al. (2020)
Review	Stem cells	Lower urinary tracts		Investigate the advantages of MSCs, ESCs, and ASCs on urological regeneration.	Davis, Cunnane, Mulvihill, et al. (2018)
Research	Cell-enriched collagen hydrogel for detrusor	Bladder	SDF-1	Tissue engineering of the dextrose muscle using smooth muscle cells and pADC in collagen hydrogels using a rat model.	Smolar et al. (2020)
Research	Using stem cells for neurogenic bladder in injured rat model	Bladder		Using a rat model to analyze the effects of SDF-1 on a neurogenic bladder using immortalized MSCs.	Zhu et al. (2020)

Continued

Table 1.1 Review and research papers related to the genitourinary system.—*cont'd*

Paper type			Growth factor	Detail	References
Research	Biomimetic scaffold (PVDF nanofibers)	Bladder tissue	TGF- β	Using adipose derived MSCs to create a nanofiber scaffold for bladder tissue regeneration.	Ardeshirylajimi et al. (2018)
Research	Spheroids of bladder smooth muscle cells	Bladder		Using 3D spheroid cell aggregation in comparison to 2D cell culture proliferation.	Gerwinn et al. (2021)
Review	Neo-urinary bladder	Bladder			Sharma and Basu (2022)
Review	Bladder regeneration	Bladder	BMP4 PDGFBTGF- α VEGF TGF β 1 KGF IGF-1 HGF KGF bFGF EGF	Explores regenerative approaches to bladder tissue engineering including natural and synthetic scaffolds, and urinary bladder bioprinting.	Serrano-Aroca et al. (2018)
Review	Pediatric urology	Urethra and bladder		Explores approaches of reconstructive urology in order to develop a neo-urinary bladder.	Casarin et al. (2022)
Research	Acellular dermal scaffold with collagen	Urethra	VEGF	Using injured beagle model for urethra reconstruction using collagen binding growth factor VEGF.	(Wang et al., 2021)
Research	Nano-laponite/PLGA scaffold	Urethra		Using PLGA scaffold strengthened with laponite for urethra tissue engineering.	(Wang et al., 2020)
Research	Decellularization	Urethra		Developing methods to decellularize a complete urethra.	Simões et al. (2017)
Research	Decellularization	Ovarian		Using a 3D decellularized human ovarian scaffold on sodium lauryl sulfate.	Hassanpour et al. (2018)
Review	Stem cells	Ovarian		Using stem cells, MSCs, in order to treat premature ovarian failure.	He et al. (2018)

Review	Patient-derived xenografts	Prostate	VEGFbFGf	Investigating the effectiveness of ureteric reimplementation.	Namekawa et al. (2019)
Review	Tissue engineering approaches for hypospadias			Investigating applications of tissue engineering for treatment of hypospadias.	Abbas et al. (2018)
Research	Decellularized porcine kidney	Kidney		Developing methods in order to decellularize a porcine kidney.	Hussein et al. (2018a, 2018b)
Research	Acellular rat kidney and AD-MSCS	Kidney		Using MSCs on a rat kidney scaffold.	Shahraki et al. (2021)
Research	Kidney decellularization of male rabbit	Kidney		Kidney decellularization methods using animal model of male rabbits.	Abou-Rabia et al. (2021)
Review	Electrospun scaffolds for kidneys	Kidney		A review on research done to investigate electrospun kidneys as a form of treatment for chronic kidney disease.	Miranda et al. (2022)
Review	Kidney regeneration and reconstruction	Kidney		Research on kidney regeneration using tissue engineering. A review of the numerous investigations on kidney tissue engineering using both scaffold-based and scaffold-free methods.	Nishimura et al. (2022)
Review	Approaches to vascularize kidney	Kidney		Investigating research into vascularization of kidneys so they are able to carry out the function of a health human.	Lim et al. (2022)

It is evident from the material's characterization that the stent's surface has the structural characteristics of an ECM. When Laponite is introduced at 1%, the nano-Laponite/PLGA composite scaffold can be used as an effective urethral repair material, although Wang suggests further study and development are needed to increase its efficacy (Wang et al., 2020).

1.7.2 Bladder

For patients who need a replacement bladder owing to oncologic, acquired, or congenital neurogenic and myogenic diseases, bladder tissue engineering is presently being developed as a potential therapy option (Chua et al., 2020). There have been numerous developments in kidney tissue engineering-related fields such as bioprinting, decellularization, recellularization, vascular casting, and organoid creation in the past few years (Smolar et al., 2020).

Due to the designed bladder wall substitute's fibrosis, clinical testing revealed insufficient development of bladder capacity and compliance. For a tissue-engineered bladder to be successful the expansion and isolation of smooth muscle cells is required. However, this has many difficulties because mature smooth muscle cells lose their ability to contract in cell culture, for expansion, resulting in limited proliferation. The study used male Wister rat bladder in order to investigate the use of smooth muscle cell spheroids, a spheroid is a 3D cell aggregation made from a single primary cell type, a multicellular combination of primary cells, immortalized cell lines. The results of the study concluded that the spheroids were viable for 14 days, and expressed α -SMA, the protein for muscle contraction (Gerwinn et al., 2021).

1.7.3 Genital tissues

Finding a safe and transplantable ovary is urgently needed due to the rise of conditions that lead to ovarian insufficiency. Researches have investigated the approach using a sodium lauryl ester sulfate (SLES)-treated human decellularized ovarian scaffold. The sample underwent decellularization methods and was washed with PBS to remove cellular components and only leave behind the structural ECM. The sample underwent a range of testing in order to determine efficient cell elimination, ECM preservation. Primary ovarian cells (POCs) grafts, from rats, were made on the decellularized scaffolds; tests concluded the scaffold's cytocompatibility. Following transplantation, the POCs replicated primordial or primary follicle-like structures within the scaffolds, in vivo evaluation validated that the POCs maintained their viability and bioactivity. Hassanpour suggests that there are still considerable obstacles in the recellularization process to complete organ scaffolds (Hassanpour et al., 2018).

Adult stem cells of the bone marrow-derived mesenchymal stem cell (BMSC) type have a low immunogenicity. They are widely dispersed throughout the bone marrow and are capable of self-renewal and tissue differentiation. They move to

the site of wounded tissue and also differentiate into particular cell types in the tissue when certain stimuli are induced to rebuild the local microenvironment. BMSCs are excellent for cell seed transplantation because they are involved in the repair of tissue by controlling the immune response and boosting the function of endogenous cells. Some positive outcomes have been attained despite the low survival rate and restricted differentiation potential following BMSC transplantation. One study found that MSCs move to injured tissue after being stimulated by cytokines but do not develop into oocytes. To promote ovarian regrowth, they secrete a number of GFs, such as hepatocyte growth factor (HGF), insulin-like growth factor (IGF), and vascular endothelial growth factor (VEGF) (HGF). By reducing oxidative stress and regulating the inflammatory response, they also maintain ovarian function (He et al., 2018).

In premature ovarian failure cisplatin in rats, BMSCs were crucial in repairing damaged ovaries. Additionally, in a mouse model of chemotherapy-induced premature ovarian failure, BMSCs reactivate folliculogenesis and restore ovarian hormone production. According to other studies, BMSCs lessen granulosa cell apoptosis brought on by cisplatin and perimenopause. These findings demonstrate that BMSCs can successfully treat premature ovarian failure models. To increase the therapeutic efficacy of BMSC transplantation in clinical trials, BMSC numbers and transplantation methods must be adjusted (He et al., 2018).

Patients develop castration resistance despite a positive initial response to androgen restriction therapy, which can cause castration-resistant prostate cancer to progress or return. In order to overcome the drawbacks of using cancer cell lines, patient-derived xenografts are a crucial preclinical cancer model. They enable researchers to get preclinical results that more closely resemble clinical responses in patients. This is due to the fact that these xenografts, which are responsible for their three-dimensional growth, are grown in immunocompromised mice, which retain the main molecular abnormalities found in patient tumors. At first, there was little success when human prostate cancer tissues were transplanted into athymic mice. Meanwhile, recent technical advancements have increased the success rate of patient-derived xenograft transplantation, including the use of immunodeficient mice, co-injection of prostate cancer tissues with ECM, and transplantation into renal capsules. Utilizing the ECM improves epithelial cells' capacity for differentiation, cellular composition, and survival. ECM layered with a liquid media was used to create organoid models of prostate cancer (Namekawa et al., 2019).

1.7.4 Kidney

Tissue engineering of the kidney will help in the excessive demand on the healthcare system, for new fully functioning kidneys. There are many researchers that have successfully decellularized animal kidneys. In 2021, five New Zealand white rabbits were used in a study where their kidney was decellularized using 0.5% sodium dodecyl sulfate. The findings of the study concluded that the decellularized kidney was successful, due to the removal of all cellular material (Abou-Rabia et al., 2021).

One of the challenges that is faced in tissue engineering of the kidney is vascularization. The provision of adequate nutrition, metabolite exchange, and oxygenation of the resultant tissue is critical to the success of any tissue engineering system. Developing a sophisticated vascular network is a limitation of tissue engineering and is an area that needs exploring. Glomerular filtration, blood pressure control, and chemical balance all depend on the blood flowing through the kidney and, more specifically, through its functional unit, the nephron. Therefore, the vascularization of the kidney scaffold is essential in order for it to fulfill its function in the human body. According to a study, human microvascular endothelial cells can be employed in combination with droplet-based bioprinting to create microvascular constructs with a confluent vessel lining that displays strong cell proliferation and cell-to-cell adhesion. Numerous developments in kidney tissue engineering have been made in the past few years in the fields of bioprinting, decellularization, recellularization, vascular casting, and organoid formation (Lim et al., 2022).

1.8 Summary

Despite the current advancements, further research is necessary before these procedures may be used in clinics (Caneparo, Sorroza-Martinez, et al., 2021). Stem cell therapies for urology are evolving quickly and have made significant strides already. Despite encouraging in vitro and pre-clinical results, stem cells will not soon be used in routine urological practice. Before stem cells are employed to replace conventional treatment approaches, more research is necessary. It is understood that cell ingrowth-required cellular matrixes are unlikely to succeed outside of very specific constraints. Human trials, such as injecting stem cells, are insufficient and further research is required in order to determine whether these methods are successful (Chapple, 2020; Davis, Cunneane, Mulvihill, et al., 2018).

Despite improvements in the clinical and pre-clinical data, many researchers have concluded that there is still more work to be done in tissue engineering of the genitourinary system. The system is intricate, has a wide variety of tissues, and a unique link to the body; all of these make it incredibly challenging to recreate the organs with no issues. The current outcomes are encouraging and create a foundation, to be built on in order to drive the advancements further. Furthermore, the process from bench to clinic is a lengthy one, with many clinical, animal, and human trials that are required before the treatment can be used.

The overall aim of tissue engineering is to be able to create organs and tissues that will be available “off the shelf,” that allow repair and regeneration of tissue similar to its original function with limited side effects.

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The progress in tissue engineering of kidney

2

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2.1 Introduction

Millions of people across the world are affected by chronic kidney disease (CKD) with the best solution being a kidney transplant even with all the drawbacks it possesses. Due to the limitation of donor kidneys, one person in the United Kingdom dies every day waiting for kidney transplant. CKD affects approximately three million people within the United Kingdom with hypertension and diabetes being the most prevalent causes for end-stage renal failure (UK, 2022). Additionally, other risk factors include kidney stones, systemic infections such as HIV, hepatitis B and C, polycystic kidney disease, or acute kidney injury which can be caused by heart attacks (Chen et al., 2019). CKD is categorized into five stages with each stage corresponding to the levels of kidney function and is done by measuring the GFR (glomerulus filtration rate) (NKF, 2021). The GFR value is taken as the percentage of kidney function and is the best overall indicator for kidney function with reduced values linked with structural kidney damage, as well as the reduction of GFR values being parallel with the decline in kidney function in CKD (Levin et al., 2013). CKD is defined by the presence of GFR of below 60% for more than three months can be very concerning as this foreshadows end-stage renal failure (stage 5) (Vaidya & Aeddula, 2018). Renal failure cannot be reversed; however, there are a limited number of options which the patient has to help get back to a normal life.

In 2017, 1.2 million people across the globe died from CKD due to lack of treatment or costs (Bikbov et al., 2020). Current renal replacement therapy (RRT) includes kidney transplants and dialysis which are both highly effective methods used and help increase life expectancy significantly. Dialysis is the process used to restore the balance of the intracellular and extracellular fluid environment of a

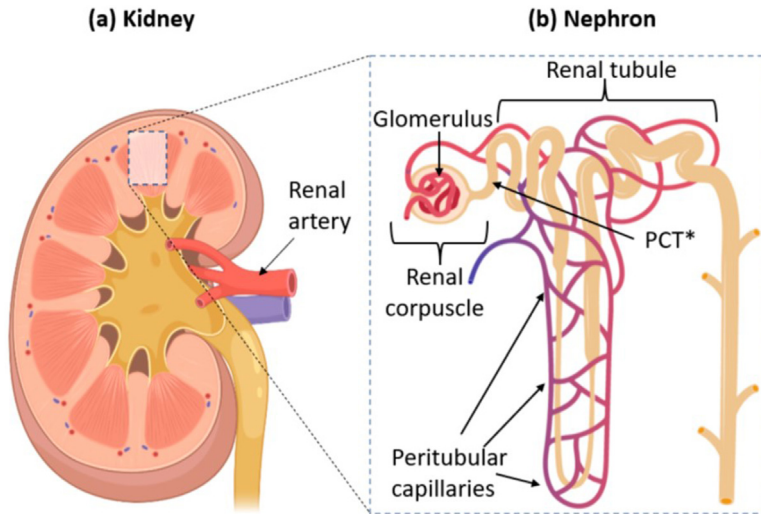
patient with stage 5 renal failure that is characteristic of normal kidney function (NHS, 2021). This generally requires the patient to visit the hospital three times a week with each session being 3–4 h. Although this method is widely used and highly effective, it can impact the patient's quality of life negatively between the regular sessions at the hospital and low potassium diet. The alternative is a kidney transplant which allows the patient to live a normal life; however, medication including anti-rejection medication must be taken daily to stop the kidney from being rejected. Although the kidney transplant can be sufficient for the rest of the patient's life, rejection can take place almost instantly even with antirejection medication. Moreover, the medication taken to help with rejection comes with a high number of undesirable side effects with high costs in addition to the cost of kidney transplants. Even if we disregard these risks, it takes an average of two and a half to three years to receive a kidney transplant from a deceased donor in the United Kingdom with the availability of kidneys being much lower than the demand (Statista, 2020). For those in countries where they are required to pay for a kidney transplant such as the United States of America, it can cost an average of \$442,500 (McQueen, 2017).

With new advances in technology and tissue engineering, we can eliminate these issues and help make kidneys more accessible to everyone diagnosed with CKD. This can be done by identifying and then applying the correct tissue engineering techniques and biomaterials. Tissue engineering uses a combination of materials engineering, cells, and biological growth factors to produce a biological substitute for tissues and organs. In order to produce tissue-engineered kidney, these factors must be evaluated and tested for their functionality and compatibility. The result of producing a kidney in vitro would positively impact the lives of millions and increase the quality of life of individuals with CKD. While reviewing the current advancements and progress being made in tissue engineering of a kidney, it is essential to look at the challenges that are still present. This report will review and evaluate the three building blocks of tissue engineering and how they may impact the production of a kidney.

2.2 Anatomy and physiology

The kidney is a complex and vital organ responsible for multiple functions within the body to help maintain homeostasis. A function the kidneys are well known for are its ability to filter out the toxic leftovers from the blood and role in the excretion of waste. Additionally, they are essential for regulating water volume, ion and salt concentrations, and pH levels along with influencing red blood cell production and blood pressure.

We can take closer look at the kidney (Fig. 2.1) and see the intricacy of this extraordinary organ. Each kidney has approximately one million filtering units called nephrons, most of which is located within the renal cortex of the kidney the rest in the renal medulla. The nephrons are responsible for processing the blood and do this via three steps: filtration, reabsorption, and secretion. The structure of a nephron can be divided into two sections: the renal corpuscle and renal tubule. The renal corpuscle consists of two parts, the Bowman's capsule (found at the start of the nephron and has a cup like structure) and the glomerulus (a network of capillaries

**FIGURE 2.1**

Anatomy of kidney and nephron.

which carry the blood to the Bowman's capsule for filtration). The blood which flows through the kidneys equates to approximately 25% of the cardiac output with the glomeruli filtering approximately 180 L of blood daily (Carroll, 2006). Afferent arterioles bring blood to the glomerulus and efferent arterioles take it away. The afferent arterioles have a larger diameter than the efferent which creates a blood pressure within the glomerulus capillaries which is higher than in the Bowman's space, allowing water and solutes to pass into the Bowman's capsule with relative ease. The rate at which the filtration of the blood plasma takes place is known as the glomerular filtration rate (GFR). The afferent and efferent arterioles have the ability to change their diameters when there is a change in the systemic blood pressure. This allows the GFR to remain at a fairly constant rate regardless of the blood pressure around the body through a process called autoregulation. Next, the glomerular filtrate makes its way into the tubule where reabsorption starts to take place.

The renal tubule is divided into three parts: the proximal convoluted tubule (PCT), the loop of Henle, and distal convoluted tube (DCT). The PCT reabsorbs Na^+ (sodium ions), Cl^- (Chlorine ions), and K^+ (potassium ions) and approximately 65% of the water in the blood at that point in addition to the glucose and amino acids (Christensen et al., 2003). The thin descending limb part of the loop of Henle is responsible for removing more water from the filtrate, with the ascending limb removing salts such as calcium and magnesium. Once the filtrate reaches the DCT, more reabsorption takes place with K^+ and Na^+ being reabsorbed and the rest of the filtrate is emptied into the collecting duct. Finally, the filtrate passes through a few other parts of the kidney which include the renal pyramids, renal papilla, calyx, and renal pelvis before collecting as urine in the bladder via the ureter.

There are two types of nephrons within the kidney. Almost 85% of the nephrons are cortical nephrons, these have a shorter loop of Henle located in the outer medulla with

the glomerulus found in the outer part of the renal cortex. The other 15% of nephrons are the juxtamedullary nephrons which have a longer loop of Henle found in the inner medulla with the glomerulus positioned in the renal cortex close to the renal medulla.

2.3 History of kidney transplants and treatments

The need for kidney transplants has always been there; however, limited technology and knowledge have stunted the progression and the longevity of transplants when they have been carried out. Over the last 70 years, there has been significant progress within the field of organ transplantation with the need for kidneys at an all-time high with around 4385 (2019–2020) people still waiting for a kidney transplants in the United Kingdom (Nordham & Ninokawa, 2022). This value will increase with those to receiving kidneys being carried over to the next year as well as those diagnosed with CKD and people who have had their new transplants rejected. Therefore, it is essential that we can not only improve the longevity of transplants but to also provided an alternative means of meeting this demand through tissue engineering.

Fig. 2.2 shows the history of kidney transplant from when the first successful transplant between two identical twins took place to how we are now able complete transplants between nonsiblings with the help of antirejection medication. Even with new and improved antirejection medication, the patients will still experience multiple unwanted side effects and the kidney can still be rejected. Although technology now allows us to identify and predict the best kidney for individuals who require a transplant, it not the same as their own kidney. With so many factors that must be considered and criteria that need to be met before a transplant, it becomes very

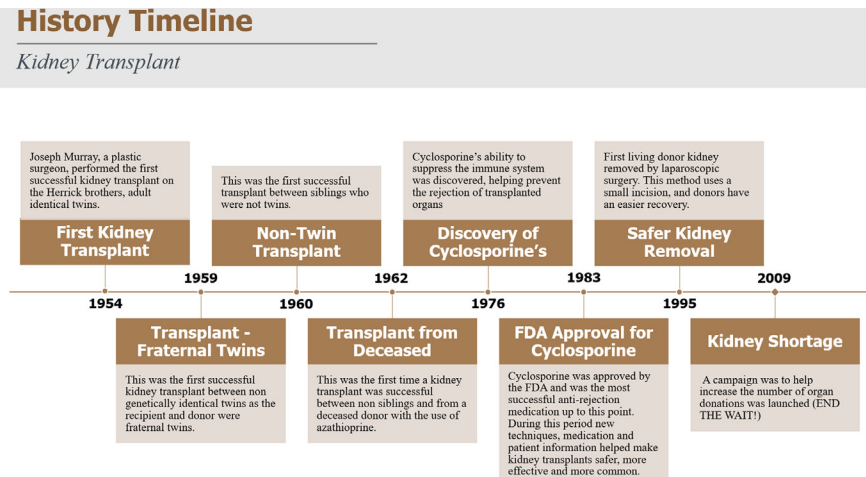


FIGURE 2.2

Kidney transplant timeline.

difficult to provide all patients with a donor kidney due to compatibility or number of donors. New and developing research regarding tissue engineering of kidneys and other organs mean we can now aim to create 100% compatible and fully functioning kidneys, essentially removing the need for kidney donors and dialysis.

2.3.1 Biomaterials and scaffolds

Biomaterials are one of three building blocks to tissue engineering organs along with growth factors and cell types. Biomaterials are used to create scaffolds which serve as a template for tissue engineering and encourage cell migration, differentiation, and proliferation in addition to maintaining cell type-specific phenotypes. Also, it is essential that the biomaterials used for creating scaffolds must be biocompatible, biodegradable with suitable pore size, and have mechanical properties which can uphold engineered tissue formation. As well as the materials used for building scaffolds, the architecture for the scaffold is just as important with the pore size major aspect for a desirable and functional scaffold, determining the success of cell adhesion and proliferation. The biomaterials and the scaffold design serve as a foundation for engineering tissues and organs, while having significant impact on the interaction with cells and growth factors of the organ being fabricated.

2.3.1.1 *Natural biomaterials*

The materials used to create a scaffold are a pivotal factor in the success of tissue engineering. Natural biomaterials are the ideal choice for tissue engineering purposes due to their inherent biological properties which include cell proliferation, migration adhesion, and differentiation (Brovold et al., 2018). The biomaterials used for tissue engineering of a kidney include collagen, fibrin, hyaluronic acid (HA), agarose, alginate, chitosan, and gelatine (Ullah & Chen, 2020). Natural biomaterials do not have any adverse effects in vivo but must provide a structure suitable for the seeded cells to form the complex 3D tubular structures of a kidney; like all scaffolds. Natural biomaterials will be discussed in further detail below with an overview of their advantages and disadvantages compiled in Fig. 2.3.

2.3.1.1.1 Collagen

Collagen is a protein and is fundamental in maintaining structural and biological integrity of the ECM (extracellular matrix). Forming 25%–35% of the protein in the entire body, collagen can be found in connective and fibrous tissues, for example, ligaments, tendons, blood vessels, and other tissues which require flexibility and strength (Parenteau-Bareil et al., 2010). There are 29 distinct types of collagen formed by different combinations of α chains, differentiating each type of collagen and its properties from the next (Parenteau-Bareil et al., 2010; Sinha & Trehan, 2003). The noncytotoxic, biocompatible, and biodegradable properties along with its structural and physical properties make collagen an ideal material for 3D scaffolds. A variety of structures can be fabricated with the use of collagen which includes beads, cross-linked films, meshes, sponges, fibers, and others, increasing

Natural Polymers							
	Collagen	Gelatin	Fibrin	Chitosan	Alginate	Agarose	Hyaluronic Acid (HA)
Advantages	✓ Present in most natural tissues ✓ Modifiable ✓ Biocompatible ✓ Can be crosslinked for increased mechanical strength	✓ Non toxic ✓ Biocompatible ✓ Helps with delivery of growth factors	✓ Biodegradable ✓ Biocompatible ✓ Support cell growth and proliferation ✓ Promote tissue regeneration	✓ Biodegradable ✓ High Absorption Rate ✓ Gel forming Ability	✓ Used with a combination of other biomaterials ✓ Biocompatible ✓ Biodegradable ✓ Good Cell adhesion	✓ Biocompatible ✓ High water absorption capacity ✓ Gel forming ability	✓ ECM like conditions ✓ Biocompatible ✓ Non toxic
Disadvantages	✗ Fast Degradation ✗ Reduced mechanical strength ✗ Poor immunogenicity	✗ Weak Mechanical strength ✗ Rapid degradation ✗ Has to be combined to other biomaterials	✗ Degradation require more research ✗ Weak mechanical Properties ✗ Difficult to collect	✗ Low Mechanical stability ✗ Poor solubility	✗ Weak mechanical strength ✗ Fast Degradation	✗ Weak Mechanical strength ✗ Rapid degradation	✗ High Cost ✗ Degrades too slow 6-7 years ✗ Weak mechanical strength
	(Parenteau-Bareil et al., 2010, Sinha and Trehan, 2003, Klimek and Ginalska, 2020)	(Parenteau-Bareil et al., 2010, Ranganathan et al., 2019, Gervaso et al., 2013, Hoque et al., 2015)	(Parenteau-Bareil et al., 2010, Li et al., 2005, Ma, 2004, Laurent et al., 1996)	(Parenteau-Bareil et al., 2010, Chircov et al., 2018, Ye et al., 2005, Li et al., 2005, Alaribe et al., 2016, Islam et al., 2017, Croiser and Jérôme, 2013, Ma, 2004)	(Parenteau-Bareil et al., 2010, Ma, 2004, Lee and Mooney, 2012, Azala and Mooney, 1997, David et al., 2016, Zarrinjal et al., 2018)	(Parenteau-Bareil et al., 2010, Li et al., 2005, David et al., 2016, Awad et al., 2004)	(Li et al., 2005, Gunatillake et al., 2003, Saini et al., 2016, Muraris and Dobois, 2016)

FIGURE 2.3
Natural biomaterials.

potential applications (Klimek & Ginalska, 2020). Additionally, the availability of collagen is considerably high with the extraction of collagen possible from almost all animals. Nonetheless, collagen for tissue engineering purposes include bovine tendons and skin, rat tail, porcine skin among others. Isolation and purification of collagen can easily be done on a large scale making it suitable for the high demands of kidney tissue engineering (Sinha & Trehan, 2003).

2.3.1.1.2 Gelatine

Gelatine is a highly suitable natural polymer for tissue engineering applications derived from type 1 collagen, with several advantages including biodegradability, biocompatibility as well as being fully absorbable (Ranganathan et al., 2019). Among its many strengths, gelatine has been widely used due to its biological nature, excellent solubility in water, and low cost, in addition to its high commercial availability in comparison to its parent protein collagen (Gervaso et al., 2013; Parenteau-Bareil et al., 2010). For gelatine extraction, collagen is usually extracted from bones, tendons, and skin of animals, but it is primarily obtained from bovine and porcine collagen (Hoque et al., 2015). However, like most natural polymers, gelatine also has a weak mechanical strength and rapid degradation in vivo when used as a scaffold (Hoque et al., 2015).

2.3.1.1.3 Fibrin

Fibrin, another protein biomaterial, can be used to form 3D scaffolds due to the fabrication of fibrin gels, which are prepared with the combination of fibrinogen and thrombin solutions (Chircov et al., 2018; Parenteau-Bareil et al., 2010; Ullah & Chen, 2020). Although fibrin degrades rapidly, studies show the use of aprotinin decreases the rate at which the fibrin degrades (Chircov et al., 2018). Moreover,

other groups demonstrate the degradation of the fibrin gel correlates with the adjusted concentration levels of aprotinin, allowing controlled degradation (Chircov et al., 2018; Ye et al., 2000). Furthermore, fibrin has several advantages which include biocompatibility, cell adhesion, nontoxic, and antiinflammatory properties (Parenteau-Bareil et al., 2010; Ranganathan et al., 2019). The human body produces fibrin when there is an injury making fibrin a biomaterial which cannot be obtained with significant ease (Ranganathan et al., 2019; Ullah & Chen, 2020).

2.3.1.1.4 Chitosan

Chitosan is a linear polysaccharide that is produced by partial deacetylation of chitin, next to cellulose, as the second most abundant polysaccharide in nature (Li et al., 2005; Parenteau-Bareil et al., 2010). Several properties that belong to chitosan include high absorption capacity, biodegradability, and an ability to form gel-like structures which can be used to form scaffolds when paired with other biomaterials such as alginate (Alaribe et al., 2016; Li et al., 2005; Parenteau-Bareil et al., 2010). Studies have demonstrated that chitosan scaffolds can facilitate the differentiation of stem cells in tissue engineering and regenerative medicine both in vivo and in vitro (Ranganathan et al., 2019; Ullah & Chen, 2020). Acetylation, molecular weight, crystallinity, and degradation are all factors that affect the physicochemical properties of chitosan. These additional benefits gained from adjusting the factors mentioned include the ability to: form gel structures when exposed to acidic environments, accelerate bone regeneration, bind to microbial and mammalian cells, chelate transitional metal ions with the addition of antiinflammatory and antitumorigenic properties (Islam et al., 2017). Not only does chitosan has the potential to build scaffolds for artificial kidneys but it also holds great promise as the membrane for artificial kidneys (Croisier & Jérôme, 2013). This is due to the fact chitosan has high intrinsic mechanical strength along with permeability to creatinine and urea. When used as a scaffold, there are a few drawbacks to the properties of chitosan which include poor solubility and mechanical stability (Ma, 2004).

2.3.1.1.5 Alginate

Alginate is another natural biomaterial which belongs to the polysaccharide family and like most other natural polymers has biodegradable and biocompatible properties (Lee & Mooney, 2012; Ullah & Chen, 2020). As mentioned previously, alginate is used in combination with chitosan to help form suitable scaffolds which promotes cell proliferation and cell adhesion (Lee & Mooney, 2012; Parenteau-Bareil et al., 2010). Alginate can be used to form hydrogels, through a variety of crosslinking methods, and other 3D scaffolds such as sponges, microspheres, foams, microcapsules, and fibers (Atala & Mooney, 1997; Parenteau-Bareil et al., 2010). Alginate is sourced from species of brown algae (Phaeophyceae) and easily processed into scaffolds while being a low-cost material (Atala & Mooney, 1997; David et al., 2016). Although alginate has many benefits, it does have a few downfalls such as weak and limited control with regards to the mechanical properties of the material (Zarrintaj et al., 2018).

2.3.1.1.6 Agarose

Agarose also belongs to the polysaccharide family and is extracted from red algae and seaweed exhibiting characteristics that are ideal for tissue engineering and scaffold structures (Awad et al., 2004; David et al., 2016; Parenteau-Bareil et al., 2010). Agarose's ability to form gel-like structures (hydrogels) allow the formation of 3D matrices in addition to supporting cell adhesion due to similarities between the ECM (Awad et al., 2004). Moreover, like other natural polymers, agarose is biocompatible, while forming thermoreversible gels and having a high-water absorption capacity (Awad et al., 2004; Ranganathan et al., 2019). Cell growth, proliferation, and differentiation thrive in these hydrophilic conditions increasing the chances of successfully regenerating new tissue and organs. Unfortunately, it also has similar weaknesses such weak mechanical strength and short degradation period.

2.3.1.1.7 Hyaluronic acid

HA is a polysaccharide found in soft connective tissue and a major component in synovial fluid and tissue (Wang et al., 2012). HA meets a majority of the requirements needed to be classified as a proficient biomaterial for building a scaffold. These properties include biodegradability, biocompatibility, noncytotoxicity, and provide support for tissue regeneration by creating ECM-like conditions (Laurent et al., 1996; Ranganathan et al., 2019). Rapid degradation and weak mechanical properties are a theme which fit with all natural polymers and to combat this issue they are cross-linked with other materials to build suitable scaffolds (Laurent et al., 1996). The ability to form a scaffold successfully is ultimately defined by the composition and manufacturing process of each biomaterial. HA can be used to make gel-like structures which are used for scaffolds like the other polysaccharides once combined with other biomaterials (Gunatillake et al., 2003; Laurent et al., 1996).

2.3.1.2 Synthetic biomaterials

Synthetic biopolymers provide an alternative to natural polymers while providing similar or better advantages for developing scaffolds. Although natural polymers are highly biocompatible and biologically active, but their lack of mechanical properties and rapid degradation limit the applications for which they can be used for (Lee & Mooney, 2012). The mechanical properties and degradation for synthetic biomaterials on the other hand can be tailored to fit the desired purpose (Lee & Mooney, 2012; Ranganathan et al., 2019; Saini et al., 2016). In addition to this, the manufacturing process for synthetic polymers can often be more cost-effective than biological scaffolding with a longer shelf life (David et al., 2016; Ranganathan et al., 2019). The synthetic polymers used for kidney tissue engineering include PGA (polyglycolic acid), PLA (polylactic acid), PCL (polycaprolactone) and PLGA (poly lactic-co-glycolic acid) (Lee & Mooney, 2012; Ranganathan et al., 2019; Ullah & Chen, 2020). In comparison with biologically derived scaffolds, a concern with synthetic polymer scaffolds is that they interact with the body less favorably with degradation resulting in inflammatory responses (Saini et al., 2016; Ullah & Chen, 2020). Fig. 2.4 provides an overview of the strengths and

Synthetic Polymers				
	Poly lactic acid (PLA)	Poly lactic-co-glycolic acid (PLGA)	Polyglycolic Acid (PGA)	Polycaprolactone (PCL)
Advantages	✓ Biodegradable ✓ Non toxic degradation ✓ High mechanical strength ✓ Flexible	✓ Adjustable degradation ✓ Modifiable ✓ Biocompatible	✓ Good mechanical strength ✓ Biocompatible	✓ Biodegradable ✓ Bioabsorbable ✓ Good mechanical strength
Disadvantages	✗ Can cause inflammation	✗ Can cause inflammation	✗ Rapid degradation in aqueous solution ✗ Can cause inflammation	✗ Slow degradation rate ✗ Can cause inflammation
	(Liu et al., 2020, Yuan et al., 2018, Pan and Ding, 2012, Bose et al., 2012)	(Li et al., 2005, Yuan et al., 2018, Boland et al., 2001, Siddiqui et al., 2018)	(Li et al., 2005, Ma, 2004, Malikmammadov et al., 2018, Burton et al., 2018)	(Gao et al., 2019, Lannutti et al., 2007, Xue et al., 2019)

FIGURE 2.4

Synthetic biomaterials.

weaknesses which the synthetic biomaterials possess, with more detail about the specific materials provided below.

2.3.1.2.1 Poly lactic acid

PLA is a thermoplastic and widely used as a synthetic biopolymer for the fabrication of scaffolds. It holds great potential for the use of biomedical and tissue engineering applications due to the numerous advantages it possesses. PLA meets many complex criteria which include biodegradability, biocompatibility, the ease and scale at which PLA can be processed, and mechanical strength (Granados-Hernández et al., 2018; Liu et al., 2020; Murariu & Dubois, 2016). However, PLA may only be used when high impact toughness (low impact toughness) and hardness (brittle) are not the requirements needed for its application such as in vivo implants, bone scaffolds, and other mechanically demanding environments (Murariu & Dubois, 2016). Due to PLA's ability to be combined with other natural and synthetic biomaterials, PLA's disadvantages can be removed thus making PLA an ideal candidate for the scaffolds for kidney regeneration (Murariu & Dubois, 2016; Yuan et al., 2018). Other disadvantages that PLA has are a slow degradation rate and a hydrophobic nature which would significantly reduce cell adhesion and proliferation (Murariu & Dubois, 2016). These properties can also be altered when the PLA is blended with other biomaterials as well as providing the ability to control these properties by adjusting the amount of the other biomaterials added to the PLA. PLA is a renewable natural

resource obtained by extracting it from potato, corn, sugarcane, and so on (Granados-Hernández et al., 2018; Liu et al., 2020; Yuan et al., 2018).

2.3.1.2.2 Poly lactic-co-glycolic acid

PLGA is a copolymer of PLA and PGA and exhibits many beneficial advantages for the use as a scaffold and other tissue engineering applications (Granados-Hernández et al., 2018). Biocompatibility, biodegradability, and high toughness are some of the relevant properties PLGA possesses which allow it to be categorized as a suitable material for tissue engineering purposes (Ranganathan et al., 2019). Even though PLGA has several disadvantages, PLGA can be modified allowing the disadvantages to become advantages. Without any modifications, PLGA has a hydrophobic nature, long degradation period as well as poor cell adhesion and inflammation upon degradation of the biomaterial (Pan & Ding, 2012; Ranganathan et al., 2019). In an attempt to modify and improve the properties of PLGA, PLGA microspheres have been used as way to improve cell adhesion (Pan & Ding, 2012). Using other biomaterials in combination with PLGA or changing the composition of the copolymers allows the degradation period to be manipulated to suit the need for a scaffold while making PLGA more hydrophilic (Bose et al., 2012; Granados-Hernández et al., 2018). To help improve the cell–matrix interaction, scaffolds can be fabricated with larger pore size for tissues with larger cell size (Ranganathan et al., 2019).

2.3.1.2.3 Polyglycolic acid

PGA is a synthetic biopolymer with biocompatible and biodegradable properties ideal for the use in tissue engineering applications (Boland et al., 2001; Siddiqui et al., 2018) and with additional properties including high initial strength, bioabsorbability, and high melting point (Siddiqui et al., 2018; Ullah & Chen, 2020). Due to the hydrophilic nature of PGA, the degradation time for PGA in vivo or aqueous solution ranges from two to four weeks by which point most of the mechanical integrity of the scaffold is lost (Boland et al., 2001; Ullah & Chen, 2020). PGA is used with a combination of other biomaterials such as PLA to help increase strength and integrity of PGA when used as a scaffold. With all the benefits PGA and the other synthetic polymers possess, a common issue that affects all of them is the degradation of the biomaterials, although they are not toxic, they may still cause inflammation (Ranganathan et al., 2019).

2.3.1.2.4 Polycaprolactone

PCL is another polysaccharide which is used and considered for tissue engineering applications of kidneys. This biopolymer is biodegradable and bioabsorbable with great mechanical strength and thermal stability (Burton & Callanan, 2018; Malikmammadov et al., 2018). It has a slow degradation rate which can be altered when merged with other materials, for example, PLGA as well as being hydrophobic (Burton & Callanan, 2018; Gao et al., 2019). Although the porosity plays a role in cell diffusion and migration, PCL does allow some cell migration and proliferation to take place on its surface. Due to PCL's ability to allow surface modification, PCL can be modified to

allow greater cell diffusion and adhesion to take place (Malikmammadov et al., 2018). This makes PCL an ideal candidate for scaffolds of a kidney.

2.3.2 Fabrication techniques

Not all materials can be manufactured using the same fabrication techniques with some suitable for multiple techniques but at the same time producing scaffolds with distinctive design, functions, and properties. Different techniques can be used to produce different scaffold designs affecting the function of the scaffold and interaction of the scaffold with the cells and growth factors. The fabrication technique used can determine the physical and mechanical properties of the scaffolds it produces. Tissue engineering of the kidney can be carried out with several techniques such as electrospinning, decellularization and 3D bioprinting, and so on.

2.3.2.1 Electrospinning

Electrospinning is a simple and cost-effective fabrication technique used to produce 2D and 3D scaffolds by generating a nonwoven mesh using nanofibers (Lannutti et al., 2007). The fibers produced to create polymer scaffolds vary in diameter, ranging from 10 to 100 μm (Xue et al., 2019). Electrospinning is carried out by the electrification of a polymer solution which is stored in a syringe pump attached to a needle (spinneret) at the end of it. At the tip of the needle, a drop of solution is allowed to gather before an electric current is passed through the drop to form a Taylor cone (a jet in the shape of a cone) (Pham et al., 2006). The syringe then fires the jet to the platform, which is grounded resulting in the solidification of then solution into fibers forming a mesh (O'Brien, 2011; Pham et al., 2006; Ullah & Chen, 2020). By altering the different parameters for electrospinning such as the size of the fibers, needle size, solution concentration, distance between the platform and needle, and the motion and shape of the platform while collecting the fibers, the properties of the scaffolds can be changed (O'Brien, 2011). These changes include porosity of the scaffolds, the thickness of the fibers, the shape and structure of the scaffold, and the orientation of the fibers within the scaffold. Natural biomaterials are commonly used to create scaffold via electrospinning with synthetic polymers used if they can be converted into an aqueous solution (Ullah & Chen, 2020). Not only do the materials used for electrospinning have to be suitable to undergo the process, but they still need to adhere to the requirements for building a scaffold such as biocompatibility, biodegradability, cell attachment, and so on (Jansen et al., 2019). To replicate the complex structure of the kidney and the specialist functions that are carried out within the kidney, by the glomerulus and tubules, research must be carried out and tests must be done to guarantee function, cytotoxicity, and longevity of each part (Jansen et al., 2019; Liu et al., 2020). Look at using the electrospinning technique to build scaffolds for PCTs that have the suitable and sustainable mechanical and physiological properties when implanted in vivo, using different concentrations of PCL as the biomaterial base. The findings of this study show electrospinning can successfully produce a scaffold with the correct physical traits while being compatible with the appropriate

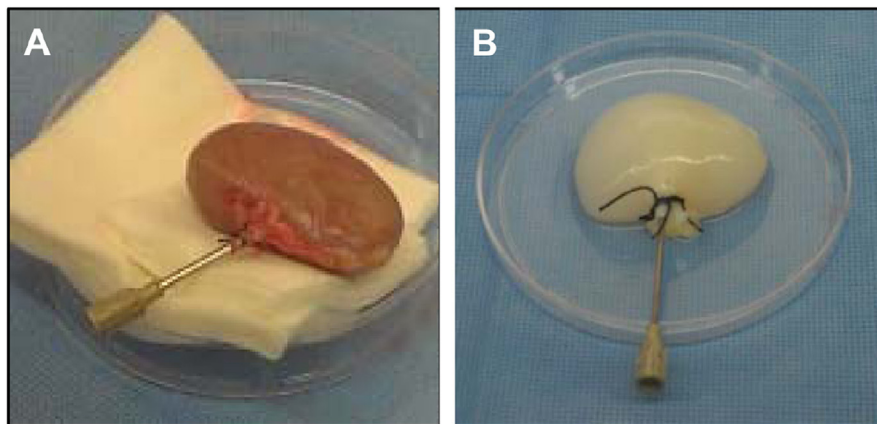
biomaterials. The scaffold produced demonstrated rapid diffusion, tight cellular monolayer formation, and great mechanical stability in comparison to existing models which use hydrogels to form the scaffolds. Due to the hydrophobic nature of PCL, it can be paired with other synthetic materials like PLGA which is hydrophilic (Burton & Callanan, 2018). The variations within the structure of the scaffolds that may be produced using electrospinning make this a very viable technique for producing scaffolds for tissue engineering a kidney. For this purpose, PLA and PGLA are both promising materials with experiments still taking place to see which materials would be best for kidney scaffolds (Batchelder et al., 2015).

2.3.2.2 Decellularization

Decellularization is the process of removing all the cellular components from a kidney or organ leaving behind a natural biological scaffold (Sullivan et al., 2012). With the aim of preserving the preexisting architecture and native ECM. This decellularized organ may be recellularized using autogenous cells and implanted into the patient without antirejection medication required postoperation. The decellularization of the kidney takes approximately 10–14 days over which it undergoes several processes removing all cellular components (Crapo et al., 2011). At the current moment in time there is no standardized process, specific chemicals, or enzymes used to decellularize kidneys or any organs.

Zwitterionic, ionic, and nonionic detergents have the ability to solubilize cell membranes as well as detaching DNA from proteins as a result it is an effective way to remove cellular material from tissues (Moon et al., 2016). However, for the decellularization of a kidney, it is becoming a common practice to use sodium dodecyl sulfate (SDS) along with Triton-X100 as detergents with the initial use of heparin to flush out any blood remaining in the kidney (Fedecostante et al., 2017; Moon et al., 2016; Sullivan et al., 2012). After the blood is cleared from the kidney, 0.25%–1% of SDS is perfused with a salt base solution (PBS or HBSS) for 24–48 h and then perfused again with just the salt-based solution for a short while (Fedecostante et al., 2017; Nakayama et al., 2013). After which the Triton-X100 is perfused into the kidney for a few more hours, removed and the salt-based solution added again for 48 h to remove any residue of the detergent (Nakayama et al., 2013). These timings are not fixed, with different studies using different timings with most waiting for the kidney to change color. Before the process is finished, DNase solution is perfused for a few hour before being washed out with the salt-based solution for the final time (Fedecostante et al., 2017). In order to establish the decellularization process is complete and the ECM and the different structures within the kidney are well preserved, further tests are carried out. Due to the early stages of the decellularization process, there are still many unknown factors which may impact the processes and integrity of the final scaffold and its vascular system. A simple method used to identify that the process is complete is by looking out for the change in color seen in Fig. 2.5.

Following the decellularization of the kidney, the next steps required are to reintroduce the renal, stem, endothelium, and other cells from the recipient of the soon recellularized kidney (Nakayama et al., 2013). After the infusion of the cells, the

**FIGURE 2.5**

(A) Kidney of monkey before decellularization, (B) kidney after decellularization—(Batchelder et al., 2015).

kidney is kept in a bioreactor (to mimic the bodies environment) providing the best suited condition for the differentiation of the cells perfused into the kidney (Batchelder et al., 2015; Crapo et al., 2011; Sullivan et al., 2012). Some cells are pipetted directly into certain locations of the kidney to help increase the cell attachment of the seeded cells as well as the repopulation (Sullivan et al., 2012). Perfusion has proved to be the more successful method for cell suspension in comparison to manually injecting the cells (Nakayama et al., 2013). Incubation period of a recellularized kidneys is upwards of three weeks and only after this is complete may the kidneys be implanted (Sullivan et al., 2012).

A lot of work is still required for decellularization to be an effective method for tissue engineering of a kidney. Though the detergents used currently work, the methods used do not always produce a complete biological scaffold with no damage to the structure or ECM. None the less the decellularization is a promising fabrication technique which has a potential to help thousands. A disadvantage that this technique fails to overcome is that organ donors are still required with CKD patients increasing at a higher rate than the number of donors. Even if this hurdle is overcome, the decellularization process is not cost-effective (Nakayama et al., 2013).

2.3.2.3 3D printing

3D printing is a rapid prototyping process and uses an additive manufacturing technique to produce the desired object through layering of the materials used for its production. With the use of this technology in tissue engineering, it can make the production of tissue engineered organs to be completed with very little complications, in a single location and relatively quickly. Moreover, the quality of the scaffold produced using this technique is unlike what can be produced manually with the ability to produce small-scale scaffolds and scaffolds with a finer resolution. Not

only does this technique produce a scaffold it can also have the cells integrated into the scaffold as it is being fabricated. Induced pluripotent stem cells (iPSCs) are used and able to form cell lines which help with the epithelial structure, tubular formation, and nephron production (Wragg et al., 2019). The biomaterials used and compatible with the 3D bioprinting technique include a combination of both synthetic and natural biomaterials (Lawlor et al., 2021). Beside the biocompatible aspect of the materials used in 3D bioprinting, they also require mechanical properties which provide structural integrity once printed while being initially stored as an aqueous solution for some 3D printing technologies (Edgar et al., 2020; Wragg et al., 2019). The main challenge for 3D bioprinting is to produce and print living biological material while creating a complex structure for ECM and the specialized cells to replicate the function of an organ. In order to achieve a replicated biological organ like the kidney through 3D bioprinting, a combination of three approaches is used.

Biomimicry is the first approach, for it to succeed physiologically accurate biomaterials and cellular and extracellular components of the organ must be used, in order to achieve identical replication of biological tissues and natural function (Liu, 2004). It is essential to understand cell biology, biomaterials, biophysics, imaging, and medicine to be successful with this approach (Edgar et al., 2020).

The second approach that is used is the autonomous self-assembly which relies on the embryonic organ development to replicate tissue. In the early stages of tissue development, cellular components achieve organ functionality by producing ECM components, autonomous organization, and cell signaling. Cells are primary drivers of autonomous self-assembly. They determine the tissues composition, functional, localization, and structural properties (Edgar et al., 2020).

The final approach is mini-tissue, which looks at the smaller functional and structural building blocks which make up the tissues and organs such as nephrons in the kidney. The idea is to fabricate these smaller functional components and bring them together to form the desired tissue or organ using self-assembly and biomimicry. Biomimicry allows accurate reproductions of the functional components which are assembled via cell assembly (Edgar et al., 2020).

Medical imaging technology such as MRI, CT and X-rays are all used as noninvasive methods of scanning, providing better understanding of the architecture and function of organs and tissues. The complex shapes and architecture can be digitized using CAD-CAM (computer-aided design and computer-aided manufacturing) tools producing an accurate model of the organ. The final aspect of 3D printing is the printing which can be done through several types of printers which are inkjet, micro-extrusion, and laser-assisted (Edgar et al., 2020; Liu, 2004).

2.3.2.4 Other fabrication techniques

Other fabrication techniques may be used for building scaffolds such as freeze drying, solvent casting, and gas forming. Freeze drying is a process in which dissolved polymer is frozen into the shape of the scaffold after which the solvent is removed via lyophilization. The rate of freezing determines the size of the pores with larger pore sizes produced with longer freezing rates. Another method, solvent casting, uses a mold dipped

in polymer solution which is then left to evaporate forming a polymeric membrane on a composite structure (mold). This composite material is then dissolved leaving behind a 3D scaffold. The final method which may be used to produce a kidney scaffold is gas forming. This process involves carbon dioxide being exposed to polymers with high porosity at high pressures. The saturated carbon dioxide results in the creation of pores which cause the polymer to expand producing a 3D scaffold (Lu et al., 2013).

2.3.3 Cell culture

Over 20 types of specialized cells make up kidney tissue. These cells are structurally organized into distinct compartments functionally and morphologically (Baskapan & Callanan, 2022). To successfully produce a tissue engineered kidney, the incorporation of these cells is necessary. Whether they are perfused into the kidney or stem cell are used with the correct growth factors to allow the differentiation of the desired cells. The cells currently used to artificially create a tissue engineered kidneys are primary renal cells (which include proximal tubule cells (PTCs)), pluripotent stem cells (PSCs), embryonic stem cells (ESCs) fetal stem cells (FSCs), and amniotic fluid-derived stem cells (AFSCs).

2.3.3.1 Primary renal cells

Primary renal cells can be extracted from kidney tissue and cultured. PTCs or proximal tubule epithelial cells are a type of renal cell and provide physiological functions to the kidney (Baskapan & Callanan, 2022). Different studies showed the successful interaction with PCL scaffolds and good adhesion and monolayer formation on the scaffold (Jansen et al., 2019; Liu et al., 2020). Other groups also used renal cells but with PLA scaffolds which presented great promise for full tissue regeneration with key markers for crucial kidney cell being present (Burton & Callanan, 2018; Lannutti et al., 2007). Majority of the renal cell population used consists of PTCs with a small percentage of distal tubular cells and podocytes present (Baskapan & Callanan, 2022). However, both the studies conclude that there is not a complete interaction with the cells and the scaffolds, hence further work must be carried out. When renal cells are seeded on to hybrid scaffolds, they able to provide an appropriate environment which support renal cells over time (Ross et al., 2009).

2.3.3.2 Stem cells

2.3.3.2.1 Embryonic stem cells

ESCs are a type of PSCs located in the inner cell mass of an embryo and are able to differentiate and self-renew (Baskapan & Callanan, 2022; Loukogeorgakis & De Coppi, 2017). With the correct growth factor, these cells can differentiate into any cell within the body including renal cells and other specialized kidney cells. Due to ethical purposes, ES cells are limited and not used as to their full potential. Recent work in the field of genetics has uncovered a technique allowing genetic modification of adult cells transforming them into pluripotent stem cells called iPSCs (induced Pluripotent Stem Cells). These cells possess the ability to self-renew and

produce renal cell lineages and have been used to produce podocytes and tubular cells (Baskapan & Callanan, 2022). A decellularized rat kidney was seeded with ESCs in a study by Ross. *E et al.* which provided some encouraging results. The ESCs successfully interacted with the kidney scaffold with proliferation occurring within tubular and vascular structures as well as the glomerular exhibiting differentiated kidney cells markers (Biswas & Hutchins, 2007).

2.3.3.2.2 Fetal stem cells

FSCs are another source of cells which may be used for regenerating a kidney; however, like the ESCs, there are ethical concerns when it comes to obtaining FSCs. AFSCs originate from FSCs and may be considered as source of cells with considerable potential with its ability to differentiate into different cell types (Biswas & Hutchins, 2007; Eberli & Atala, 2006). AFSCs are able to differentiate into fibroblasts, osteocytes, and adipocytes which are particularly important for any organ. Amniotic fluid is found around the fetus providing protection, nutrients, and mechanical support (Eberli & Atala, 2006). With high potential and low risk associated with the use of AFSCs, they should be strongly considered for tissue engineering of a kidney.

2.3.3.2.3 Adult stem cells

Unlike the ESCs and FSCs, there are no ethical issues with obtaining adult stem cells and usually extracted for bone marrow, muscle, bone, adipose tissue, and skin (Baskapan & Callanan, 2022; Rookmaaker et al., 2004). Adult stem cells share similar qualities as the other stem cells with the ability to differentiate into other cells as well as self-renew. It can be difficult to isolate adult stem cells when extracting and finding enough stem cells to extract thus limiting the use adult stem cells (Rookmaaker et al., 2004).

2.3.3.3 Progenitor cells

Progenitor cells trace back to stems cells as all cells do; however, the progenitor cells have limited ability when it comes to differentiation and can only differentiate into one cell type (Murphy & Atala, 2014). With its self-renewal capability, progenitor cells have been used to repair the kidney rather than assist with the regeneration of it although it may be capable of doing so (Nakayama et al., 2013). Endothelial progenitor cells (EPCs) can be found in the kidney and helps with the restoration of the endothelium cells found in the kidney such as the glomerular endothelium and peritubular capillary endothelium (Murphy & Atala, 2014). Tubular and fibroblast progenitor cells also exist and if these can be isolated and extracted, they may be utilized in the application of tissue engineering of a kidney.

2.3.4 Growth factor

2.3.4.1 Hepatocyte growth factor

HGF (hepatocyte growth factor) is found within the kidney with an increase in the production of HGF occurring due to CKD or acute kidney injury (Igawa et al., 1993). They positively impact and play a key role in regeneration and repair of renal tubules

while preventing chronic renal fibrosis initiation and progression (Liu, 1999). HGF also neutralizes transforming growth factor (TGF)- β 1 expression in different cell types in the kidney as well as inhibiting apoptosis in endothelial, renal tubular, and podocytes cells (Bach & Hale, 2015; Igawa et al., 1993; Matsumoto & Nakamura, 2002). A study on the kidneys of mice demonstrates HGFs ability to promote myofibroblasts production and the regeneration and proliferation of tubular epithelial cells (Bach & Hale, 2015). Additionally, HGF is responsible for simulating the growth of endothelial cells and accelerating repair and regenerating glomerular capillaries (Igawa et al., 1993). Being the crucial factor in the repair and regeneration of the kidney, the conclusion can be made that HGF would be a successful growth factor in tissue engineering a kidney.

2.3.4.2 Insulin-like growth factor 1

IGF-1 (insulin-like growth factor 1) is responsible for the growth, development, structure, and function of the kidneys and is found within all sections of the kidney (Farooq et al., 2021). Promoting cell proliferation and differentiation in the kidneys is an advantage which IGF-1 provides (Oh, 2012). Along with the ability to increase GFR and renal blood flow in humans and animals, IGF-1 has also proved to encourage glomerular and renal hypertrophy (Farooq et al., 2021; Oh, 2012). IGF-1 plays an important role in the reabsorption of both phosphate and calcium by mediating the growth hormone which simulates the reabsorption process in the proximal tubules. Although evidence suggests that IGF-1 does not improve kidney function significantly in humans with acute kidney injury, it increases and accelerates the regeneration of PTCs in rats. The use of IGF-1 with mesenchymal stem cells has increased the migratory response of these cells as well as improving the function of the kidney with endogenous synthesis of IGF-1 and proliferation of the stem cells in the peritubular region. The use of IGF-1 with mesenchymal stem cells has increased the proliferation of the stem cells in the peritubular region as well as an increased migratory response of these cells with the addition of endogenous synthesis of IGF-1 resulting in an improved kidney function (Oh, 2012).

2.3.4.3 Fibroblast growth factor

FGFs (fibroblast growth factors) are involved in several biological processes and cellular functions including cell differentiation, proliferation, embryonic development, regeneration, and tissue repair (Asghari et al., 2017; Villanueva et al., 2014). FGFs can be effective for the proliferation of various cells such as smooth-muscle cells, endothelial cells, neurons, chondrocytes, and adipocytes (Villanueva et al., 2014). In addition to this, the proliferation of renal cells occurs when exposed to FGF which can result in higher number of functional tubular cells and glomeruli. When tissue engineering a kidney, FGF can be added exogenously accelerating tissue repair and initiate production of repairing proteins. Tubulogenic, epitheliogenic, and angiogenic proteins are responsible for kidney regeneration which are all produced as result of FGF (Asghari et al., 2017). The use of FGFs could prove crucial in the regeneration of a kidney.

2.4 Tissue engineering of a kidney

The process of engineering a complete organ must take into consideration the factors covered above with various combinations of these factors having potential to providing a viable kidney. The complexity of the kidney structure and distinct segments presents a challenge most organs do not. Through the successful isolation of specific renal cells, the specialized components of the kidneys such as the tubules, nephrons, and glomerulus may be guaranteed long-term survival and differentiation (Atala, 2009). Researchers were able to demonstrate the use of renal cells seeded to collagen-coated polycarbonate to produce successful tissue engineered renal units with evidence of reabsorption, filtration, and secretory functions. There was also evidence showing the presence of self-assembled tubule like structures and glomeruli which consisted of various cell types (Atala, 2004). With new advance within the tissue engineering field, a number of in vivo studies have been conducted with regards to the tissue engineering of the kidney. Research shows kidney cells from a donor rabbit were used to tissue engineer a kidney which was then implanted into athymic mice. Glomeruli, proximal tubule, and distal tubule cells were all cultured separately until adequate expansion was achieved before being seeded onto a PGA scaffold ready for implantation (Atala, 2009). The results of this study demonstrated nephron formation within the artificial scaffolds while showing renal cells can successfully attach to the biodegradable polymers.

According to some studies, the progress made with the regeneration of a kidney using human PSCs which try to focus on different segments of the kidney (Geuens et al., 2020). With the most promising work being carried out by Taguchi et al., using a number of different cell types which include progenitor cells, human PSCs, and endothelium cells to produce a kidney organoid. However, this tissue engineered organ did not produce the desired end goal due to its instability for long-term use. This study produced promising results with the organoid consisting of ureteric trees enclosed with nephrons (Taguchi et al., 2014).

Producing all the renal structures together as a single organ has yet to be accomplished successfully for kidneys. A method which may be utilized and mentioned within the fabrication techniques is decellularization. Decellularized kidneys would provide a natural tissue matrix with preexisting kidney architecture which would allow more cells to be transplanted producing an increased volume of renal tissue (Atala, 2009). The existing structure would provide designated areas for the different renal cells providing sufficient space for the cells to differentiate in comparison to the cells having to be cultured individually before being merged.

2.5 Discussion and conclusion

After reviewing all three elements required to produce a viable tissue engineered kidney, it can be concluded that more research and testing is still required. Although a great deal of progress has been made regarding the development of a kidney within

the field of biomaterials, cell culture, and growth factors, the complexity of the kidney still proves a great challenge to overcome. Looking back at each section as a whole it can be seen that the compatible materials required have been identified but the correct combinations and concentration are yet to be found.

The biomaterials used for tissue engineering play a huge role in the success of tissue engineered organs and tissue. The choice of materials used dictates how other biological substances will interact with the scaffold as well as the techniques that are compatible with that biomaterial. The properties which were identified as necessary are biocompatibility, biodegradability, good cell adhesion, degradation, and suitable mechanical strength. All the natural biomaterials considered met most of the criteria required for tissue engineering bar the fact they share a common disadvantage of natural biomaterials. Mechanical instability and poor degradation period limit the use of the natural polymers for larger tissues and organs shifting the focus for biomaterials on to synthetic biomaterials. The ability to manipulate the mechanical properties as well as the degradation period make synthetic biomaterials a good replacement for natural biomaterials. Their biocompatible and biodegradable properties only cement this idea further. The issues with synthetic biomaterials can be credited to the lack of cell adhesion and proliferation with an inflammatory response as the polymer degrades. Even after evaluating this large group of materials, a perfect material is yet to exist. However, almost all of the problems that plague the synthetic biomaterials for the use in kidney regeneration can be solved. This can be done through crosslinking and copolymerization and coating the scaffold with hydrophilic materials. Different combination of synthetic with natural polymers can be used, each producing different properties suited for a range of tissue engineering applications. PLA crosslinked with collagen and copolymerized PCL and PLGA have been used for kidney scaffolds in studies and show great promise for tissue engineering of a kidney.

Three of the most popular and well-suited fabrication techniques for kidney scaffolds have been discussed with electrospinning already in use for other tissue engineering applications. Electrospinning is a popular fabrication technique proven to produce successful scaffolds using biomaterials such as PLA, PGA, PCL, etc. With its ability to produce a variety of different 2D and 3D scaffold structures, electrospinning can be used to produce the complex structures required for kidney scaffolds. Decellularization is a technique which has the similar limitations as a kidney transplant with organ donors still required to use this technique. This method shows great promise as it allows the patient to live a normal life without the use of any anti-rejection medication or worry about the kidney being rejected while providing a perfect ECM and scaffold. The final technique discussed for the production of a tissue engineered kidney was 3D bioprinting. This exciting new method provides a base for producing the whole kidney in a single attempt contained in a bioreactor. It also provides great potential for producing kidneys on large scale with fewer process and complications. All these fabrication techniques are capable of producing kidney scaffolds but not all are currently capable of doing so. With research still being carried out for the other factors involved (biomaterials, cells, and growth factors), it

provides adequate time to complete the research and testing required for the successful use of these fabrication techniques.

The kidney is a complex organ with multiple cells carrying out different functions within the organ. Renal, stem, and progenitor cells are all involved in the regeneration of the kidney and lead to the formation of tubular, vascular, and glomerular structures. When these cells are used with the correct growth factors, tissue engineering of a kidney can be achieved. However, the cells are only effective if used with the right biomaterials which allow good cell adhesion and differentiation.

The growth factors discussed play a key role in allowing the kidney to regenerate and help the cells to differentiate, proliferate, repair, and regenerate. HGF, IGF-1, FGF all promote the production of tubular cells and proteins in turn producing majority of the kidney and its main functioning systems.

Tissue engineering of a fully functioning kidney is yet to be achieved. Even though good progress has been made, the optimum combinations of the tissue engineering factors are yet to be discovered. With different structures of the kidney such as the tubules, nephrons, and glomerulus now producible, the next goal is to engineer the whole organ with all the individual structures functional. As for now using a decellularized kidney is very promising and will produce results quicker than tissue engineering the whole kidney.

The progress in tissue engineering of a kidney has come a long way in the last few years with more work still required to produce a viable kidney. This report has shown what the essential materials and techniques are and what technology is available and being used in the application of tissue engineering of a kidney. With the combination of the right biomaterials, cells, and growth factors, the aim of producing a kidney in vitro can be achieved.

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The progress in tissue engineering of bladder

3

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3.1 Introduction

Bladder disease is a condition where the bladder is unable to function normally and will result a person to, in some cases, urinate uncontrollably, frequently, and urgently than normal. Each year, almost 500,000 people worldwide are diagnosed with bladder cancer or illness. Understanding the physiology of the bladder's muscular, vascular, and neurological structures, as well as the illnesses and current treatments connected with them, can provide a wealth of information. Understanding these factors will enable us to detect problems and develop cutting-edge remedies using tissue engineering. Many methods and techniques, such as cystoplasty and bladder auto-augmentation, are currently available to replace damaged sections and relieve patients of symptoms associated with certain bladder diseases, such as urinary incontinence. However, these techniques have many limitations and drawbacks, including being very invasive and causing a host immune response in the patient. This chapter examines how tissue engineering has innovated to overcome these drawbacks, as well as what future technology tissue engineering is developing, such as the use of stem cells and intestinal submucosa to repair and replace damaged bladder sections, and scaffolds that can be used for bladder tissue engineering.

Tissue engineering is the use of biological components such as cells and growth factors in conjunction with engineering concepts and the use of biomaterials to create biological alternatives that can heal diseased and damaged organs and tissues while also maintaining their normal function (Petrovic et al., 2011). It uses a variety of cell transportation and biological techniques, such as injecting functional cells into nonfunctional void places both inside and outside the body to stimulate regeneration in skin-related diseases like burns (Shpichka et al., 2019). Biomaterials can be made synthetically as well as from natural resources, and these are referred to as

biological scaffolds. Scaffolds aid in directing the proliferation of new cells and enhance the inherent ability of cells to repair themselves (Atala, 2011).

3.1.1 Bladder physiology and structure

The bladder is part of the renal system which is a combination of the kidney, ureter, bladder, and urethra all working in conjunction with each other to form the urinary tract. It is necessary to understand the anatomy, vascularity, and nephrology of the bladder to be able to identify the reasons why bladder diseases occur and why certain treatments and procedures are implemented (Chmielewski, 2003). The bladder is a hollow muscular organ located in the subperitoneal area of the body that is held together by ligaments attached to the ureter and the pelvic bone. The bladder's main function is to act as a reservoir for temporary urine storage and to assist in the expulsion of urine. It can do so because of its muscular and robust structure, which allows the bladder to store up to 400–600 mL of urine in healthy adults, but this capacity is significantly reduced in people with bladder disease (Shermadou et al., 2018).

The shape and structure of the bladder depending on the volume of urine in the bladder, which is oval when full and flat when empty. The position of the bladder also shifts depending on how contained it is, since it is located within the smaller pelvis when empty and stretches into the abdominal cavity for adults when fully contained. The bladder, on the other hand, is located within the belly in children until adolescence, at which point it slips into the pelvis (Shermadou et al., 2018).

There are four main parts that constitute the bladder including:

- Apex—The apex is located the top of the bladder and is connected to the ureters above by the umbilical ligament.
- Body—This is the main part of the bladder hence named the “body” and is the largest area of the bladder which is located between the apex and the base.
- Base—Located at the bottom of the bladder, hence the name of “base” and this section of the bladder is pointy shaped.
- Neck—This is the most bottom part of the bladder and is the bit that is constricted to allow for higher during expulsion for urine, the neck is also the section of the bladder that conjoins with the urethra (Shermadou et al., 2018).

3.1.1.1 Muscle structure

The bladder consists of detrusor muscle which is the main functional muscle of the bladder that enables contraction during micturition and makes up the muscular wall surrounding the bladder. This is a smooth muscle that is composed of urothelium and transitional epithelium; transitional epithelium is stratified tissue which has round shaped cells that are able to reconfigure their shape without damaging the epithelial lining (Hickling et al., 2017; Shermadou et al., 2018).

The urothelium is made of specialized tissue lining the urinary tract and consists of large and rounded cells that change shape in accordance with the volume of urine

present within the bladder. These cells become flat as volume increases and return to their rounded shape when the bladder is empty to increase urine holding capacity and decrease pressure within the bladder walls (Birder et al., 2012).

The bladder also has sphincter muscles that act like valves to allow entrance and exit of fluids and substances; the ureter consists of two sphincters that link to the bladder called the internal urinary sphincter muscle and the external urinary sphincter muscle and these carry slight variation in males and females. The internal urinary sphincter muscle is there to prevent seminal regurgitation during ejaculation in males and consists of smooth circular fibers that are under autonomic control and in females no sphincter muscle is present and it is thought to be a functional sphincter and is formed by the bladder neck and proximal urethra. The structure is the same for both genders in the external urethral sphincter and is under voluntary control (Hickling et al., 2017; Shermadou et al., 2018).

However, blood replenishment differs slightly between males and females, with males receiving arterial supply from the inferior vesicle artery and females receiving arterial supply from the vaginal arteries. The vesical venous plexus is where venous drainage occurs, and in males, the vesical venous plexus is related to the prostate venous plexus; however, for females this differs because females do not have a prostate (de Treigny et al., 2017).

3.1.1.2 Nervous structure

The bladder consists of a very complex nervous structure as it receives input from both the somatic and autonomic branches of the nervous system. The autonomic is the section of the nervous system that enunciates the internal organs and blood vessels and regulates various process that occur in the body such as rate of breathing and blood pressure and operates automatically without the need for any human conscious effort. Whereas the somatic nervous system is voluntary and requires human consciousness to operate for the control of muscles and movement (Fowler, 2008).

The autonomic nervous system can be divided into two divisions called the sympathetic and parasympathetic. The sympathetic system prepares the body through neural signals for activities that induce stress often called the “fight or flight response” and the parasympathetic system helps to retain and return the body to its normal functioning environment often being called the “rest and digest” response. These two divisions of the autonomic nervous work in harmony and simultaneously to maintain homeostasis and efficient internal body environment (Jones, 2020).

In terms of the bladder system, the sympathetic nervous system acts on the detrusor muscle and results in the relaxation of this muscle to promote urine retention in order build up storage and not frequently have the need to urinate. The parasympathetic fibers from the sacral spinal nerves (S2–S4) also acts on the detrusor muscle and causes the contraction of it to invigorate the excretion of urine. The somatic nervous system is also involved within the bladder in renal system as it supplies the external urethral sphincter with nerves providing voluntary control over urination which can be deduced by the contraction of the skeletal sphincter muscles to increase the pressure and frequency of urination (Sam et al., 2021).

The bladder walls consist of receptors called mechanoreceptors that respond to stimuli which in the case of the bladder wall is the stretching of detrusor muscles. Through nerve fibers called general visceral afferent (GVA) located within the hypogastric and pelvic splanchnic nerves, the mechanoreceptors upon detecting stimuli send sensory information to the central nervous system (CNS).

The superior and inferior part of the bladder both follow different nerves where the superior follows the sympathetic nerve, and the inferior follows the parasympathetic nerves. The speed of the nerve impulse due to the stimuli is according to the size of the stimuli or the amount of pressure and swelling in the bladder; therefore, a higher pressure in the bladder would result in a faster stimulus sent back to the CNS. Which is why the urge to urinate becomes increased when the bladder gets filled (Fowler, 2008).

Stimulation of the detrusor muscle by both nerves results in the contraction and relaxation to allow for the release of urine where the parasympathetic stimulates the muscle to contract for emptying and the sympathetic to constrict the internal sphincter and relax the muscle for filling up of the bladder (Sam et al., 2021).

3.1.2 Bladder function

When blood is filtered by the kidneys, the by-product urine passes through the ureters and the ureterovesical junction into the bladder. As the bladder fills with urine, sensory nerves within the bladder send signals to the CNS which then communicate with conducting somatic and autonomic nerves to control the release of urine through contraction and relaxation of the detrusor muscle, with simultaneous relaxation of the internal and external urethral sphincters. The bladder is very efficient and well adapted to its main function of storing and expulsion of urine due to the complex muscular, vascular, and nervous structure of the bladder (Shermadou et al., 2018).

Muscle and fat make up the bladder wall, which may relax and expand to hold urine and contract and flatten to discharge urine through the urethra. The bladder's main function is to temporarily store urine, and due to its structure and folded internal lining called a rugae, it can carry greater than 300 mL urine in normal adults. The bladder also aids in the expulsion of urine by contracting the bladder's musculature during micturition while the sphincters relax (Hickling et al., 2017).

3.2 Bladder disease

3.2.1 Urinary incontinence

A very common type of bladder disease is known as urinary incontinence and it is the lack of voluntary control over urination which results in the uncontrollable leaking of urine and is a symptom of major bladder disease and if untreated can carry the risk of bladder removal (cystectomy). There are four types of urinary incontinence which

are listed in with description about the common causes, treatments, and common symptoms (Haylen et al., 2010).

In females more than male, urinary incontinence symptoms are highly prevalent and have a serious effect on the quality of life (Saarni et al., 2006). There are two main types that effect women greatly and these are stress incontinence and urge incontinence and they become even more common with progression of age and obesity (Hannestad et al., 2000).

3.2.2 Interstitial cystitis

This is also known as the painful bladder syndrome caused by the irritation and inflammation of the bladder walls and this results in reduced bladder capacity due to stiffening of the bladder wall; symptoms consist of pain, discomfort, and pressure in the bladder and pelvis and frequent urination which results in a reduced quality of life due to interference in social situations and activities. This condition also tends to increase the urgency to urinate as the passageway to the ureter becomes more constricted due to inflammation which can be seen on Fig. 2.1, and this causes a high pressure to form at the internal sphincters which could lead to further damage and infection into the bladder walls which could increase the risk of cystectomy (Davis et al., 2015).

3.2.3 Bladder cancer

Bladder cancer is caused by the mutation and damage of bladder cells and is a diverse disease with 70% of patients acquiring superficial tumors which aren't directly life threatening however do tend to recur and the 30% consist of muscle invasive disease which can become life threatening and result in death. Symptoms of this consist of hematuria which in some cases can be painful and a lot of cases not painful and increased frequent urination frequency. There are also other factors that can result in this change of cells are diet, untreated bladder disease, and smoking as chemicals taken in by smoking pass into the bloodstream and are filtered by the kidneys into the urine, which is stored in the bladder; for some instances, these toxic substances and particles are able to cause irritation within the bladder walls causing an involuntary response to occur which can result in inflammation and bleeding (Kaufman et al., 2009).

When patients are suspected of cell carcinoma within the bladder, certain diagnosis techniques and assessments are carried out that consist of cystoscopy, cytology, and medical radiodiagnostics for tumor detection within the bladder. Cystoscopy and cytology are invasive methods and currently the noninvasive method used is urine cytology which is the examination of a single cell type found in fluid specimens. There are four stages of bladder cancer, and the cancer generally forms with the internal lining of the bladder (McDougal et al., 2008).

Treatment of bladder cancer currently is a different series of chemotherapies such as neoadjuvant chemotherapy which is aimed to target and shrink a cancerous

tumor using drugs before moving on to invasive surgeries and other treatments (Mieog et al., 2007). Due to failure rates of 30%–45%, there has been lots of interest in neoadjuvant chemotherapy (Black et al., 2006).

In 2003, a study by Grossman HB et al. utilized a large randomized trial to assess the effect of this type of chemotherapy on patients that had been treated by bladder irradiation and radical cystectomy and results of this clinical trial suggested that neoadjuvant chemotherapy of drugs such as vinblastine, doxorubicin, cisplatin, and methotrexate is better than cystectomy (bladder or part of bladder removal); however, this study loses some of its reliability as it took more than 10 years to acquire all the patients for the trial and within his time period a lot of changes were made in the medical industry for better or worse to patient care, diagnostic standards, and supportive care during chemotherapy (Grossman et al., 2003).

There are other different diseases relating to the bladder. All of bladder disorders have been summarized in Table 3.1 below:

Table 3.1 Diseases/disorders relating to the bladder.

Disease	What it is?	Symptoms	References
Bladder cancer	- Occurs when bladder cells start to grow out of control and form tumors - Generally, starts in the inner lining of the urothelium Different types of bladder cancer: 1. Urothelial carcinoma—most common 2. Squamous cell carcinoma 3. Adenocarcinoma 4. Sarcoma	- Blood in urine - Painful and frequent urination - Lower back pain - Bone pain - Swelling in legs - Burning sensation when urinating - Unexplained weight loss	Antoni et al. (2017), Kaufman et al. (2009)
Interstitial cystitis	- Chronic inflammation of the bladder walls - Cause of it is not known yet; however, research is still being done - Also, can be known as painful bladder syndrome (PCB)	- Painful and frequent urination (up to 60 times a day) - Sudden urges to urinate - Harsh and strong pelvic pains - Pain during sex	Sant (1997), Vij et al. (2012)
Urinary incontinence	- The loss of bladder control - The individual cannot prevent urine from leaking - Occurs when the detrusors contract and the sphincter cannot close	- Pressures and spasms in pelvic region - Urinating during sleeping - Frequent urination	Aoki et al. (2017), Bardsley (2016)

Table 3.1 Diseases/disorders relating to the bladder.—*cont'd*

Disease	What it is?	Symptoms	References
Urinary stones	efficiently due to them not being strong enough. Results in the leaking of urine - Small, mineralized crystals that are formed in urine and get trapped in the bladder - Crystals are formed when urine stays in the bladder as the minerals clump together to form a stone - Can lead to urinary tract infections and urination issues	- Blood in urine - Painful and frequent urination - Intense stomach pain and pressure - Clouded urine - Loss of urine control	Miano et al. (2007), Shafi et al. (2016)
Urinary retention	- Gradual inability to not empty the bladder of urine fully - Small amounts of urine are left in the bladder - Can be due to blockages, medications, or nerve problems - Blockages in men are primarily due to prostate gland blocking the urethra	- Inability to urinate - Weak urine flow - Frequent urination in small amounts - Urge to urinate but no urine is released - Pressure in bladder and needing to urinate again just after urinating	Billet and Windsor (2019), Kowalik and Plante (2016)
Urinary tract infection	- An infection occurs through the urinary system affecting the bladder, kidneys, and urethra - Occurs via microbes entering the body	- Burning sensation while urinating - Bloody urine - Strong urine odor - Increased frequency and urgency of urination - Pelvic pain for women too	Gupta et al. (2017), Vasudevan (2014)

3.3 Biomaterials for bladder regenerative medicine

Biomaterials are any materials used to cooperate with a biological system. Biomaterials can be either biological or synthetic and can be used to facilitate the procedure of creating tissues and organs (Table 3.2). Due to the complex anatomy and the functions of the bladder, it is of extreme importance that the materials are suited to the surroundings. As the bladder acts as a base for storing urine while maintaining pressure, the regenerated organ must accommodate the original functions and should contract to release the urine outside the body to prevent the retention of it (Orabi et al., 2013).

Table 3.2 Biomaterials used within bladder engineering.

Biomaterial	Advantages	Disadvantages
Poly(lactic-co-glycolic acid) (PLGA), polyglycolic acid (PGA), polylactic acid (PLA)	Biodegradable, easily molded and shaped with electrospinning and gas foaming due to thermoplastic nature to give a desired microstructure for 3d scaffolds.	Lack of bioactive factors of natural biological tissues. May also reduce pH levels and result in a foreign body response.
Acellular matrix scaffold	Natural scaffold made from the subject's submucosa. Reduced risk of foreign body response. Consists of many biofactors for the efficient proliferation of cells.	Still being tested as many trials have seen cases of infection, ruptures, and stone formation.
Composite polymer scaffold of collagen and PGLA	Strong mechanical resilience and stable within internal environment. Highly moldable. Multiuse cell scaffold. Strong microstructure and degradable.	Can produce acidic by-products that can result in activating an immune response. Still under trial and early development stage.
Silk fibroin	Provide good mechanic endurance and stability comparative to biological scaffolds. Degradable.	Lack of bioactive factors of natural biological tissues. Can produce acidic by products within the body.
Natural acellular scaffold derived from BAM and SIS	Like the native tissues in terms of composition, mechanical properties, and microstructure. Degrade after implantation without causing an immune response. Maintain in cell ingrowth and differentiation for regeneration of bladder wall.	Degrades very slowly after implantation. Requires subject cells.

3.3.1 Scaffold

Within tissue engineering, the scaffold can be considered as a 3D template and is very important for the whole process of tissue repair to work efficiently as scaffolds are support structures designed to facilitate cellular growth and proliferation, its vital that the scaffold used mimics the properties of biological tissues to work effectively. However, efficiency is not everything that a scaffold must hold, as anything foreign introduced into the body can cause a reaction which may induce inflammation or other immune responses; therefore, the biological scaffold must be engineered to be biocompatible with the patient's body to avoid any unwanted reactions ([Horst et al., 2013](#)).

Biological scaffolds should promote cell interaction and growth and allow easy transportation of substances that cells will require to grow such as oxygen and nutrients. Any suitable scaffold that is going to be used within the body should have strong mechanical properties to survive the body's internal conditions such as temperature and pH levels and be degradable after cell regeneration (Chan & Leong, 2008).

For bladder tissue engineering, two main types of scaffolds have been utilized called the acellular matrix which are made from donor cells and polymer scaffold which consist of composite scaffolds that are made using two or more different polymers, the reason why these two materials are preferred is due to the study carried out by Pariente et al., in 2002 where biocompatibility of the materials was tested in environments of inside and outside (artificial) of the body and both proved to be very good. The most desirable scaffold for the bladder should be able to support the urothelial cells and the smooth muscle cells (SMCs) (Pariente et al., 2002).

3.3.1.1 Polymer scaffold

Polymers to be used within the human body and for medical use must be approved by the FDA and a variety of synthetic polymers have been approved such as polylactic-co-glycolic acid (PLGA), polyglycolic acid (PGA), polylactic acid (PLA), polyanhydrides, and polyortho esters, which are biodegradable can also be used to make scaffolds for tissue (Horst et al., 2013).

The reason why these polymers are used as medical products is because they can easily be molded and shaped with various techniques such as electrospinning and gas foaming due to their thermoplastic nature to give a desired microstructure for 3d scaffolds.

A study by Mauny et al., in 2011 tested the use of silk fibroin within bladder tissue engineering for bladder augmentation and results showed that the silk fibroin polymer biomaterial had excellent results and that it could be used as alternate biomaterial as it overcame the issue of foreign body rejection (Mauney et al., 2011).

Another study conducted in 1999 by Oberpenning et al. discusses the use of transplantable urinary neobladders that were created in vitro from smooth muscle and urothelial cells grown in culture and seeded on to polymers that were shaped as the bladder organ. The need for this study arose from the complications arisen by human organ replacement complications and the risks associated with it such as shortages of organs due to less donors, rejection of the organ by the patient's body due to the organ being foreign, and tissue compatibility. The bladder neoorgans molded into shape were made from PGA meshes and surface coated with PGLA and were replaced with the native bladders and evaluated for 11 months and successfully demonstrated normal bladder elasticity and the ability to retain urine to normal capacity. This study successfully illustrated for the first time that by methods of tissue engineering it is possible to reconstitute an autonomous hollow organ (Oberpenning et al., 1999).

3.3.1.2 Acellular matrix scaffold

This type of scaffold is made from bladder acellular matrix and is extracted using decellularization of tissues. Decellularization is a process used to separate residing

cells (dermal and epidermal) from the extracellular matrix of a tissue where only the extracellular matrix (ECM) scaffold remains of the original tissue, and this scaffold can then be used with various techniques of tissue engineering to help in tissue regeneration, wound healing, and production of artificial organs; two methods of this technique exist called chemical and mechanical decellularization which both leave behind a natural scaffold. The main advantage of this type of scaffold is that even after decellularization they maintain their biological activity (Gilpin & Yang, 2017).

The scaffold originated from small intestinal submucosal tissue and some of the bioactive factors have been identified in both small intestinal submucosa (SIS) and bladder acellular matrix such as FGF, laminin, fibronectin, type IV collagen, and VEGF as well as the bioactive factors which play a very important role as they regulate differentiation and migration of cells, proliferation, and cell adhesion (Shi & Ronfard, 2013).

In 2006, a study by Zhang et al. uses the bladder acellular matrix and small intestinal submucosal scaffold to reconstruct the bladder by seeding the SMCs of the bladder and urothelial cells. This method of bladder reconstruction using acellular matrix was tested in larger animals such as dogs and pigs however was unsuccessful due to issues that transpired which included incomplete muscle development, ingrowth of fibroblasts causing graft shrinkage, and inadequate vascularization (Zhang et al., 2006).

However, in another study conducted in 2007 that tested both seeded and unseeded matrices in smaller animals such as rats and mice by Sievert et al., had promising results which were successful in complete bladder regeneration dependant on the amount of damage and healing required (Sievert et al., 2007).

3.3.1.3 Composite polymer scaffolds

To overcome the inconsistencies and disadvantages of normal polymer scaffolds, two or more different polymers can be combined to create a composite polymer system scaffold which will combine the properties of each polymer used to produce it including the advantages. A popular hybrid of collagen matrices has been elaborated on in a study by Nakanishi et al., in 2003 where collagen was combined with a PGLA mesh to become a gel or sponge and very interesting results were received where the SMCs of the bladder tissue were only able to proliferate on the gel mesh and not on the sponge mesh; however, the opposite was seen with the urothelial cells where they were able to proliferate on the sponge and not on the gel which meant that the composite polymer scaffold was able to facilitate both cells at once (Nakanishi et al., 2003).

3.3.2 Cells

Autologous cells from the bladder are typically extracted and planted on to scaffolds to enhance tissue regeneration and proliferation. Autologous cells are employed to avoid inflammatory responses produced by foreign cell recognition and immune

system rejection. Smooth muscle and urothelial cells are collected, cultured, and then reimplanted into the patient in the event of bladder reconstruction; however, if bladder cells cannot be obtained, stem cells are employed (Serrano-Aroca et al., 2018).

Stem cells are cells that have excellent regenerative potential and can renew and develop into specialized cell types in response to the adhered stimuli and environment presented to them. In tissue engineering, these stem cells are utilized with bio-materials to enhance their regenerative and differential properties (Mashayekhan et al., 2013).

There are two types of stem cells used in tissue engineering called pluripotent and multipotent stem cells. Pluripotent stems can develop into most cells within an organism such as bladder cells and multipotent stem cells can develop into specialized cell types. However, these two types of stem cells are known to be embryonic stem cells and therefore have many ethical issues surrounding them as they must be harvested from early stage embryos; therefore, adult stems cells can also be used in tissue engineering and these cells can be harvested from various tissues such as the adipose tissue (fat), striated muscles, bone marrow, testicles, and skin (Mashayekhan et al., 2013).

Mesenchymal stem cells are a type of adult stem cells that have been studied in 2005 by Song et al. (2005) by placing them in vivo and surprisingly found that they do not form tumors and concluded by testing the regeneration ability of the mesenchymal stem cells in smooth muscle bundles and found that these adult stem cells could in fact differentiate into bladder SMCs.

The use of bladder ECM has been explored by Aitken & Bägli, 2009. They elaborate on bladder ECM interaction with its constituent cells in a dynamic way through several ECM receptors, which are important for differentiation and proliferation. The study found that the bladder ECM increased amounts of collagen III, matrix metalloproteinases, and tissue inhibitors which were found to be critical in providing the bladder with structural support which if further explored could assist in treating urinary incontinences that are caused by a lack of structural support (Aitken & Bägli, 2009).

An alternative for embryonic stem cells was explored in 2009 by Jack et al., 2009 where they demonstrated that adipose stem cells could also differentiate into functioning contractile bladder SMCs which help in the repair of specific form of urinal incontinence. Furthermore, the availability of adipose stem cells makes them ideal for trial and testing in the future of bladder tissue engineering comparative to other available stem cells (Jack et al., 2009).

3.4 Tissue engineering for bladder

Bladder damage can be caused by a variety of factors, including cancer, trauma, infection, congenital disorders, inflammation, and other genitourinary system conditions, and the damage caused by most of these factors will necessitate reconstructive

surgery of the bladder, which may include cystectomy of parts or the entire bladder, which can result in complications and a poor quality of life. This is where the importance of tissue engineering arises and we will discuss the current techniques and innovations that are being developed, as tissue engineering of the bladder (Fig. 3.1) may be able to restore damaged muscles and tissue and allow for certain conditions, such as urinary incontinence, to be better treated and managed, and aspire to provide a better quality of life for patients (Adamowicz et al., 2013).

For the replacement and repair of bladder tissue, gastrointestinal (GI) segments were most used. However, there are and have been some drawbacks with using GI segments because every organ and tissue is designed its specific function, and bladder tissue is designed for the temporary storage and expulsion of urine, whereas GI tissue is specifically designed to absorb specific solutes, and thus multiple problems arise when GI tissue is used to replace bladder tissue (Kaefer et al., 1997).

Problems associated with using GI tissue consisted of perforation, infection, increased mucus production, and malignancy (McDougal, 1992). Various treatments and procedures for bladder replacement and repair, such as ureterocystoplasty and autoaugmentation, have been trialled and proposed because of many of these issues. It has also spurred the use of regenerative medicine, such as cell transplantation and tissue engineering, to heal damaged portions of the bladder rather than reconstructing the complete organ, which is more intrusive (Atala et al., 2006).

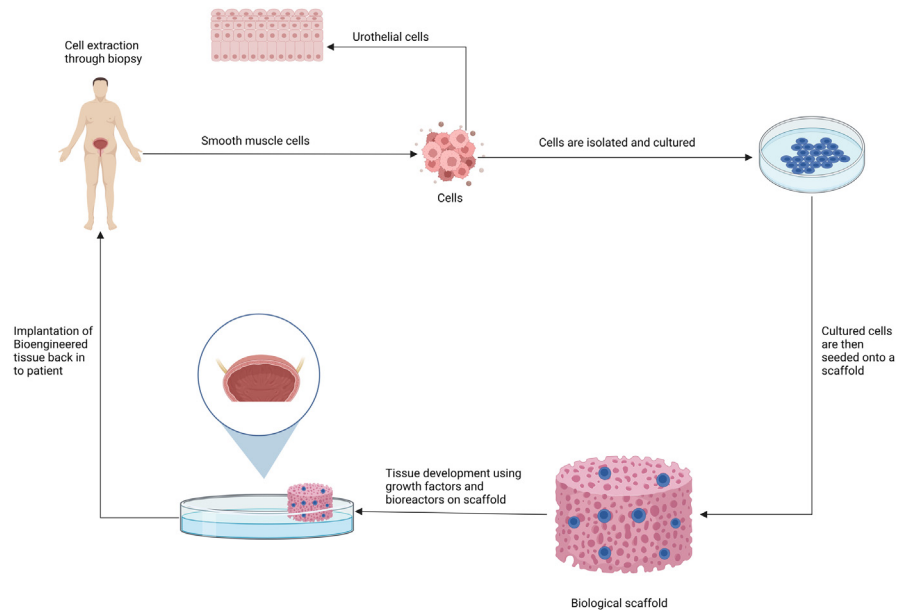


FIGURE 3.1

Bladder tissue construction cycle.

In bladder engineering, there are two common approaches called the cellular and acellular approach; the acellular approach aims to enhance the body's natural growth to regenerate and repair itself via the use of synthetic and natural biomaterials (Kim & Mooney, 1998). Whereas the cellular method, on the other hand, divides the separated donor tissue into individual cells. After that, the functional cells are either grown in culture before being seeded on a suitable medium to be implanted in the nonfunctioning section of the bladder, or they are immediately implanted into the host (Atala et al., 1992).

3.4.1 Engineering for stress incontinence

Treatment of urinary incontinence has seen the use of silicone prosthesis and synthetic materials by the implementation of artificial urinary sphincter and detachable balloon system by Levesque et al. In 1996, this specifically targets the stress-related incontinence, and it was found to be a durable long-term successful solution in children (Levesque et al., 1996).

In 2006, an investigation of seven patients was carried out within the ages of 4–19 who were all in need of cystoplasty due to being diagnosed with end-stage bladder disease, conducted by Atala et al. The group decided to use each patient's urothelial and smooth muscle cells via bladder biopsy and seeded them after seven weeks of culture. Each patient was suffering from a condition called myelomeningocele which is a type of spine bifida. The patients were then each supplanted with a neobladder created with their bladder cells, and three different neobladders were constructed by Atala et al. (2006).

Three types of constructs including: (1) homogenous decellularized bladder submucosa scaffold, (2) cell seeded collagen matrix implant coated with omentum, and (3) a collagen-PGA hybrid composite implant wrapped with omentum and fibrin glue were prepared by Atala et al. Results indicated that all of them were successful in their roles of declining the pressure level within the bladder. Type 3 had the greatest bladder capacity out of all three types and this is due to the enhanced elastic nature of the PGA and collagen hybrid matrix. Over time, patients who had their bladders repaired using engineered bladder tissue made with cell-seeded PGA-collagen scaffolds and omental covering had higher compliance, lower end-filling pressures, higher capacities, and longer dry periods. Therefore, this trial for a neobladder construct proved to successfully tissue engineer the restoration of a damaged section of the bladder (Atala et al., 2006).

3.4.2 Tissue-engineered cystoplasty treatment for neurogenic bladder

A neurogenic bladder is when a person has less control of their bladder due to the CNS which includes the brain and spinal cord getting damaged by a spinal cord injury (SCI) followed by spinal shock. This may induce the person to involuntary empty their bladder or cause them to urgently and frequently excrete (Taweel & Seyam, 2015).

The most common way to treat neurogenic bladder is through cystoplasty and bladder augmentation (Zhang & Liao, 2014). In cystoplasty, a segment of the abdomen, such as the large intestine, is taken and made into a patch to cover the damaged bladder that has been cut open for enlargement. This increases the capacity of the damaged bladder and relieves the patient of a few symptoms of bladder disease; however, incisions are made in the abdomen and bladder walls resulting in this method to be very invasive, and complications such as excess mucus production, malignancy, stone formation, and intestinal disorders can occur postsurgery (Turner et al., 2007).

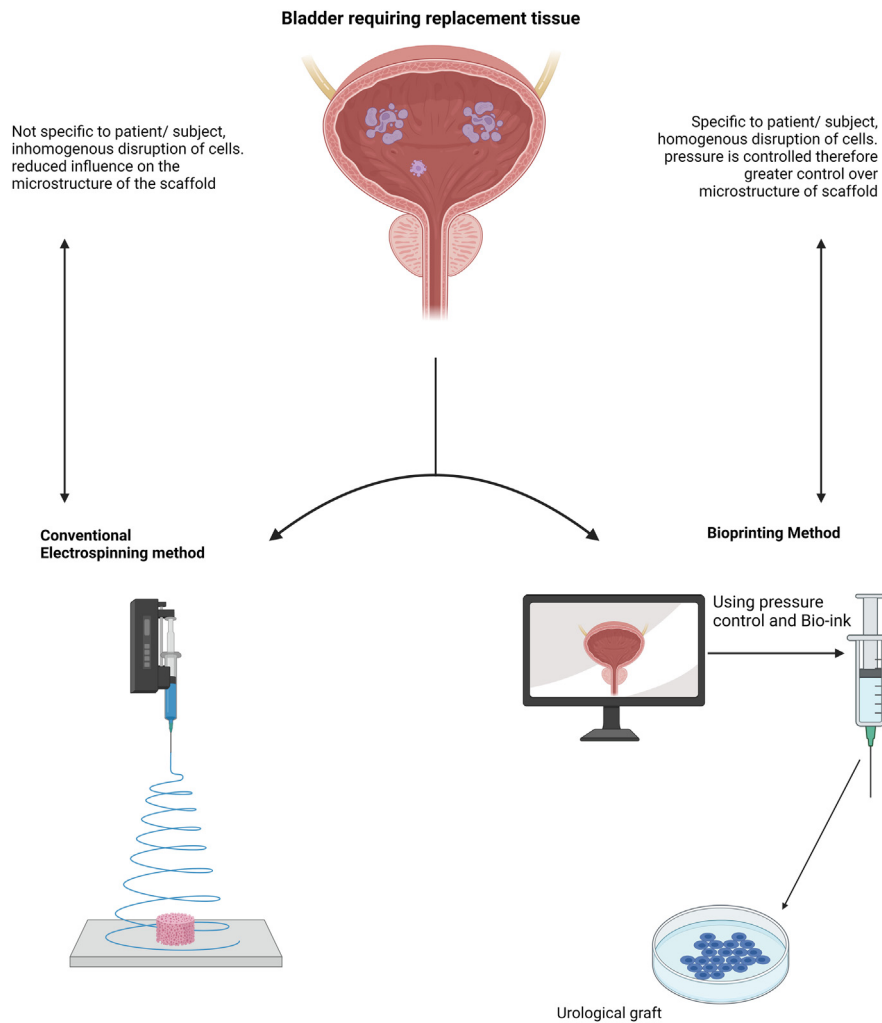
In 2014, Zhang and Liao conducted a study using small intestinal submucosa as a biological scaffold for reconstruction as an alternative to augmentation and cystoplasty for treating neurogenic bladder. Their findings showed that the maximum bladder capacity of the candidate's postoperation was significantly larger, and there was a decrease in maximum detrusor pressure at 7-month and 12-month follow-ups and that bowel function had returned swiftly. Furthermore, no unwanted metabolic implications were noted, and renal function was also preserved. This study proved that a functional bladder could be rebuilt by using SIS as a biological scaffold and that tissue engineering can be a potentially practicable technique in the treatment of neurogenic bladder condition (Zhang & Liao, 2014).

3.4.3 Urinary bladder bioprinting

Conditions such as bladder cancer and inflammation require augmentation of the bladder, which is the invasive method of removing tissue from the intestinal tract and then adding it on to the tissue of the bladder. Augmentation is performed to make the bladder larger; however, this method can result in many postsurgery complications (Cleveland Clinic, 2024).

The European Union funded a project by the name of UroPrint to engineer bladder tissue using laser-induced forward transfer (LIFT) bioprinting technology for the autologous transplantation of bladder tissue (Cordis.europa.eu, 2021). This would resolve the problem of the patient's body triggering a host immune response as the urothelial cells would be of the same person. The bioprinting technique (Fig. 3.2) has seen clinical success for simpler organ such as the bronchi which are much simpler than the bladder (MacChiarini et al., 2004).

In 2004, a bioprinted bladder engineered by Dr. Antony Atala was implanted into a child patient suffering from birth spina bifida, the condition caused the patient to have a defective bladder which resulted in the failure of the patient's kidneys. The patient would have to be put on to a dialysis machine for the rest of their life; however, the engineered tissue to be successful in allowing the patient to avoid dialysis and live a normal life. Since then, this bioprinted bladder has proved great clinical success as the tissue continues to function even after 14 years (Atala, 2011).

**FIGURE 3.2**

Grafting techniques: bioprinting and electrospinning.

3.5 Conclusion

The focus and goal of tissue engineering within the bladder organ has been to develop technology that can aid in the regeneration of diseased and damaged sectors within the bladder without having to compromise on the quality of life of the patient as the general treatments which include removal of the bladder include such compromises. However, through the advancement of time and technology not only have there been successful integration of polymer and acellular scaffolds that can aid

in the recovery of the damaged bladder walls (Sievert et al., 2007), which are the main part of the bladder that get affected by disease; there has been a breakthrough of artificially producing a whole bladder called the neobladder (Atala et al., 2006) that has shown a lot of promise within patients that may have had the need to have their bladder fully removed (cystectomy). However, there are still many issues that need to be resolved through trials and studies as there is a lack of data and research on using newly derived scaffolds and biomaterials on larger animals successfully where although in smaller animals have proven to be successful (Sievert et al., 2007) in a study, and another study proved to be unsuccessful (Zhang et al., 2006) causing various reactions and inflammatory responses within the test subject. The development of an ideal biomaterial has seen a lot of promise too as we now know that different polymers can be combined to create a stronger and advantageous composite polymer with enhanced properties.

Furthermore, knowing what the desired biomaterials are really gives direction to what can be achieved in the future as a paper by Horst et al., in 2012 discusses the desired environment for proliferation of different bladder cells that should be integrated into a biomaterial. Such as the luminal side of the biomaterial should provide for the urothelial cells and the visceral side should cater for the SMCs while also being a preventative barrier for urine leakage (Horst et al., 2013). One of the main concerns around embryonic stem cell use has been issued an alternative using adult stem cells which can be harvested easily and available in large quantity is still under clinical trial and experimentation, the breakthrough of this would result in a greater potential use of tissue engineering of the bladder (Lo & Parham, 2009).

The most promising approach to bladder tissue engineering in recent years has been bioprinting of tissues, and we believe that as bioprinting equipment and technology improves, it will become the main alternative to bladder augmentation due to its benefits of eradicating the problems associated with it, as successful transplantation has already been achieved in previous studies and has shown to be durable and functional while being specific to the patient.

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The progress in tissue engineering of urethra

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4.1 Introduction

A multidisciplinary field called tissue engineering (TE) integrates components of biology, material science, medicine, and engineering to provide novel strategies and treatments for tissue and organ regeneration (Pastorek et al., 2021). It discusses the utilization of scaffolds constructed from synthetic materials or extracellular matrix (ECM)-derived constituents for tissue reconstruction (Sengupta et al., 2014). Scaffolds in respect to biomedical engineering refer to designed supporting structures which expedite cellular growth and propagation through favorable biological contact, in addition to producing a biochemical and physical microenvironment that is comparable to natural tissue (Qiu et al., 2013).

The development of tissue engineering and regenerative medicine technologies provided a refreshing and prospective solution to address the lack of autologous tissues and resolve difficulties associated with employing conventional grafts (Xue et al., 2016). Fundamentally, tissue engineering is heavily dependent on three principles. (i) The successful seeding of cell onto manufactured scaffold. (ii) Scaffolded biomaterial to function as designed. (iii) The effective employment of growth factors to assist with environmental parameters such as cytokines.

Urinary bladder, urethra, and urinary sphincters compose the lower urinary tract, which is accountable for the major role in urinary continence mechanism through storage and expelling of urine. The male urethra in reference to mentioned tasks also functions as the transporter of seminal fluid from the verumontanum to the external urethral opening (Mousa et al., 2015). This organ consists of two significant functional tissues, urothelium layer and a smooth muscle coat, prohibiting reabsorption of toxic remnants within urine through its passage of urethra due to specialization of the urothelium layer.

Currently numerous urethral abnormalities exist which may necessitate substantial urethral regeneration, particularly procured hereditary conditions including hypospadias, epispadias, and strictures (Chung et al., 2014). If left unattended, these

abnormalities may aggravate into impediments causing pain, urinary tract infections (UTIs), or even kidney disease (Rashidbenam et al., 2019). Correspondingly leading to greater complexities of treatment impeding possible options for individuals, regardless of the abundance of urethral stricture repairment options available for noncomplex abnormalities. If urethral stricture is examined and determined as noncomplex, a simple anastomosis methodology is highly feasible for treatment. However, a greater complexity of ailment demands repair through more novel complex methods, still under examination such as incorporation of grafts through tissue engineering (Raya-Rivera et al., 2011).

4.2 Etiology and anatomy

The urethra is anatomically connected to the reproductive system. Hence, the property of male urethra is distinctly dissimilar from the female counterpart. An adult male's urethra is approximately 7–8 inches in length, accountable for both the transportation of semen from seminal vesicles into ejaculatory ducts interconnected from either side in addition to the emission of urine from the bladder.

Numerous disease disorders that impact the urethra can decrease quality of life or potentially result in organ dysfunction. Prevalently occurring urethral irregularities include acquired urethral disorders like urethral strictures and congenital birth impairments such as hypospadias occurring approximately (1 in every 1000 men) (Mundy, 2006) and (1 in every 300 births) (Pierik et al., 2002), respectively. The urethra is segmented into various sections, each of which experiences strictures on a variable frequency. The urinary bladder exiting into the urethra neck is initially encountered by the posterior urethra, a tissue collectively composed of the prostatic urethra and membranous urethra. Whereas the anterior urethra located further down encompasses the bulbar urethra, a subdivision of a mass of tissue surrounding urethra named spongy urethra. The section where strictures occur most frequently (approximately 50%) (Tritschler et al., 2013). Recorded cases of stricture in the posterior region are infrequent and generally occur due to traumatic urethral rupture or cancer treatment therapies (Jordan et al., 2006).

Majority of urethral strictures in adulthood occurs because of scarring, remodeling the corpus spongiosum culminating into ischemic spongiofibrosis. Gradual degradation of structure due to scar tissue formation diminishes the lumen inevitably impeding lower urinary tract (Davis et al., 2016).

The entrance of the female urethra is positioned between the labia and is implanted into the vaginal wall. The female urethra is just 4 cm (1.5 inches) length, substantially shorter than the male. Consequently, resulting in stricture occurring less frequently in females (Ryu & Kim, 2001).

4.3 Tissue engineering of urethra

The advancement of regenerative medical technologies has proven to provide a modern and promising solution to address the issues of urethra treatment. This is

made possible through a fusion of the three main components of regenerative medicine which are the cells that are seeded on the chosen scaffold, the biomaterial that serves as a substratum, and the surrounding stimuli including cytokines and growth factors. These three essential elements, collectively known as the “tissue engineering triad,” are fundamental to the production of an effective tissue engineering construct for urethral repair. The application of the right signals and growth factors, a pertinent cell collection, and a scaffold constructed from biocompatible material are essentially all required to function in conjunction to coordinate the formation of new tissue (Almouemen et al., 2019).

Regenerative medicine of the urethra follows two basic approaches for replacing the ureter; using cell-seeded matrices and noncellular matrices (Xue et al., 2016).

However, more emphasis has been placed on the cell source employed and various studies have laid credence to this fact as it is noted that the cell source is an essential component needed for successful urethral tissue engineering.

Additionally, research suggests that in cases of urethra defects, the use of tissue regeneration and restructuring of the urethra, in the event of short, simple urethral strictures, can easily solve the problem through the end-to-end anastomosis. The lengthy and intricate urethral deformities, on the other hand, are treated with transplants or a fold at the injured area (Raya-Rivera et al., 2011).

4.3.1 Cell sources

The success of tissue engineering is greatly influenced by the cell source. Cells suitable for tissue engineering can be categorized as autologous (patient’s native), allogenic (human sourced not from patient), and xenogeneic (cell of animal origin). Autologous cells currently stand as the best suitable source for tissue engineering. As opposed to allogenic and xenogeneic cells which are probable in insinuating an immune response making them immunogenic, often requiring immunosuppressive therapy. However, an issue remains with cell-derived autologously and in their challenge of extracting adequate cells particularly when a patient is elderly or has a serious illness (Ikada, 2006). Cells utilized in urethral reconstruction can include human urothelial cells (HUCs), urethral epithelial cells, and fibroblast displayed by Table 4.1.

Currently a comprehensive range of cell sources for urethral reconstruction have been explored and evaluated. Commonly, two classifications have been recognized as the predominant cell types for the utilization of urethra engineering, progenitor cells, and stem cells (Rashidbenam et al., 2019).

4.3.1.1 Progenitor cells

Progenitor cells are stem cell successors that undergo additional cell type differentiation for specialization of their role. The human anatomy contains a variety of progenitor cell types where each progenitor cell only undergoes differentiation into cells from the same tissue or organ, which have been greatly utilized in tissue engineering. These include epidermal cells derived autologously

Table 4.1 Display of previously utilized cell sources and cases in which they occurred.

Case type	Utilized cells	Scaffolding type	Proceeding study duration	Method of measure	References
White rabbit	Epithelial cells derived autologous	Modified PLLA	2 days	Electron microscope	Fu et al. (2011)
White rabbit	Fibroblast and keratinocytes	Poly(lactic-co-glycolic acid) (PLGA)	2 months	Electron microscope/microanatomy analysis	Selim et al. (2011)
Rabbit	HUCs	Tubular adipose-derived stem cell-based self-assembled scaffold	2 weeks	Electron microscope/microanatomy analysis/microscopic analysis using fluorescence dyes	Cattan et al. (2011)
Beagles	Urothelial cells	Silk fibroin	5 months	Electron microscope/microanatomy analysis/analysis in conjunction with Trypan blue dye	Xie et al. (2013)
Human patients	Fibroblast and buccal mucosa autologous	Sterilized donor de-epidermised dermis	3 years	Cystoscopy analysis of urethra	Bhargava et al. (2008)

(Rashidbenam et al., 2019), autologous keratinocytes (Bhargava et al., 2008), epithelial cells obtained autologously (Raya-Rivera et al., 2011) in addition to smooth muscle more however infrequently.

An experiment conducted by Rogovaya et al. examined the feasibility of utilizing living skin equivalent (LSE) composed of fibroblast and keratinocytes in regeneration of the urethra. Cells were acquired from rabbit keratinocytes cultured upon collagen gel for the formation of desired LSE. The findings demonstrated that all experimental rabbits were able to undergo independent urination rather swiftly as 4–7 days following grafting keratinocyte seeded LSE on de-epithelialized urethra, with no evidence of urethral stricture or constriction further developing. The urothelium at the afflicted region was completely operational and structurally recovered three months following implantation, indicating that LSE may be utilized as a viable graft in urethroplasty with its procedure provided by Fig. 4.1, in addition to removing issues related to hair development in the neourethral lumen (Rogovaya et al., 2015).

Successful reconstruction of anterior urethra was again demonstrated through utilization of progenitor cells, directly sourcing epidermal cells derived through rabbit foreskin. Regeneration of 1.5×1.0 cm urethra was accomplished by implanting 5-bromo-2'-deoxyuridine epidermal cells derived autologous onto tabularized acellular collagen matrix accurate in nature to original tissue, in which results expressed restoration of urethral mucosal defect in six months by establishing a multilayered epidermal cells structure. To demonstrate effects of cell-seeded scaffolding, control groups were treated with nonseeded scaffolding in which localized migration of cells occurred, which progressively led to urethral stricture. Demonstrating the importance in seeding progenitor cells onto scaffold prior to implantation (Fu et al., 2007).

Tubularized substitution urethroplasty

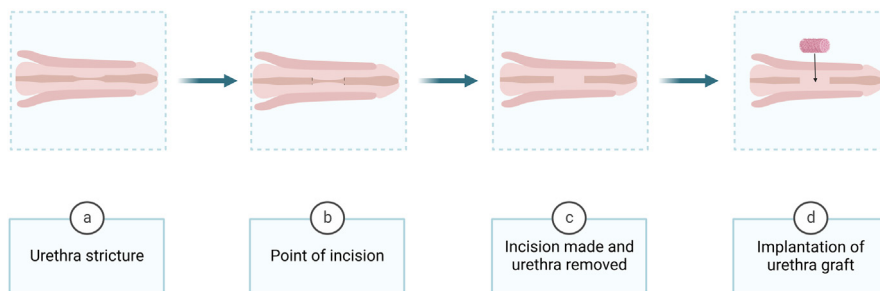


FIGURE 4.1

General proceeding and steps which occur for a tabularized substitution urethroplasty. Where incisions are made at either end of the stricture site, followed by its removal. A manufactured graft is then implanted at the site of incision to assist in reconstruction of urethra.

Other researchers have aimed to prove and investigate the feasibility of the tissue engineering approach in treating urethral strictures. Raya-Rivera et al. employed autologous urothelial and smooth muscle cells from patients to evaluate the efficacy of tissue engineering urethras. To replicate the microscopic structure of the urethra, tabularized polyglycolic (PGA) scaffold was seeded by surface muscle cells in addition to luminal (urothelial cells). The scaffolds fabricated were used with intention for urethral reconstruction, on five patients which portrayed previous experiences of unsuccessful posterior urethral repair and complete urethral strictures. Results were analyzed following a three months waiting time, which displayed a now conventional composition of tissue engineered urethra with no abnormal histological deviations reported (Raya-Rivera et al., 2011). These excellent results were used as evidence of successful application of tissue engineering methodology in treating strictures, providing great encouragement into its implementation in a commercialized setting within a near future.

4.3.1.2 Stem cells

Stem cells are undifferentiated cells which have the capacity to develop into urothelial cells, have displayed potential in their employment to construct the urethra (Mousa et al., 2015). Application of stems cells as a substitution to progenitor cells could be due to a variety of reasons, progenitor cells scarcity or derivation from unhealthy or malignant origins that are inappropriate for tissue regeneration, then stem cells might be regarded of as an alternative (Subramaniam et al., 2011). Adult stem cells and embryonic stem cells have both been investigated in relation to applications; however, the use of embryonic stem cells is constrained because of limitations related to their biological recognition and rejection (nonautologous embryonic stem cells), ethical implications, and their propensity to develop tumors such as teratomas when transplanted in vivo (Mahfouz et al., 2013).

The potential use of stem cells regarding urethral reconstruction was demonstrated by Takahashi and Yamanaka in 2006, through gene reprogramming of adult cells to produce pluripotent stem cells. Their study demonstrated the possibilities of a two-step differentiation procedure to derive definitive endoderm further proceeding into urothelial cells using pluripotent stem cells and the appropriate defined medium (Takahashi & Yamanaka, 2006). Stem cells obtained using this procedure originating from human CD34⁺ hematopoietic stem cells in conjunction with human foreskin produced favorable results of UP1a (64.9%) and UP1b (58.7%) protein expression (Osborn et al., 2014).

4.3.2 Biomaterial scaffolds for urethral regeneration

Just as previously mentioned regarding careful consideration of cell types to use for urethral reconstruction, the same is also paramount when considering biomaterial with varying composition for use in scaffolding. A scaffold must be topographically sound to enable localization of native cells and allow for reconstruction with sufficient arrangement and composition of cells to form neo-tissue in addition to

supplying high surface area for efficient cell attachment. The ideal scaffold must be adequately porous to grant diffusion of biomolecules efficiently but not contain such porosity where the mechanical structure is compromised (Wen et al., 2007). It should pertain linked micropores to permit for the seeding of several cells, their migration into the interior, their expansion, and their proliferation, as well as supplemented with enough nutritional levels. Key features that require careful consideration when constructing a viable scaffold include surface parameters such as roughness (Fu et al., 2011), porosity in terms of volume and size (Stachewicz et al., 2019), in addition to mechanical attributes such as elasticity and tensile strength (Figallo et al., 2007). Furthermore, as the scaffold is intended for biological application, a scaffold should permit standard sterilization without enduring detrimental effects on its ability (Selim et al., 2011). The kinetics of the scaffold's absorption, which is dependent on the tissue that must be rebuilt, are also very significant. The scaffold is only required to persist within body postimplantation for approximately one month in tissue reconstruction comparative to bone which requires a longer duration for its degradation. The scaffold may potentially impede rather than encourage tissue regeneration if it is employed for a longer period than is ideal. This suggests that the success rate of tissue engineering will be significantly impacted by the scaffold material's absorption kinetics.

As it stands, currently three classifications of biomaterials have been applied in tissue engineering of genitourinary tissue which will be explored in further detail. Classifications include (i) decellularized tissue matrices such as decellularized urethra (Simões et al., 2017), (ii) naturally derived biomaterials such as collagen (Pinnagoda et al., 2016) and silk fibroin (Chung et al., 2014), and (iii) synthetic polymers which include poly-L-lactic acid (PLLA) (Fu et al., 2011), poly-ε-caprolactone (PCL) (Zhang et al., 2015), and polylactic acid (PLA) (Wang et al., 2015). A list of biomaterials previously utilized in urethral reconstruction are displayed by Table 4.2.

4.3.2.1 Decellularized matrixes

Porcine bladder smooth muscle (BSM) derived through decellularization of submucosal layer of bladder has demonstrated capabilities in its utilization for urethral regeneration (Simões et al., 2017). This biomatrixes is collagen based in principle and xenogeneic. Pertaining many key features mentioned previously for a viable scaffold, such as its biocompatible and biodegradability in addition to its nonimmunogenic reaction. The process of decellularization is exhibited by Fig. 4.2 where native cells are removed while simultaneously maintaining preserves a tissue's native environment, elevating the success rate of implantation void of immunogenic responses.

Attempts have been made to apply this biomaterial while seeking to consolidate its effectiveness which incorporated BSM with urethral muscle collected autologously from rabbits for treatment of stricture. In this investigation, results following a 12-week therapy session postapplication displayed a standard urethral lumen reconstruction. Demonstrating the successful possibility in biomaterial as well as

Table 4.2 Displays a diverse range of biomaterials utilized in urethral reconstruction and their rate of success in amending strictures.

Biomaterial employed	Surgical approach	Position	Proceeding analysis duration	Results	References
Small intestinal submucosa PLGA	Dorsal onlay graft anterior urethroplasty	Anterior urethra	91 weeks	The study retrieved results displaying 85% success in patients	Palminteri et al. (2007) Raya-Rivera et al. (2011) Barbagli et al. (2018) Mandal et al. (2020)
	Posterior urethroplasty	Posterior urethra	308 weeks	A success rate of 100% repair was achieved	
Oral mucosa from MukoCell®	Dorsal onlay	Stricture in bulbar	239 weeks	82% repair in patients was reported	
Bovine pericardial patch	Dorsal onlay	Urethral stricture within anterior region	34 weeks	89% success in stricture repair was achieved	

The Decellularization Effect on the Extracellular Matrix (ECM)

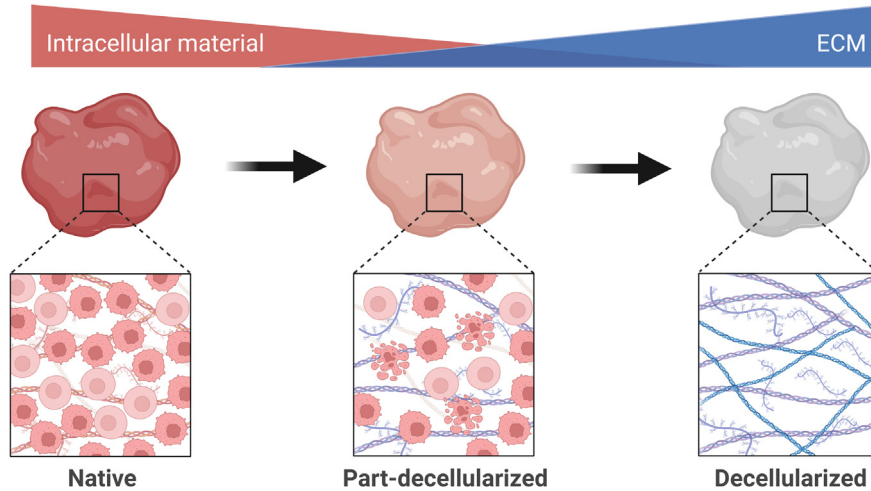


FIGURE 4.2

Exhibiting the process in which decellularization of ECM is achieved, in which key components are maintained within to a greater likelihood of urethral reconstruction.

enhancement of its application. In addition, epithelialization, and permeation of blood vessels onto graft was clear, further indicating its success (Chun et al., 2015).

Small intestinal submucosa (SIS) is also an example of decellularized matrices. The aim of this study was to compare SIS and BSM on their regenerative abilities when seeded with HUCs. Urothelial cells were sourced from human bladder biopsy and applied in a vitro investigation. Both SIS and BSM have been utilized and established thoroughly in previous investigation; however, BSM displayed greater proficiency in proliferation and viability in accommodating HUCs. This illuminates that a possible distinction in SIS and BSM matrix components exists which cause behavioral adjustments on seeded cells (Davis et al., 2011).

4.3.2.2 Natural material for scaffolding

Naturally produced scaffolds are regarded as a secure alternative for tissue engineering applications due to the potential for disease transmission and ethical concerns related with the usage of animal-derived scaffolds (Mousa et al., 2015). An example and a popular material for naturally derived scaffold is silk fibroin, derived through *Bombyx mori* cocoon with its manufacturing process displayed by Fig. 4.3. This biomaterial pertains approximately 90% glycerine, serin, and alanine amino acids and exhibits total degradation through proteolytic enzymes which develop naturally. Furthermore, the biomaterial expresses excellent biocompatibility and minimal inflammatory attributes, as well as promoting adequate proliferation, hence its

Silk Scaffolds: Fabrication and Application

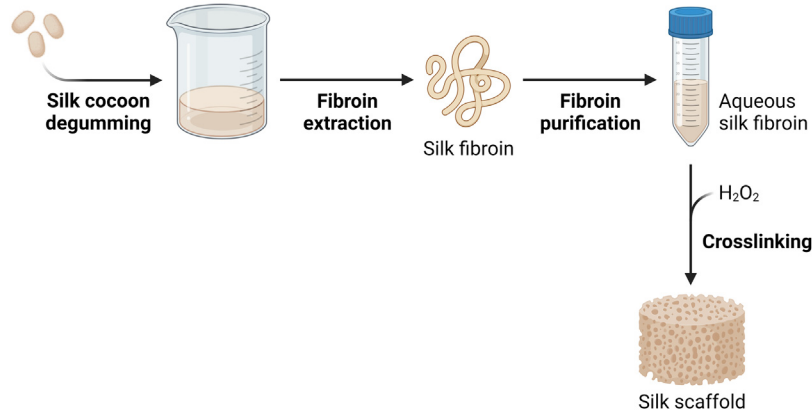


FIGURE 4.3

Process in which silk are applied to manufacture porous scaffolds that maintain sufficient rates of degradation.

elevated volume in application for urological defects. However, a setback in its application is its comparably low mechanical strength when woven into matrices. Another group attempted to amend this issue by extending the material within 90 vol% ethanol solution and developing its mechanical strength. The extension occurred at a rate of 0.1 mm/s for a full $1.4\times$ ratio extension. This method aimed to enhance the order of alignment between the molecules and assist in increasing parallel lining of the fibers to one in injunction with pathway of extension. The completed fiber was implanted in beagle dogs following seeding of urothelial cells with scaffold displaying high porosity. In vitro results contained a multilayer of urothelial cells tightly arranged within the surface of the scaffold. Examination of test in vivo revealed promising results, with no evidence of urethral contraction or ulceration occurring following six months postsurgery, even forming an epithelium. The control group which received no treatment eventually generated strictures and fibrosis (Xie et al., 2013).

4.3.2.3 Synthetic material for scaffolding

Synthetic polymers are frequently employed in applications of tissue engineering, due to their efficiency in providing great biocompatibility as well maintaining adequate mechanical strength to enable the construction of constructs such as scaffolds with hollowed structures. The capacity of the scaffold to enable cell adhesion, development, and proliferation as well as maintaining sufficient porosity (Fig. 4.4) have a significant influence on the success of creating a desired tissue (Fu et al., 2011).

Bulk Properties of Scaffolds for Tissue Regeneration

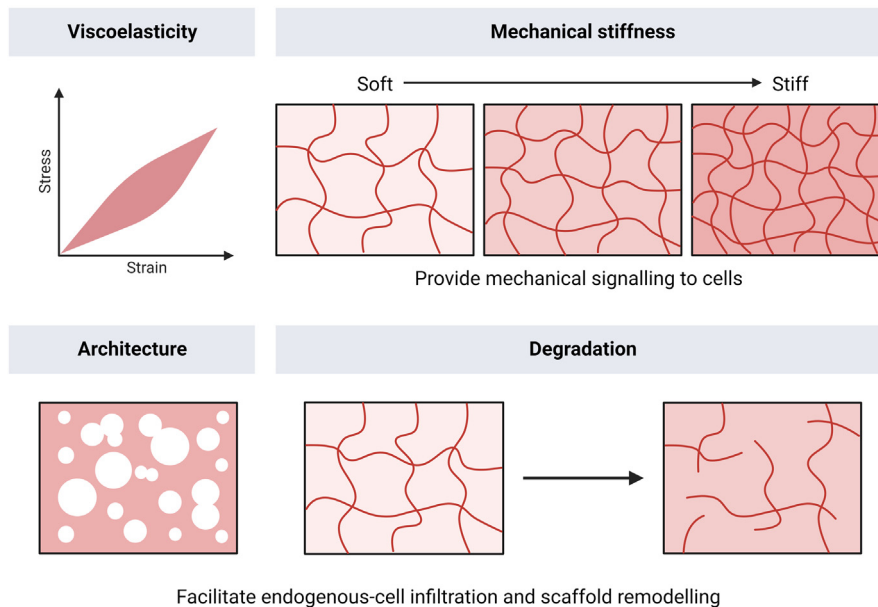


FIGURE 4.4

Bulk properties of scaffolds, all scaffolds follow this general rule. Where parameters such as mechanical properties and rates of degradation as well as rate proliferation are interlinked as dependent variables.

Sartoneva and colleagues studied the capacity for acellular human amniotic membrane (hAM) derived from human placenta and poly-L-lactic-co-e-caprolactone (PLCL) in promoting proliferation and aid in viability of HUCs. Upon examination of PLCL and hAM post 3 days seeding by urothelial cells, confluency and viability of PLCL and hAM displayed great retainability. However, as time progressed to 14 days postseeding results emerged which revealed an intriguing product. PLCL scaffold still maintained confluency and its viability in opposition to hAM in which cell apoptosis had occurred in volume. Furthermore, continuous assessment only reaffirmed the findings, whereupon cell population on PLCL further increased whereas hAM scaffold displayed no growth in cell confluency from a 7-day period onwards. Therefore, PLCL had demonstrated its superiority comparative to hAM as more supportive concerning proliferation ([Sartoneva et al., 2011](#)). Moreover, assessment of cytokeratin status of uroplakin III pronouncement and CK19 demonstrated the ability of PLCL to conserve phenotype of HUCs within the 14-day culture duration, unlike hAM where phenotype expression was not perceived. Therefore, PLCL performance appears to be a promising synthetic biomaterial for the construction of scaffold regarding urothelial tissue reconstruction ([Rashidbenam et al., 2019](#)).

In order to be effective in cell-based tissue engineering, a synthetic polymer must be capable in enabling cell adhesion. However, establishing the ideal environment for cell attachment and proliferation is an intricate process to perfect, as surface chemistry of scaffolds constructed synthetically may not produce compatibility with specific cells. An attempt to amend this intricate issue by enhancing synthetic polymer parameters which established a novel modification to PLLA surface to promote superior adhesion and proliferation. In this assessment, a tabularized scaffold was constructed using a PLLA solution which was further encompassed using nonknitted PLLA filaments. To further ameliorate the constructed scaffold for adhesion, 2% chitosan from shrimp shells was utilized and administered followed by fibronectin. The evaluation of derived results disclosed a porosity volume greater than 90% on the manufactured scaffold, which expressed exceptional biocompatibility, proliferation, encouragement in adhesion of rabbit urethral epithelial cells. The exterior nonknitted filaments supplied sufficient friction to prevent the scaffold from becoming dislodged from the initial location of implantation, whereas the interior hollow component of the scaffold facilitates drainage of urethral discharge, hence reducing the probability of infection (Fu et al., 2007).

The viability of synthetic biomaterial and its reconstruction abilities in tissue engineering applications were assessed by Naji et al. In this study, a hybrid scaffold was woven using the electrospinning method with a combined solution of PLLA and PCL, seeded by urothelial cells. However preceding seeding of cells onto scaffold, the PLLA/PCL construct was initially remedied utilizing oxygen plasma to achieve greater wettability, by transmuting the extremely hydrophobic nature of the biomaterials into a more considerable hydrophilic nature which was achieved, acquiring 100% wettability. The findings indicated that the PLLA/PCL construct could maintain urothelial cells survival and proliferation without compromising their phenotype since after 14 days of cell culture, an interconnected and consistent layer of urothelium was established (Naji et al., 2015).

4.3.3 Growth factors

There exist several distinct proteins essential for cell growth and differentiation. These proteins are autonomously released by cells (autocrine) or as a consequence of interactions with neighboring cells in the body (paracrine). These proteins are an essential component of tissue engineering and are responsible for stimulating proliferation and communication between cells and structures, named cell growth factors or plainly growth factors. These proteins have experienced a large volume of investigations latterly in tissue engineering assessments. The frequently utilized growth factors include vascular endothelial growth factor (VEGF), fibroblast growth factor (bFGF or FGF-2) in addition to transforming growth factor- β (TGF- β) (Ikada, 2006). The extraordinary capacity for influence by growth factors is that the presence of bFGF solely can instigate regeneration even with the absence of scaffolds or appropriate cells. In compared without the employing growth factors, the inclusion of the appropriate growth factors to a cell-scaffold structure must continue to enhance

tissue regeneration (Ikada, 2006). Studies that have been publicized in the recent years have demonstrated and displayed the capacity in which inclusion of growth factors can modify biomaterial to encourage proliferation of stem/progenitor cells and develop regeneration (Chan et al., 2020). VEGF is crucial for angiogenesis. It can encourage the growth of new blood vessels and relocation of endothelial cells. A few investigations also revealed that, in pathological circumstances, VEGF decreased cell death and increased cell viability. However simultaneously the sudden excretion of VEGF from manufactured scaffold will actually lessen the therapeutic benefits and have unfavorable side effects (Zhang et al., 2009). Therefore, Wang et al. attempted reconstruction of urethra with a modified collagen-binding VEGF based on this projection. The study employed three different groups for cross-comparison of results which included urethral injury control group, bovine acellular dermal matrix implanted group as well as bovine acellular dermal matrix possessing VEGF implantation group. In order to evaluate urethral regeneration and urination performance urethrography, urodynamic testing, and urethral histological examination were conducted as means of measure to assess viability following a 6-month postsurgery process. Physiological examination exhibited a greater arrangement of blood vessels among beagles administered with bovine acellular dermal matrix possessing VEGF comparative to control group which saw escalation in strictures as well as group implanted with scaffold containing no growth factors, which experienced improvement in strictures however not proportionally to VEGF group. It was concluded that the inclusion of a growth factor such as VEGF optimizes and elevates reconstruction of tissue while simultaneously preserving urethral function (Wang et al., 2021).

The method of delivery for growth factors to the desired site of activity deserves analytic contemplation due to its critical importance. Conventional bolus injection doesn't provide the optimum circumstance for solution, as the hormones diffuse rapidly from site of injection. Therefore three distinct approaches to delivery have been attempted. An exciting methodology is delivery through utilization of DNA plasmids which encode the preferred growth factors. Upon injection and recognition by the body, growth factors are biosynthetically produced by plasmids and excreted to site by cells (Aguilar et al., 2019).

The second strategy involves the application of gene technology. Alternatively, genes possessing programs for growth factor development are directly transported to urothelial cells, or cells deemed appropriate for urethral reconstruction through deployment of vectors. These altered cells are relocated to the site of operation for chosen tissue and allowed to diffuse and interact for reconstruction, given that engineered genes remain within host (Vo et al., 2012). Finally, the last method, the one most accosted and frequently examined, involves the direct transmission of growth factor to desired site in conjunction with a carrier. In this approach, nature of carrier employed has tremendous influence on growth factor released kinetics. The prerequisites for a carrier are to enable growth factor molecules to undergo denaturation to which quaternary structure is lost as minimally as possible during their integration or confinement, for ideal absorption (Qian et al., 2019).

4.4 Clinical issues and discussion

Tissue engineering was enthusiastically received in the late 20th century but didn't gain its current wave of popularity until later. To illustrate, publication analysis on Pubmed upon search of key words such as "tissue engineering" revealed over 4000 studies attempting reconstruction in 2010 onwards. As opposed to fewer than 400 publicized studies under identical key word search before the start of the 21st century (Fisher & Mauck, 2013). This research study sought to compile the developments in urethral tissue engineering during the last 2 decade. Discovering a setback in the development of tissue engineering prospects is the lack of consensus within tissue engineering on optimum parameters for urethral engineering.

Surgical effectiveness isn't dependent solely on the abilities of the surgeon but more importantly on the accessibility of the appropriate tissue. A diverse range of tissue have been examined, such as the bladder and buccal mucosa as well as skin grafts, which have seen relative success. Nevertheless, in comparison to urethral tissue derived autologously, alternatively employed sources of tissue maintain drawbacks that could cause difficulties for instance graft failure (Ramsay et al., 2016). Particularly appealing approaches are tissue-engineered frameworks that include ECM in conjunction with autologous cells. However autologous derivatives and the volume of tissue collection are limited for clinical application, particularly in extensive defects, which places further restraint for progress to grasp sustainable clinical applications. Furthermore, the introduction of stem cells provides alluring possibilities especially by a study which demonstrated the possibility in extracting stem cells directly from urine. Nevertheless, the quantity of collected stem cells is frequently insufficient for therapeutic applications meaning the collected cells must be multiplied through cell culture. Unfortunately placing them in a difficult situation, due to their often-poor proliferative capacity (Bharadwaj et al., 2011).

The substantial culture durations needed for tissue engineering models could prohibit their capacity for application in a clinical setting, but as reconstructive urethral procedures are often conducted on a discretionary basis, this would only be a minor inconvenience. Numerous attempts have been regulated to stimulate neo-vascularization utilizing different growth factors in order to provide the cells involved in tissue regeneration with adequate nutrition. However, growth factors are relatively expensive, the reduction in costs and wide availability of growth factors will experience an accelerated utilization worldwide. The key disadvantages of the approach are its high manufacturing costs and lack of readily available off-the-shelf components; however, these issues may be avoided with cautious commercialization and suitable scientific research.

Another challenge with tissue engineering possibilities is that employing TE models solely in situations with significant complexities will render it difficult for tissue engineering to demonstrate its value. Furthermore, although several triumphs in animal trials had been documented, relatively few research ever proceeded on to the clinical phase (Moysidou et al., 2021).

Overall enhancement of tissue engineering generally in opinion upon research on circumventing its major disadvantages would be in achieving an increase in rate of vascularization, prevention of inflammatory response by native tissue in addition to attaining tissue engineering materials such as the aforementioned triad more commonly (scaffolding, cultured cells, growth factors).

In conclusion, tissue engineering with aims of urethral reconstruction is a methodology yet in development, where a substantial effort worldwide is still applied to gaining a better understanding and control over its parameters. The constant development of biomaterials and the introduction of stem cells provides exciting possibilities; however, the field is still far away from commercialization to achieve reconstruction in a clinical setting reliably.

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The progress in ureter tissue engineering

5

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5.1 Introduction

Over the last few years, technological improvements have accelerated the development of numerous medical procedures in urethral tissue engineering (Adamowicz et al., 2012; Caneparo et al., 2021). Using stem cell proteins for regeneration proposes is the most promising but ethically challenging. These cutting-edge procedures have been helped by improvement in ethical acceptability of these procedures and the use of animals, whereas previously the research indicates general principles of ureteral injury reconstruction (Png & Chapple, 2000).

It was not always this promising for urethral injuries in women. Ureter was known to be frequently injured during pelvic, gynecological, and abdominal operations, but was not a huge risk (Lezin & Stoller, 1991), but ureteral injuries during procedures do increase morbidity rates (Elliott & McAninch, 2006). For example, on such injury, ureteral fistula treatment of 303 women from 1970 to 1985 showed the majority treated by gynecological surgical intervention, and stem cell culture was unheard of (Lee et al., 1988).

Medical industry has therefore used technological advancements in biotechnology to simplify surgical procedures. An example is the increased use of endourology for ureteroscopy and arthroscopic procedures, recorded highest globally during the previous decade. Complications, however, with these procedures are frequently underreported, possibly due to a gestation period and low postoperative follow-up. Among these uterine complications, “narrowing” happens with an occurrence rate of up to 3.5% during ureteroscopy and is more prevalent as compared to “avulsion” or significant puncture. Around 1% of gynecological procedures (Dandolu et al., 2003; Dowling et al., 1986; Johnson et al., 2012; Vakili et al., 2005) involve ureteral lesions, but add up to 73% of all ureteral lesions—thereby indicating

a major source of occurrence. Laparoscopic surgery injuries to the ureter are also part of the mix ([Ostrzenski et al., 2003](#)) with 77 injuries in 2491 incidences between 1966 and 2003. Apart from these iatrogenic injuries, ureteral lesions can be caused by trauma too ([McDougal et al., 2015](#)) with database between 2002 and 2006 of 582 patients ([Siram et al., 2010](#)).

5.2 Ureters anatomy

5.2.1 General anatomy

The ureters are about 25 cm and 6 mm long and wide, respectively, and there are two present in the body one attached to each kidney doing the same thing which is carrying urine to the bladder. Ureters are continuation of the renal pelvis and begin at L₂ level and the distal end of the ureter ends at the bladder. They exit the abdomen to enter the pelvis at the pelvic brim. Ureters then curve medially to open into the posterior wall of the urinary bladder. This curvature has a clinical importance as it prevents backflow of the urine into the ureters when increased amount of urine causes the bladder wall to stretch ([Ellaine et al., 2017](#)).

5.2.2 Microscopic anatomy

The walls of the ureter are divided into three layers which are mentioned below:

- **A mucosa:** This layer helps in stretching which is made up of transitional epithelium so that when urine passes through the ureter it can stretch to allow easy flow of the urine down to the bladder. Ureters also consist of a lamina propria which is made up of a stretchy, elastic connecting tissue which has rare pieces of lymphoid tissue.
- **A muscularis:** This middle wall of the ureter is divided into two different layers an inner longitudinal layer and an outer layer make up of smooth muscles. A third layer also appears for this middle layer, an external longitudinal layer which can be seen in the inferior third of the ureter.
- **An adventitia:** This wall of the ureter is usually made up of connective tissue ([Ellaine et al., 2017](#)).

5.3 Physiology

The ureters are supposed to carry urine from both kidneys to the bladder so the waste products can be removed from the body. It can be thought that gravity might play a big role in transporting the urine down to the bladder, but in this case, urine would not be transported to the bladder when the person is laying down. In fact, urine is even transported to the bladder even when the person is standing on his head as the ureters play an active role in the transportation of urine to the bladder. The urine

is pushed down the bladder by the contraction of the smooth muscle cells (SMCs) that squeeze the urine down the ureter into the bladder known as peristalsis. Once the urine is emptied into bladder, a valve which is made up of like folds of bladder mucosa prevents the backflow of urine into the ureter ([Ellaine et al., 2017](#)).

5.4 Diseases of ureter

5.4.1 Congenital anomalies

Congenital anomalies of the ureters are not that common but are found in 2%–3% of all autopsies. Even this low number is not that dangerous, but in some of the cases, this might lead to blocking of urine which would be dangerous and of clinical significance. Different types of congenital anomalies of the ureters are mentioned below.

5.4.1.1 *Double and bifid ureters*

This happens due to an error or defect when the ureter is being developed. In double ureter, the ureter splits into two different ureters that carry urine probably using different routes but from the same kidney and transport it to the bladder entering from different positions. This has no clinical significance as it does not alter or block the flow of ureter.

Bifid ureter is a congenital anomalies disease where two different ureters originate from the kidney but unite before entering the bladder. This has no clinical significance but as urine still flows from the kidney to the bladder in a proper manner ([Kumar et al., 2014](#)).

5.4.1.2 *Ureteropelvic junction obstruction*

In this case, there is a blockage at the point where the kidney drains into the ureter leading to less urine being transported to the bladder and in return increasing the pressure inside the kidney known as hydronephrosis. This is more common in females than males and usually it effects one kidney at a time. Ureteropelvic junction obstruction can be due to the abnormal organization of SMCs at the ureteropelvic junction or could be due to the collagen accumulation in between the smooth muscle bundles or another reason could be the constriction of the ureteropelvic junction by the renal vessels ([Kumar et al., 2014](#)).

5.4.2 Tumors and tumor-like lesions

Around 6%–7% of cancer in the ureter originates from the renal pelvis. A tumor named as fibroepithelial polyp is present in mostly children is lesion that presents as a mass projecting as a lumen. The ureters primary malignant tumors are somewhat similar to the tumors present in the kidney, bladder, and urethra. The most common tumor of the ureter is called the urothelial carcinomas and usually occurs in the later stage of life ([Kumar et al., 2014](#)).

5.4.3 Ureter stones

Stones present in the ureter might have been formed in the kidney and would have traveled with the urine and got stuck in the ureter or could have been made in the ureter itself. The stone in the ureter is made up of a mineral mass. A stone is formed when the little particles of minerals in urine stop flowing and start crystalizing to form a mass and eventually into a stone (Papadoukakis et al., 2006).

5.5 Ureter tissue engineering

Between 2002 and 2006, overall, research held in the US (United States) revealed that 2.6% of the genital injuries occurred in the ureter. This is most probably due to the ureter's distinctive physical attributes, like segmental vascular supply, are effectively harmed, and the absence of nearby tissue in the organ prevents surgical ureteral fix. When anastomosis is not practicable in a uterus, many procedures such as ureteroneocystostomy, board flap, ilea interposition, and renal auto transplantation are introduced to repair lengthy ureteral abnormalities. Finally, if surgical skill is unavailable and the mentioned treatments fail, undesirable nephrectomy is the sole choice.

The goal of ureteral reconstruction is to ensure that urine is safely transported from the kidney to the bladder. Constructs must have an efficient circulatory system, a functioning smooth muscle, and a urothelial barrier in order to operate well since the absence of these layers might lead to stent-related complications. Ureteral tissue engineering has made significant progress in the last several years. Tissue engineering of the ureter has benefited from advances in cell source, implantation method, and novel biomaterials (Fig. 5.1). Biopolymers, natural proteins, and other substances may improve the performance of decellularized tissues and scaffolds. A variety of naturally derived and manmade materials have been studied for their qualities and efficacy.

In the last decade, there is vast amount of work done in the field and separate attention has given to each vital organ or tissue in human body. Tubular small intestine submucosa (SIS) without cellular preseeding has been the primary focus of most investigations. Collagèn (Tachibana et al., 1985) and Gore-Tex (Baltaci et al., 1998) were also used. SIS and collagen are similar, but not actually Gore-Tex, were shown to be able to aid in the regeneration of urothelium and muscle cells. Fibrosis, on the other hand, was the most common outcome.

When it comes to designing treatment solutions to restore, heal, maintain, and/or increase tissue function, the fundamental assumption of tissue engineering is that using the system's inherent biology will lead to higher success. As well as preserving the original tissue's anatomical and functional characteristics, the biomaterials used must also have the necessary mechanical and structural qualities. They should also be able to sustain particular cell types in their microenvironment. Cells will be able to distinguish and repair tissues in accordance with the tissue from which they came

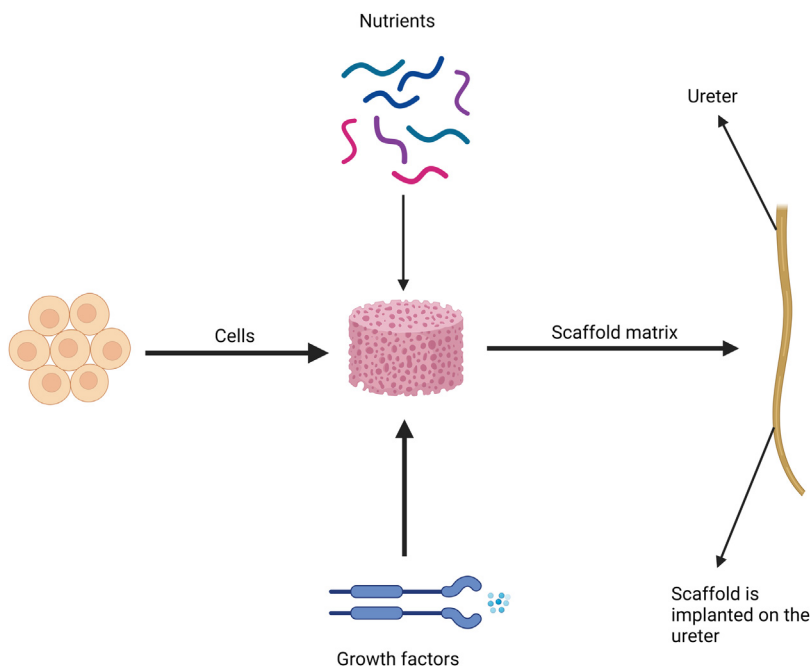


FIGURE 5.1

Different things needed in tissue engineering.

Drawn by author.

under these circumstances. The original scaffolding is eventually replaced by new extracellular molecules produced by cells over time, resulting in a healing and regeneration process. Biocompatibility and the prevention of transplant rejection make autologous cells the preferred method of treatment. So far, ureteral tissue engineering has received little attention in the area of stem cell therapy. [Table 5.1](#) summarizes the ureteral tissue engineering research conducted to far, including the kind of construct employed, animal studies, and length of the scaffold.

Table 5.1 Summary table of ureter reconstruction over the years using the approach of tissue engineering.

Biomaterial	Animal	Length	Authors
Collagen	Dog	5 cm	Tachibana et al. (1985)
Collagen	Rat	0.3–0.8 cm	Dahms et al. (1997)
Gore-Tex	Dog	5–8 cm	Baltaci et al. (1998)
Gore-Tex	Dog	10 cm	Sabanegh Jr et al. (1996)
AM	Dog	3 cm	Osman et al. (2004)

Continued

Table 5.1 Summary table of ureter reconstruction over the years using the approach of tissue engineering.—*cont'd*

Biomaterial	Animal	Length	Authors
SIS	Pig	7 cm	Liatsikos et al. (2001)
SIS	Pig	2 cm	Smith III et al. (2002)
AM, SIS	Mini pig	1.5–2 cm	Shalhav et al. (1999)
SIS	Pig	2 cm	Duchene et al. (2004)
SIS	Pig	2 cm	Sofer et al. (2002)
SIS	Dog	4 cm	El-Assmy et al. (2004)
SIS, AM, DSB	Pig, dog	3–5 cm	El-Hakim et al. (2005)
PLLA	Rat	1 cm	Xu et al. (2012)
PLLA and collagen	Mice	NA	Shi et al. (2012)
PLLA and collagen	Mice	1–1.5 cm	Fu et al. (2012)
Autologous graft	Dog	3 cm	Zhang et al. (2012)
VECM	Rabbit	3 cm	Zhao et al. (2012)
BAM	Rabbit	4 cm	Liao et al. (2013)
AM	Dog	3 cm	Salehipour et al. (2013)
Collagen Vicryl VEGF bFGF	Pig	5 cm	de Jonge et al. (2018)
AAM, PLCL	Rat	1 cm	Kloskowski et al. (2014)
SIS	Pig	8 cm	Xie et al. (2000)
SIS	Rabbit	11 cm	Jaffe et al. (2001)
Collagen	Dog	5 cm	Derham et al. (2008)
PTFE	Dog	10 cm	Varady et al. (1982)
PTFE	Dog	5 cm	Peters et al. (1996)

AAM, *acellular aortic arch*; AM, *acellular matrix*; AM, *amniotic membrane*; BAM, *bladder acellular matrix*; DSB, *decellularized small bowel*; PLCL, *poly(L lactide-co-caprolactone)*; PLLA, *poly(L-lactic acid)*; PTFE, *polytetrafluorethylene*; SIS, *small intestinal submucosa*; VECM, *vessel extracellular matrix*

5.5.1 Repair of a ureteral dysfunction

A ureteral defect system must be used to assess the effects of tissue-engineered structures on ureteral regeneration. Even yet, the scientists opted not to use tissue-engineered structures in an induced ureteral lesion model in recent investigations. Subcutaneous implantations in rats and mice were instead conducted by the researchers (Fu et al., 2012; Shi et al., 2012). For regenerated tissues, biomaterials manage the microenvironment and create a three-dimensional (3D) space. The qualities of the biomaterials utilized in tissue engineering have an impact on the functionality of scaffolds. As a result, in addition to serving as drug carriers for nutrients and wastes, biomaterials must also have the ability to promote cell growth, division, and adhesion. Ideally, biomaterials should be able to degrade in a regulated manner without causing any harm. To rebuild the urinary system, many biomaterial scaffolds were employed (Zou & Fu, 2018). Studies have shown that preimplantation can be used to replace a ureter, but nothing is known about how the construct

would behave when used as a ureteral substitute. Constructs are subjected to the harmful effects of urine and diverse mechanical stresses in the intracorporeal environment. Spiral poly(L-lactic acid) (PLLA) stents were the backbone of all three investigations. It was discovered that the body may function as a natural bioreactor for the development of tissue fleeces around the PLLA spiral stent that was implanted in subcutis (Xu et al., 2012). Primary bladder urothelial cells (USs) were transplanted back into the freshly created tissue after it had been decellularized. When compared to SIS, cell growth was comparable. To enhance cell adhesion and proliferation, the same PLLA stent was used as electrospun collagen. The final constructions were cultured with human adipose-derived stem cells (hADSCs) or human urothelial cells prior to subcutaneous injection in athymic mice. After 2 weeks, both studies were able to discover live human cells, indicating that the cells may survive the process. Prevascularized autologous tubular tissue may be generated in the subcutis prior to implantation, which may help reduce ureter fibrosis (Fu et al., 2012; Shi et al., 2012).

5.5.2 Different biomaterials used

There are generally three types of biomaterials that are used in tissue engineering (Table 5.2) which are mentioned below:

- Ceramics
- Manmade polymers
- Natural polymers

Each of the set of polymers has its own specific advantages and disadvantages and are used for different purposes; to deal with these problems, the use of composite scaffold has increased over the previous years. Ceramic scaffolds such as hydroxyapatite and tricalcium phosphate have been used for bone regeneration techniques. Some of the scaffolds such as polystyrene, PLLA, polyglycolic acid (PGA) and poly-dl-lactic-co-glycolic acid (PLGA) have been successfully been made from synthetic polymer by the right technique. But these post a lot of problems such as a chance of rejection. Natural polymers on the other hand have properties such as promotes cells adhesion and growth.

Table 5.2 Different types of biomaterials and their properties (O'Brien, 2011).

	Ceramic scaffold	Manmade polymers	Natural polymers
1.	High mechanical stiffness	Can be made to desired functions	Biologically active
2.	Low elasticity	Bioresorbable	Excellent cell adhesion
3.	Hard brittle surface	Readily available	Excellent cell growth

5.5.3 Sources of regenerative cells

Even though the function and need of cell sowing in tissue architecture has just been challenged morally, once evaluated, a variety of options exists. Some people have issues with regenerative cell items derived from porcine subjects, which have promising results in regeneration (Shukla et al., 2008) and even tissue culture that requires extensive blood use (Swindle et al., 1998). Stem cells from embryo are highly contentious due to their harvesting origins and potential of forming tumors in grafts. There is evidence to show that biopsies of such tissue can lead to differentiated primary cells or multipotent stem cells such as mesenchymal (Anumanthan et al., 2008). When autologous cells are available, a safer and less contentious choice is to use them.

Earlier experiments in ureteral tissue engineering used simple scaffolds which mostly resulted in fibrosis, indicating the requirement for cellular stocking, which is now part of the procedure. This is consistent with prior researcher's claims that cellular seeding is necessary for substantial defects to enhance tissue regeneration and stop scarring (Chen et al., 2000; Fu et al., 2012).

By extracting the major uterine cells from those patients who are suffering from disease named as nephrectomy, were transplanted on PLLA stent and placed subcutaneously in a mice model for 2 weeks in this method. Following this technique, a tissue which was thin and narrow in size was generated and later, it preserved the viability of those used cells. This effective approach is relatively straightforward, less medically contentious, and time-taking, as it is shown by studies that this procedure gets completed in almost a month (4 weeks) till the uterus develops. Those calls used there were enhanced for 14 days for making the preimplantation for another 14 days (Tian et al., 2010).

A more expedient method would be to implant simply the cell-seeded construct directly. To test this method, 0.5% highly porous tubular (5.0-cm long type I) collagen constructs were implanted into 11 landrace pig to correct a full urethral lesion. This research data were unpublished though. In order to detect the basic urothelial and smooth muscles cells, the urinary bladder of pig was used in the research. After being planted equally dispersed with urinary bladder SMCs, the scaffolds were seeded with urine sac. A pivot incision was used to access and activate the right ureter, and a 5 cm part of the ureter was eliminated and reconstructed with scaffolding of the same size. After a six Fr (double J) valve was implanted to improve urine flow, the animals were evaluated for 4 weeks. Following a urine leak, stomach inflammation was observed in 7 of the 11 experimental pigs after 2–3 weeks. Regardless of the existence of the stent, the other animals experienced strictures and hydronephrosis, which is a common side effect of such surgery.

Similarly, a five Fr (Double-J) stent was used in the stent intervention surgery of six pigs using a small intestine mucosa patch and the animal survived for 6 weeks (Liatsikos et al., 2001). Apparently, SIS patch worked well with porcine subjects for a maximum of 9 weeks (Smith III et al., 2002).

Following the investigation, it became clear that the urine leak could be linked to the collagen scaffolding's poor mechanical strength, resulting in scaffold ruptures or

dissections. When scaffolding remained patent in animals, it was covered by one layer of UCs. The principal disadvantage of primary cells, particularly UCs, is safe extraction in the event of a malignant tumor. Although data research on existing literature on ureteral carcinoma has revealed no such trend in Europe (Rouprêt et al., 2013). Additionally, adequate tissues for isolation are not always available.

As a result, alternative cellular sources are being investigated, with particular emphasis on mesenchymal stem cells (MSCs) (Drzewiecki et al., 2011) and adipose stem cells (ASCs) (Zhao et al., 2012). Without the greater risk for developing a tumor, these extracted cells can be segregated into many cell lines, including muscle and epithelium. Additionally, they have antiinflammatory capabilities and the capacity to release various cytokines linked with normal wound scarring. During the rebuilding process, mixed adipose cell populations do not usually cause any cellular discomfort (Blaber et al., 2012).

Zhao et al. differentiated rabbit adipose-derived stem cells (ADSCs) into an SMC phenotype before seeding and cell implantation. To generate an avascular extracellular matrix, cells were injected onto a decellularized rabbit abdominal aorta (EM). Applying cellular scaffolding, a 3.0 cm long tumor in the hare's ureter was treated. After 16 weeks, the defect resembled native tissue in terms of a correct muscle layer and stratified urothelial. There was also no evidence of narrowing or hydro-nephrosis (Zhao et al., 2012).

Acellular matrix allograft (BAMA) on porcine bladder was studied for 22 weeks and repopulation did occur at the fringes but not at the center (Brown et al., 2002). Human lipoaspirate has multifunctional cells that may be an alternative source of stem cell to the more ethically cumbersome bone marrow derivatives (MSC) (Zuk et al., 2001). Human umbilical cord has already been shown to be a proper substitute of regenerative MSCs (Lee et al., 2004). There is also research to suggest human amniotic membrane (AM) also has reconstructive use in ureteral wall defects, because of its significant advantages (Barski et al., 2017; Koziak et al., 2007).

5.5.4 Preimplantation seeding

In this procedure, the cells are placed into a liquid cell medium and allowed to scatter around. The cells get the nutrients and oxygen it needs from the medium to grow and multiply as seen in the figure. The upper most layer of the cells use most of the available nutrients and oxygen, which leads to restricted number of pioneering cells migrating to the scaffold (Mozafari et al., 2019).

Without a functional urothelium and appropriate vascularity, sterility and fibrosis might develop, as urine can readily destroy regenerating tissue. Preimplantation enhances vascularization and aids in the maintenance of in-vivo stocked cell viability. Numerous preimplantation locations, including the omentum, peritoneum, and by-products, have been employed in a range of purposes (Liao et al., 2013; Zhang et al., 2012). Prior to correcting a ureteral anomaly, significant preimplantation enhancements were made. Before implantation, a pipe-shaped scaffold was produced from the fiber structure that developed within the peritoneal space (Zhang et al., 2012).

Furthermore, omental preimplantation was employed as an in-vivo bioreactor to boost neovascularization in the structure. It also permits the generation of a monolayer epithelial structure able to preserve the structure after implanting in a potentially harmful urine condition, which is a significant benefit (Liao et al., 2013).

It is crucial that new blood vessels remain intact during the harvest of preimplanted material. A mobile preimplantation organ, such as the bigger omentum near the ureteral lesion, may be suitable for this, as the blinks are usually easily migrated. While preimplantation procedures appear to be encouraging, they are time-consuming and need further surgical treatment, which complicates the success rate. As a result, they are not always suitable for acute problems or unanticipated treatments, as is frequently the case with ureteral damage.

5.5.5 Decellularized tissue and synthetic polymers in scaffolding

Urinary diversion and correct sterilized scaffolding are huge challenge in the ureter injury repair discussion (Casarin et al., 2021). In earlier procedures, scaffolding was constructed using various materials. In recent decades, most investigations have used decellularized tissue rather than artificial scaffolding and there is a reason for that. The incorporation of tissue-related growth factors and other molecules, as well as the preservation and maintenance of tissue architecture, are two important characteristics of decellularized tissue. For this surgery, SIS has always been the most preferred tissue. The results/outcomes were dissatisfying.

Salehipour et al. had to use an AM as a biomaterial to reconstruct lengthy ureteral anomalies. The antiinflammatory benefits of AM are well known. In dogs, a small portion of the size of 3 cm from the ureter was eliminated and replaced with decellularized tubular AM. The method had one drawback: from every seven animals, two of them died as a direct result of urine leakage. Severe pelvic problems and fibrosis of the regenerated site, as well as lymphatic and granulocyte intrusion, were seen in the rest animals (Salehipour et al., 2013).

While an experiment revealed promising results when the AM was employed as a transplant, scientists determined that this material did not work well in full-on abnormalities (Koziak et al., 2007). This result was comparable to one found in prior research on the same topic (Osman et al., 2004).

Even though decellularized blood arteries and bladder acellular matrices have been reported to be effective before (Chen et al., 2000), this result might be related to the utilization of stem cells and preimplantation procedures, which have not before been used in the context of SIS for urethral tumor.

Apart from decolorized fabrics, it is possible to use artificial scaffolding in medical applications as indicated. Scaffolds result in increased plasticity, improved mechanical characteristics, and well-defined materials use for experiments involving biosubstances. Majority of materials used in scaffolding can be molded to any shape, including flat, film, tube, or custom size, and can incorporate various bioactive proteins and compounds, for example, spiral PLLA stents with collagen inside tubular scaffolding. Where mechanical strength enhancements are required, such artificial scaffolding can simply be applied to decelerated tissue for robustness.

A new composite stent of PLLA-collagen endoluminal used in conjunction with centrifugal technique for equidistribution of reconstruction cells provided a promising solution when it worked for 2 weeks in the laboratory (Fu et al., 2012). Additionally, it is indicated that when hADSCs are transplanted subcutaneously in shaved mice, they not only survive but also keep their phenotypic for at least 2 weeks. This demonstrates the existing technological capacity to use these cells for their stimulant qualities over longer time (Shi et al., 2012).

MSCs are another possibility in the same sphere. ADSC and MSC share many of the same characteristics, but MSCs are isolated from bone marrow (Tian et al., 2010). In contrary to ADSCs, removing these cells is challenging, the discriminating capability declines with time, thus there are fewer segregating cells. MSC with SMC injected into the cell matrix were utilized to rectify the problem of a 4 cm ureteral damage in rabbits (Liao et al., 2013). To construct tubular tissue grafts that would not be rejected, BAM was seeded on one side with MSC obtained from the bone marrow and on the other side with MSC derived from the bladder. For 2 weeks, tissue-engineered tracheal graft (TETG) was tabularized around an inserted catheter and preimplanted in rabbit omentum. MSC developed and formed a monocellular epithelium during this preimplantation period. After 16 weeks, the ureteral lesion was healed, revealing a multilayer urothelium with core neo vasculature. There was no evidence of tightness or hydronephrosis, even when the catheter from ureter was removed 6 weeks following the operation. Without the use of MSC, attempted ureter repair resulted with severe hydronephrosis and scarring. The researchers found that by creating the single-layer epithelium at preimplantation point, the surrounding tissue may have been protected from urine.

In order to regenerate the urinary system, UCs are utilized which are scraped from a urine tract. A technique in which the urothelial tissues are being withdrawn from the bladder is called bladder biopsy. Romagnoli et al. successfully corrected posterior hypospadias using urethral epithelium obtained from a urethral mucosa biopsy. These procedural regimens, however, are intrusive since they need surgical work and cause stress to the bladder and to the urethra (Romagnoli et al., 2017). Some researchers have also managed to isolate UCs from urine and bladder washes, which need further research. The ureter cells exhibit the properties of bladder cells and may hold promise for urethral restoration. Furthermore, epidermal cells can be collected with small donor morbidity (Smith III et al., 2002). Because epithelial cells derived from autologous oral tissue have a structure comparable to that of urothelium, they are similarly suitable for urinary tract substitution (Shalhav et al., 1999).

Buccal mucosa is composed of a stratified squamous epithelium that is nonkeratinized and exceptionally resistant to mechanical friction. Oral keratinocytes are isolated physically from the epidermis, and expansion of oral keratinocytes can be facilitated using a typical feeder-dependent technique. Additionally, an extra biopsy can result in donor morbidity, which is a downside of the procedure. However, as compared to alternative epithelial cell sources from the urethra and bladder, this approach is less invasive (Duchene et al., 2004). SMCs help to keep flexibility

and in order to prevent the wall contractures and collapsing, they are being withdrawn from the ureter (Sofer et al., 2002). SMCs can also induce angiogenesis by secreting TGF-1 (transforming growth factor-1) (El-Assmy et al., 2004). Then it has been demonstrated that dermal fibroblasts can release various cytokines and collagen, which enhance the mechanical characteristics of the grafts and encourage keratinocyte proliferation (El-Hakim et al., 2005). Proximal tubular cells (PTCs) are essential for sustaining renal function in kidney tissue engineering, and PTCs may be isolated from healthy or sick kidney tissue (Tachibana et al., 1985). Immortalized cell lines lack the features of PTC cells produced from primary cells. Fundamental kidney cells, on the other hand, endure passaging and have a limited proliferative capacity during the process (Dahms et al., 1997). A collagen synthetic polymer conduit showed 80% survival rate in porcine and can be considered for future research (Geutjes et al., 2012).

There is even discussion on the use of biodegradable elastomers instead of polymers for scaffolds like carbonate urethane that mimic porous organ walls for muscle cell access (Sivaraman et al., 2019). Also fibronectin-gelatine (FN-G) on carbonate urethane adding SIS further helps in structural integrity, fibrous contractures, and loss of volume with time (Da et al., 2017; Uchida et al., 2016). The relative degradation rates of these biodegradable substances have also been evaluated (Hong et al., 2010). Acellular dermal biometric from bowel linings worked well in bladder reconstruction but failed in porcine environment (Akbal et al., 2006), so further research will be needed here. BAMG (bladder acellular matrix graft) was successfully manufactured and utilized for partial restricted organ replacements in rabbit (Zhu et al., 2011).

5.5.6 Repair of full-circumference ureteral defects

Neotissue development is time consuming and that enhances risk, a better option research says is a hybrid scaffolding (Janke et al., 2019). When an entire piece of the ureter must be rebuilt, major ureteral reconstructions are required. Repairing onlay plugins is frequently impossible and may be performed only in stringent repair. As a result, it's not unexpected that most research focuses on repairing long-length rather than whole length ureter tube construction flaws. Zhang et al. employed 3 cm long connective cells derived from dog's peritoneal cavity following silicone tubing transplantation. The researchers were able to maintain sufficient vasculature and generate a new tissue that resembles the ordinary ureter by maintaining one-third of the natural urethra. Within 12 weeks, the tubelike structure was completely coated with multilayer urothelium and exhibited an ordered muscle layer and mucous creases. While these findings are encouraging, one must keep in mind that 1/3 of a vigorous ureter is unlikely to be present in a clinical scenario, such as extensive adhesions or long-term injury. It is acceptable that after 1 cm, tissue infiltration becomes more difficult and is probably related to fibrosis; so, the authors avoided this obstacle by keeping the ureteral section (Zhang et al., 2012).

5.5.7 Models using animals

Rabbits, dogs, and, in our instance, pigs (unpublished) were employed in ureteral replacement animal investigations, whereas rats or mice were used in subcutaneous implantation research. Pigs are the favored model because their abdominal structure is comparable to that of humans and ureter functions including kidneys resemble humans (Dalmose et al., 2000).

The ultimate animal's abdominal size and structure should be equivalent to that of humans, it should be fully matured, and it should have wound scar properties like those of humans. Goats, sheep, cattle, and horses are examples of possible animals. Typically, random control trials are required, pave the way by advanced toxicity assessments, before new technologies are extensively employed in clinical settings. Conducting safety studies on healthy people is unethical in tissue engineering, and there is frequently a paucity of reference treatments. As a result, selecting the most appropriate animal models is acute for accurately predicting the expected clinical result (Janke et al., 2019). The autologous UCs are obtained from urinary ladders, which are frequently utilized to reconstruct the urinary tract. The bladder biopsy procedure is performed to remove UCs from the bladder. In a study, posterior hypospadias was corrected utilizing urethral epithelium obtained from a urethral mucosa biopsy. These methods, however, are intrusive due to surgical procedures and bladder and urethral damage (Png & Chapple, 2000). Several researchers have isolated UCs from enemas of the urine bladder. Urine cells have the same characteristics as normal bladder cells and can be utilized to regenerate the urethra (Elliott & McAninch, 2006; Liatsikos et al., 2001; Simaioforidis et al., 2013). Skin cells collected rather simply from the foreskin with minimal donor morbidity are possible. When sown to an acellular matrix, these cells are easily incubated and sufficient in quantity (Smith et al., 2002). Due to their structural similarity to urothelium, autologous oral epithelial cells are also adaptable to urinary tract replacement (Shalhav et al., 1999).

The buccal mucosa is composed of a stratified squamous epithelium that is extremely resistive to mechanical abrasion. You obtain oral keratinocytes by mechanically isolating the epidermis. A typical diet-dependent method may also promote oral keratinocyte growth. Additionally, biopsies can result in donor morbidity. It is the most less invasive technique as compared to all other epithelial cells techniques which are available in urethra and bladder (Duchene et al., 2004). SMC can be extracted from the bladder, and by rapidly forming the muscular layer, these cells rise the caliber's pliability and stop wall adhesion and subsidence (Sofer et al., 2002). Additionally, modification of the TGF-1 released by SMCs may promote angiogenesis (El-Assmy et al., 2004). Derma fibroblasts may trigger the release of cytokines and collagen, which boost the qualities of the graft and encourage keratinocyte proliferation (El-Hakim et al., 2005). There is a possibility to extract SMCs from the bladder, and by rapidly forming the muscular layer. SMCs increase the caliber's elasticity and prevent wall adhesion and subsidence (Sofer et al., 2002), which is a desirable trait. Additionally, modification of the TGF-1 released by SMCs may promote angiogenesis (El-Assmy et al., 2004).

Derma fibroblasts may synthesize and secrete collagen, which improve the properties of the transplants and enhance keratinocyte production (El-Hakim et al., 2005). In kidney tissue engineering, the existence of proximal tubular cells (PTCs) is essential for retaining kidney function, and PTC may be separated from healthy or ill kidney tissues (Tachibana et al., 1985). PTC primary cultures are more typical of immortalized cell lines than immortalized cell lines are. Prime kidney cells lose some expression of critical genes during passage and have a restricted capacity for proliferation (Dahms et al., 1997). Because of their potential to develop several times, Embryonic stem cells (ESCs) are ideal for cells in theory. Nevertheless, a fundamental difficulty, mainly the origin of their embryos and the recognized risk of tumor growth, is prolonging implantation (Sabanegh Jr et al., 1996). When ESCs are treated with all transretinoic acid in vitro, they can develop into UCs (Sloff et al., 2014). The ability of ESCs to differentiate into UCs in vitro has substantial implications for urinary tract repair (Xu et al., 2012). The availability of PTCs is critical for maintaining kidney function in the area of kidney tissue engineering, and PTC may be isolated from healthy or sick renal tissues (Tachibana et al., 1985). Basic renal cells have a limited potential for growth and lack some development of important genes throughout migration (Dahms et al., 1997), which is not optimistic.

Moreover, implantation is being limited by a severe issue, namely the source of their embryos and the possibility of tumor progression (Sabanegh Jr et al., 1996). Furthermore, when exposed with completely trans-in vitro retinoic acid, ECs may distinguish between urothelial and epithelial cells (Sloff et al., 2014). The ability to integrate ESC into UCs in vitro has substantial implications for urinary tract restoration (Xu et al., 2012). The application of ESCs is a matter of debate. Therefore, variant examination is needed before they may be used in reformative medicine. Adding to this, the characteristics that are more common in between the induced pluripotent stem cells (iPSCs) and ESCs, such as the ability of spreading and duplications. Also, different groups produced a successful in vitro initiation approach for converting iPSCs to urothelium, setting up origin of biomodified urothelium, by producing required features in autologous mature somatic cells using just a viral vectorial technique (Osborne, 2006; Shi et al., 2012; Xu et al., 2012).

5.6 Discussion and challenges

The goal of ureteral reconstruction is to ensure that urine is safely transported from the kidney to the bladder. Structures must have an efficient circulatory system, a functioning smooth muscle, and a urothelial barrier in order to operate properly since the absence of these layers might lead to stent failure and hydronephrosis. The field of ureteral tissue engineering has made significant strides in the last several years. Tissue engineering of the ureter has benefited from advances in cell source, insertion method, and novel biomaterials. However, only stem cells were used to study the performance of decellularized tissues or scaffolds containing endogenous proteins and other compounds. As ureteral tissue engineering advances, reported studies on this

topic remain sparse. Numerous recent research fails to examine how the structures behave in a ureteral replacement environment. Future research should strive to repair a whole ureteral defect in order to learn more about the effects of various materials, cell sources, and implantation procedures. Ureteral regeneration may be improved by using stem cells, originating from mesenchymal and adipose tissue could be grown on any bioactive substance. As an additional benefit, preimplantation of these constructions in the omentum may increase vascularization and stimulate stem cell differentiation, which might lead to better outcomes. Because of this, when employing the human body as an in vivo bioreactor, ureteral healing may be difficult. There should be a thorough evaluation of several preclinical lab animals before clinical studies begin to avoid vertebrate bias in the results (de Jonge et al., 2018).

Urinary tissue bioengineering is expected to be closer to reality as a result of the current focus on developing clinically viable, biodegradable urothelium utilizing adequate scaffolds for neotissue construction. It is possible that in vivo investigations of UC proliferation, differentiation, and morphogenesis will provide information that is missing from in vitro research when exposed to urine and the biomechanical constriction and extension of the bladder. The reactions of transdifferentiated or reprogrammed or epithelial cells from various sources to biomechanical functions and the ability of urothelium regeneration in vitro and in vivo must thus be studied. Structural materials that closely mirror the chemical structure and physical characteristics of urine are needed to attain this goal. To comprehend and restore the functional components of bladder filling and voiding, mechanical signaling and neuron transduction are needed; hence, the field will ultimately advance toward more complex neurogenic bioengineered prostheses (Singh et al., 2018).

To understand the effects of various materials, cellular sources, and implantation methods, future studies need to focus on a complete ureteral defect repair, as many previous studies have failed to do so. According to current studies, the ideal technique for ureteral regeneration is to use mesenchymal and fatty stem cells preserved on any proper bioactive media.

Furthermore, implantation of a scaffold in the omentum might improve the ultimate outcome by boosting vascularization and initiating stem cell differentiation. When the body is used as a bioreactor in vivo, the prolonged incubation time may be an issue for ureteral healing. Lastly, several preclinical animal systems should have been evaluated before clinical studies started to establish out species-related outcome bias.

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The progress in tissue engineering of penile corporal tissue

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6.1 Introduction

One of the two sponge-like areas of erectile tissue that keep furthermost of the blood in the penile throughout erection is termed the corpus cavernosum. A comparable corpus can be contrasted to the female corpus cavernosum clitoridis, which is the core of the clitoris that comprises the erectile tissue in two corpora cavernosa through anatomically apparent connections. Three expandable erectile tissues, or corpora cavernosa and corpus spongiosum, run the length of the penis besides fill with blood throughout penile erection. From the pelvic bones to the top of the penis, where they converge, the two corpora cavernosa are located along the penile chute.

These structures are made of trabeculae, which are asymmetrical blood-filled pockets lined by endothelium and divided by the penis septum. Instead of vestibular bulbs, which are found in the female anatomy, males have a corpus spongiosum, a lesser area along the bottommost of the penis that houses the urethra and creates the glans penis. In some situations, nitric oxide release occurs before the muscles in the corpora cavernosa and corpus spongiosum relax, a procedure like female stimulation. A blood supply from veins running the length of the penis fills the spongy tissue. A small amount of blood arrives in the corpus spongiosum; the mainstream expands the corpora cavernosa, which lengthens and widens to retain 90% of the blood intricate in a erection.

6.2 Disease/disorders related to penile corporal tissue

Lack of native penile tissue frequently restricts the range of reconstructive techniques that attempted to retain functional and cosmetic features. These current techniques require extensive surgeries and may involve the use of autograft implantations and prostheses, but the function of corporal tissue is not restored. There are many kinds of penile disorders. Some are minor and may not pose a major

threat, whereas others are serious medical problems that require immediate attention or surgery. Congenital penile conditions are those that are present from birth; otherwise, they can arise over time. The following are a few of the highly prevalent illnesses and diseases that affect the penis.

6.2.1 Balanitis

Infection of the glans (head) of the penis is known as balanitis. Balanoposthitis is the medical term for the condition if the membrane is also irritated. Balanitis symptoms might include penile pain, swelling, and itching, as well as a rash on the penis and a foul-smelling discharge. Balanitis is most usually brought on by uncircumcised guys' inadequate hygiene. Inflammation can develop around the glans if the penis beneath the foreskin is not thoroughly cleaned of bacteria, perspiration, dead skin cells, and debris (Motta et al., 2022). The likelihood of inflammation increases in uncircumcised males who have phimosis and are unable to clean behind the foreskin.

6.2.2 Epispadias

A urethra that will not fully form causes epispadias, an unusual birth disorder that makes it difficult for someone to properly evacuate urine from their body. Epispadias can occur in both boys and girls at birth. Boys who have it generally have an abnormally bent, small, broad penis at birth. The urethra may not open at the end of the penis but instead may open on the upper side or even the entire length of the penis (Yu et al., 2018).

6.2.3 Hypospadias

A congenital condition known as hypospadias causes the urethra's opening to form on the base of the penis rather than the tip. Depending on where the opening develops, the ailment might be mild or severe. The urethral entrance is frequently located close to the penile head. Rarely, males might be born with the urethral opening below the scrotum. Some boys are born with an opening in the mid of the shaft or the base of the penis (Yu et al., 2018). An aberrant urethra opening, chordee, irregular spraying during urine, and foreskin abnormalities that give the penis a hooded appearance are all indications of hypospadias.

6.2.4 Penile cancer

Penile cancer, widely known as penile cancer, almost always starts in the penis' skin cells. Squamous carcinoma, melanoma, basal cell cancer, carcinoma, and sarcoma are the five main kinds of penile cancer. Squamous cells, which are flattened skin cells, are the main source of almost 95% of all penis malignancies. Squamous cell carcinoma is a cancer that arises from squamous cells. Squamous cell cancers often have a modest rate of growth and are curable if discovered early. Squamous

cell penile cancer is diagnosed at its earliest stage, known as carcinoma in situ, or CIS (Hoffman et al., 2019).

6.2.5 Peyronie's disease

When plaque, or scar tissue, builds up inside the penis, Peyronie's disease, a penile disorder, results in painful, curved erections. There are no issues with the modest curve that many men's erect penises have. However, when it hurts or the bend is considerable, it can cause erectile dysfunction and even prevent sexual activity (Hoffman et al., 2019). Peyronie's disease can be identified by the presence of scar tissue beneath the skin of the penis, a considerable bend or curve in the penis, trouble achieving or maintaining an erection, pain in the penis, and shortening of the penis.

6.2.6 Phimosis and paraphimosis

The condition of phimosis makes it challenging to retract the penis' foreskin. Repositioning the foreskin is challenging when you have paraphimosis. Uncircumcised boys and men can develop either condition. Due to the tightness of their foreskin, babies frequently experience phimosis (Tam et al., 2022). As males become older, it typically goes away on its own. Boys may require treatment if they are unable to retract their foreskin by the time, they are teens. If a steroid cream is ineffective, further treatments include circumcision.

6.2.7 Priapism

A prolonged erection that lasts for longer than four hours and is not released by orgasm is referred to as priapism. This disorder can cause painful erections that are not always related to sexual activity. Medication, drug and alcohol addiction, spinal cord issues, and specific blood illnesses are among the common reasons. You should visit an emergency department for care if your erection has persisted longer than four hours. Blood is typically removed from the penis as part of treatment (Terlecki & Boes, 2022). There may also be a need for blood vessel shrinkage medications. Rarely will surgery be required to resolve the issue and avoid doing long-term harm to the penis.

6.3 Materials used for transplantation for penile corporal tissue

Based on physical location, the urethra in males is split into three sections. The membrane or penial urethra extends along the length of the chute of the penis over the corpus spongiosum, the prostatic urethra passes over the prostate gland, the membrane-forming urethra, which is the thinnest segment of the urethra, passes over the external urethral sphincter. The bladder mucosa is mainly continued in the urethral epithelium,

which is made up of transitional cells. The mucosa of the urethra is composed of stratified columnar cells as it approaches the meatus, and stratified squamous cells as it reaches the meatus. In the field of urology, congenital and acquired urethral abnormalities continue to be problematic (Wang et al., 2022). Strictures, trauma, congenital abnormalities, and cancer are just a few of the situations that can frequently result in major voiding problems that need additional tissues for rebuilding and repair. The practice of tissue transplants from additional bases is frequently necessary for proper repair since there is frequently insufficient native urethral tissue.

Small periurethral glands throughout the urethra's length secrete mucus that helps to shield it from the urine's acidity. Acellular supports for urinary tract repair, acellular collagen, and polyglycolic acid-based scaffolds produced from the small intestine and urethra have also been employed in animal models to help in urethral tissue regeneration. Using cell-seeded scaffolds to repair the urethra is another option however, it has been established that tabularized urethral upkeeps using noncellular shells are not practical, because natural cells cannot properly integrate into the circular scaffold for lesions larger than 0.5 cm.

6.4 Physical bodies

The intricate biology of the penis is serene of three dissimilar cylinders. The preponderance of the penis is made up of two paired cylinders termed the corpora cavernosa, which are the main organs responsible for erectile function. A link of sizable intravenous fistulae connected by trabeculae impenetrable connective tissue makes up the corpora cavernosa. Blood enters the corpora during an erection as of the vital vein through the helicine veins, and then it travels to the sinuses. Later, smaller veins that discharge into the dorsal vein drain these sinuses. The urethra is in the third cylinder, which is known as the corpus spongiosum. A substantial fibrous coating known as tunica albuginea covers each of the three corpora. These three cylinders are encased in Buck's fascia, a substantial membrane (Abbas et al., 2019). The entire structure is covered by Collie's fascia, which continues down the abdominal wall. Patients with congenital malformations, cancer, or trauma may require extensive reconstructive treatments affecting the corporeal bodies. These surgical techniques are frequently staged and frequently use congenital tissues as a grafting source due to the lack of autologous tissues. Due to contamination, donor site illness, and graft failure, these handlings carry a high risk of complications. Due to the distinctive functional manner of the corporal forms, the expansion of substitute resources is a significant barrier to priapic restoration.

6.5 Albuginea tunica

Due to the commemoration foundation and penis curving, patients with penis abnormalities, mainly Peyronie's illness, may have severe pain and practical issues.

Currently, disorders like Peyronie's are treated by removing the plaque or by infiltrating the tunica albuginea with materials like fascia, pericardium, and small intestine submucosa. All these products have limitations, such as limited accessibility, donor site morbidity, and insufficient efficacy. Most of the drawbacks of present methods might be solved by using a bioengineered tissue, which would also enable long-lasting outcomes. The seeded matrices were subsequently grown for 21 days in a bioreactor under constant multiaxial stress. Controls were static values that were not subjected to mechanical stress (An et al., 2020).

After seven days, fibroblast cultures that had been mechanically strained exhibited a uniform multilayer matrix penetration, regular cell orientation during the straining axis, and enhanced synthesis of extracellular matrix proteins likened to the static control. Many organs can currently be treated with or replaced with the help of tissue engineering. Most of these activities took place in the last 10 years. However, more research is required to guarantee the safety and effectiveness of these novel materials before these engineering approaches may be used on people. Recent developments imply that cell therapy and modified urologic tissues may soon be useful in the clinic.

6.6 Latest and previous tissue engineering approaches for penile corporal tissue

The bulk of the penis is made up of the corporal bodies, which are essentially smooth muscle cells (SMCs) surrounded by collagen, nerves, and blood vessels. The penis' capacity to get erect is a result of its complex but highly structured architecture. Therefore, superstructure materials must be both robust, with the capacity to expand and contract. One of the biggest difficulties in creating a completely working penis is replicating this extremely specific structure, which necessitates a designed scaffold. The essential mechanical characteristic that permits sexual functionality is penile stiffness. It is a fluctuating rigidity produced by the corpus cavernous tissues' hydraulic principles (Sharma & Gupta, 2019). Intracavernosal pressure, urethral geometry, tunica contractility, cavernosal expandability, and cavernosal bulk modulus all affect penile rigidity. The cavernosal bulk modulus quantifies how difficult it is for erectile tissue to enlarge as intracavernosal pressure rises. They used a mathematical model to determine the cavernosal bulk modulus for two constant standards of cavernosal extensibility (x) at 0.1 and 0.05 mmHg.

Though, an examination of the literature revealed scant support for these biomechanical features. Choosing materials that approximate natural tissue function becomes challenging as a result. Everybody experiences penile stiffness under different pressure. Rigidity typically occurs at pressures between 60 and 90 mmHg. However, rigidity may appear in some people at pressures as low as 40–50 mmHg or may not appear till compressions are greater than 100–120 mmHg. The surrounding soft tissue must be able to endure significant pressure without buckling. Inadequate penile stiffness is correlated with poor penile expansion in diseased, fibrotic

penile tissue. In corporeal tissue engineering, a variety of materials have been employed as scaffolds, including synthetic polymers, natural elements, and acellular matrices (Gupta et al., 2022). They have been studied for corporal regeneration and the restoration of appropriate erectile function after being seeded with a variety of cell types. Following their *in vivo* transplantation into the rat cavernosum, mesenchymal stem cells (MSCs) successfully differentiated into muscle tissue and endothelial cells, according to studies.

MSCs are a desirable choice for the restorative restoration of penile that occurs when a blood tissue is because of their adaptability, accessibility, and distinctive repeatability. Several research teams have successfully infused regular human corporeal SMCs and human endothelial cells into poly(lactic-co-glycolic acid) (PLGA) scaffolds (EC). When transplanted into athymic mice, they were able to develop organized, perivascular cavernosal muscle mass that demonstrated capillary development. Despite being organized, the tissue's structure was different from that of the original bodily tissue. Human corpus cavernosal muscle cells and endothelial were then seeded into acellular corporal collagen matrices made from processed donor rabbit corpora (de Kanter et al., 2022). There was a development of structured elastin and smooth muscle as well as neovascularized sinusoidal gaps. The ability of newly generated tissue to contract in reaction to electrical stimuli was assessed for functionality.

The cells were capable of constricting, but they were unable to show a persistent erection. Full inclusion of the complete cross-section of modified corpora into animal studies has been demonstrated in later investigations. On collagen matrices, muscle tissue and endothelial cells were planted. Kwon et al. used human corpus occurs when blood SMCs and endothelial cells are transplanted onto acellular matrices to replace a cross-sectional piece of both corporal bodies in rabbits. They displayed successful, long-lasting penises and ejaculation, which proves that functional tissue has formed. Due to the low concentration of SMCs, the intracavernosal pressure was, however, only about half of the typical pressure. To get beyond this restriction and create better corporeal tissue morphologically and biochemically, consider dynamic seeding (Porzionato et al., 2018). To construct fully functional erectile tissue, their lab is now researching the substitution of anterior corporal tissue. Cells are sown onto the matrix during the first 48 h to facilitate adhesion. Quantifying the number of cells, the identical number of cells per volume are sown onto each scaffold. When creating artificial body tissue, dynamic seeding in bioreactor systems enables the simultaneous seeding and mixing of several cell types. Additionally, it offers more cellular content. The researchers noted that cellular levels reached 71% of the typical corporal tissue, as opposed to 39% when utilizing alternative seeding techniques.

6.7 Cell/stem cell used in penile corporal tissue

Currently, three methods embryonic stem cell (ESC) extraction, bodily cell fissile transmission, and bodily cell reprogramming can be used to obtain pluripotent stem cells. First identified in the internal mass of cells of the mouse rudiment in

1981, undifferentiated ESCs are a type of stem cell. The possibility of their use in stem cell transplantation and regenerative medicine was greatly encouraged by their capacity to differentiate into a variety of cell types. But as was already mentioned, ESCs have ethical implications. New stem cell knowledge, like reprogramming and the production of pluripotency through somatic cell nuclear transmission, has been developed to get around this restriction. The successful creation of induced pluripotent stem cells (iPSCs) in recent years has given rise new hope for regenerative therapeutics (Gur et al., 2018). Now, numerous studies have demonstrated that tissue samples can be reprogrammed by overexpressing diverse combinations of a small number of transcription factors in cells that have traits in common with ESCs. There are now several unique reprogramming techniques that have expanded the range of cell categories and species that can be used to create iPSCs. iPSCs exhibit the same capacity for cell-type differentiation as ESCs.

The level of molecular homology among iPSCs and ESCs has not yet been fully clarified, yet. It is still unclear if these minor variations are the consequence of interexperiment variation or whether reprogramming corporal cells results in ESC-specific state formation. iPSCs appear to have the potential for regeneration medicine, which was previously only associated with ESC. Adult stem cells (ASCs) are receiving more attention as a means of preventing ethical conundrums. ASCs are typically self-renewing, tissue-specific cell populations that could develop into cell types related to the organ system in which they are found. Despite once being thought to be just multipotent, an ever-growing body of research now contends that some ASCs may be capable of pluripotent differentiation (Ozinko, 2021). ASCs can be quickly found in tissues with a high rate of cell turnover. In the past 10 years, numerous ASC niches have been found in a variety of tissues, including the blood, eye, brain, liver, skin, skeletal muscle, intestinal system, adipose tissue, and pancreas. The MSCs, which are referred to as (multipotent) MSCs, are among them and are the most striking exception to the identifying specific of ASCs.

If injected into a blastocyst, they can differentiate in vitro into a variety of mesenchymal/mesodermal cells and differentiate developmentally. MSCs are currently being tested in clinical trials that have been approved by the US Approved for the handling of myocardial injury, stroke, knee ligament injury, limb ischemia, and immune deficiency disorders. They have undergone extensive testing and be effective in preclinical studies. MSCs are known to generate a wide variety of growth factors and cytokines that have both endocrine and autophagy effects on injured tissues; however, the exact mechanisms are still unclear (Yi et al., 2020). Bone marrow, adipose tissue, muscle hypertrophy, dental pulp, and cord blood have all been used to isolate MSCs. Bone marrow-derived stem cells (BMSCs), which are generated from bone marrow stroma, are among them and are the most researched. Due to their capacity for self-renewal and the capability to differentiate into multiple cell types, BMSCs were initially thought to oversee bone marrow restoration. Adipose tissue-derived stem cells (ADSCs), or MSCs produced from the stromal vascular portion of adipose tissue, have recently emerged as a plentiful and convenient source of stem cells.

6.8 Conclusion

The corpus cavernosum is one of the two sponge-like sections of erectile tissue that maintain the preponderance of the blood in the penile throughout erection. The female corpora cavernosum clitoridis, which is the center of the clitoris and comprises the penile tissue in two corpora cavernosa with physiologically obvious connections, can be used as a comparison corpus. The corpora cavernosa and corpus spongiosum, two stretchy erectile materials that run the span of the penis, fill with blood through the penial formation. The two corpora cavernosa are positioned along the penial chute from the lumbar bones to the top of the penis, wherein they converge.

Trabeculae, which are amorphous blood-filled pockets bordered by endothelial and divided by the septum of the penis, are the building blocks of these structures. Nitric oxide release, which is like female arousal in some circumstances, happens even before muscles in the corpora cavernosa and corpus spongiosum relax. The spongy tissue is filled with blood from arteries that span the lengths of the penis. Most of the blood engorges the corpora cavernosa, which elongates and expands to retain 90% of the blood implicated in an erection. Only a minor amount of blood reaches the corpus spongiosum.

The goal of tissue engineering is to help with penis restoration after (partial) penectomies caused by penile cancer, traumas, or congenital penile deformities. However, to prove its relevance in clinical research, testing in large animal models which have a penis size and ICP range close to human penis is essential. When conventional medical and surgical treatments fail to treat related disease, the ongoing advancements in tissue engineering approach may provide a long-term or permanent remedy.

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The progress in tissue engineering of prostate

7

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7.1 Introduction

The prostate is a tubuloalveolar gland that forms part of the genitourinary system and serves a vital role in the male reproductive system. It produces prostatic fluid which forms part of the seminal fluid which nourishes and protects sperm from the acidic conditions of the urethra and vagina. During ejaculation, the prostate is responsible for mixing the prostatic fluid and the other secretions of the seminal vesicles with sperm to form semen. This gland has the general shape of a walnut with an average mass of 15–20 g in grown men and is one of the most cancer-prone parts of the male anatomy with over 1.1 million cases of prostate cancer (PCa) diagnosed worldwide every year (Pereira et al., 2019). Apart from prostatic carcinoma, the other major condition that affects the prostate is benign prostatic hypertrophy (BPH) and it is mostly prevalent in the aging population of men (over 50 years).

Despite being plagued by so many disorders and the prostate itself being a vital organ in terms of reproduction, it has been forgotten in terms of advancements in treatments of these disorders. The medical treatments that are currently used in treating diseases like PCa results in complications such as erectile dysfunction (ED) and urinary incontinence. There are currently several methods being researched to alleviate these side effects brought by PCa and its treatments. Tissue engineering has opened a new venture that promises to reduce the severity of PCa and the associated effects of PCa treatment. Stem cell therapy has shown promise in rebuilding prostate tissue damaged by prostatectomy as well as reducing postprostatectomy effects such as ED and urinary incontinence.

The currently new research showing whole prostate tissue regeneration to help individuals who have gone through whole prostatectomy. Although this research is showing great promise, there are still a lot of avenues to venture into to produce a complete tissue engineering technique that can be applied in practice to help people. This is in part because the application of tissue engineering for the prostate being a new field.

7.2 Anatomy and histology of the prostate

The human prostate is a walnut-shaped gland that is located between the urinary bladder and the base of the penis. It sits in front of the rectum and posterior to the symphysis pubis. The prostate surrounds the urethra as it passes from the bladder to the penis. This gland is characterized into four zones: The anterior zone, the peripheral zone, the central zone, and the transition zone.

The anterior fibromuscular zone is mostly composed of smooth muscle fibers and connective tissue. The peripheral zone contains the glandular tissue that produces prostatic fluid, and it is located at the back of the gland. It makes up around 70% of the prostatic tissue mass and the most prone to prostatic cancers. This zone is the one that is palpated by the doctor when conducting a digital rectal examination (DRE). The central zone is the one that surrounds the ejaculatory gland as it enters the prostate from the seminal vesicles. The urethra is surrounded by the transition zone which makes up around 5% of the prostate mass and it grows throughout the life of a man. This leads to hyperplasia which is incident with age and the zone is significantly enlarged by BPH ([Ittmann, 2018](#)).

7.3 Conditions of the prostate

7.3.1 Prostate cancer

PCa is the second most prevalent cancer in men after lung cancer and the fourth most common cancer overall with 1,276,106 new cases recorded in 2018. It was also recorded that the rate of prevalence is significantly higher in developed countries such as France and cases of PCa tend to be twice as like in African American men than in white men. They also tend to have a more aggressive form of PCa than white males. Many cases of PCa are slow-growing and generally confined to the prostate gland where they are unlikely to cause substantial damage ([Giona, 2021](#); [Rawla, 2019](#)). These types normally require minimal to no treatment as, however, some forms tend to be more aggressive and even metastasize to other parts of the body.

7.3.1.1 Stages of PCa and treatments at each stage

- **Stage I:** Malignancy is only detected in the prostate in stage I. The cancer is discovered by lumbar puncture or after ablation for other circumstances in a

small amount of tissue (such as BPH). The prostate-specific antigen (PSA) level is below 10 and the Gleason score is 6 or less. It is present in 50% of one ventricle of the prostate or less. It is not detectable by radiology and cannot be sensed during a mechanical rectal inspection. They normally grow gradually and do not exhibit any signs while they are immature. During a rectal assessment, the cancer is very minimal and entirely within the prostate gland, which feels ordinary. Radiation therapy is a type of treatment that uses an elevated X-ray beam to kill PCa cells or prevent them from dividing and increasing fast (Sandhu et al., 2021).

- **Stage II:** The cancer is more progressed in stage II than in stage I, but it has not spread outside of the prostate. It can spread beyond the prostate if not handled. Stage II is broken into two parts: IIA and IIB. The PSA level is less than 20 but greater than 10, and the Gleason score is 6–7 in stage IIA malignancy. It is discovered in a little piece of tissue after operation for other purposes or by lumpectomy. Cancer at stage IIB: This type of cancer can be seen on both sides of the prostate. The Gleason score can vary from 2 to 10 and the PSA might be any amount. In certain situations, the tumor has not migrated beyond the prostate, and such tumors are not detectable on a mechanical rectal check or by imaging. The PSA level must be 20 or greater, and the Gleason score must be between 2 and 10 (Hernandez & Thompson, 2004).
- **Stage III:** The malignant cell has grown exponentially at this point. The PCa tumor has gone further than the prostate gland, but not to the bladder, lymph nodes, rectum, or any other organs. When it comes to seniors, doctors sometimes choose to use the cautious waiting method. This is only recommended if the patient is suffering from several serious problems or another sickness. The Gleason score can vary from 2 to 10 and the PSA might be any amount. External radiation therapy combined with hormone replacement, as well as a complete prostatectomy in some situations, are all options for treatment. Many times, the lymph nodes in the pelvis are also removed. This operation may be followed by radiation therapy. In other cases, hormone therapy is the sole option (Mazhar & Waxman, 2002).
- **Stage IV:** This is the cancer's ultimate stage, and it is usually incurable. It has spread to the lymph nodes or another region of the body, or it has grown into the bladder or rectum. It has the potential to expand to distant locations such as the bones, liver, or lungs. The problem will not be solved by eliminating any part of the body in this scenario. It is preferable to wait and observe for the elder. Hormone treatment, chemo, external beam radiation (in some situations), surgeries like transurethral resection of the prostate (TURP), which helps to lessen symptoms such as bleeding, and other soothing treatments are also done to mitigate symptoms such as bone pain are all options for treatment (Rodney et al., 2014).

7.3.1.2 Treatment of PCa

When treating PCa, there is normally a few criteria that need to be assessed before the type of treatment is decided. Treatment depends on type of tumor and its size, the

stage of the cancer and its grade, the general health of the patient as well as whether it has metastasized or not. There are normally five treatment methods used to tackle PCa.

- **Surgery:** The mostly commonly used form of surgery for treating PCa is the partial or complete removal of the prostate gland. This measure is a drastic one since it can result in impotence and ED (Kaler et al., 2020).
- **Radiation:** The bombardment of the cancerous prostate cells with high energy radiation to kill them. This method has a disadvantage as the cancer can come back, and usually in a more aggressive form than before (Gay & Michalski, 2018).
- **Immunotherapy:** This is used in more older patients who are deemed that the cancer will not threaten their life before it comes to a natural end. It slows down the growth rate of the tumor (Wang et al., 2022).
- **Hormone therapy:** This form of treatment is utilized to restore the testosterone-estrogen balance to also slow down the rate of progression of the cancerous growth. Its method of action is to shrink the PCa growth to prevent it from spreading to other parts of the body such as the bladder (Belkahla et al., 2022).
- **Chemotherapy:** This method of treatment is mostly applied when the PCa has spread to other parts of the body. It is highly exhaustive procedure for patients as it weakens the general immunity of the patient. Some of the sign of a patient on chemotherapy is the loss of hair and paleness of the skin (Kaler et al., 2020).

7.3.2 Prostatitis

Prostatitis is a prostate infection or inflammation that is most felt in and around the pelvic area. It can affect males of any age. It affects 50% of all men at a certain time in life. In men under the age of 50, it is the most prevalent urinary tract problem. It is the third most prominent among males above 50 (Dickson, 2013). Every year, well over 2 million men seek treatment for prostatitis symptoms. Urinary tract infections are connected to two forms of prostatitis (urinary tract infections (UTIs)); however, other types are not (Kontos et al., 2011). Prostatitis is a harmless condition (not cancerous). It has no effect on the risk of PCa. Prostatitis-related inflammation, like PCa, can boost the number of PSAs in the blood. Additional tests can be used to figure out what is triggering the increased PSA levels (Potts & Payne, 2007).

7.3.2.1 Types of prostatitis

- **Acute bacterial prostatitis.** A prostate infection is caused by an UTI. Fever and chills are common symptoms. Urination may be uncomfortable and frequent, or you may have difficulty urinating. Acute bacterial prostatitis necessitates medical attention right away (Ritter et al., 2022).

- **Chronic bacterial prostatitis:** Bacteria become lodged in the prostate gland, resulting in challenging to treat recurring UTIs. The symptoms appear gradually and linger for three months or longer ([Vaidyanathan and Mishra, 2008](#)). These may include blood in the urine or sperm, ill-scented urine, urination pain or burning, ejaculation discomfort, bowel movement pain, pain in the following areas:
 - **Lower back**
 - **Between the genitals and anus**
 - **Pubic bone**
 - **Testicles**
 - **Lower abdomen**
- **Chronic prostatitis/chronic pelvic pain syndrome:** An infection of the prostate or the neurons that run through this area. CP/CPPS pain can linger for weeks, months, or even years. This pain could also be caused by pelvic muscles. Although this is not an infection, symptoms may be similar to those seen in individuals with CBP ([Krieger et al., 1999](#); [Murphy et al., 2015](#)).
- **Asymptomatic inflammatory prostatitis:** This condition causes the prostate organ to become irritated yet no indicators. One might learn they have this condition after getting tests to track down the reason for different issues. For instance, a semen investigation for barrenness might identify asymptomatic incendiary prostatitis. This type need not bother with treatment ([Nickel et al., 1999](#); [Zlotta et al., 2014](#)).

7.4 Need for prostate tissue engineering

Tissue engineering provides a way to get beyond the constraints of existing management methods for prostate conditions as well as the problems brought about by current medical treatments. The biotechnological capability of mesenchymal stem cell extraction, in vitro propagation, and insertion, which has been developed in recent years, has become the foundation for novel cell-based treatments aimed at inducing adult prostate tissue regeneration. Healing is a complex defense mechanism that protects the body from infection and additional harm. Average adult lower urinary tract restoration is controlled primarily by a chronic progressive response that results in engraftment. Through tissue engineering applications such as injection stem cell therapy, normal function of the prostate can be restored.

Moreover, the current treatment methods used in tackling prostate conditions have been found to leave patients with adverse side effects. The side effects for PCa treatment are unacceptable and tissue engineering is aimed at finding better treatment procedures that do not have bad side effects and treating the side effects already present due to the harsh and widely practiced treatment methods. The 80% incidence rate of urinary incontinence postprostatectomy, ED, impotence, and reoccurring PCa are why there is a great need for a tissue engineering treatment approach to be developed to tackle these ([Adamowicz et al., 2021](#)).

7.5 Tissue engineering approach

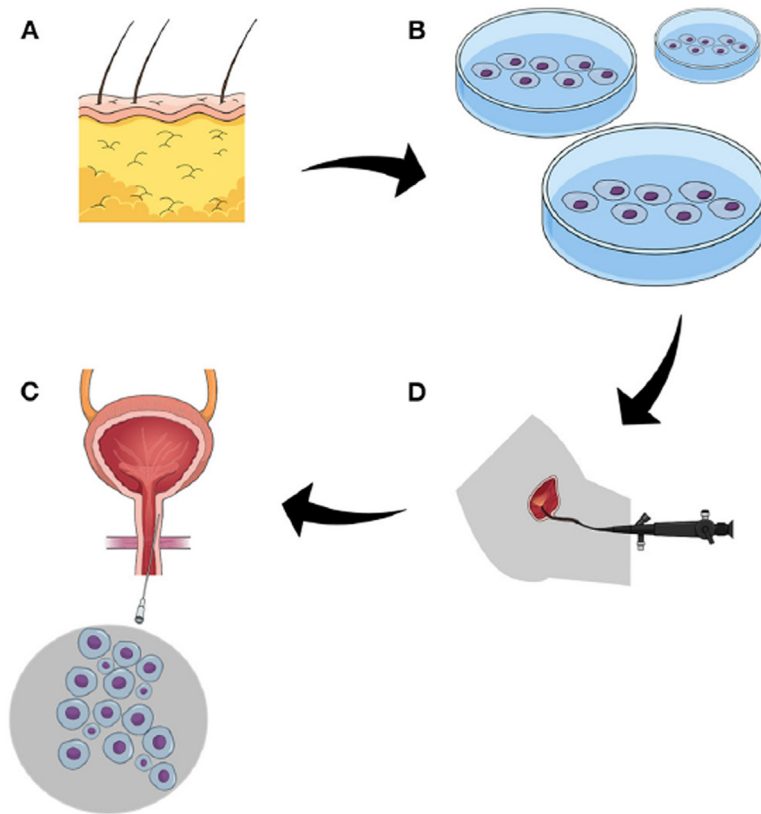
Prostate tissue engineering is a relatively new field of interest despite PCa being a high ranked disease among men and prostate problems associated with many conditions that affect men such as ED and urinary incontinence. In this section, the current research will be discussed based on using tissue engineering techniques and methods for treating the various conditions of the prostate and the side effects brought about by the current medical treatment processes.

7.5.1 Injection stem cell therapy

Recent years have given rise to the use of invitro cultivated autologous mesenchymal stem cells aimed at tissue regeneration in postprostatectomy patients (Adamowicz et al., 2021). This stem cell therapy was used in treating prostatectomy incontinence and ED. Reported cases of incontinence can be as high as 80% postsurgery despite advanced surgery techniques and equipment. This is because prostatectomy is a highly invasive procedure and results in damage to mechanisms responsible for continence such as damage to the proximal urethral sphincter and continence mechanisms are highly dependent on the rhabdosphincter (Hoyland et al., 2014). The main cells of choice in the investigation by Adamowicz et al. were the bone marrow-derived stem cells because of their high proliferation rate and their tendency to boost wound healing through reepithelialization and neovascularization by producing growth factors (Adamowicz et al., 2021).

Some of the growth factors that were observed to be released by BMSCs were vascular endothelial growth factor (VEGF), hepatocyte growth factor (HGF), and basic fibroblast growth factor (bFGF) which promote angiogenesis. However, this study has shown that the sole use of stem cells without a 3D biomaterial to aid as a scaffold to deliver the cells to the target location and to fully heal the damaged area is still lacking. The acquisition of BMSCs is an invasive and expensive procedure so a cheaper and less invasive alternative has been researched and was discovered as the adipose-derived stem cells. These cells have been investigated to have the same differentiation capability as BMSCs and how they can differentiate in vitro when cultured next to myoblasts (Zuk et al., 2002). It was also recorded that ADSCs can be a source for smooth muscle cells that can be used to repair damaged tissue (Fakhrieh et al., 2019).

The process of the stem cell therapy is to acquire ADSCs as illustrated in A which are the purified and propagated as shown in B (Fig. 7.1). The complex of the sphincter is assessed through the use of an endoscope before they are administered to the target area for tissue regeneration by endoscopic periurethral injection. This method has shown an 88% improvement of urinary incontinence and ED for postprostatectomy patients (Adamowicz et al., 2021).

**FIGURE 7.1**

Injection stem cell therapy (Adamowicz et al., 2021).

7.5.2 Tissue generation

Hepburn et al. made a study in which they showed that they can regenerate an entire prostate gland in vitro and in vivo in nude mice. The regenerated tissue was made from a coculture of human embryonic stem cells (ESCs) and rodent urogenital sinus mesenchyme (UGM). The result showed full prostate histology in the mice after harvesting the organs after a 12-week period. The results obtained are shown in Fig. 7.2 (Hepburn et al., 2020).

In A on Fig. 7.2, induced pluripotent stem cells are cocultured with UGM as a xenograft in nude mice. After a period of 12 weeks since this coculture was injected into the mice, the mice were sacrificed and the resulting tissue from the culture was studied. The resulting tissue in C is similar to the in situ human prostate

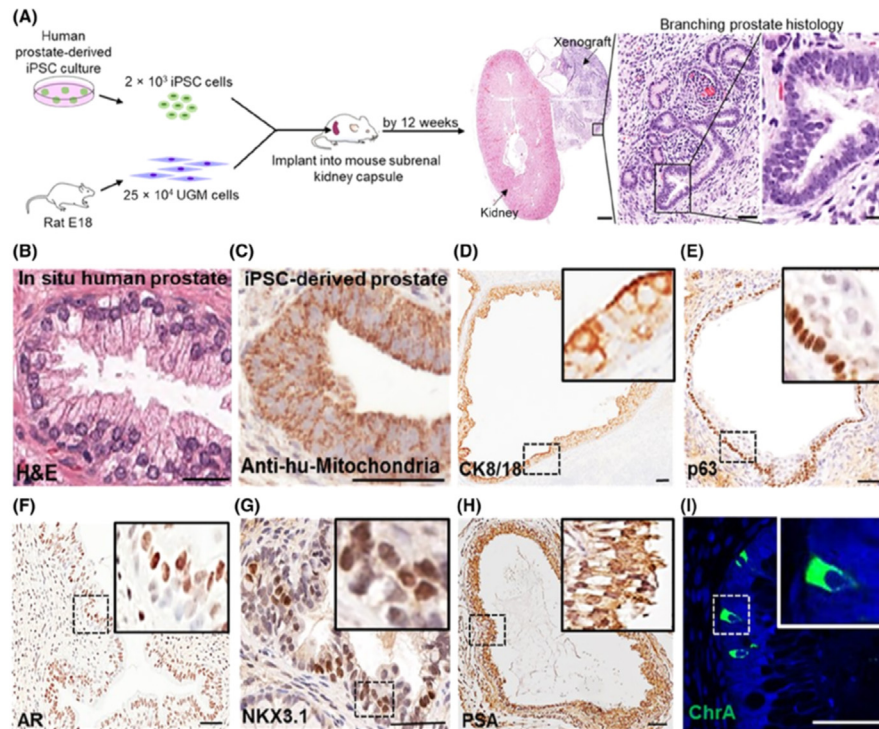


FIGURE 7.2

Generation of prostate tissue in vivo (Hepburn et al., 2020).

in B. Anti-Hu mitochondria was used to mark and confirm that this was indeed human tissue that was generated and the prostate epithelium was confirmed by using CK8/18 and p63 in D and E. The fully differentiated prostate tissue was confirmed using the NKX3.1 marker in G and this was supported by the presence of PSA as well.

This is a promising study for further research as it could enable the in vivo generation of the prostate gland in patient who have gone through prostatectomy. This study was also conducted in vitro and it showed similar results where a fully differentiated prostate histology was achieved (Hepburn et al., 2020).

7.5.3 Biomaterials

Currently there are no scaffold materials used for treating prostate problems. The bulk of the research is on the use of scaffold in culturing PCa cells in order to study them and understand the disease better. This is brought about by the shortage of PCa samples as they are usual ethical issue on acquiring samples from patients or cadavers.

7.5.3.1 Porous chitosan alginate

This biomaterial is usually used for culturing bone cells, but there has been a recent avenue to culture PCa cells with a 3D porous version of it. It is acquired from shells of crustaceans such as crabs. This material has been found to promote the independent growth of PCa cells as well as help them express themselves phenotypically (Xu et al., 2019). This property can be extended to normal prostate tissue to enhance the regeneration of the tissue and its growth.

7.5.3.2 Collagen

Collagen is another biomaterial that has been successfully used to culture PCa cells. A mixture of collagen and glycosaminoglycan or nanohydroxyapatite has been shown to enhance the growth of PCa cells and this natural biomaterial with its composite can be applied in prostate tissue engineering and regeneration (Fitzgerald et al., 2015).

7.6 Conclusion

The tissue engineering applications for prostate tissue regeneration and generation have come a long way in a short period of time. The current studies underway such as the one conducted by Hepburn et al. show great promise that offers hope to men across the world that they do not have to live with prostate problems for long. Despite the limitations of neuroregeneration with damaged prostate tissue, a lot of research is being poured into finding solutions and tackling these limitations. A topic for future prospect would be to tackle most of the nerve damage associated with prostate problems as well as to safely move the research that are at animal testing stage to human trials. There is also a need to research how biomaterials can be utilized in prostate tissue regeneration to act as scaffolds for stem cell therapy. This could lead to opening a new avenue which could see an even greater development of more complete prostate tissue encompassing a fully regenerated neural network.

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The progress in tissue engineering of testicle

8

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8.1 Introduction

There have been many extensive investigations to demonstrate the anatomy, biomechanics, functions, and problems associated with testicles (testes) in recent decades. The possible solutions from a tissue engineering point of view, using biomaterials, to solve specific anatomical problems of the testicles have been implemented since the early 1940s. This coursework is to prepare a review paper on testicles. As a functional part of the male reproductive system, the testicle is the male reproductive gland responsible for producing sperm and hormones called testosterone (Suominen et al., 2015). Testosterone is the hormone that is mopped up into the bloodstream to oversee the physical development of the body, such as male sex organs, facial and body hair, muscle mass, etc., especially in both the young and adult stages. Unfortunately, certain factors like testicular cancer, testes malformed at birth, injury or torsion, and infection can cause a person to lose the testes (Rubin, 1980, p. 665). To this end, testicle repair and regeneration are of distinct interest to the healthcare community, especially the tissue engineering department to arrive at the various challenges that patients face when they lose or destroy their testicles (Turner, 1915). Even though testicle replacement is of foregoing interest, the research community is striving hard to achieve novel solutions and approaches to treat testicular difficulties. According to the American Cancer Research Society report for 2016–2018, 1 in every 250 males can develop testicular cancer in their lifetime. In every year, it is estimated that about 6% and 8% of testicular cancer occur in children and males who are over 55 years old, respectively, out of the 9470 cases recorded in the USA. Unfortunately, about 400 people die of testicular cancer every year in the USA. In the

United Kingdom, testicular cancer is prevalent in males between 30 and 34 years old, but the main reason why is not known. The testicular cancer rate has increased to about 7% from 2006 to 2018, compared to 27% since 1990. According to UK cancer research, testicular cancer can rise to 12% in the UK by 2035, but the survival rate is 95% if treated on time ([Cancer Research, UK, 2021](#)).

8.2 Anatomy and physiology of the testes

The testes are a pair of male reproductive gonads enclosed in a sac called the scrotum. The normal testis is about 4–5 cm long, 2–3 cm wide, and 2.5–3 cm thick. The two testes are separated by a scrotal septum. The main function of the testes is sperm production and the secretion of a sex hormone called testosterone ([Hussain & Gilloteaux, 2020](#)). The testicles have both the superior and inferior ends, which are slung by the spermatic cord and scrotal ligaments, respectively. Generally, the right testis is a bit longer than the left testis ([Gartman, 1957](#)). The testicle is also covered with a lasting fibrous called tunica albuginea, extended from tunica vaginalis. The tunica albuginea sends septic to separate the testes into lobules. Each lobule contains about four seminiferous tubules ([Mathur et al., 2007](#)). The transparency of the tunica albuginea enables the gender of the child to be identified if an ultrasound is performed. At the end of the testes are a mesh-like structure called the rete testis, which receives the sperm produced and migrates them into the epididymis for maturation and motility via the efferent ductile. The sperm produced by the testicle leaves the epididymis via the vas deferens and continues to the seminal vesicles and prostate. The fluid produced by the seminal vesicles mixes with sperm to form semen, which is discharged through the penis during ejaculation ([Sakhnini & Dairi, 2007](#)). The length of the epididymis allows it enough space to store and mature the sperm. Follicle-stimulating hormone and luteinizing hormone control the levels of sperm and testosterone, respectively. The tunica wraps the entire testes, excluding the junction where the superior and posterior ends separate to attach the spermatic cord and epididymis. The epididymis covers the side and back of the testicle. The tunica is very close to the testes, epididymis, and deferens. When viewed from the posterior lateral surface, there is a gap between the testes and epididymis. The testis is connected to the head of the epididymis, which receives the sperm produced by the testes ([Lee & Houk, 2013](#)). The ductus deferens received the matured sperm and transports it to the male urethra. The testicles move down into the scrotum during the early stages of life ([NIE et al., 2009](#)). The absence of this transformation will lead to the injection of hormones, or surgery may be recommended to move the testicles into the appropriate position in the scrotum. The tunica albuginea covers the testes ([Ravikanth, 2018](#)). The seminiferous tubule is the factory for sperm production in the male reproductive system. The innermost part of the seminiferous tubule is the lumen, which is surrounded by epithelial cells. Two cells cover the seminiferous tubule ([McElhinney & Raja, 2015](#)). These are germ cells and Sertoli

cells. The germ cells, usually called spermatogonia, are nourished by the nutrients produced by the Sertoli cells. The capillaries and Leydig cells are outside the tubule. The capillaries supply blood to the testes, while the Leydig cells produce testosterone to develop the sperm.

8.3 Pathophysiology of the testicles

There are lots of testicular pathologies in andrology, but the most common one is germ-cell cancer. Germ cell cancer can be classified into two types, namely seminoma and nonseminoma. This includes both benign and nonbenign diseases, which can lead to the implementation of a tissue engineering approach. Some of the diseases that affect the testicles can be left without seeking medical attention and will heal by themselves with time. Risk factors that can cause testicular cancer include cryptorchidism, hydrocele, testicular microlithiasis, and testicular cancer (Colombo et al., 2015).

8.3.1 Cryptorchidism

Statistics show that about 3%–4% of babies undergo full-term care, whereas 21% of premature babies are diagnosed with cryptorchidism (undescended testicles). Usually, one testicle delays descending into the scrotum, but in a case where both testicles are not descending, the delay is 10%. It is a condition in which one or both testes are not able to slant into the scrotum at the stage where the fetus is developing (Okeke & Osegbe, 2001; Samm & Dmochowski, 1996). The formation of testicles starts from the abdomen during gestation and descends into the scrotum via the inguinal canal, usually when the baby is in the eighth month of pregnancy (Berko-witz et al., 1993). The undescended testicles after 3 months can be identified by feeling them during a physical examination (palpable). About 80% of the testes are not descended during the gestation period. The testis is mostly found in the inguinal canal in the scrotum. The inguinal canal transports the spermatic cord to the scrotum and penis (Burge, 1989). Symptoms of cryptorchidism include abdominal pain and a high temperature, which can destroy the quality of the sperm (Lee, 2018). Orchiopexy is the medical procedure used to bring down the undescended testicles into the scrotum. The procedure is usually done before the patient is more than 7 years old, as delay can cause more risks to the spermatozoa cells (Abo-lyosr, 2006).

8.3.2 Testicular microlithiasis

Testicular microlithiasis is a testicular disease in which a cluster of calcium forms around the testicles. According to the research conducted by Suominen microlithiasis has a relationship with testicular cancer since the actual cause of this condition is not known (Pedersen et al., 2016; Suominen et al., 2015). Microlithiasis prevalence

in the UK ranges between 0.68% and 5.6%, subject to the time of detection. Sometimes the body heals itself (La Vignera et al., 2012). This condition is detected between the 10th month and 74 years (Corut et al., 2006). The intratubular germ cells produced by microliths develop a tumor in the testes when exposed to excess calcium (Cast et al., 2000). According to the table below, there is a greater chance of being affected by testicular microlithiasis when you are below 40 years old. The table shows an increase in the prevalence of spermatocele as the risk factors are attributed to the cause of testicular microlithiasis (Peterson et al., 2001; Vegni-Talluri et al., 1980).

8.3.3 Hydrocele

This is a disease that affects the testicle when there is too much fluid in one of the testes. Mostly, this happens at birth, injury, or inflammation. There are basic symptoms of obvious swelling around the testes, scrotal heaviness, and aches. This can lead to a hernia if left untreated. Surgical operations are performed in more severe cases, but they mostly don't need treatment (Dandapat et al., 1990; Kapisiz et al., 2011).

8.3.4 Epididymitis

The epididymis is a condition where the epididymis dilates and expands the scrotum. There is no pain associated with varicocele, but it can lead to a low sperm count, thus infertility (Mentessidou & Mirilas, 2020). It is an infection of the epididymis, which is usually caused by sexually transmitted diseases, injury, or vasectomy. Epididymitis can affect men who are between 14 and 35 years old. The symptoms include swelling, pain, nausea, fever, and vomiting. The epididymis is mostly treated with scrotal support, antibiotics, and antiinflammatory medication (Robaire et al., 2006).

8.3.5 Varicocele

Varicocele is another disease that can affect the testes. This is a condition where the varicose or scrotal vein enlarges due to excessive blood flow through the vein. Statistics indicate that 25% and 15% of adolescent and adult males, respectively, are of higher prevalence. This condition retards the development of the testicles, which can lead to infertility, low sperm production, and low-quality sperm if left untreated (Akbay et al., 2000).

8.3.6 Spermatocele

It is also called a spermatic cyst, a fluid that fills the epididymis. This disease is painful when the cyst enlarges and can cause heaviness in the affected testicle. Surgery is the only option for spermatocele. There are no known factors that contribute to spermatocele. Diethylstilbestrol is a drug to prevent complications and miscarriage

during pregnancy, thus contributing immensely to spermatocoele. Due to the side effects of this drug, such as vaginal cancer and spermatocoele, the FDA banned it from the market in 1971 ([Rioja et al., 2011](#)).

8.3.7 Testicular torsion

This is a condition when the testicle alternates in the scrotum. Testicular torsion changes the arrangement of the spermatic cord, thereby affecting the transport of sperm and blood supply. Pain in the abdomen and testicles, as well as swelling, are common symptoms of testicular torsion. In some severe cases, vomiting, frequent urination, and nausea can occur. Factors that can lead to testicular torsion include too much exercise, injury, and a cold temperature. The most common treatment is by hand to reposition it ([Mukendi et al., 2020](#)).

8.3.8 Hypogonadism

It is a condition when your testicles make fewer hormones (low levels of testosterone). This can result in a lot of testicular complications, especially when the hormones produced are not enough to stimulate the brain. Although injury or infection can cause hypogonadism, most of the victims were born with it. Sometimes the genitals will be there, but they are very small to identify. This condition leads to infertility, loss of body hair, erectile malfunction, and loss of bone density. This condition is treated by replacing the hormone produced either by the brain or testicle ([Seftel, 2006](#)).

8.3.9 Testicular cancer

This is a condition when cancerous cells grow around one or both testicles. It usually starts from the testicle internal structures that are responsible for sperm production ([de Kretser, 2004](#)). A lot of research conducted has not confirmed the exact cause of testicular cancer, but it has been attributed to untreated lumps in the testicles, fluid, or heaviness in the scrotum, testicular, or abdominal pain ([Nielsen, 1997](#)). The testicular cancer cells can be treated, depending on their nature (seminoma or non-seminoma), by chemotherapy or radiotherapy ([Wu et al., 2007](#)). In some cases, total or partial removal of the testicle(s) is also recommended, depending on the nature of the cancer cells around the testicle(s), and the process is called orchidectomy ([Baird et al., 2018](#)). [Table 8.1](#) represents the current treatment for testicular cancer.

8.4 Biomaterials used for testicular tissue engineering

The main remedy for testicular cancer is either total or partial removal of the affected testicle(s) and replacement with a compatible implant from a biomaterial. Testicular implants started in the early 1940s, and different materials have been

Table 8.1 Current treatment of testicular cancer with their respective benefits and side effects (Goodman, 2003).

Current treatment	Benefit	Side effect
Pain killers	Painkillers such as Bleomycin Sulfate, Cisplatin, Cosmegen, Dactinomycin, Etopophos, Etoposide, Ifex, Ifosfamide, Vinblastine sulfate cause pain relief	Nausea, drowsiness, dizziness, sweating, depression, a weakened immune system, tolerance, addiction
Orchidectomy (testes surgery)	Decrease of testosterone production and suppression of cancer growth	Pain, bleeding, fever, urination difficulty, hematoma, loss of feeling around your scrotum, osteoporosis loss of fertility, hot flashes, depression, erectile dysfunction
Chemotherapy	It stops the cancer cells from multiplying, growing, and dividing	Nausea, tingling in the hand feet, and hands, vomiting, fatigue, numbness, high pitch loss, and damages to the blood cells
Sperm banking	To monitor genetically transmitted disorders, hypogonadism	Infections, sperm contamination, and exposure of the spermatozoa to inappropriate weather conditions
Testosterone replacement therapy	It restores the shape and appearance of the testes	Inflammation reaction and infection

used to treat testicular cancer. Examples of synthetic biomaterials used include a metal alloy, Lucite, marbles, directly injected gel foam, Plexiglas, polyethylene, solid and gel-filled silicone, PLGA, PCL, PLA, Dacron, etc. Conversely, due to the numerous side effects of the synthetic biomaterials used to find a remedy for testicular cancer, natural materials such as collagen, alginate, and chitosan have also been used, but [Tables 8.2 and 8.3](#) detail the advantages and disadvantages of the materials used previously for a testicular implant. This procedure enables the scrotal sac to appear the same as normal when viewed from the outside, but is not able to produce.

8.4.1 Extracellular testicular matrix

In mammals, the extracellular matrix is the base membrane located between the epithelial cell and the connective tissue. The tubular wall in the testes is the main membrane that lies between connective tissue and the seminiferous tubule. The tubular wall is made up of fibronectin, collagen, and laminins ([Baert et al., 2019](#)). The extracellular matrix proteins support the various connective tissues and tubular walls in the testes. The components of the extracellular matrix of the tubular walls control the growth factors, especially growth factor beta 3. During spermatogenesis,

Table 8.2 Advantages and disadvantages of synthetic and natural biomaterials used for testicular tissue engineering.

Material	Source	Advantages	Disadvantages
PP	Synthetic	Inert, biocompatible, good mechanical properties, supports tissue ingrowth	High inflammatory response, high stiffness
PLGA	Synthetic	Biocompatible, good for drug delivery, low degradation rate	Loss of mechanical stability
PLA	Synthetic	Biocompatibility, tissue ingrowth, low biodegradable rate	Deprivation of tensile strength, development of harmful oxidative species
PLC	Synthetic	Biocompatible, FDA-approved, long degradation rate, easy processibility, cost-effective	Poor antimicrobial properties, possible mismatch of mechanical properties
PET	Synthetic	Cytocompatibility, tissue ingrowth, minimal inflammatory reaction	Erosion and infection
PTFE	Synthetic	The good mechanical property, biologically stable	Poor tissue ingrowth, erosion
PGA	Synthetic	Low inflammatory reaction, resistance to adhesion	Lack of tensile strength
Dacron	Synthetic	Cheap, recyclable, high longevity	Low elasticity, nonrenewable, nonbiodegradable, and low porosity
Lucie	Synthetic	It is ductile, water-repellant, translucent, recyclable	Expensive, prolonged production time
Alginate	Nonsynthetic	Enhance cell growth, biocompatible, biodegradable, hydrophilic, hemostatic properties	No hydration properties, rapid degradation
Chitosan	Nonsynthetic	Biodegradable, good for drug delivery, biocompatible, and antibacterial properties	Low mechanical properties, short half-life, limited solubility, biostability, and solvents
Collagen	Nonsynthetic	Biodegradable biocompatible, enhancement of fibroblast activity	Expensive, fast degradation rate, fragile, rapid swelling

peptides, collagen, and laminins contribute to the permeability of blood to the testis barrier. The microenvironment of the testes is because of the decellularized testicular matrix (DTM), which enhances cell attachment and proliferation (Siu & Cheng, 2004).

Table 8.3 Different mechanisms for decellularization and their respective advantages and disadvantages.

	Technique	Advantages	Disadvantages
Physical technique	Supercritical fluid dissolution	Easy to operate	It is difficult to remove thicker tissues
	Freezing and thawing	No chemical reagent is needed to decrease the reactions to the body	
	Pressure gradient	It has a low response to the immune system	There is an effect on the mechanical properties due to the physical destruction of the ECM
	Mechanical oscillations	Removal of cells is effective	
	Sonication		
Chemical technique	Electroporation	Less residual cells are efficiently removed	It reduces some tissue and organ growth factors and the bioactivity of the extracellular matrix The chemical residual reagents increase the negative immune response thereby reducing biocompatibility
	Acids and bases		
	Tonic and nontonic detergents		
	Hypotonic and hypertonic solutions		
	Zwitterionic detergents		
Biological technique	Collagenase	It helps to remove specific cells and antigen components	It affects protein microstructures
	Thermolysis	It does not cause much damage to the bioactive components	There is no specific time for enzyme treatment
	Nucleases		
	Trypsin		
	Dispase		
Combination technique	This comprises a combination of biology, chemistry, physics, and other methods	It reduces the immune response It reduces the damage to ECM mechanical properties	A better combination technique is required to meet the specific requirement for different organs or tissues

8.4.2 Decellularized testicular matrix

Testicular tissue decellularization is a process that can be achieved by either a physical, chemical, or enzymatic reaction to isolate cells and cellular materials without causing damage to the cell. The extracellular matrix has been accepted as a prospective biomaterial suitable for a testicular implant but needs decellularization to remove or kill any infection that can be transmitted. Sodium dodecyl sulfate is used to obtain DTM from human testicular tissue (Steinberg et al., 2015). The extracted testicular stem cell (TSC) is cultured on the decellularized scaffold. The DTM contains factors needed for cell proliferation and differentiation, such as proteins, growth factors, nutrients, and other conditions to enhance cell attachment and growth. The purpose of decellularization is to mimic the testicular environment to reproduce cells capable of producing sperm. This technique is ideal to solve male infertility challenges (Guler et al., 2018).

8.4.3 Natural biomaterials

A biomaterial selected for a testicular implant must be biodegradable, biocompatible, and have good mechanical strength to suit the location of the native tissue. So many different natural polymers have been used in testicular implants, such as chitosan, collagen, hyaluronan, silk, and alginate. All these natural biomaterials have been used, and their advantages and disadvantages have been outlined in Table 8.2 below.

8.4.3.1 Collagen

The easiest protein located in the extracellular matrix is collagen. Collagen supports both intracellular communication and the behavior of the cell. The porosity of the collagen scaffolds is made biocompatible by mimicking tissue integrity. The 3D collagen scaffold integrates well with new tissues due to its flexibility and tensile strength with proper adhesion (Hansen et al., 2013).

8.4.3.2 Alginate

Alginate is a polysaccharide that is biodegradable, biocompatible, and provides good adhesion to cells. Alginate is one of the natural biomaterials that are suitable for the survival of seminiferous tubules and spermatogonia cells. Sodium alginate cross-linked with calcium is suitable for biological applications, especially porous structures of 200 μm diameter. The low-toxicity nature of alginate hydrogel permits differentiation of male germ cells, which resembles the extracellular matrix (da Silva et al., 2013). According to Fan et al. alginate can create tubular channels through positively charged polyelectrolyte complex chains. According to Zhao et al. (2020), alginate improves the growth of germ cells and the microenvironment of the testes to save them from disruption of the spermatogenic process (Zhao et al., 2020).

8.4.3.3 Silk

Spiders and silkworms produce silk, a natural protein. It is biocompatible and supports cell adhesion, locomotion, differentiation, and cell proliferation. Silk is tough, has good mechanical strength, is elastic, and is ductile. Silk can be fabricated into powdered forms, sponges, hydrogels, and fibrous materials. Despite the advantages of silk in testicular tissue engineering, it has a fast degradation time for revascularization (Zhang et al., 2019). The electrospun silk scaffold provides an environment like the extracellular matrix to guide spermatogonia stem cell (SSC) proliferation. SSCs were isolated and encapsulated on a silk scaffold for 2 weeks. After 2 weeks, it was discovered that the scaffold was the appropriate substrate to multiply the viability of spermatogonia sperm stem cells and proliferation during the cell culturing period (Narimanpour et al., 2020).

8.4.3.4 Fibrin

Fibrin is mostly used as a substitute for collagen. Mostly, fibrin is extracted from blood and used as a scaffold. It has high porosity and regulatory degradation of about 1- μ m-diameter pores. Fibrin and collagen matrix gel have commonly been used as protective mediums for testicular deficiencies. Fibrin scaffolds enhance the survival of spermatogonia cells and angiogenesis (Seol et al., 2005).

8.4.3.5 Hyaluronic acid

This is a polysaccharide located in the extracellular matrix of animals. Hyaluronic acid functions as homeostasis and permeability regulation. It is biodegradable, non-immunogenic, and biocompatible (Jensen, 1954). It has a viscoelastic property that enables it to create a unique microenvironment for the attachment of cells and the binding of sperm receptors and motility (Watanabe et al., 2011).

8.4.3.6 Chitosan

Chitosan has been used in diverse ways in testicular tissue engineering. It is a deacetylated type of chitin. Its physicochemical property creates many opportunities for it to be used to design different desired scaffolds (Ardila et al., 2017). Chitosan can be combined with other biomaterials, such as cells, nanoparticles, etc., for testicular implants (Vermeulen et al., 2017). It has specific biological and mechanical characteristics to mimic the nonsynthetic extracellular matrix. Similarly, chitosan is biodegradable, biocompatible, has tensile strength, and has a high antimicrobial characteristic against bacteria. These properties of chitosan make it a perfect choice of natural biomaterial for testicular tissue engineering as it promotes tissue integration, encapsulation, cell adhesion, proliferation, and differentiation, which are fundamental requirements for a biomaterial for testicular tissue engineering (Poels et al., 2016).

8.4.4 Synthetic biomaterials for a testicular implant

In selecting synthetic biomaterials for a testicular implant, various properties such as elasticity, degradation ability, and mechanical stability are of prime consideration.

There are a lot of synthetic biomaterials in the market for testicular engineering as compared with nonsynthetic ones (Table 8.2), but they may potentially cause immunogenicity and inflammation after transplantation. According to Peng et al research, some synthetic biomaterials such as PGA, PLA, PCL, and PLGA have been tried on the reproductive system and have achieved limited success (Peng et al., 2017). PCL, as a synthetic biomaterial, has high porosity and is suitable to mimic the structural characteristics of testicular engineering. A 3D-printed PCL scaffold has been used on porcine to keep the follicle architecture (Guvendiren et al., 2010). Studies have shown the limitations of synthetic biomaterials outweigh their advantages as compared with natural biomaterials.

8.5 History of biomaterials used for testicular prostheses

In 1940, there were massive changes and developments after the Second World War. The affected healthcare, engineering, and other industrial sectors began to reconsider different ideas to sustain life. Given this, the first testicular prosthesis was designed from an alloy of chromium and molybdenum called Vitallium in 1941 (Girsdansky & Newman, 1941). According to Sudhakar, different tests, such as microhardness tests and tensile tests, were conducted on the material to check the mechanism of fracture, and the analysis concluded that the implant was susceptible to corrosion (Sudhakar, 2005). As part of developing the vitallium for the testicular implant, a high-quality acrylic plastic resin material called Lucite was introduced into the market in different sizes for the testicular (Rea, 1943). Other materials, such as Gelfoam were used in the 1950s following intracapsular orchidectomy. The material is injected into the tunica to repair prostate cancer (Baumrucker, 1957; Merrill, 1991). The FDA rejected the product due to a lot of side effects, such as fever, blood clots, reactions, and infections (Barbolt et al., 2001). Most of the other materials before 1960, like Plexiglass and Dacron, were not successful due to failures in mechanical testing, structural stability, elasticity, and toxicity (Raya-Rivera et al., 2008). Given this, an ideal testicular prosthesis that can resist mechanical strain is sterilization-friendly, noncarcinogenic, and structurally stable was proposed by Latham et al. (1974; Merrill (1991). As a result, different prostheses were developed with silastic and solid silicon rubber for testicle implants in 1960 (Turek & Master, 2004). The implementation of the saline-coated product won the market and became the best biomaterial, but less research has been conducted on it to find its side effects. The product was mainly used for breast and testicle implants, but the FDA halted it due to autoimmune disorders, mechanical stability, cancer development, and risks to connective tissues (Latham et al., 1974; Robinson et al., 1995). In 1995, silicone-filled gel breast implants faded out of the system due to concerns raised about the various side effects on the users. Saline-filled testicular prostheses were introduced as a substitute for the silicone-filled gel

in the US. Studies conducted by [Turek and Master \(2004\)](#) to assess the safety of the Mentor saline-filled prosthesis, taking into consideration the connective tissues, indicate that the implantation is associated with infection and structural instability.

8.6 Tissue engineering approach

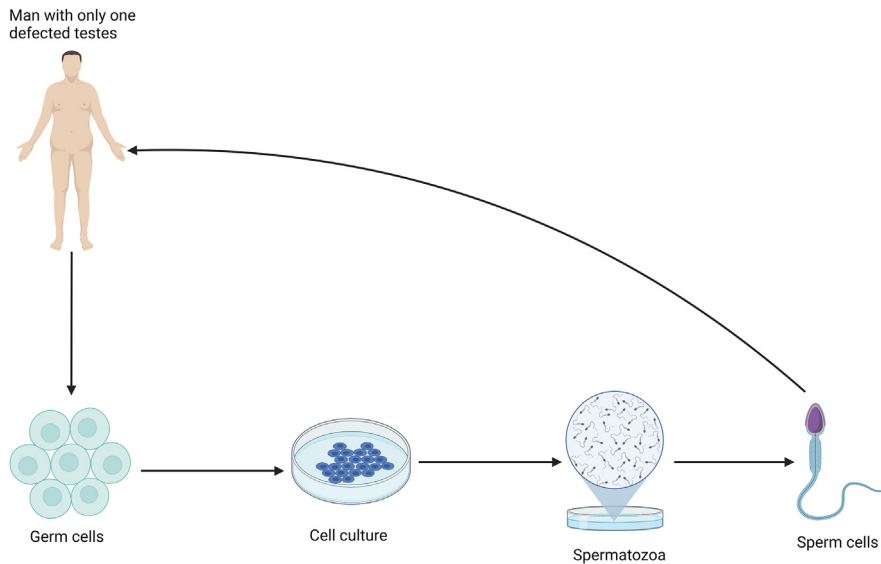
A lot of factors have led to the further implementation of a new approach to mimic problems associated with testicular cancer. Some of these factors include lethargy, low libido, infertility, erection difficulties, fatigue, reoccurring cancer after chemotherapy, destruction of blood cells, etc. ([Hart, 1995](#)). Several tissue engineering approaches have been implemented to calm down the problems associated with the male reproductive system, especially the testes. Such techniques include cell-based and scaffold-based ([McGlynn & Trabert, 2012](#)).

8.6.1 Cell-based tissue engineering

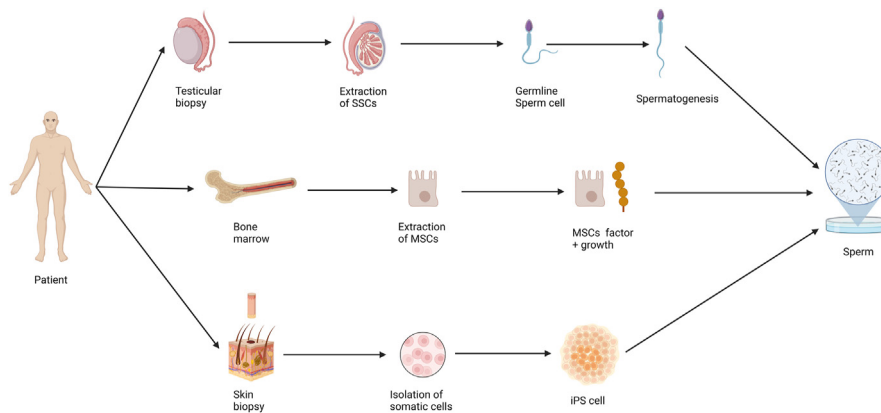
This technique is based on isolating cells from the same site of the affected tissue or organ from the same person (autologous) or a donor (allogeneic). The isolated cell is decellularized and cultured before transplanting. There are four different types of cell-based tissue engineering approaches. The transplanted cell is either isolated from (1) local connective cell progenitors, (2) autologous cell progenitors, (3) cultured or modified cell progenitors, or (4) fully formed progenitors ([Muschler et al., 2004](#)). In autologous cell-based tissue engineering, the cells are isolated from the same person. If the isolated cells are defective, they undergo decellularization and are cultured at a certain control to direct the growing process to suit the location of transplantation. Some of the growth factors include fibroblast growth factors (FGFs). There are 23 members of FGFs, but there are six (6) of them that are suitable for testicular cell-based tissue engineering techniques. These are FGF-1, FGF-2, FGF-4, FGF-8, FGF-9, and FGF-21 ([Yoshioka et al., 2016](#)). The tissue engineering cell-based approach has proven that natural biomaterials underpin cell proliferation and adhesion; however, synthetic biomaterials are fabricated into any shape of choice to support biocompatibility ([Brehm et al., 2012](#)).

The primary cells involved in testes include Leydig cells, Sertoli cells, and spermatogonia cells. The germ cell that differentiates into spermatozoa is isolated from the patient and cultured ([Oh et al., 2012](#)). The necessary temperature and growth factors are employed in the process to produce a potential sperm to function normally, as shown in [Fig. 8.1](#).

The diagram below ([Fig. 8.2](#)) shows how different stem cells are isolated from different parts of the body to produce sperm.

**FIGURE 8.1**

Cell-based tissue engineering.

**FIGURE 8.2**

Cell-based tissue engineering approach.

8.6.2 Scaffold-based tissue engineering

There has been an immense advancement in developing biomaterials to assist transplanted cells. This is a technique used to treat, replace, or augment part(s) of the body that is destroyed. The scaffold should be biocompatible, biodegradable, stimulate native ECM protein synthesis, and nontoxic to the body after degradation to the

physiological structures around the targeted site (Das et al., 2013). The scaffold can be synthetic biopolymers (PLA, PLGA, PGA, PCL, etc.) or natural polymers (collagen, alginate, chitosan, etc.). Porosity is very important in considering scaffolds to enhance cell attachment. There is a greater need for cell biology, specifically cellular differentiation, extracellular matrix formation, and cell-to-cell interaction formation, in tissue engineering to help scaffold fabrication (Nazarpak & Pourasgari, 2014).

8.6.2.1 Scaffolding techniques used for testis cell culture

- Extracellular matrix scaffolding technique
- Porous polymer scaffolding technique
- Electrospinning scaffolding technique
- 3D scaffolding technique

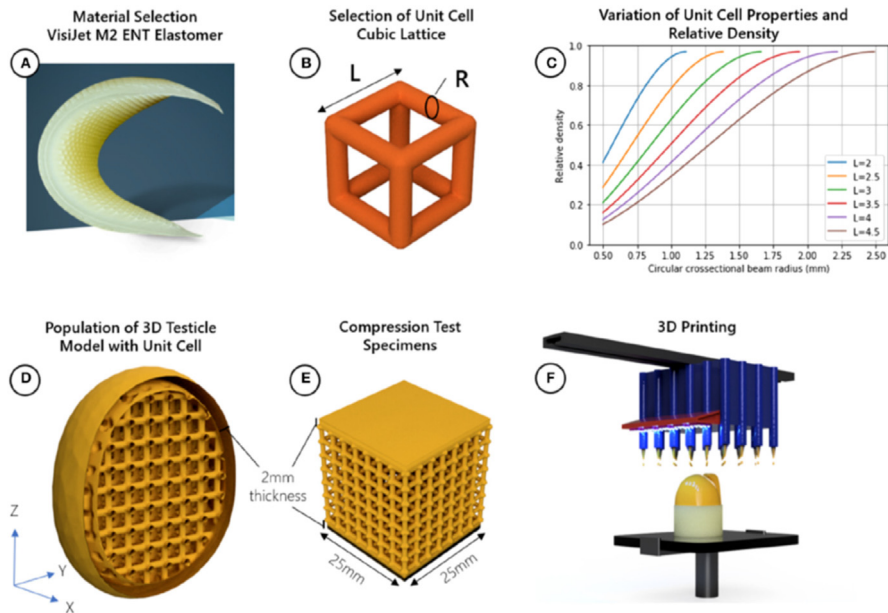
8.6.2.1.1 3D printing of testes scaffolds

Testicle prostheses become an option after orchidectomy for testicular cancer due to undescended testes, torsion, or other risk factors. The current prostheses are made from silicone filled with gel or fully solid. According to Turek and Master (2004) report, silicone-filled implants cause auto-immune disorders and ruptures; therefore, there is a need for an alternative soft-design implant. Silicone implant products in the market come in only small, medium, and large sizes, which makes it difficult for patients who do not fall within these categories. The implantation of odd sizes leads to dissatisfaction and regret among patients for going for the implant as it affects their sexual and other physical activities. The implant is sutured against the scrotum at the top, leaving the remaining parts to swing in the scrotum (Hayon et al., 2020).

8.6.2.1.2 Manufacturing of 3D testicular prostheses

A lot of complaints, such as size, firmness, position, and shape, have been identified with the use of silicone biomaterials for testicular prostheses. The 3D testicular prosthesis mimics the natural stiffness, size, and shape by using a lattice infill structure. The 3D-printed prosthesis is anatomically accurate compared with other prostheses on the market.

The 3D printer for testicular implants is called Project MJP 2500. The printer uses an inkjet printing head that receives the liquid photopolymer and dispenses it in droplets onto a building surface. The droplets are hardened by UV light. Any void in the printed prosthesis is amended with a wax support structure, as shown in Fig. 8.3. The X, Y, and Z resolutions are set to 1600 by 900 by 90 DPI, respectively, with a minimum feature size of 25 μm . The thickness in the Z direction depends on the DPI. The printer is set to factory default settings irrespective of the temperature, speed, time, and thickness of the material. With the help of a 3D printer for testicular implants, the correct size of the testis can be designed for implantation.

**FIGURE 8.3**

3D testicular saline prostheses (Skewes et al., 2021).

8.6.2.1.3 Electrospinning technique

Electrospinning is an ultrafine fiber production technique that employs electric force to pull charged threads of polymer solution into polymer fibers at high voltage. There are two sections involved during the process: the ohmic flow section and the convective flow section (Sharma et al., 2016). The ohmic flow section is where the molten polymer is injected into the syringe and subjected to a high voltage. When the voltage reaches a threshold, the electric charge overcomes the surface tension created by the repulsive force and generates a jet of charged materials. The accumulation of the jet-charged materials forms the fiber polymer. The connective flow section is where the jet-charged materials are. The fibers are accelerated to the target electrode, which is negatively charged (Tong & Wang, 2013).

8.6.2.1.4 Extracellular matrix scaffolding technique

The extracellular matrix is a noncellular structure that controls cellular functions. During pathological or normal conditions, with the help of enzymes, the ECM undergoes several remodelings (Chaput et al., 1996; O'Toole, 2001). The ECM is composed of biomolecules such as fibrous proteins (e.g., collagen, vitronectin, elastin, laminin, and fibronectin), glycosaminoglycans, and proteoglycans (Kusindarta & Wihadmadayatami, 2018). ECM is the basic factor needed during the formation of tissues in the body. Given this, ECM is used in several mechanisms, such as wound healing and testes prostheses. Wound healing depends on the

migration and proliferation of fibroblasts in the affected site. Fibroblast affects the process of reformation of epithelia and type III collagen to cover the wound (Qiang et al., 2019).

8.6.2.1.5 Decellularization

Decellularized materials are extracted from natural organs or tissues and undergo a decellularization process to remove antigenic components, maining only the structures of the extracellular matrix. Decellularized materials are gaining more attention due to their native structure, biodegradability, bioactivity, and less immunogenicity, which cannot be accessed by synthetic materials (Chu & Chao, 2011). Decellularized materials have great potential to outweigh the autograft disadvantages. Decellularized materials have been a promising material for tissue engineering. The mechanisms for decellularization include biological, physical, chemical, and combi-national methods, as shown in Table 8.3, with their respective advantages and disadvantages.

8.6.3 Stem cell used in tissue engineering approaches for testes

Stem cells are raw materials from the body that can be used to generate all other cells. They can be categorized into three namely: adult stem cells (somatic stem cells), induced pluripotent stem (iPS) cells, and embryonic stem cells. The purpose of these stem cells is to renew or restore damaged cells. The common stem cells extracted from the testes include epiblast stem cells, embryonic germ stem cells, multipotent germline stem cells, and multipotent adult germline stem cells (Wu et al., 2008).

8.6.3.1 The embryonic stem cells

The embryonic stem cells are extracted during the early blastocyte stage and can be segregated into many other cell types called pluripotent. Under normal circumstances, the inner cell mass grows to form the tissue and organs of the body, so when isolated from the body and growing in the laboratory condition will retain the embryonic stem cell properties. They have unlimited capacity to renew themselves and differentiate into other cells in the body (Blum & Benvenisty, 2007; Dang et al., 2004).

8.6.3.2 Adult stem cells

Adult stem cells are undifferentiated cells extracted from fully developed tissues in the body. The primary function of the stem cell is to repair damaged cells in the body. They are usually located in a microenvironment called the stem cell niche. Both intrinsic and extrinsic signals are produced by the niche to regulate and renew the cells (Prochazkova et al., 2015). Adult stem cells can be categorized into different types, which include neural stem cells, mesenchymal stem cells, skin stem cells, hematopoietic stem cells, epithelial stem cells, etc. Bone marrow was first discovered in 1961 as the adult stem cell that can provide multilineage,

hematopoietic colonies in the spleen (Hansen et al., 2013). Certain factors, such as age and cryptorchidism, affect the growth and regeneration of the testes, blood, and adipose tissues that help in sperm production and ejaculation and may lead to male infertility.

8.6.3.3 The induced pluripotent stem cells (iPS)

These cells are acquired from adult somatic cells in the laboratory to form embryonic stem cells via the expression of genes and growth factors. In 2006 and 2007, the Yamanaka lab reported the first iPS cells for mice and humans, respectively, isolated from their fibroblasts (Sui et al., 2013; Yu et al., 2007). iPS cells are generated from adipocytes, peripheral blood, oral mucosa, amniocytes, skin fibroblasts, and dental pulp cells (Panopoulos et al., 2011).

8.6.3.4 Testicular stem cells

These are mainly for the restoration and preservation of the testes. During sperm production from normal testes, the germ cell, Leydig cell, and Sertoli cell contribute to the overall sperm aggregate. Given this, cells such as mesenchymal stem cells and SSCs, together with scaffolds from both synthetic and nonsynthetic materials, are used as part of the techniques to restore male fertility with the help of fibroblast and hormonal growth factors.

8.6.3.5 Spermatogonia stem cells (SSC)

SSCs are in the basic membrane of the seminiferous tubules. The basic function is to ensure a smooth process of spermatogenesis and fertility. They are few but undergo mitotic divisions to multiply the germ cells and contribute to the genes for the next generation (Chiarini-Garcia et al., 2003). There is less understanding of the impact of SSC in stem cell therapy due to the complexity of multiplication in its differentiation properties and function in vitro (Valli, 2014). Since SSCs are scarce to isolate from the seminiferous tubule, it is crucial to develop an efficient and effective in vitro culturing protocol for cell- or scaffold-based test prostheses under optimal conditions to restore fertility (Sheng et al., 2009).

8.6.4 Fibroblast growth factors

The male reproductive system ensures the passage of hereditary traits to generations. It functions through endocrine and signaling pathways. This is achieved by gametes and hormones. A couple of cellular processes such as proliferation, motility, cell survival apoptosis, and cell division are signals cascaded by FGF receptors (Dereka et al., 2006).

The FGFs are hormonal factors that play a major role in some tissues, organs, and organ systems in the body. Regulations of the FGF signals serve as an antidote to the growth of numerous cancerous cells or tumor angiogenesis to enhance the development of the reproductive system, like testes. Statistics show that the hindrance of FGF signals can cause antiproliferation effects in vivo and in vitro (Brooks et al.,

2012). There are feedback mechanisms that regulate the FGF signals to induce proteins and cause receptor dephosphorylation. There are 23 members of FGFs, but only nine of them (FGF-1, FGF-2, FGF-4, FGF-5, FGF-8, FGF-9, FGF-13, FGF-18, and FGF-21) are applied as the basis to model and treat diseases associated with testes (Cotton et al., 2008).

8.7 Conclusion

The application of silicone has become the commonest and latest biomaterial used in so many industries, such as biomedical, pharmaceutical, and many others, because of its chemical stability, biodegradability, excellent mechanical properties, biocompatibility, and interrelation with host tissues. Silicone polymers are adaptable and can be constructed into distinct material types depending on the desired application. Apart from its thermal stability, it is suitable for therapeutic applications. However, due to inflammation reactions, structural deformity, leakage, and infection after its half-life, there is a need to resort to other tissue engineering approaches to restore male fertility and testicular malfunctions.

There has been an achievement in in vitro spermatogenesis in rodents using soft matrix culture and organotypic systems. Nevertheless, all the cells were embedded in a thick layer. Furthermore, a new culture system that engages 3D bioprinting and alginate-based hydrogel has also been used to produce testicular constructs, but the pores of the interstitial cell-laden scaffold were not able to enhance cell attachment. However, immature seminiferous tubules are sensitive and flexible enough to adapt to other environmental conditions, which makes it difficult for cell attachment in old age. A novel culture method is required to maintain in vitro testicular cell attachment. In the future, early detection of testicular cancer emanating from the seminiferous tubule and culture technique can assist cell attachment and the provision of environmental conditions to suit that of the in vitro before being used in clinical settings.

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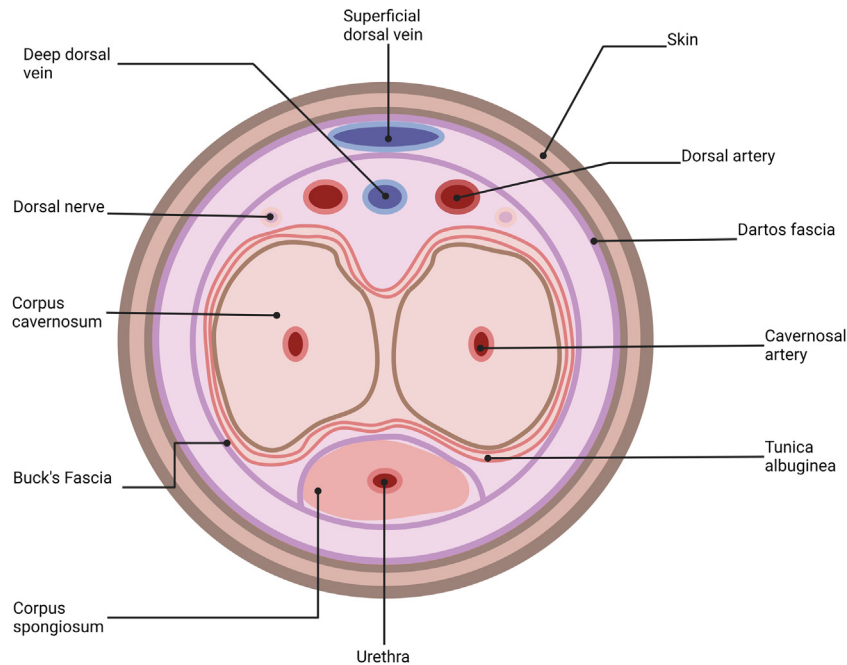
Tissue engineering applications for erectile dysfunction

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9.1 Introduction

Among all the sexual dysfunctions in men, the most common one is erectile dysfunction (ED) which is the incapability to attain and hold an erection long enough to engage in sexual activity (Shamloul & Ghanem, 2013, p. 381). One in 10 men permanently suffers from ED at least once in their lifetime (Cleveland Clinic, 2019). The graph of males suffering from ED is in increasing order. In 1995, almost 152 million men globally were victims of ED, and with an increment of nearly 170 million, it is forecasted that in 2025 worldwide, 322 million males will suffer from it (Ayta et al., 1999). ED is caused due to insufficient blood supply to the penis, damage to cavernous tissue, and various psychological and physical causes. Although currently there are several therapies available for ED, like phosphodiesterase type-5 inhibitors (PDE5i), intracavernosal injections (ICIs), penile prosthesis, and penile revascularization, these therapies are not always effective, have side effects, and there is a high probability of reoccurrence of ED (Chung, 2019). Hence, a different approach to treating ED has been studied, and the most promising area is tissue engineering (TE). In this strategy, the engineering of cavernous tissue is achieved using both native and stem cells and a variety of scaffolds. TE aims to treat ED by encouraging endothelial revascularization, protecting fibrosis in cavernosa from occurring, and restoring erectile function (Chung, 2019). This chapter reviews the developments done in the TE of corpus cavernosum (CC), provides an overview of cell types and scaffolds used, illuminates the milestones achieved, and casts light on the potential of TE to treat ED.

**FIGURE 9.1**

Cross-sectional anatomy of the penis.

9.2 Anatomy and physiology of penis

The penis has a complicated structure, and it comprises three cylindrical bodies enclosed by a tenacious layer known as Buck's fascia (Fig. 9.1). Among three, two pairs of cylinders are called corpora cavernosa and are responsible for the erectile function of the penis. Dorsally, the CC consists of a significant portion of the penis and is enclosed by a sturdy fibrous layer called tunica albuginea. The two cylinders are separated by a medial septum which is also the source of their communication. The corpus spongiosum (CS), the third cylindrical tissue, on the other hand, is surrounded by a skinny covering of tunica albuginea. Ventrally, the CS surrounds the urethra, prolongs from the penis to the tip, and expands to form the glans penis. The CC and CS are both known as erectile tissue (Carson & McMahon, 2008).

9.2.1 Anatomy of corpus cavernosum

The main composition of CC is smooth muscle cells (SMCs) and endothelial cells (ECs) (Cleveland Clinic, 2022). It consists of hollow sinusoids surrounded by smooth muscle lined by endothelium. These structures are composed of muscle, collagen, and elastin (The Editors of Encyclopaedia Britannica, 1999). The TA surrounding CC provides structural support and consists of different sinusoids, allowing

blood flow to maintain rigidity during an erection (Panchatsharam et al., 2022). The CC is made from a combination of SMCs and ECs, which is why they are an essential factor in the structural and physiological function of the penis (Xie et al., 2017). Unlike the CC, CS has larger sinusoids consisting of less space which does not allow it to become enlarged during erection as it has a continuous blood supply (The Editors of Encyclopaedia Britannica, 1999).

9.2.2 Mechanism of erection

Erection begins with sensory and mental stimulation by local nerves and the brain. In response to that stimulus, the muscle of CC relaxes, allowing the blood to flow through cavernosal arteries and helicine arteries and to the cavernous sinusoids. The pressure from blood expands the trabecular walls, resulting in the tunica albuginea walls also broadening. The TA walls compress the draining veins making the penis expand and creating an erection. This mechanism is called the veno-occlusion mechanism. The erection is reversed after the smooth muscle contraction, which opens the blood outflow channels and stops the inflow of blood (Cleveland Clinic, 2020; Wagner & de Tejada, 1998).

9.3 Risk factors and diseases associated with ED

ED occurs because of physical or emotional issues, which reduce the blood flow in the penis (Urology Care Foundation, 2018). ED can be the underlying factor for other diseases. According to diagnostic methods, there is evidence that men suffering from ED have hidden vascular disease (Carson & McMahon, 2008). Hence, experiencing ED can be the discovery of coronary artery disease (CAD) (Shamloul & Ghanem, 2013, p. 381). Various factors can lead to the occurrence of ED, such as lifestyle choices, aging, cardiovascular disease, hypogonadism, hypertension, diabetes mellitus, depression, and high cholesterol (DeLay et al., 2016; Hyde et al., 2012; Shamloul & Ghanem, 2013, p. 381). The causes of ED are classified according to their nature of origin, as shown in Table 9.1.

Table 9.1 Causes of ED.

Causes	Examples	References
Psychogenic	Depression, performance anxiety, relationship conflicts	Carson and McMahon (2008)
Neurogenic	Spinal cord injury, diabetes mellitus, radical pelvic surgery, alcohol	Carson and McMahon (2008)
Vasculogenic	Microangiopathy, failure of the veno-occlusive mechanism, fibrosis	Shamloul and Ghanem (2013)

Continued

Table 9.1 Causes of ED.—*cont'd*

Causes	Examples	References
Endocrinological	Hormonal deficiency, hypogonadism, testosterone deficiency	Carson and McMahon (2008); Shamloul & Ghanem (2013, p. 381)
Drug-induced	Antihypertensives, antidepressants, recreational drug use	Shamloul and Ghanem (2013)
Local penile factors	Cavernous fibrosis, peyronie's disease, penile fracture	Shamloul and Ghanem (2013)

9.4 Current treatments

The origin and development of ED are based on various psychological and physiological factors, and their modifications due to neural, hormonal, and vascular functions make it challenging to treat ED (Chung, 2019). Hence, different treatment approaches have been used recently. Although some therapies show effective results, many challenges, such as short-lasting results and possible side effects, must be addressed to achieve the required aim (Protogerou et al., 2021).

9.4.1 Phosphodiesterase type-5 inhibitors (PDE5i)

Phosphodiesterase type-5 inhibitors (PDE5i), like mirodenafil, avanafil, and lodenafil, are the most commonly used oral drugs to treat ED. It is very efficient and safe to use (Chung, 2019). But it stops responding in around 40% of patients despite being successful initially, and it is suggested to use this therapy along with other therapies to increase its effectiveness (Jannini et al., 2012).

9.4.2 Intracavernosal injection

ICI is an effective therapy that provides gratifying results and has negligible side effects compared to other therapies (Belew et al., 2015). But it is an invasive therapy that requires the patient to learn the procedure of self-injecting (Chung, 2019).

9.4.3 Testosterone therapy

The testosterone therapy (TT), combined with other therapies like PDE5i, is given to patients suffering from ED due to low blood levels of testosterone (Urology Care Foundation, 2018). Although TT assists in the functioning of erection, it has a finite role in treating ED (Shamloul & Ghanem, 2013, p. 381).

9.4.4 Intraurethral drugs

Drugs like alprostadil are used, a single-use drug applied by an applicator in the urethra (Chung, 2019). Despite having the flexibility to use and efficiency in result, it has side effects like urethral irritation and urine burning (Raina et al., 2005).

9.4.5 Penile implants

There are two types of implants available in the market; they are noninflatable and malleable penile implants. It has pair of rods that are flexible enough to bend in up and down directions according to the requirement (Chung, 2019). Among the two, the inflatable prosthesis is advised to imitate normal erectile function by being rigid and flaccid like a natural penis (Chung, 2017).

9.4.6 Shockwave therapy

LI-ESWT aspires for one to have a natural erection. It includes passing focused shocked waves to the affected tissue, which generates trauma in underlying cells. As a result, angiogenic factors are released, improving the vascularization and blood supply to the penis (Gruenwald et al., 2013).

9.4.7 Penile revascularization

This treatment involves surgery; hence, considering its complications, it is not suggested to older patients and is offered only to patients younger than 55 years who do not suffer from diabetes or CAD (Chung, 2019). The surgery involves a conjunction of the inferior epigastric artery with the dorsal penile artery or vein and resembles a bypass surgery done for the heart (BUMC, 2003).

The development of these therapies has helped to treat ED to some extent, but the long-term result has not yet been achieved. The unresponsiveness of oral drugs (Jannini et al., 2012), several complications such as pain, erosion, fibrosis, and infection (Sooriyamoorthy & Leslie, 2022), and risks of mechanical failure and optic neuropathy (Wolfe, 2005) demand for other possible solution for the treatment of ED. The application of TE can effectively provide a promising and permanent cure for ED. The TE approach uses the combination of different cells and scaffolds made from different biomaterials to create engineered tissue. Thus, the engineered cavernous tissue to regenerate or replace the damaged tissue can provide a lasting solution by enhancing vascularization and restoring desired erectile function.

9.5 Biomaterials used for CC scaffolds

TE is accomplished to create a biological replacement of damaged tissue by using different kinds of biomaterials as scaffolds. These biomaterials act as artificial ECM by mimicking the native cell ECM biologically and mechanically (Patel & Atala, 2011). Biomaterials provide 3-D space for seeded cells to grow, adhere, proliferate, and form new tissue with the required function and structure and act as a delivery mechanism of different bioreactors to the needed sites of the body (Kim & Mooney, 1998). Biomaterials used for the engineering of CC can be categorized into three types (Table 9.2): synthetic polymers like PLA, PLGA, and PGA, biological polymers like collagen, and biological scaffolds like acellular tissue matrices (Patel & Atala, 2011).

Table 9.2 Comparison of biomaterials used for engineering of CC.

	Biomaterials	Advantages	Disadvantages	References
Biological polymers	Collagen	Biodegradability and adhesive properties.	Poor and vulnerable mechanical properties	O'Brien (2011)
Biological scaffold	Acellular matrices	Provides 3D structure and mechanical properties similar to native tissue and low immune reaction	Inadequate source, difficulty in fabrication, and low vascularization ability	An et al. (2020) ; Park (2021)
Synthetic polymers	• Poly-lactic-acid (PLA) • Poly-glycolic-acid (PGA) • Poly-lactic-co-glycolic-acid (PLGA) • Polycaprolactone (PCL)	High mechanical strength can be produced on a large scale with desired modification in required function and structure	Risk of rejection	Park (2021)

Biological polymers have biodegradability and adhesive properties so that cells can proliferate rapidly. They allow the host cells to replace degraded scaffold by assisting them in creating their own ECM. However, the poor and vulnerable mechanical properties are the most significant limitations in their application and fabrication (O'Brien, 2011). Biological scaffolds provide an architecture similar to the native tissue-protecting elements of ECM (Vocht et al., 2017). Hence, due to the similar structure of various proteins in ECM as native tissue, biological scaffolds mimic mechanical properties analogous to them (Dahms et al., 1998). The advantages of using these biological scaffolds are their low immune reaction and 3D structure replicating natural tissue, but they have limitations of the inadequate source, difficulty in fabrication, and low potential for vascularization (An et al., 2020; Park, 2021). Synthetic polymers can be manufactured widely and modified according to the desired function and structure. They possess very high mechanical strength but also have rejection risks (Park, 2021). ACM and biological materials are biocompatible and have less chance of rejection, whereas synthetic polymers can be produced with desired mechanical properties but lack biological recognition (Patel & Atala, 2011).

9.6 Desired properties of scaffolds

Scaffolds are the foundation of TE. They are essential for cell growth and proliferation. Hence, specific biomaterials properties are required to form needed scaffolds to obtain successful engineering of CC. This includes biodegradability, biocompatibility, mechanical properties, and appropriate pore size (Fig. 9.2). Biocompatibility is the crucial property that must be considered most while selecting biomaterials for the CC scaffold. The scaffold should provide an environment for cells to adhere, grow, and migrate without complications. And there should be a minimal immune reaction by the implantation of scaffold or engineered tissue to prevent the risk of rejection and enhance healing (O'Brien, 2011). The ultimate goal of TE of CC consists of implanting scaffold in vivo. Therefore, biodegradability is another critical property required to fulfill that aim. Scaffolds are supposed to degrade with time and get replaced by the formation of new tissue (O'Brien, 2011). Hence to allow the development of cells' own ECM, the implantable scaffold must be biodegradable (Babensee et al., 1998). As the penile structure is complex and possesses mechanical properties like rigidity and flaccidity, the selected scaffold must contain the same mechanical properties. In addition, its mechanical properties should correspond to that of the targeted anatomical location, and it must be strong enough for the surgical procedure for its implantation to be successful (O'Brien, 2011). Pore structure and its size are another important property considered for scaffolds. It is needed for cell seeding and growth, delivery of nutrients, and transportation of waste (Pluta, 2015). Hence, the scaffold should have a pore size that is big enough for cells to transport into the required space but small enough for efficient cell binding to the scaffold (O'Brien et al., 2005).

**FIGURE 9.2**

Desired properties of CC scaffolds.

9.7 Cells used

Constructing engineered CC requires two types of primary cells: SMCs and ECs (Patel & Atala, 2011). SMCs are in charge of the contractile activity of the vessel wall, responsible for controlling blood inflow and outflow to achieve and keep a penile erection, which is why it is the most essential for the engineering of CC (Dean & Lue, 2005). ECs help vascularize and develop new blood vessels (Sukmawati & Tanaka, 2015). Hence, two types of cells are mainly used for engineering CC: stem cells and native progenitor cells, which can differentiate into required ECs and SMCs (Patel & Atala, 2011).

Stem cells can turn into specialized cells and have self-renewal properties (Zhang et al., 2012). Based on their differentiating capability, stem cells are classified into four categories: totipotent, pluripotent, multipotent, and unipotent (Keller, 2005).

Embryonic stem cells, despite having the most potential because of ethical reasons, are not preferred, and thus, different adult cells are used as an alternative (Yamzon et al., 2008). Generally used stem cells in TE of CC are mesenchymal stem cells (MSCs), muscle-derived stem cells (MDSCs), and adipose tissue-derived stem cells (ADSCs). MSCs can be easily harvested and proliferated using required culture conditions and are excellent sources of cells in corporal TE (Song et al., 2007; Sun et al., 2017). Maintaining tissue integrity is another vital trait of MSCs (Pittenger et al., 1999). Like MSCs, the primarily other used stem cells are MDSCs because of their various properties like rapid proliferation, differentiation capability into several lineages, and low risk of carcinoma and immune response (Urish et al., 2005; Usas & Huard, 2007). Likewise, ADSCs are used because of their easy and quantitative harvesting quality with minimal intrusion into the donor site (Si et al., 2019). They have high renewal capacity and cell viability (Maqsood et al., 2020). Progenitor cells can differentiate into only one type of specialized cell (Young & Black, 2003). The native progenitor cells are harvested from the organ that needs to be regenerated. As a result, there is no risk of rejection, and it can be efficiently utilized to construct CC (Patel & Atala, 2011). Progenitor cells used in CC TE are SMCs and ECs. Using native cells poses less risk of immune response; however, obtaining them is a bit complex because of the surgical procedure involved. There's a limited source of native cells, whereas stem cells can be harvested quantitatively.

9.8 Tissue engineering approach used

There has been various development in the field of TE of CC to date (Table 9.3). Different approaches are used to obtain desired tissue, such as the acellular approach, which involves implanting scaffolds without cells letting the body act as a natural bioreactor for cell growth and new tissue formation, and the cellular approach, which involves cell-seeding in scaffolds before implantation for regeneration of tissue (Shokeir et al., 2010). The mostly cell-seeded approach has been used on the animal model to construct cavernous tissue.

The cell-seeded TE approach involves several steps: (1) harvesting cells and expanding them with standard culture conditions, (2) fabricating scaffolds and seeding them with cells, (3) implanting scaffold into the corporal space, and (4) analysis of retrieved engineered tissue (Fig. 9.3).

Cells are harvested from different sources such as cavernous space (Chen et al., 2010), umbilical artery and vein (Oh et al., 2019; Song et al., 2009), bone marrow (Laks et al., 2015), limbs of rabbits (An et al., 2020) and human. Different fabrication methods are used to develop the scaffold, such as electrospinning, cell-lysis technique, decellularization, and 3D printing according to the requirement (Kajbafzadeh et al., 2017; Kwon et al., 2002; Oh et al., 2019; Park et al., 1999). Numerous growth factors like vascular endothelial growth factor (VEGF) are added to cultured cells to stimulate cell growth and proliferation, which are seeded into the grafts and interposed to target locations. Afterward, various analyses are conducted, such as histological, SEM,

Table 9.3 Overview of milestones achieved in the engineering of CC.

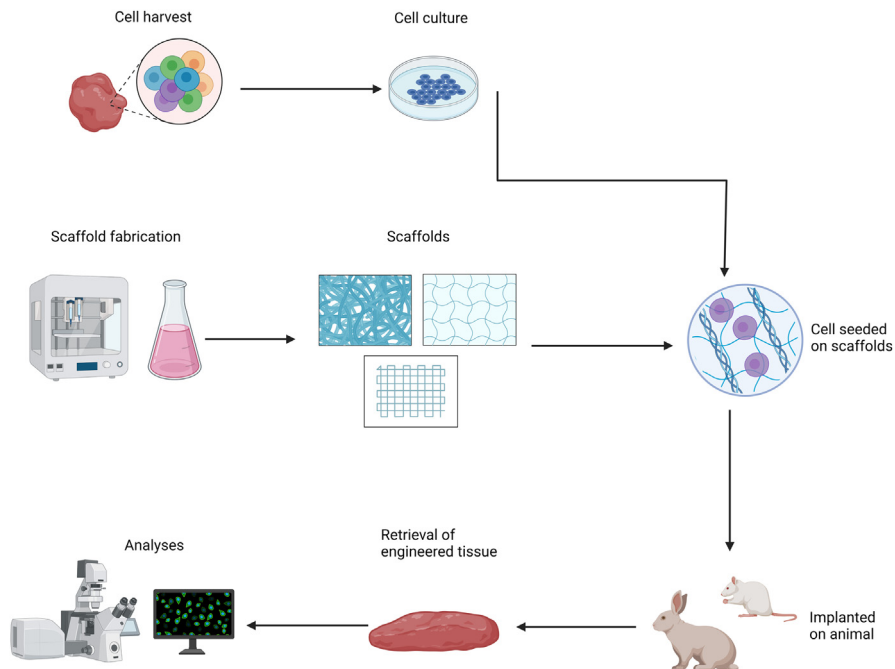
Year of study	Cell used	Scaffold used	Scaffold fabrication process	Number of scaffolds used	Analyses done	Results	References
1999	Autologous chondrocytes obtained from rabbit ears	PGA coated with PLLA	N/A	18 in total	Histological analysis	Formation of cartilage structure and degradation of the implant by 2 months.	Yoo et al. (1999)
2002	Chondrocytes obtained from human ears	PGA	N/A	N/A	Histological analysis	Formation of cartilage structure with mechanical properties like flexibility and elasticity.	Kim et al. (2002)
1999	Human SMCs and ECs	PGA	N/A	80 in total 60 seeded and 60 unseeded	Immunocytochemical and histological analysis	Scaffold seeded with cells formed vascularized structures of CC. Pioneer of development of capillary using EC.	Park et al. (1999)
2002	SMCs and ECs	Rabbit ACCM	Cell-lysis technique	26 in total 18 seeded and 8 unseeded	Histological and western blot analysis	The formation of intact cavernous tissue and erectile function was observed.	Kwon et al. (2002)

2009	HUASMCs	Rabbit ACCM	Decellularization	18 in total 9 seeded and 9 unseeded	Histochemistry, immunohistochemistry analysis, and SEM	Formation of structure similar to native cavernous smooth muscle with a contractile response but no relaxation.	Song et al. (2009)
2010	SMCs and ECs	Rabbit ACCM	Decellularization	24 in total 12 seeded and 12 unseeded, and 3 negative control group	Organ bath studies, histological and immunohistochemical analyses	Formation of the entire CC with contraction and relaxation response.	Chen et al. (2010)
2010	MDSCs	Rabbit ACCM	Decellularization	24 in total 12 seeded and 12 unseeded	Histochemistry, immunohistochemistry, and SEM	Development of tissue similar to native CC and MDSCs turns into SMCs and ECs.	Ji et al. (2010)
2013	MDSC with VEGF	Rabbit ACCM	Decellularization	32 in total 24 seeded and 8 unseeded	Histochemistry, immunohistochemistry, SEM, and western blot analysis	Increased capillary density and cell content, and erectile function were observed.	An et al. (2013)
2015	MSCs	Rabbit ACCM	Decellularization	16 in total 6 seeded, 6 unseeded IV MSCs and 4 unseeded	Histological and immunohistochemistry analysis	Partial construction of tissue similar to native tissue, and no notable differences were observed.	Laks et al. (2015)
2017	MSCs	Rabbit ACCM	Decellularization	30 in total 20 seeded and 10 unseeded	Histological analysis and transmission electron microscopy	MSCs differentiated into ECs and SMCs.	Xie et al. (2017)

Continued

Table 9.3 Overview of milestones achieved in the engineering of CC.—*cont'd*

Year of study	Cell used	Scaffold used	Scaffold fabrication process	Number of scaffolds used	Analyses done	Results	References
2017	No cells	Human ACCM	Decellularization	and 10 control DCC was interposed into the omentum of 4 rats	Histological, immunohistochemical, and SEM	Formation of cavernous smooth muscle with appropriate cell architecture.	Kajbafzadeh et al. (2017)
2019	haSMCs and HUVECs	PCL	3D-bioprinting	3 PCL (strand thickness/period). One 300/150, one 300/300, one 300/450	SEM, porosity measurement, and immunofluorescence analysis	Formation of spongy and distinguished architecture in grafts. haSMCs and HUVECs adhered efficiently.	Oh et al. (2019)
2020	HIF-1 α mutated MDSCs	Hydrogel coated with heparin	3D-printing	3 blank scaffolds, 9 seeded scaffolds with MDSC variation,	Immunofluorescence and histological analysis	Effectively repaired the injured corpora cavernosa and restored penile erection in vivo within 4 months.	An et al. (2020)
2022	ADSCs	Rabbit ACCM	Decellularization	4 groups SSI, SSS, u-SSI, and u-SSI	Histological and immunofluorescence analysis	ADSCs are differentiated into SMCs and ECS. SSI showed tissue formation with similar architecture to native tissue without deformation and fibrosis.	Cao et al. (2022)

**FIGURE 9.3**

Cell-based tissue engineering approach used to construct CC.

histochemistry, immunohistochemistry, and western blot analysis, to analyze the success of required tissue formation (Laks et al., 2015; Xie et al., 2017).

9.9 Engineered prosthesis

The history of TE for ED goes back to using cartilage as a penile prosthesis. Two studies used autologous chondrocytes to create engineered prostheses (Yoo et al., 1999; Kim et al., 2002). During the first study, autologous chondrocytes were used by collecting them from rabbit ears. The cells were cultured for expansion in vitro and seeded into PGA polymer coated with PLLA. Scaffolds were interposed in the corporal space of 10 rabbits, and further analyses were done between 1 and 6 months of implantation. The result showed that the cartilage structure was formed, the scaffold degraded in 2 months, and no immune reaction was observed (Yoo et al., 1999). The subsequent study used chondrocytes isolated from the human ear and seeded on cylindrical-shaped PGA. For 1 month, seeded scaffolds were kept stirring in bioreactors and interposed into athymic mice. After 2 months, the polymer was degraded, showing the development of a cartilage rod of identical structure and size as the original implant with mechanical properties like flexibility, elasticity,

and ability to resist high tension and compression (Kim et al., 2002). These approaches were not further developed as there was no erectile function observed.

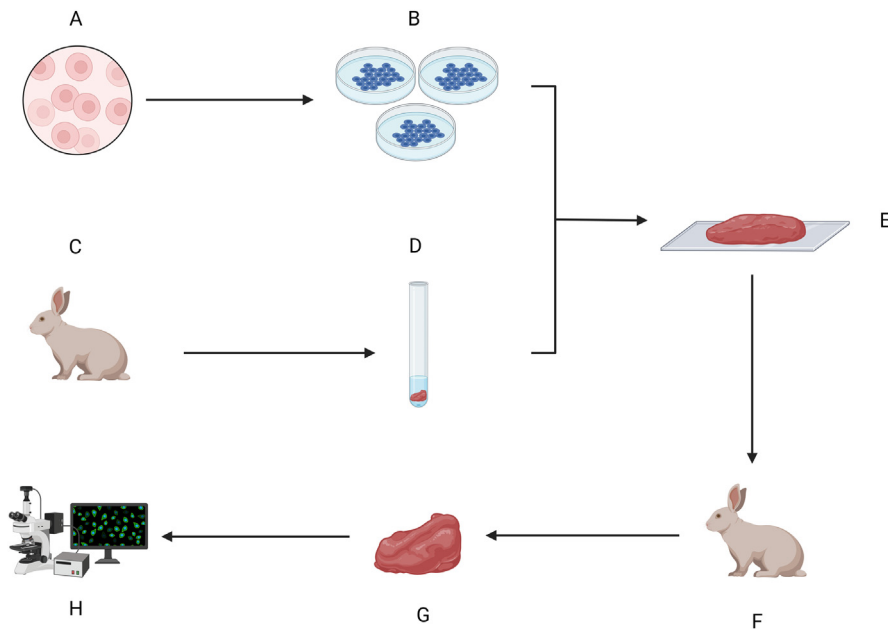
9.10 Engineered corpus cavernosum

To construct functional erectile tissue and regain physiological erection, further studies were conducted using SMCs and ECs as they comprise the normal erectile tissue. Human ECs were obtained from the umbilical cord, and human CC SMCs were obtained from the specimen of tissue retrieved from biopsy during penile surgery with patients' consent. The cells were seeded on PGA and subcutaneously implanted in athymic mice. Histological analysis was done in retrieved structures, and the formation of vascularized cavernous muscle structure was observed. The presence of antipancytokeratins and α -smooth muscle actin was proved by immunocytochemical staining. This study was foremost to develop capillary in the TE field in vivo using ECs (Park et al., 1999).

Although synthetic polymer like PGA was a suitable choice, due to the complex structure of CC, most studies use ACM. ACCM harvested from donor rabbits' penile tissue was used in the study by Kwon et al. (2002). The cells used were autologous SMCs and ECs, which were characterized on rabbit ACCM and interposed into the corporal area of rabbits, and further analysis was done (Fig. 9.4). Structural integrity was observed along with the erectile function; however, tissue scarring was also detected (Kwon et al., 2002).

Harvesting pure cavernous cells without fibroblasts is pretty challenging. Their results are unclear because of the differences among cavernous SMCs and fibroblasts regarding their proliferation and migration properties (Filippi, 2003; Pilatz et al., 2005; Rees et al., 2002). Hence, to determine the feasibility of other cell types rather than cavernous SMCs, another study using SMCs from the human umbilical artery was done (Song et al., 2009). Isolated HUASMCs seeded on rabbit ACCMs were interposed into athymic mice. The result illustrated the formation of cavernous smooth muscle similar to native tissue with smooth muscle contraction response, but a relaxation response was not demonstrated (Song et al., 2009). The first functional CC in an animal model was developed using autologous SMCs and ECs (Chen et al., 2010). Cells were seeded on the rabbit ACCM obtained by decellularization process using a multistep procedure. The grafts were implanted into the corporal space of rabbits, and they were analyzed for up to 6 months. The result showed the formation of the entire CC with contraction and relaxation response and the presence of the erectile function, which was proved by cavernosography and cavernosometry (Chen et al., 2010).

In 2010, a study to evaluate the feasibility of MDSCs to construct CC was done (Ji et al., 2010). Cells were harvested from the limbs of rabbits and seeded on rabbit ACCM, cultured for 5 days, and subcutaneously implanted into the corporal space of rabbits. The engineered structure was examined 2–6 months after implantation. The result demonstrated the development of cavernosal sinuses by differentiating

**FIGURE 9.4**

Engineering of CC using rabbit ACCM. (A) Autologous SMCs and ECs harvest. (B) Cell culture and expansion. (C) Rabbit cavernous tissue harvest. (D) Corporal tissue processed to extract ACCM. (E) Cell seeded on ACCM. (F) Implantation of seeded ACCM. (G) Engineered tissue retrieval. (H) Analyses such as histology and western blot analysis are performed.

MDSCs into SMCs and ECs. This was evident with the finding of α -smooth muscle actin cells during histological analysis. The use of MDSC is beneficial because of its minimal invasive harvesting process. This study has one drawback because of the complex structure of ACCM, which makes delivering nutrients and oxygen to MDSC in lower zones quite challenging (Ji et al., 2010). Hence because of the limitation of the previous study, another study was done to improve vascularization and oxygen supply to cells by transfecting MDSC with VEGF (An et al., 2013). MDSCs were obtained from rabbits and seeded on rabbit ACCM, and implantations in four sets were done varying the presence of cells and growth factors. The result showed that the group implanted with MDSC transfected with VEGF illustrated better growth and increased cell contents, and capillaries were formed. It also showed a better contractile response with an ICP of 60% of the normal group (An et al., 2013).

Further studies were done exploring more stem cells. In another study, MSCs were used to construct segments of CC (Laks et al., 2015). Compared to other SCs, MSCs can be easily extracted from bone marrow (Dominici et al., 2006). MSCs were extracted from rabbits, characterized on ACCM, and interposed in the cavernous region, whereas another group of rabbits was directly infused with

MSCs intravenously. After analysis, partial reconstruction of tissue similar to native tissue was observed, though a similar result was noticed in the two groups (Laks et al., 2015). A similar study used MSCs to develop CC in vitro (Xie et al., 2017). Through the process of bone marrow aspiration, MSCs were harvested. They were seeded on rabbits ACCM and incubated. ACCM, after incubating for 14 days, provided an excellent environment for cells to grow and differentiate. The histological analysis showed the differentiation and growth of SMCs and ECs in engineered CC (Xie et al., 2017).

Most of the studies used animal ACCM for the engineering of CC, but one study was done to generate whole human DCC for cell seeding in vivo (Kajbafzadeh et al., 2017). DCC was extracted from human CC tissue during SRS with informed consent and implanted without cell seeding in the omentum of Sprague–Dawley rats, utilizing rats' bodies as a natural bioreactor. The analysis was done in 1, 3, and 6 months. The result showed the formation of decellularized ECM similar to native tissue with suitable structures for cell seeding in vivo. Structural integrity could not be analyzed due to the small size of grafts used, which is a limitation in this study (Kajbafzadeh et al., 2017).

Another study was conducted using the coculture of haSMCs and HUVECs using 3D PCL scaffolds to determine their feasibility as CC grafts (Oh et al., 2019). The scaffolds were fabricated through a 3D-bioprinting fabrication process. Firstly, haSMCs seeded grafts were incubated for 5 days. After that, HUVECs seeded grafts were cultured with haSMCs for 7 days. The result demonstrated the construction of an exemplary structure with uniform pores mimicking native CC, and seeded cells showed adhesion, differentiation, and proliferation after 7 days. Since this study was observed in vitro, the feasibility of using PCL grafts seeded with haSMCs and HUVECs in vivo still needs to be evaluated (Oh et al., 2019).

Reconstruction and repair of CC are challenging because of their complex structure and function (Schultheiss et al., 2004). Thus, varieties of scaffolds have been used and tried to obtain the best possible result. Most studies used ACCM, but it also has its limitations, such as low porosity, risk of the immune response, and finite sources (Chen et al., 2010). Hence, another study was conducted exploring other biomaterials for effective results. In one study, hydrogel was used for its mechanical properties like strong, well-defined porous structure and because it provides feasible physical properties to upgrade cell adhesion and growth (Bertlein et al., 2017; Ouyang et al., 2016). This study was done to repair injured CC using HIF-1 α mutated MDSCs (An et al., 2020). MDSCs were isolated from the muscle of rabbit legs and mutated with HIF-1 α by lentiviral transfection to improve angiogenesis. The scaffolds were developed by 3-D printing and coated with heparin to expedite neovascularization and implanted in the injured cavernous area of rabbits. The analysis was done after 4 months of implantation. The result showed differentiation of MDSCs into SMCs and ECs through the positive stain of α -SMA and vWF, respectively, in histological examination. The injured CC was successfully repaired, and erectile function was restored (An et al., 2020).

The latest study used ADSCs to construct engineered CC using ACCM extracted from rabbits' penile tissue (Cao et al., 2022). The ADSCs are harvested from rabbits' necks and seeded into ACCM. Four groups were formed based on the site of implantation and the presence of cells. ACCMs seeded with cells and without cells were interposed into rabbits' intramuscular and subcutaneous space, and analyses were done postimplantation between 1 and 6 months. The result demonstrated that seeded grafts into intramuscular space formed engineered CC tissue with architecture similar to native cavernous tissue, whereas grafts into subcutaneous space illustrated deformed architecture with fibrosis (Cao et al., 2022).

9.11 Challenges

Although studies accomplished till now have shown promising results, some hurdles still need to be addressed. The most significant issue faced by TE applications is vascularizing of developed tissue. Supplying nutritional factors for cell growth and oxygen to the deeper region in the volume of cells is challenging (Ikada, 2006). Insufficient vascularization can lead to fibrosis, creating the possibility of risk of implant rejection. Due to the complex anatomical structure of CC, ACMs are preferred; however, the fabrication process to develop those leads to poor reproduction and deformation of natural ECM (Vocht et al., 2017). Research and studies are mostly performed in rabbit and rodent models. Because of the anatomical and physiological differences between them and humans, applying obtained results to humans is challenging (Andrew et al., 2019).

9.12 Conclusions

The increasing number of men suffering from ED worldwide has become a severe matter of attention. The ineffectiveness of current treatment methods and their side effects is the prime reason for an alternative method for ED. Engineering cavernous tissue to replace and regenerate damaged erectile tissue has shown great potential in treating ED. CC with functional structure along with its sinusoids is crucial for the physiology of penile erection. TE combines the cells and scaffolds with bioreactors to generate the required tissue. Different scaffolds produced using various biomaterials are used, which provide architecture and a suitable environment for the cell to adhere, proliferate, migrate, and differentiate. Synthetic polymers are used because they can be developed in a wide range, and their structure and properties can be altered according to the need, but there's a risk of infection. Due to structural differences in physiology required to achieve erectile function, ACM of corporal bodies is preferred as the scaffold. Specific properties from scaffolds are required to meet certain physiological conditions and the complex structure of cavernous tissue, such as biodegradability, biocompatibility, mechanical strength,

rigidity, and high porosity. The construction of CC is achieved by using both native progenitor and stem cells on various scaffolds. Using native cells makes the risk of rejection minimal, but its source is limited; thus, stem cells are used as they can be extracted easily and quantitatively. Growth factors like VEGF have been used for cell signaling to regulate cellular function.

TE strategy using ACCM as a scaffold seeded with SMCs, ECs, and SCs has shown feasible and favorable results in the engineering of CC. Although promising results are shown, some challenges, such as selecting feasible biomaterial for scaffolding, vascularizing engineered tissue, and fibrosis, need to be addressed. Because of the difference in anatomical size, composition, and physiological performance between animals and humans, further research, study, and trials must be done before using engineered CC in humans to treat ED.

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Tissue engineering strategies to treat male infertility

10

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10.1 Introduction

Infertility is a disease associated with the reproductive system in which a couple can't achieve pregnancy due to some reasons. It affects over 146 million people globally, being mature males around the 50%. The comprehension of the physiology of the male reproductive system and the diseases associated with will give us some insight, revealing not only the cause of the problem but also finding a solution that best suits the patient. We can classify the current treatments in: hormonal treatment, surgical procedures, and gamete donation; however, there are several side effects associated with these treatments, reason of why we will review and discuss some of the tissue engineering (TE) techniques that are being developed to treat infertility. The use of autologous cells combined with a good selection of biomaterial for building the matrices will be the reference point for these studies, as we are looking for a method that won't unleash an immune reaction when implanted and will grant a good microenvironment for the seeded cells so they can proliferate and generate new tissue, giving the organ its function back; so far there have been good results using collagen as base biomaterial for urethra reconstruction. The combined use of corpus cavernosa cells plus epithelial cells in a decellularized matrix is promising, as the results of some studies show, the new engineered penis can attain an erection when implanted in the subject of study. Stem cells therapy is another technique that seeks to stimulate spermatogenesis and production of testosterone proliferating spermatogonial stem cells (SSCs) in vitro and then implanting them back into the patient.

10.2 Male reproductive system physiology

It's a conjunct of organs integrated by the pennis, testicles, prostate, vas deferens, and epididymis (Orestres, 2016). The good comprehension and understanding of the anatomy and physiology of the male reproductive system is vital to identify the different processes involved in the production of spermatozoa, and how some structures are involved in the reproductive stage (Universidad Autónoma Metropolitana, 2014).

10.2.1 Testicles

They have the function of producing spermatozoa and testosterone, consists of numerous lobules formed by the seminiferous tubules, which at the same time keep bonded together to form bigger conduits, called the stretch conduits. These conduits integrate the testicular network which transports seminal fluid from the testicles to the epididymis (Molina, 2013).

Sperm production is regulated by a "hormonal axis" which starts in the hypothalamus with the segregation of gonadotropin-releasing hormone (GnRH). This will stimulate the hypophysis to release two hormones, luteinizing hormone (LH) and follicle-stimulating hormone (FSH), the first one tells the Leydig cells to start producing testosterone, the FSH, in the other hand, stimulates the Sertoli cells to start the maturation process of spermatozoa (Bustos, 2016).

The sperm production process is stopped once testosterone has reached its peak level, as this hormone has a depressive effect on the hypothalamus, which will stop producing GnRH and therefore, hypophysis will stop producing LH and FSH as well (Arrondo, 2017).

Testosterone is directly associated with the amount of libido produced, and it's also responsible for interacting with other enzymes such as nitric oxide synthase (NOS) which is necessary to ensure an adequate erectile process (Vignozzi et al., 2005).

10.2.2 Pennis

It's a cylindric organ, composed of two structures; the corpus cavernosa that allows the flow of blood inside the pennis, strengthening it during the erection phase; and the corpus spongiosum, which is the column where the urethra is contained. Both are covered and shaped by the tunica albuginea (Boskey, 2021).

The main function of the penis is to ensure the sexual act, whether it's for reproductive purposes or just pleasure. Contractions during orgasm are responsible for expulsing sperm out of the body; a mature male can ejaculate between 5 and 15 mL of semen (Bustos, 2021).

An erection may be produced by one of these three factors: psychogenic erection that is induced by audiovisual stimulations or fantasies. Reflexogenic erection is provoked by physical stimulation in the genital organs, giving way to

sensorial perception. And finally, nocturnal erection which may be induced by the activation of the cholinergic neurons during the REM sleep phase (Martínez-Salamanca et al., 2010).

10.2.3 Urethra

It's a membranous tube (between 18 and 22 cm long in males) that travels inside the corpus spongiosum, from the base of the bladder to the urinary meatus (Azucas, 2021). Its function is to transport urine from the bladder to the outside of the body; in the male, it also drives the semen during the ejaculation (Guerra, 2013).

10.2.4 Prostate

It consists of a series of glandular structures linked by a stroma in the shape of a nut, and this gland is traversed by the urethra (Hermabessière, 2002). The prostate plays an important role in reproduction, as it's responsible for releasing secretions (mainly fructose) that sperm needs to live longer; these secretions are blended with the sperm in the urethra forming the semen (Quinta, 2012).

10.2.5 Vas deferens

It's a thick-walled tube structure (35–45 cm long) that runs from the base of the epididymis to the ejaculatory duct, located just behind the prostate. Each of the deferent conduit ends in an ampulla, where sperm is stored prior ejaculation (Briannica, 2018). Its main objective is to transport sperm during ejaculation emission; it also produces a fluid that facilitates sperm transport and can reabsorb the sperm that wasn't expelled (Flannigan, 2018).

10.3 Male infertility

Between 12% and 15% of all sexually active couples are infertile. If analyzed by gender, 50% of this infertility problems can be identified with the male component (Parekattil, 2012). Globally, it is estimated that about 146 million individuals and 48 million couples, in reproductive age, live with some infertility issue (World Health Organization, 2020).

Most of the male infertility causes are treatable; however, the main objective of a specialist shouldn't be limited just to retrieving sperm, but to optimize the couple's reproductive potential using less invasive techniques (Parekattil, 2012). Time also plays an important role when treating infertility; it is strongly recommended to consult a reproductive specialist whenever a couple has been trying to get pregnant without success for at least 12 months. It has been demonstrated that couples with an infertility period less than three years have a higher probability of conception than those with longer periods (Rosas, 2007).

10.3.1 Causes of male infertility

There are a lot of different pathologies that can cause male infertility; for practical purposes, we will classify them in: *pretesticular*, *testicular*, and *posttesticular*.

Understanding the mechanisms of the different main causes of infertility will help us to take the next step on developing new techniques of TE to help these patients. [Table 10.1](#) enlists the most common causes of male infertility, and a short description of how it is related to it.

10.3.2 Evaluation of the infertile male

It is strongly recommended to undergo a clinical evaluation if pregnancy is not achieved after a year of holding sexual relations, earlier if there are infertility risks for male infertility such as advanced age or cryptorchidism ([Jonathan Jarow et al., 2010](#)).

The basic evaluation of the patient consists in the three following steps:

1. **Clinical and sexual anamnesis:** Its objective is to identify possible risk factors or behavioral habits that could have repercussions on patient's fertility. It's important to ask about previous/current diseases associated with the reproductive system (cryptorchidism, sexual transmission diseases, etc.). The reproductive history of the couple must be known by the specialist as well, including current sexual habits, in order to identify libido alterations, erectile dysfunction, and environmental risk factors (radiation) ([Stahl et al., 2012](#)).
2. **Physical exam:** Consists of the general inspection of the patient, the specialist must search for alterations in secondary sexual characters, presence of gynecomastia and scars from past surgeries. A penis examination is mandatory to determine any signs of sexually transmitted diseases (STDs), and testicles must be palpated in order to evaluate shape, size, and consistency. It's also important to confirm the presence of both deferent conducts ([Palma, 2014](#)).
3. **Seminal analysis:** In order to complete the basic evaluation of the patient, it's necessary to take a sample of the patient's semen. The lab studies will determine the seminal volume, total count of spermatozooids (this also includes the total mobility, vitality, and morphology), pH, and leukocyte count ([Jungwirth et al., 2012](#)).

Depending on the results of the evaluation, further examinations may be necessary to complete the diagnosis of the infertile male patient, some of which are listed below:

- Endocrine evaluation
- Urine postejaculation analysis
- Transrectal echotomography
- Testicular echotomography
- Genetic tests
- Specialized tests (DNA integrity, antibodies, etc.)

Table 10.1 Causes of male infertility.

Causes of male infertility			
Classification	Name	Relation with infertility	References
Pretesticular	Hypogonadotropic hypogonadism	This condition affects fertility in several levels. There's lack of testosterone that negatively affects sperm production. It also produces erectile dysfunction, compromising the sexual function.	Fraietta et al. (2013)
	Elevated prolactin	High levels of prolactin produce erectile dysfunction and loss of libido.	Parekattil (2012)
	Pharmacologic	The use of some substances such as marijuana and ethanol can reduce the testosterone levels. Strogen and progestins may also affect the production of testosterone. Hypogonadotropic hypogonadism can be produced by using some medications.	Sharpe (2000)
Testicular	Varicocele	It is a malfunction of the venous valves of the internal spermatic vein, the causes of infertility related to varicocele aren't clear enough yet, but the most accepted theory suggests that it causes a disruption of the countercurrent heat exchange mechanism in the tests, leading to hyperthermia.	Practice Committee of the American Society for Reproductive Medicine (2008)
	Testicular cancer	Testicular tumors secrete hormone that may stop sperm production. Infertility also can be associated with the	Parekattil (2012)

Continued

Table 10.1 Causes of male infertility.—*cont'd*

Causes of male infertility			
Classification	Name	Relation with infertility	References
	Cryptorchidism	inflammatory reaction the tumors provoke. Cancer treatments may also decrease semen quality. Depending on the severity of the cryptorchidism, will be the severity of infertility, in which worst case would be bilateral cryptorchidism, as both testes lose their function.	Hutson et al. (2010)
	Chemotherapy	As chemotherapy targets rapidly dividing cells, this will cause a major damage to the germinal epithelium. Acute chemotherapy is directly related to a decline in spermatogenesis.	Dohle (2010)
	Environmental factors	There are a lot of different factors that are related to the decrease of semen quality (sperm density, morphology, and motility). Constant exposure to high temperatures (hyperthermia), ionizing radiation, tobacco use, and sedentary lifestyle.	Paparella et al. (2017)
	Testicular injury	Normally, testis trauma will result in the closure of the tunica albuginea and the loss of seminiferous tubules, the fact that some of them may be blocked is considered one cause of decreased fertility.	Li Canessa and Madrigal Ramírez (2015)
	Sertoli cell-only syndrome	This syndrome is related to a lack of migration of the germ cells to the seminiferous tubules during embryologic development; therefore, those patients won't be able to produce sperm on adult stage.	Koc et al. (2019)

Table 10.1 Causes of male infertility.—*cont'd*

Causes of male infertility			
Classification	Name	Relation with infertility	References
Posttesticular	Antisperm antibodies	A rupture in the blood –testis barrier causes immunity against the germinal epithelium and spermatozoa. This can be caused by infections or inflammation.	Parekattil (2012)
	Absence of the vas deferens	Some problems during the embryologic development may result in a congenital unilateral/ bilateral absence of the vas deferens.	Oates and Amos (1994)
	Ej DO/seminal vesicle dysfunction	Ejaculatory ducts can be blocked due to some calcifications, infections, or even urinary calculi. These blockades can also damage the seminal vesicles when present for a long time. Seminal vesicle dysfunction can be related to other illness such as diabetes, multiple sclerosis, or spinal cord damage.	Parekattil (2012)
	Vasectomy	Vasectomy is a surgical procedure that is intended to produce infertility. However, some patients may regret and attempt a vasectomy reversal, in which vas deferens are reconnected; the rate of success depends on how much time has passed since the vasectomy. Three years or lower times have shown a high success in recovering fertility.	Patel and Smith (2016)
	Nerve injury	Lesions in spinal cord or brain damage may affect ejaculation; the severity of the ejaculatory	Master and Turek (2001)

Continued

Table 10.1 Causes of male infertility.—*cont'd*

Causes of male infertility			
Classification	Name	Relation with infertility	References
	Medications	dysfunction is directly related to the level and extension of the injury. The use of some antipsychotics has negative effects on many different neurotransmitters (such as dopamine, norepinephrine, acetylcholine, and serotonin), provoking ejaculatory dysfunction.	Just (2015)
	Resection of the prostate	Surgery of the prostate (removing of the proximal prostatic urethra) may cause retrograde ejaculation.	Parekattil (2012)
	Coital	Abnormal coital practices may play a role in infertility when interfering with the semen deposition in the vagina. The use of lubricants has shown to negative affect fertility.	Parekattil (2012)

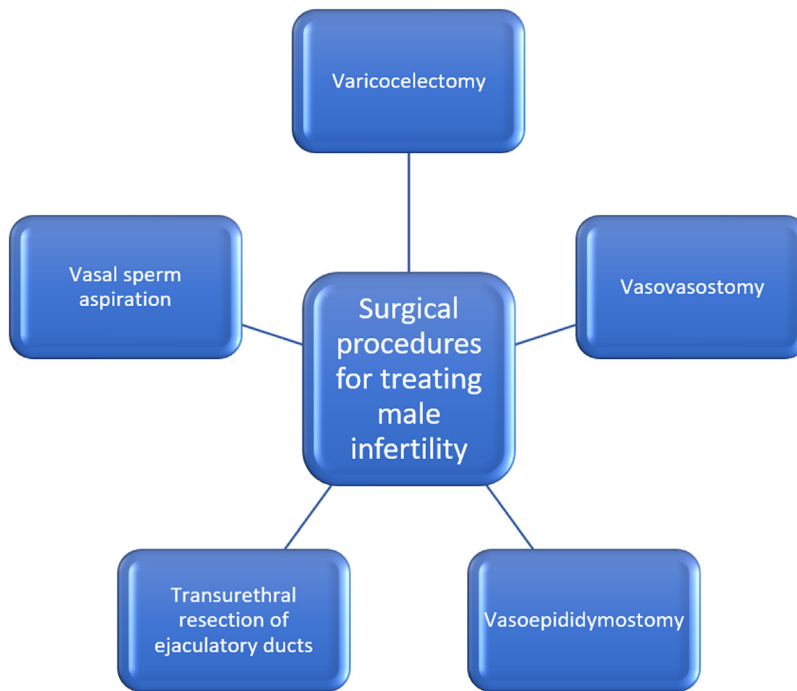
A successful diagnostic not only will determine the best option for treating the infertility but also will significantly improve the pregnancy rates of the couple ([Palma, 2014](#)).

10.4 Current treatments

10.4.1 Surgical treatment for male infertility

For some patients, and depending on the nature of the disease, infertility can be “reversed” when undergoing a surgical procedure ([Fig. 10.1](#)). This option must be evaluated by the specialist and is a good option when medication is not enough; most of the procedures are performed with success, and men may become fertile again ([Zambrano Serrano & Carvajal Obando, 2020](#)).

The use of microsurgery increased the success in the reconstruction of the reproductive tract, and it also reduced considerably the recovery time after surgery.

**FIGURE 10.1**

Surgical procedures for treating male infertility.

Own source, with information of Zambrano Serrano, C. A., & Carvajal Obando, A. (2020). Surgical treatment for male infertility. *Actas Urológicas Españolas*, 44(5), 314–320. <https://doi.org/10.1016/j.acuro.2019.10.012>.

Another viable technique is the intracytoplasmic sperm injection (ICSI), which demonstrated that the spermatozoa retrieved from either the epididymis or the testis were capable of fertilization and pregnancy (Parekattil, 2012).

However, the big problem with surgical treatments is that they are too invasive, sometimes patients may face some complications during the recovery time, leading to having to undergo a second or third procedure.

10.4.2 Hormonal treatment

Due to anatomic/hormonal anomalies, and after a medical examination, some patients are candidates to undergo a hormone therapy; between 46% and 50% of these cases are idiopathic (Ring et al., 2016). However, there has not been a consensus about how long the therapy will be needed, and in some cases, there will be secondary reactions, such as headache, chest pain, sudden changes in weight, body mass, etc. (Zambrano Serrano & Carvajal Obando, 2020). Fig. 10.2 shown current hormonal treatment for male infertility.

Clomiphene citrate	Human chorionic gonadotropin	Prolactin	FSH therapy
• Improves sperm count, motility and morphology. • Increase the testosterone levels. • Side effects associated: dizziness, alopecia, gynecomastia	• Analog of LH in Leydwig cells. • Improves testicular volume. • Side effects associated: Hyperglycemia, headache, depression.	• Regulates steroidogenesis. • Possitive results in terms of libido, ejaculation and orgasms. • Side effects associated: Nausea, headache, fatigue, runny nose.	• Regulates the concentrarion of androgens in the seminiferous tubes. • Improvements in cases of ogliospermia, however, no pregnancy rates associated yet.

FIGURE 10.2
Current hormonal treatments for male infertility with information of [Behre \(2019\)](#) and [Khourdaji et al. \(2018\)](#).

10.4.3 Gamete (sperm) donation

Due to its demonstrated effectiveness, this is one of the most used methods to engender a child. It consists of using a third party sperm to achieve pregnancy, as a result genetic relation with the parent is lost, however there are some couples that decide to try this method to avoid transmitting genetic disorders or diseases to their future children ([American Society of Reproductive Medicine, 2014](#)).

There are two approaches to use the donor’s sperm:

1. **Intrauterine insemination:** In which a catheter is used to introduce the semen inside the uterus during the most fertile days of the woman. There’s a slight risk of infection after the procedure.
2. **In vitro fertilization:** In this case, female’s eggs are taken into a laboratory so they can be fecundated. When ready, fecundated eggs are implanted back to the uterus. The whole process takes around three weeks to complete and is very expensive.

In both cases, the donor’s sample must be “washed” to separate the normal and most active sperm from the rest ([Clinic, 2021](#)).

[Table 10.2](#) includes the other main and most known treatment methods, while mentioning what each method can do and what area it can target, or what condition it can treat.

10.5 Tissue engineering approaches for treating male infertility

TE is the method of using biomaterials, cells, and growth factors (substances that the cells need to grow, regenerate, and survive) to create biocompatible and functional tissues ([Reis, 2019](#)). TE has a huge range of application as a variety of artificial

Table 10.2 Common solutions and methods to treat male infertility, with a brief description and treatment area.

Common solutions and methods to treat male infertility			
Method	Description	How can it help/what does it treat	References
Testicular sperm extraction (TESE)	Extracting sperm cells directly from the testes.	1. Azoospermia 2. Severe oligospermia 3. Cryptozoospermia 4. Sperm with high DNA fragmentation	Leaver (2016)
Medication	The use of medication to treat male infertility.	1. Gonadotrophin injections, for hypogonadism 2. Pseudoephedrine, which closes the opening for the bladder, to treat retrograde ejaculation 3. Various medicine to assist with erectile dysfunction	Agarwal et al. (2021); Katz et al. (2017)
Medical devices	The use of medical devices that can stimulate the penis or the ejaculation process to retrieve semen.	1. Erectile dysfunction, using a vacuum constriction device (VCD) 2. Anejaculation	Parekattil et al. (2012)
Surgery	Removing, altering, or reversing the current condition of the tissue or organ, using surgical means.	1. Removal of blockages/ obstructions in the tubes, treating obstructive azoospermia 2. Vasectomy reversal 3. Varicocele, treating varicoceles	Agarwal et al. (2021); Leaver (2016)
Prolactin management	Normalize the levels of prolactin in the body, which is used for spermatogenesis.	1. Helps with erectile dysfunction 2. Hypogonadotropic hypogonadism	Ali and Mirza (2021); Hair et al. (2002); Saleem et al. (2018)
Hormone therapy	Injecting? Hormones directly to the body, to increase the levels of specific hormones, or to decrease/block them if they are increased.	1. Hormonal therapy for prostate cancer treatment: treats cancer by stopping or slowing down the growth or easing the symptoms in prostate cancer patients. This	Agarwal et al. (2021); Basaria et al. (2015); Katz et al. (2017)

Continued

Table 10.2 Common solutions and methods to treat male infertility, with a brief description and treatment area.—*cont'd*

Common solutions and methods to treat male infertility			
Method	Description	How can it help/what does it treat	References
		happens due to the decreased amount of testosterone that reaches the cancer cells, which need testosterone to grow in case of prostate cancer. 2. Gonadotropin treatment or menopausal gonadotropin treatment: treat azoospermia, oligospermia, high testosterone levels, or hypogonadotropic hypogonadism by supplying the body with gonadotropin, gonadotropin-release hormones (*GnRH), or chorionic gonadotropin (hCG) 3. Testosterone replacement therapy to treat hypogonadism	

human tissues can be created, and then used to treat various issues, from skin burns to corneal reconstruction ([Mozafari et al., 2019](#)).

Similarly, TE can be used for the treatment of MRS, by regenerating tissues, such as, testicular tissue regeneration, which will have similar properties of the original tissue, or can act as a structure, for the already existing tissue and cells, to grow on. TE can also be used for “In-Vitro Spermatogenesis,” a process of creating the male gametes, outside of the body, and to understand in depth about the causes of male infertility, while working on potential treatment methods ([Ibtisham & Honaramooz, 2020](#)).

A tissue, in biology, is the collection of large amounts of structurally similar cells and their extracellular matrix, which is “on-cellular three-dimensional macromolecular network composed of collagens, proteoglycans/glycosaminoglycans, elastin, fibronectin, laminins, and several other glycoproteins,” and all of them acting

together to complete specific functions (Mozafari et al., 2019). The tissues used in TE consists of stem cells which can be gathered/sourced by four grafting techniques: autografting, allografting, isografting, and xenografting:

1. Autografting—The technique of acquiring tissue from one part of the body and using it to another part of the same body.
2. Allografting—The technique of acquiring tissue from one body and using it on another body, of the same species.
3. Isografting—The technique of acquiring tissue from one body and using it on a genetically identical body. This technique is used only on twin siblings.
4. Xenografting—The technique of acquiring tissue from one body of one species and using it on another body of a different species.

10.5.1 Stem cells

Stem cells are unique cells, as they are infinite and are undifferentiated, and thus have the ability to differentiate to any kind of cell. In male infertility treatments, the mains stem cells used are SSCs, germ-line stem cells (GSCs), and/or mesenchymal stem cells (MSCs), also known as adult stem cells (Fig. 10.3). The first two types are testicular stem cells, meaning they are isolated from the testis, whereas MSCs can be isolated from a variety of tissues.

SSCs and GSCs are widely used as they play a very important role in the spermatogenesis process. SSCs are the most primitive spermatogonia and have the ability to self-renew thus making them a great source of cells and carry genes and

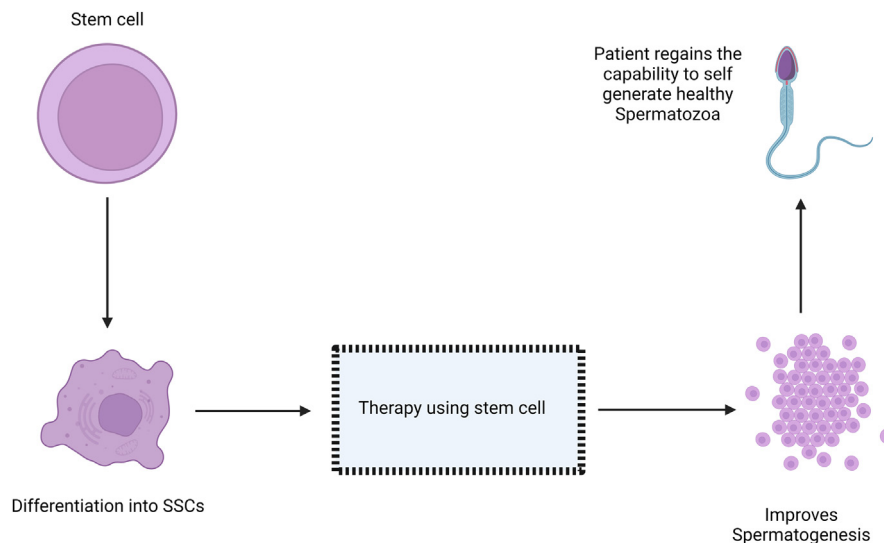


FIGURE 10.3

Basic idea of the use of stem cells for treating male infertility.

genetic information to the next generation (Ibtisham & Honaramooz, 2020). GSCs have similar properties to the SSCs as they can differentiate into various germ stem cells which then can differentiate in more specialized cells and are also self-renewable (Cheng et al., 2022). MSCs, because they can be found in a variety of tissues, can differentiate in almost any kind of tissue, making them another great source of undifferentiated cells (Ghanbari et al., 2020).

In some cases, the use of different stem cells at the same time was beneficial. For example, cotransplanted SSCs with MSCs in order to increase the colonization efficiency of the stem cells could improve the recovery of the endogenous SSCs and increase the homing efficiency of transplanted SSCs, and suggest more preclinical studies to be conducted in the future (Kadam et al., 2018).

Another interesting technique, using stem cells, was the use of cryopreservation and specifically, human cryoconserved immature testicular tissue (ITT), which was then xenografted into mice, to assess the survivability and growth of spermatogonia and Sertoli cells, which are the primary, undifferentiated sperm cells (Kubota & Brinster, 2018). Another study showed that although the survival rates of such cells is not very high, it has high potential for future practices and after careful meta-analysis and reexamination, as well as improvement in the grafting methodology, this technique could be used for humans (Wyns et al., 2007). Moreover, the cryopreserves cells could be used in various methods and ways in the future, to the extent of creating artificial testicular tissue or testicles, or boosting the spermatogenesis of an individual, achieving natural fertility (Del Vento et al., 2018).

Cryopreservation of the SSC niche was successful as a procedure, and the micro-environment was “well-preserved.” SSC niche is the specific microenvironment that stem cells live. This environment is responsible for the stem cell behavior, such as its cell cycle, differentiation, or other mechanisms (Jhala & Vasita, 2015).

Del Vento makes a useful point about the use of TE methodologies, to preserve, maintain, and transplant ITT and SSC, and create a bioenvironment similar to the original organ or tissue, which could be used to develop “transplantable artificial testis,” in the future. That could be achieved by using an artificial scaffold, explained later, made by either natural or synthetic materials, or by the decellularization of the tissue. Therefore, the combination of TE and cryopreservation seems a very prominent method that could be used in the future for testicular tissue regeneration, using cryopreserved cells (Del Vento et al., 2018).

10.5.2 Scaffolds

Scaffolds, which were mentioned before, are the structure on which these cultured cells mentioned before can be attached. Scaffolds are used to mimic the natural extracellular matrix of the cells in the human body, providing a 3D structure and the appropriate mechanical properties that would help the cells and tissue to grow and regenerate (Bhaskar et al., 2022).

These scaffolds are made by biodegradable materials (materials that can be degraded in the body) or bioresorbable materials (materials that can be absorbed

by the body) and can be either natural or synthetic. It depends on the application on which material to use; however, natural biomaterials are biologically active, allowing and stimulating cell adhesion, differentiation, growth, and movement making them a great candidate. However, the natural biomaterial-based scaffolds lack in morphological and structural properties, as well as fabrication consistency, meaning it is hard to manufacture and to keep their properties intact throughout their use. Lastly, [Bhaskar et al. \(2021\)](#) pinpoint that the methodology is still new and require further development and “fine-tuning,” as most 3D scaffolds developed to treat male infertility and specifically to be used for in vitro spermatogenesis have failed and there are only a few successful examples ([Bhaskar et al., 2021](#)).

On the other hand, synthetic biomaterial-based scaffolds excel in mechanical property tuning, controllability of the degradation rate and structural integrity, and although there are various applications where synthetic biomaterials were used for scaffolds, in vivo applications and current studies are still limited, and would require further researching and experimenting on the topic ([Bhaskar et al., 2021](#)).

The three synthetic polymers mentioned above have been used in various TE applications over the years. However, not a lot of research has been done using pure synthetic materials, but a combination of natural and synthetic materials together, such as PGA (poly glycolic acid) and collagen ([Matsumoto et al., 2000](#)), PLA with chitosan ([Hoveizi et al., 2015](#)), PLGA with gelatin ([Mehrasa et al., 2015](#)), PCL with chitosan ([Saderi et al., 2018](#)), or PCL with HA ([Entekhabi et al., 2016](#)), which were all studies on TE. All those studies are examples in which synthetic and natural biomaterials were used in conjunction but are not targeted on the reproductive system and treating male infertility; therefore, we can assume that not a lot of work has been done to support these kinds of materials, in male infertility treatment applications, meaning there is still room for more research and experimentation.

Porous scaffolds are according to [Bhaskar et al. \(2021\)](#), the best option for testicular TE, due to their porosity, allowing cells to enter, attach and grow, making strong intracellular linking and also, the similarity to the porosity of the seminiferous tubules. Porous scaffolds can be created using methods that are using natural materials, synthetic materials or using “tissue-derived scaffolds through decellularization” ([Bhaskar et al., 2021](#)).

10.5.3 Decellularized extracellular matrix

This method is a very interesting method of way of creating scaffolds as decellularized extracellular matrix (dECM) is used. Decellularization is the method of removing all the cells from a tissue/organ, but preserving the ECM components such as collagen, glycosaminoglycan, fibronectin, etc. ([Bhaskar et al., 2022](#); [Gholami et al., 2022](#)). The process of decellularization will not be discussed in this review; however, it is important to mention the reason why this method is used. The main advantage of dECM scaffolds is that the tissue’s morphology is preserved in a very good level, leaving a perfect structure behind which can be used as a scaffold. This structure is almost the same as the native tissue, making it the great 3D

environment for the cells to start grow and multiply. dECM is a biocompatible and biodegradable, natural scaffold, which at the same time, already has the appropriate growth factors and porosity, for cells to grow making it a great candidate for TE. There are multiple studies in which dECM has been used as a method for treating male infertility, such as in vitro spermatogenesis and testicular TE (Gholami et al., 2022). dECM can be used together with hydrogels, to create a three-dimensional structure for testicular cells to grow (Kanbar et al., 2021). Moreover, dECM was used to arrange testicular cells and tissue in a similar structure of that in the seminiferous tubules, and to also rearrange testicular cell suspension (Sakib et al., 2019). However, all the studies were not necessarily successful, meaning that the use of dECM is still under research and further experimentation, but has great potential in the future of male infertility.

10.5.4 3D bioprinting

Lastly, there are various ways to create artificial tissue, like electrospinning, which will not be mentioned in this review, but one of the most interesting one is the newly invented fabrication method called “3D Bioprinting” or simply “Bioprinting.” Bioprinting is a similar fabrication technique to the additive manufacturing technique of 3D printing, in which an object is created using layer-by-layer discharge of material (Bhaskar et al., 2022). Similarly, a tissue can be created using layer-by-layer disposition of cells, while discharging cells, growth factors, or biomaterials, or the mixture of all. For example, biomaterial can be released together with cells, making a complex geometry, that looks like the native tissue, with a micrometer resolution and accuracy. However, the main drawback of 3D bioprinting, apart from being a newly emerging technique, which will be improved in the future, is the fidelity of the bioink which can then affect the viability of the cells. Further investigation was suggested on the bioprinting properties such as the printing speed and multichannel printing to allow for more complex structures in the future (Bhaskar et al., 2022; Zhang et al., 2019).

Bioprinting can be used in testicular TE and in vitro spermatogenesis to control and create the desired scaffold, using alginate-based hydrogel and testicular tissue cells from mice (Baert et al., 2019). Another study was done to “assess the utility of bioprinting technology,” while trying to recreate testicular tissue, which would also be assessed for its spermatogenesis ability (Robinson et al., 2021, p. 447126). Both studies indicating good results, which act as a proof of concept for the feasibility of using 3D bioprinting for the treatment of male infertility.

10.6 Conclusion

Male infertility is a problem that affects millions of people globally that want to engender a child. Although there are many different treatments for it, not all of them will grant a 100% fertility recover; in the case of hormonal treatments, there are several side effects associated with the use of certain medications. As for the

surgical procedures, they could be quite effective and new less-invasive techniques are being developed; however, not all infertility issues can be treated with a surgery, and in the case of a major trauma, normally the organs involved doesn't recover completely their original function.

Gamete donation is one of the most explored techniques since it is considered a safe practice and gives good results in terms of achieving pregnancy; however, the genetic relation with the parent is lost.

The study of specific stem cells such as the SSC or MSC is the key for reversing infertility associated with the lack of spermatozoa and deficiency in testosterone production; as a matter of fact, the studies discussed in this work show us that proliferating SSC in vitro and then implanting the back into the subjects of study could develop new Sertoli and germ cells overtime, leading in the stimulation of spermatogenesis.

There are techniques that aim for organ reconstruction using autologous cells; for this, it is important to consider the proper use of a scaffold that is biocompatible and at the same time grants an optimal microenvironment for the seeded cells for proliferation. Urethra reconstruction shows good cell proliferation and no immune response when using a collagen-based matrix.

TE techniques seek to use self-cells to repair organs and to recover the reproductive function; in this way, we avoid immune reactions that may compromise the patient; however, more research is needed before clinical practice.

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The progress in tissue engineering of uterus

11

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11.1 Introduction

Although the reproductive system of a woman brings life into being, it can also end it. Infertility in women may be caused by a variety of genetic and acquired gynecological defects. Female infertility is a condition where women are unable to conceive after 12 months of frequent coitus (Brownell et al., 2022). As women approach the age of 35, their chances of being infertile increase significantly, with one study reporting that 12% of women in their 32s and 20% of women in their 38s experience infertility (Van Rooode et al., 2015). The demands for assisted reproduction technologies (ARTs) have increased due to modern lifestyles that emphasize education and career development for young women (Abrao et al., 2013).

Several factors contribute to female infertility, including endocrine and structural problems, ovarian cysts, congenital ovarian malformations, and hysterectomy (Kuo et al., 2017). It is important to take into consideration congenital malformations and psychological implications as well as infertility complications, since women with gynecological complications often suffer from distress relating to their identity and inability to be “natural” mothers despite current reproductive techniques such as in vitro fertilization (IVF) and surrogacy. Gestational surrogacy, metroplasty, hysteroscopic repair, and hormone therapy are all currently used to treat women suffering from congenital and acquired uterine factor infertility (UFI)

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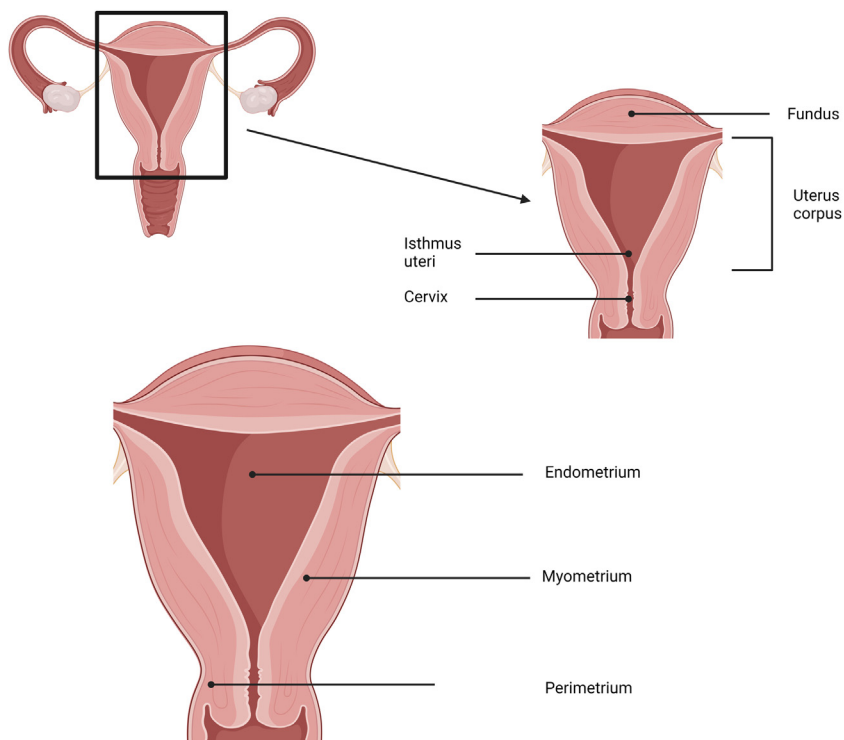
(Brownell et al., 2022). Although these approaches have some advantages, they are not without their drawbacks. For example, gestational surrogacy and its ethical implications are very expensive, and surgical methods may not be appropriate for everyone (Brännström, 2013; Brännström et al., 2015). A hormonal therapy that is designed to treat premature ovarian failure can often provide unphysiological levels of hormones.

Therefore, a number of great achievements in bioengineering approaches have been made in assisted reproductive technologies in the past few decades. For example, UFI is a challenging condition and uterus transplantation (UTx) offered a successful treatment option (Brännström, 2013; Brännström et al., 2015). While this is a major breakthrough in reproductive medicine and organ transplantation, there are still risks associated with live donor surgery and adverse side effects associated with immunosuppressive treatments following uterine engraftment. These limitations may be overcome by novel tissue engineering techniques. It might be possible to create a patient-specific uterus graft with patient-derived cells populated on a scaffold. Living donors or multiorgan donors would not be needed for such bioengineered grafting materials (Brännström, 2013). Furthermore, donors and recipients would not need to be matched or immunosuppressed (Brännström et al., 2015). It may also be possible to develop bioengineered uterine tissue to treat various conditions associated with sub-fertility through the use of uterus reconstructive surgery. In order to increase implantation success and to decrease the risk of uterine rupture during a future pregnancy, uterine tissue grafts may be used to minimize scarring of the uterine wall after repeated surgeries. Therefore, a bioengineered uterus tissue patch could serve as an intermediate objective in the development of therapies for partial uterine repair in a UTx setting as well as a longer-term goal of developing a bioengineered uterus that will replace a donor (Brännström et al., 2015).

11.2 Uterus anatomy and physiology

The uterus is present in the pelvic cavity of the torso. The uterus comprises four anatomical regions: the uterus corpus (body), fundus, isthmus, and cervix. The fundus is a rounded portion superior to the farthest opening. The largest anatomical region of the uterus is the uterus corpus. The uterus extends laterally to a constriction called the isthmus. Inferior to the isthmus is the cervix, which concludes the structure of the uterus (Fig. 11.1).

The dimensions of the uterine wall are highly variable. The uterine wall has three main layers: perimetrium, myometrium, and endometrium (Tortora & Derrickson, 2013). Perimetrium is the outer serous membrane composed of epithelial cells. The middle, thick layer of the uterus known as the myometrium accounts for 90% of the uterus. Myometrium consists of smooth muscle cells, also known as myocytes. The endometrium, is a thin inner and epithelial layer, makes up the remaining 10% of the uterus. It consists of basal cells, supportive mesenchymal cells, and luminal and glandular epithelial cells (Nishino et al., 2021).

**FIGURE 11.1**

Anatomy of the uterus ([Campo et al., 2017](#)).

11.3 Uterus physiology

Menstruation, pregnancy, and labor are functions of the uterus. The uterus also produces hormones and gametes. Implantation of the zygote takes place in the uterus. The uterus' vital functions include feeding and safeguarding the embryo. In later stages, it also removes wastes for the developing embryo. The uterus oversees the contractions and evacuating of the fetus at birth ([Jabbour et al., 2006](#)).

Perimetrium produces fluids to reduce friction and hold the structural integrity of the uterus. The smooth muscle tissue known as myometrium is what generates the force required to transport the fetus from the uterus to the vagina. The endometrium contains vascular and glandular tissue that supports the fetus's nutritional and physiological needs. In the event that fertilization is unsuccessful, the menstrual flow, which includes endometrium, blood, and mucus from the cervix and vagina, leaves the body through the vagina during a period ([Tortora & Derrickson, 2013](#)).

11.4 Common uterine diseases and disorders

The inability to conceive a child is referred to as infertility and can result from various disorders, including endometriosis, advanced age, and other uterine conditions. Infertility affects more than 7.5 million women of reproductive age worldwide (Charoensombut et al., 2022).

11.4.1 Genetic disorders

Several genes participate in aberrant and typical uterine development. Unbalance in the expression of genes leads to congenital diseases of the uterus. The patterning and differentiation of the uterus depend on the expression of Hoxa10 and Hoxa11 (Gendron et al., 1997). The uterus's structure and operation are interfered with when genes are out of balance or are not expressed. There are certain hereditary disorders in the female reproductive system, such as uterine Müllerian abnormality (Connell et al., 2013).

11.4.2 Advanced age

In recent decades, in many modern societies, it has been a popular misconception that new reproductive technology can compensate for the age-related loss in fertility. This reassurance is in vain. However, age is the most significant factor affecting fertility (Balasch, 2010). The percentage of women who put off having children until age 35 has significantly risen (Hellström et al., 2017).

11.4.3 Diseases and illnesses

The sixth most frequent malignancy is endometrial cancer. Women of reproductive age frequently encounter uterine fibroids. While a third of fibroids may develop during the first trimester, the majority do not change in size while a woman is pregnant. The risks of spontaneous miscarriage, premature birth, abruption of placenta, malpresentation, caesarean delivery, and postpartum hemorrhage are all increased by uterine fibroids. In addition, according to the majority of the literature, many women may experience decreased uterus function due to past caesarean sections or uterine fibroid removal (Lee et al., 2010).

11.4.4 Accidents and traumas

Premature labor, tissue scarring, labor that doesn't proceed, and postpartum hemorrhage, which occurs after delivery when the uterus is unable to adequately contract, are obstetric issues (Taggart et al., 2007). When the endometrium is seriously damaged in a woman of childbearing age, uterine scarring and a lack of a functional endometrium frequently follow, with an increased risk of infertility or miscarriage (Ding et al., 2014).

11.4.5 Absolute uterine infertility

The condition of absence or total malfunction which renders conception and pregnancy maintenance is called absolute uterine infertility. The UFI refers to a condition with either no uterus at all or a uterus that is not functioning for various reasons. The exact prevalence of it is unknown ([Sallée et al., 2022](#)).

11.5 Uterine histoarchitecture

In general, the uterus is a pear-shaped organ measuring 7.6 cm in length, 4.5 cm in width, and 3 cm in thickness. Among the secondary sexual organs of the gynecological tract, the uterus is the most hormone-responsive. In embryogenesis, the uterus forms by fusing two Müllerian ducts to form a single, hollow organ in the pelvic area behind the bladder and in front of the rectum. Uterus fundus is connected to both fallopian tubes, which lead to both ovaries, while the lower part, the cervix, opens to the vagina ([Brownell et al., 2022](#)). Besides feeding and supporting embryos, the uterus regulates hormones and menstruates. Endometrium, myometrium, and serosa make up the three layers that make up the corpus ([Rosner et al., 2019](#)). Menstruation alters the thickness and structure of the endometrium, which is the inner lining of the uterus. An embryo's attachment to the endometrium and the uterus' hormonal responsiveness depends on it. Endometrium histology varies greatly in response to sex hormones, as it is composed primarily of epithelial, stromal, and glandular cells along with a significant degree of vascularization. An embryo may attach to the uterus as a result of the ovaries' secretion of estrogen and progesterone. These hormones stimulate proliferation, vilification, and hormonal changes by affecting the epithelium. In the aftermath of the menstrual cycle, tissue and blood are expelled due to necrosis caused by a lack of estradiol and progesterone ([Ferenczy & Bergeron, 1991](#)). In addition to the diversity of cells and extracellular matrix (ECM) molecules in the uterus, the uterine stratification of layers such as the endometrium, myometrium, and perimetrium makes the histoarchitecture of the uterus highly complex ([Almeida et al., 2022](#)).

In addition to collagens I, III, IV, and fibrillin, elastin, tenascin C, and laminin, the myometrium is an epithelial layer that contracts and expels the fetus at birth. It also contains proteoglycans, including hyaluronic acid (HA), fibronectin, heparin sulfate, and other types. Because the endometrium regulates uterine receptivity for the attachment of embryos, hormone production, pregnancy maintenance, and protection against external agents, it is a more refined and complex layer of the uterus. When it comes to cellular components, the endometrium consists of epithelial cells, glandular epithelial cells, endometrial glands, fibroblasts, endothelial cells, mast cells, T cells, and B cells ([Almeida et al., 2022](#)).

The stromal layer is surrounded by ECM; the endometrium contains endothelial cells, and mast cells are the most numerous immune cells. There is a layer of tissue that covers the outer surface of the peritoneum, which is directly in contact with the

peritoneal cavity. Cell differentiation and morphogenesis are regulated in the implanted embryo by the uterine ECM during embryonic development and establishment of the maternal–fetal interface. When implantation occurs, placentation occurs, or if pathological conditions exist, the uterine matrix adapts. It is believed that the increase in P4 levels during the secretory phase, following ovulation, triggers decidualization, during which the glandular and stromal cells differentiate for pregnancy, accompanied by characteristic morphological and ultrastructural changes. Uterine milk proteins such as glycodefin and osteopontin are secreted by the glands as glycogen accumulates in the subnuclear cytoplasm. A fall in P4 levels caused by involuting corpus luteum (CL) triggers menstruation in the absence of implantation. Several disorders of pregnancy (e.g., preeclampsia and miscarriage) may occur if a defect in the decidualization process occurs (Woodward Davis et al., 2021).

As the level of E2 rises following menstruation, the functional layer is regenerated and expands considerably. Regeneration occurs in the basal glands, which do not shed. These basal glands express markers similar to those found in other stem cells/progenitors, such as SRY-box transcription factor 9 (SOX9), stage-specific embryonic antigen 1 (SSEA1), and CDH2. A combination of congenital factors and acquired factors, such as hysterectomy or intrauterine adhesion, can cause UFI, which affects 3%–5% of the population. Currently, these patients can only have children through adoption or gestational surrogacy.

A current bioengineering approach to uterine tissue restoration can be divided into uterine transplants and tissue engineered constructs for partial uterine regeneration. To develop tissue engineering techniques for uterus reconstruction, it is very important to understand the composition of uterine tissues (Atala, 2012). As a result, substantial efforts are being made to develop bioengineering approaches to reviving fertility, categorized into two broad categories: (i) transplant of fresh or cryopreserved organs transplanted into the host; and (ii) tissue engineering approaches that leverage the body's inherent ability to regenerate/repair reproductive organs by combining cells, growth factors, and biomaterials. Different types of cells are often used in tissue engineering (Kuo et al., 2017).

11.6 UTx animal research

A number of animal models have been used in UTx research, including rodents (mouse, rat), large domestic animals (sheep, pig), and nonhuman primates (baboon, macaque) (Table 11.1). Since a large number of inbred rat/mouse strains are commercially available, syngeneic transplantations can easily be used in rodent experiments. In larger animals, autologous transplantation is used to exclude the potential harmful effects of rejection and immunosuppression, but the same animal undergoes both organ recovery surgery and transplantation surgery, which results in a much longer surgical time than it would normally experience during a normal transplantation process (Brännström et al., 2012). Studying the mechanisms of

Table 11.1 Current and possible materials for 3D scaffolds.

Natural scaffolds			Synthetic scaffolds		
Material of scaffold	Expressed properties	References	Material of scaffold	Expressed properties	References
Collagen	Provided structural durability to the cell	Heidari Kani et al. (2017)	Poly(lactic-co-glycolic acid) (PLGA)	Retains cells for a prolonged period	Xiao et al. (2019)
Matrigel	Proper cell differentiation	Adissu et al. (2007)	Polyvinyl alcohol (PVA)-g-starch	Improved porosity and structural stability	Castillejo et al. (2012) ; Heidari Kani et al. (2017)
Decellularized tissue	Simulated contractile activity	Dallot et al. (2003)	Hydrogel (cryogel) macroporous	Elasticity	Astashkina et al. (2012) ; Heidari Kani et al. (2017)

rejection, finding immunosuppressants, and studying pregnancy in an environment that is similar to clinical UTx, will all benefit from studying allogeneic animal models. In the mouse model, C57BL/6 recipients were given uteruses from BalbC donors and the time course of rejection was examined using BalbC mice as donors (El-Akouri et al., 2006). After transplantation, swelling and necrosis developed within the first two days of the procedure, followed by major swelling and necrosis by day 15. Invading macrophages and neutrophils are the first leukocytes to invade uterine allografts (Groth et al., 2009).

A UTx setting often involves the use of allograft tissue. There are, however, complications associated with this approach. It is inevitable that transplanted patients will experience immune suppression-related side effects and tissue rejection. There are a number of important challenges that remain in the field of regenerative medicine and tissue engineering when it comes to the survival of the grafts. To overcome these challenges, novel and promising concepts for functional organ- or tissue replacement have emerged (Hellström et al., 2017).

11.7 UTx human research

The uterus transplant procedure has been proposed as a clinical treatment option for a limited number of women who are unable to use surrogacy or adoption in conjunction with IVF and have absolute UFI (Brännström & Wranning, 2007). An unrelated 46-year-old donor was used as the first live donor to provide a uterus with attached oviducts to a 26-year-old patient 12 years ago in Saudi Arabia, who had previously undergone a peripartum hysterectomy due to life-threatening bleeding (El-Akouri et al., 2006). An initial surgical procedure to remove bilateral ovarian cysts was part of the donor's elective surgery. In order for a human UTx trial to take place, there are several issues that need to be addressed both in regard to the uterus donor and the transplant patient. In renal and partial liver transplants, uteri can be donated by a living donor or one who has been brain-dead and whose heart is beating. Human UTx offers the advantage of avoiding a surgical risk for a second party when a deceased donor is used. When an organ is donated by a deceased person rather than a living one, a major systemic inflammatory reaction occurs at brain death that may adversely affect the survival of the graft. This inflammatory response is related to the time interval between brain death and organ recovery (Segev et al., 2005). If a uterus is being donated from a living or a deceased donor, several uterine-associated pathological conditions must be ruled out before the transplant. This may take a considerable amount of time, which may be disadvantageous when it comes to deceased donations. There are a few conditions that can be ruled out with appropriate testing, such as human papillomavirus infection, cervical dysplasia, leiomyoma, and endometrial polyps. Efforts in modern transplantation surgery have reduced the importance of tissue type matching in both types of uterus donations (Fabrizii et al., 2005).

Data indicate mild prematurity, decreased birth weight, hypertension/preeclampsia, but no increased rates of congenital malformations among women with solid organ transplants. There are more than 14,000 births among women with solid organ transplants who have been reported in the literature. In accordance with international recommendations for transplant patients, we recommend IVF treatment to ensure fertility in the couple prior to undergoing UTx and to store embryos for subsequent transfer trials (Brännström et al., 2012). Uterine tissue bioengineering could be a real possibility for reproductive medicine if these findings are supported by new concerns about uterus transplants (Almeida et al., 2022).

11.8 Tissue engineering

As a means of supplying organs to a rapidly increasing population, tissue engineering is capable of rebuilding tissues in vitro. Replacement of damaged organs is possible by combining biomaterial scaffolds with the different types of cells. A valuable tool for such applications is stem cells, since they can differentiate. As defined in 1987, a biomaterial is “a non-viable material that interacts with a biological system in a medical device” (Williams, 1987).

A variety of medical conditions involving structural and functional impairments can be treated with organ tissue engineering (OTE). OTE typically involves preparing a scaffold of supporting cells and tissues, implanting the cells in the scaffold, repopulating, remodeling, and then grafting the scaffold into the patient as an organ or tissue-like scaffold. In alternative treatments, cells may be transplanted directly into the patient without being repopulated and remodeled. An acellular scaffold is used to support cells that migrate from the recipient’s body, supports their proliferation and differentiation, and eventually produces the regenerated tissues and organs (Yoshimasa & Maruyama, 2021).

In OTE, three fundamental steps are involved: selecting the right source of stem cells, processing them, and choosing a scaffold that will support the cells. With regards to cell sources, stem/progenitor cells are ideal because they are capable of self-renewal and multilineage differentiation, which leads to the generation of desired tissues and organs. Cells for OTE are likely to be derived from adult stem cells (ASCs), embryonic stem cells (ESCs), and induced pluripotent stem cells (iPSCs) (Willerth & Sakiyama-Elbert, 2019).

11.9 2D models and 3D models

Research into uterine function and the mechanics of labor has been impeded by the lack of suitable models. The examination of complicated uterine activities, such as cell connections, connectedness, and contractile behavior, is limited by the ineffectiveness of conventional culturing techniques. Additionally, traditional cell culture

methods have drawbacks (Yamamoto et al., 2014). Cell culture models in 2D are cultivated on surface of a flask. The control and monitoring of the model are simple. However, cell adhesion and dissemination are limited to a flat surface in these models, which do not duplicate the *in vivo* environment (Yoshii et al., 2011). Research has focused on examining how many soluble factors affect the regulation of lineage development in 2D culture systems. Even though these findings aid in understanding interactions, it is crucial to assess 3D configurations to comprehend spatiotemporal effects, examine the reorganization of different dividers and organ development, and create systems that can be applied in clinical settings (Lawrence & Madihally, 2008). The three-dimensional architecture of the ECM and the membrane, a structurally compact part of the ECM, is more complex than that of the tissue culture plate's flat surface (Schindler et al., 2006). 3D models use cell seeding onto scaffolds to develop the cell at various levels (Table 11.1). The 3D model can simulate the *in vivo* environment and be utilized for organ transplantation. However, it isn't easy to analyze, image, and optimize 3D systems (Heidari Kani et al., 2017). 3D model applications in cell culture and regenerative medicine hold considerable potential for using current techniques to create synthetic environments that can give the necessary structural and chemical information. The difficulty of optimization of cell lines and scaffolds is present in the 2D and 3D models.

11.10 Cell source

In the last decade, bioengineering of uterine tissues has made great progress. The efficiency of recellularization *in vitro* and *in vivo* has been determined to be the major limitation of several promising scaffold designs. *In vitro* 3D culture conditions may be inadequate, or scaffold homeostasis may not be sufficient to maintain the equilibrium between hypoxia, repopulation, and revascularization of the construct. Cell type may also influence the outcome of these factors; therefore, more studies are needed to explore the culturing conditions and the different cell sources (Hellström et al., 2017).

11.10.1 Adult stem cells

As an ASC, mesenchymal stem cells (MSCs) show promise for regenerative medicine and OTE applications. In addition to bone marrow and adipose tissues, MSCs could be isolated from a wide range of tissues. Clinical trials can be conducted after they have been cultured.

In order to achieve the desired therapeutic effects, for example, the treatment of autoimmune diseases, a suspension of MSCs may be administered intravenously or locally, either from a patient or an immunocompatible donor's tissues enriched with MSCs. It has been shown that MSCs are unable to engraft and differentiate to repair tissue, but instead communicate via paracrine signals that stimulate angiogenesis and immunomodulation through cell contact instead (Fitzsimmons et al., 2018).

Surgical implantation of MSCs can also be performed using osteoblasts, chondrocytes, or adipocytes first differentiated into a desired cell type (e.g., osteoblasts, chondrocytes, and adipocytes) before differentiation into a desired cell type occurs. A failure of engraftment was observed in the early animal studies using MSC-derived chondrocytes. Thus, MSCs are unlikely to be used in practical applications for OTE ([Markstedt et al., 2015](#)).

MSCs are not the only stem cells that might be suitable for OTE. Tissue-specific stem cells are also a possibility. Tissue-specific stem cells produce differentiated cells that function as a part of their specific tissues and organs and also govern the maintenance of their tissue of origin. Because tissue-specific stem cells have the capacity to differentiate and regenerate, it makes sense to use them for OTE. Nevertheless, there are several limitations in that; (1) there is no clear identification of tissue-specific progenitor cells in all organs and tissues, (2) they are often inaccessible and difficult to isolate and handle even if identified, and (3) they can be difficult to expand in vitro and in vivo ([Yoshimasa & Maruyama, 2021](#)).

11.10.2 Embryonic stem cells

Cells derived from the blastocyst stage of embryos are known as ESCs. Despite their pluripotency, which means that they can differentiate into many different types of cells, they are capable of proliferating indefinitely in culture. Thus, using ESCs for regenerative medicine, including OTE, has been considered a successful therapeutic strategy for a long time. It has been made possible by the creation and availability of ESCs in both mice and humans. OTE based on ESCs has limited potential due to tumorigenic risks, immune rejection risks, and ethical issues. Furthermore, despite the pluripotency of ESCs, the efficiency of induction of differentiation into the desired cell type is less than 100%. It is possible that even after differentiation induction, tumors will arise from a small fraction of residual undifferentiated cells due to the inefficiency. It has been limited by these limitations that ESC research has not been translated into clinical practice ([Cossu et al., 2018](#)). A limitation to widespread use of hESCs for cell therapy and tissue engineering is the tendency of these cells to form teratomas in vivo, as it is difficult to control the tumorigenicity of these cells in vivo, and this has led to a number of safety concerns regarding their use as a therapeutic tool. Human embryos must also be destroyed in order to obtain human ESCs (hESCs), which is an ethical dilemma. Furthermore, some countries, including the United States, have banned the use of hESCs in tissue engineering, so their use is infrequent there at the moment ([Atala, 2012](#)).

11.10.3 Induced pluripotent stem cells

iPSCs, which have emerged as a promising alternative cell source for regenerative medicine, have overcome the limitations of ESCs, such as immune rejection and ethical concerns associated with using human embryos. After being reprogrammed through forced expression of factors for maintaining the defining properties of ESCs,

iPSCs can acquire ESC-like pluripotent stemness. Although iPSCs can be generated through reprogramming, their efficiency remains low. A lack of established protocols for inducing differentiation and heterogeneity in iPSCs also contributes to the low differentiation efficiency of iPSCs into specific types of cells. In addition, iPSC-based therapies have the potential to generate tumors. While autologous or HLA-matched iPSCs are not required for preclinical and clinical trials and there are no or very few risks of immune rejection, they have greatly facilitated basic research and preclinical studies. There have been or are being conducted over 70 clinical trials. Patients with neovascular age-related macular degeneration were treated with a sheet of retinal pigment epithelial cells derived from autologous iPSCs (Yoshimasa & Maruyama, 2021).

11.11 Scaffolds

The scaffold is a biomaterial that provides a 3D structure that enables cell engraftment, tissue growth, and differentiation to occur efficiently during regeneration and reconstruction of organs and tissues. To meet these requirements, scaffolds must be immunogenic, biocompatible, nontoxic, biodegradable as soon as possible, and have appropriate biomechanical properties. There are three types of scaffolds available today: natural materials, synthetic materials, and decellularized ECM materials (Yoshimasa & Maruyama, 2021).

11.12 Natural and synthetic materials

Synthetic polymers and natural polymers (hydrogels and scaffolds) are the most common biomaterials used in reproductive tissue engineering (Gargus et al., 2020). A natural biomaterial contains macromolecules found in the ECM. A variety of materials can be found in the production of these devices, including collagen, gelatin, HA hydrogels, fibrin, glycosaminoglycans (GAGs), alginate, matrigel, silk, hydroxyapatite, and others. The materials are characterized by specific mechanical and adhesive properties similar to the natural ECM. Their biocompatibility is enhanced by their less immunoreactive nature and limited ability to initiate signals. Some shortcomings of these materials include batch variability, a short degradation time, difficulty purifying, and quality management challenges. Polymers, ceramics, metals, and graphene are now recognized as four types of synthetic scaffolds that overcome the obstacles associated with natural biomaterials. Despite their different properties, synthetic scaffolds are generally easy to modify, have a wide range of properties, and are strong. Their poor biocompatibility, poor biological signals, and poor cell adhesion are, on the other hand, characteristics they are characterized by. The bioresorbability of synthetic materials can either be beneficial or detrimental to OTE depending on the nature of the material. There are several types of synthetic

materials; polylactic acid (PLA), poly(lactic-co-glycolic acid) (PLGA), polycaprolactone (PCL), polyethylene glycol (PEG), polyhydroxy ethyl methacrylate (PHEMA), and polyvinyl alcohol (PVA) are the most prevalent (Xu et al., 2019).

The eighteenth century brought the invention of lactic acid polymers, which are now widely used for a variety of applications. As far as biocompatibility, biodegradability, bioresorbability, immunogenicity, and toxicity are concerned, PLA and PLGA are superior to other synthetic polymers. In various medical fields, such as dentistry and plastic surgery, PLA and PLGA are popular 3D scaffolds. Synthetic materials can be used individually as well as combined with or without bioactive substances to improve scaffold properties, which improves OTE through facilitation of cell proliferation, differentiation, and fabrication. The mechanical and thermal properties of engineered tissues were improved by mixing PCL with PLA (Place et al., 2009).

11.13 Decellularized matrices

Scaffold fabrication and testing are significant processes. The majority of natural scaffolds come directly from the uterus tissue the basic steps are illustrated in Fig. 11.2.

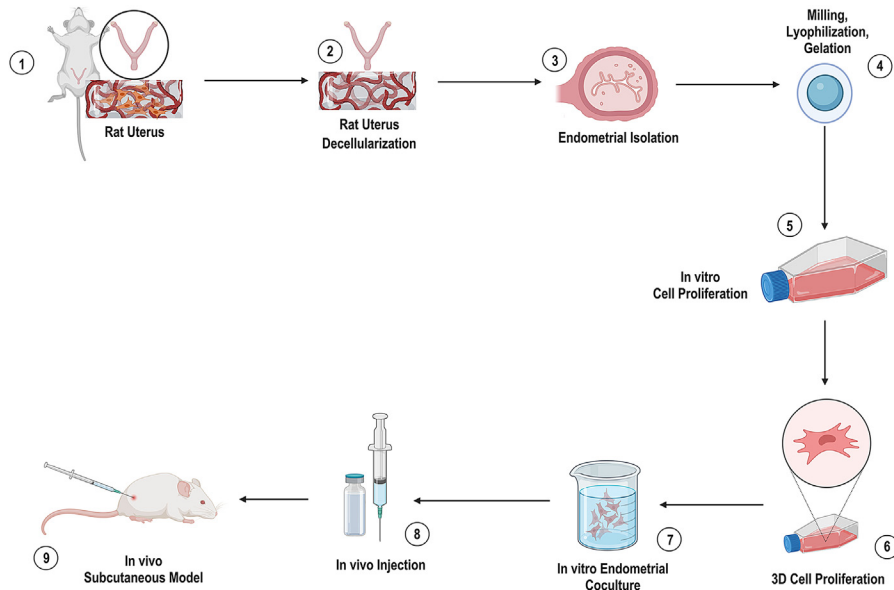


FIGURE 11.2

Preparation of in vivo subcutaneous injectable scaffolds.

A decellularization procedure involves removing viable cell components from organs or tissues from humans or animals in a series of steps. Various growth factors and proteoglycans, including collagen, elastin, microfibrils, proteoglycans, and GAGs, are retained in the scaffold, as well as their macrostructure and microstructure (Choudhury et al., 2020). It depends on the origin and property of the tissue being processed, whether decellularizing methods involve chemical, physical, or enzymatic treatments. In addition to acids and bases, hypotonic and hypertonic solutions are also available, as well as detergents like Triton X-100 and Triton X-200, sodium dodecyl sulfate (SDS), sodium deoxycholate, sulfobetaine-10 and -16, as well as solvents like alcohol, acetone, ethylenediaminetetraacetic acid, and trityl phosphate. Frozen-thawed cycles, direct pressure and force applications, and electroporation are some examples of physical methods. There are three types of enzymes used in enzyme treatment: nucleases, trypsin, and dispase. Biological and mechanical properties of the final product may be affected by the reagents and methods used for decellularization (Crapo et al., 2011). A critical aspect of the method selection is the choice of reagents.

Besides 3D ECM scaffolds prepared by whole organ decellularization, 3D ECM scaffolds derived from decellularized tissue have also been explored in tissue engineering and regenerative medicine. Dermal and urine bladder cells, small intestine cells, mesothelium, pericardium, and heart valves are some of the tissues used for ECM-based clinical products. Some of these cells are available commercially as well (Crapo et al., 2011).

11.14 Uterine tissue engineering

A review is presented here that focuses on exogenous cells and uterine tissue engineering. In order to study the mechanisms underlying the differentiation and implantation of embryos, various 3D uterine tissue culture systems have been developed and used (Hellström et al., 2017). A tissue culture system is also discussed.

11.14.1 Cell source for bioengineering of the uterus

It is most likely that MSCs, ESCs, or iPSCs will be used to repopulate the uterus with exogenous cells (Yoshimasa & Maruyama, 2021).

11.14.1.1 Uterus-specific stem cells

In the past few decades, stem/progenitor cells have been isolated and identified from uterine tissue in varying types. In the uterus, the endometrium and myometrium play the most important roles. Consequently, stem/progenitor cells can be obtained from these tissues (Yoshimasa & Maruyama, 2021). It has been identified that endometrial mesenchymal stem/progenitor cells, such as CD140b⁺ CD146⁺ or SUSD2⁺, are transplantable, that is, prospectively isolatable endometrial stem/progenitor cells. It is also thought that N-cadherin⁺ endometrial epithelial progenitor cells and

side population (SP) cells are heterogeneous populations of endothelial cells (Cousins & Gargett, 2018). There are several properties that distinguish SP cells from stem/progenitor cells. In the original tissue and organ, SP cells are present at low frequencies, which makes obtaining enough SP cells for OTE very difficult (Masuda et al., 2010). Moreover, endometrial SP cells need a functional microenvironment and supportive cells, such as a niche, to function optimally. Transplanting SP cells alone into immunodeficient mice results in low endometrial tissue regeneration in vivo. However, the activity increases when they are cotransplanted with whole endometrial cells (Miyazaki et al., 2012). The same types of myometrial stem/progenitor cells have been identified as those in the endometrium: SP cells, CD34+/CD49f+ cells, CD44+/Stro-1+ cells, and CD140b+/CD146+ or SUSD2+ cells. Only 3% of myometrial stem cells or CD34+/CD49f+ cells are located in the SP of the myometrium. Further, these stem cells alone have limited ability to reestablish myometrial tissue (Ono et al., 2007, 2015). In spite of the potential for bioengineering the uterus with endometrial and myometrial stem/progenitor cells, several issues remain, such as the difficulty of expanding them in vitro and in vivo, that make them difficult to use clinically (Yoshimasa & Maruyama, 2021).

11.14.1.2 Bioengineering of the uterus with ESCs and iPSCs

In addition to their ability to proliferate indefinitely, ESCs and iPSCs can maintain their stemness for as long as they are in culture. It is theoretically possible to overcome these problems by using these cells. It would be useful to use ESCs and iPSCs for bioengineering the uterus if a suitable method of differentiating them into each component of the uterus was developed. Endometrial cells can be generated in vitro and in vivo from human ESCs. Furthermore, Miyazaki et al. in molecularly defined embryoid body culture conditions, using specific hormonal treatments, human iPSCs were successfully differentiated into endometrial stromal fibroblasts through intermediate mesoderm, coelomic epithelium, and Müllerian ducts (Miyazaki et al., 2018).

11.14.1.3 Bioengineering scaffolds for the uterus

It has been studied in both basic and clinical studies of the bioengineering of the uterus, whether natural, synthetic, or decellularized ECM materials can be used. The bioengineering of the uterus has been explored since the early 2000s using synthetic materials, natural materials, and cell sheets. Natural materials containing collagen have been used in most studies. Humans and rats have been among the targeted species. In most studies, the endometrium is the target tissue. In vitro culture, scaffold repopulation, and in vivo transplantation have been performed with stem cells, including MSCs and ESCs (Yoshimasa & Maruyama, 2021). The use of synthetic materials and natural materials has been explored in numerous basic and clinical studies. A bioengineering of the uterus would require determining which material is optimal. Biomaterial properties, in vivo characteristics, and pregnancy support are all critical parameters. Moreover, utilizing larger animals, ideally primates, would enable the materials and methods to be characterized in a more

detailed manner (Yoshimasa & Maruyama, 2021). It may therefore be helpful to refer to the study by Magalhaes et al. in this context. The rabbit's uterus was restored by seeding it with autologous cells on a PGA/PLGA scaffold. A scaffold seeded with autologous endometrial and myometrial cells was used to reconstruct rabbits who had undergone a subtotal uterine excision. A two-layered myometrium and luminal/glandular epithelium were developed in the engineered uterus 6 months after implantation, along with stroma and vascularized mucosa. During the reconstructed segment of the uterus, four of 10 rabbits had normal pregnancies and lived to term and delivered (Magalhaes et al., 2020).

11.15 Decellularization and recellularization strategy

The first in vitro uterine tissue engineering was performed in 2013 (Young & Goloman, 2013). The decellularized rat uterus scaffold was demonstrated to be an effective supportive material for both in vitro and in vivo regeneration of functional uterine tissue (Miyazaki & Maruyama, 2014). Aortic perfusion with detergents such as SDS was used to decellularize rat uteri into acellular ECM scaffolds and perfusable vascular architectures. The scaffold was recellularized in vitro with rat uterine cells and MSCs, followed by aortic perfusion in a bioreactor to regenerate and maintain uterine-like tissues up to 10 days in vitro. Furthermore, recellularization and regeneration of uterine tissues were induced and pregnancy was almost as successful as in an intact uterus after the placement of an acellular scaffold on a partially excised rat uterus (Miyazaki & Maruyama, 2014). In the same study, disorientation of the scaffold was used in order to regenerate uterine tissue on a partially excised rat uterus; however, ectopic glands and abnormal smooth muscle layers were observed. An ECM scaffold was also prepared from rat small intestine, but it lacked the supportive capacity of the uterine scaffold. As a result of these findings, the ECM and architecture of the uterine scaffold still function and determine the orientation and topology of the regenerated uterine tissues (Miki et al., 2019).

Decellularized uterine scaffolds have been isolated most commonly using chemical treatments involving ionic detergents like SDS. A nonionic detergent like Trion X-100, however, minimizes the denaturing of ECM proteins since it is milder than ionic detergents (Padma et al., 2018). In order to bioengineer the uterus, it is necessary to determine which protocol should be used and which biomaterials are appropriate. In selecting the appropriate protocol and biomaterial, decellularized scaffolds' in vivo characteristics are more important than their in vitro characteristics. Further, the characterization and validation of decellularized scaffolds in terms of clinical applications would be improved with animals larger than rats and mice, ideally primates. However, decellularized scaffolds preserve the ECM and architecture, which may help regenerated uterine tissue follow the orientation and topology of the original. Decellularized scaffolds formed from uteruses should be used in clinical trials for bioengineering methods in order to simulate the complex structure of the uterus by transplanting them at a proper orientation.

11.16 Clinical research for tissue engineering of the uterus

Various methods have been used throughout the years to combine biocompatible materials and cell lines to create tissue that resembles the uterus' in vivo environment. Few experiments have been successful in developing a portion of the uterus, while some have focused on the organ as a whole. The basic standard operations for tissue engineering remain the same. The species, cell type, scaffolds, experimental evaluation, system, and results obtained are summarized in [Table 11.2](#).

The most popular scaffold is decellularization since it most closely resembles in vivo circumstances. Decellularization can be accomplished using either chemical or mechanical agitation. Collagen, hydrogel, and other natural scaffolds like matrigel have also been employed extensively due to biomimicking properties. Another approach, known as cell sheet engineering, has been implemented in addition to normal operations. In preclinical research, biodegradable scaffolds have been successfully investigated. Below table shows a few of the methods **adopted**.

11.17 Summary

Researchers have used natural, synthetic, and decellularized ECM scaffolds in bioengineering studies of the uterus. Clinical approaches to repairing defective uteri may be suggested by these studies. Despite this, there are still several obstacles to be overcome. The (whole) uterine scaffold must be repopulated in vitro for regeneration of a whole uterus, a process that is difficult in vitro. In order to exhibit maximal stem cell functions, including self-renewal, expansion, and differentiation into one or more lineages, many types of stem/progenitor cells, including endometrial SP cells, require an appropriate microenvironment (a niche) ([Masuda et al., 2015](#); [Morrison & Spradling, 2008](#)). Consequently, sufficiently mature and/or differentiated uterine cells capable of supporting stem/progenitor cells are required for the full repopulation of the scaffold and maintenance of the resulting regenerated uterus. The iPSCs and ESCs are necessary for generating enough of such cells, particularly the ESCs. In spite of several studies addressing this issue, no proper method for differentiating ESCs and iPSCs into uterine cells exists ([Miyazaki & Maruyama, 2014](#)). Several years ago, a new manufacturing process involving 3D bioprinting was developed ([Norman et al., 2017](#)). By printing living tissue-like structures layer-by-layer with biocompatible materials such as cells and growth factors, living tissue-like structures can be created. Bioengineering and regenerative medicine are exploring this approach for fabricating complex biological constructs. By using bioprinting, some of the limitations of repopulation can be overcome and the merit of scaffold-based uterine tissue engineering can be substantiated. In this approach, biocompatible materials such as cells and growth factors are used as “inks” to print living tissue-like structures layer-by-layer. This approach has emerged as a new strategy for the fabrication of complex biological constructs in the field of tissue

Table 11.2 Different scaffold application.

Methods	Specie	Scaffold	Evaluation/testing	Cell type	System	Result	References
Decellularization	Sheep	Decellularized uterus	Histology ECM quantification, mechanical properties evaluation	Sheep fetal bone marrow derived stem cells	Decellularization of uterus 0.5% sodium dodecyl sulfate (protocol 1) 2% sodium deoxycholate (protocol 2) or with a sequential perfusion of 2% sodium deoxycholate and 1% Triton X-100 (protocol 3).	The undifferentiated phenotype and markers for cell proliferation persisted in the cells. As a result, this study presents three useful decellularization techniques for use in upcoming in vivo sheep uterus bioengineering studies. The best treatment was uteri perfusion with sodium deoxycholate (protocol 2).	Tiemann et al. (2020)
	Mice	Decellularized uterus	Histochemical studies, Raman spectroscopy, Scanning electron microscope	Mesenchymal stem cells from menstrual blood	Decellularization using hypo- and hypertonic salts or by freezing and thawing, then immersing in solutions containing Triton X-100 and sodium dodecyl sulfate.	A good option for uterine tissue engineering is menSCs. The extracellular matrix of the decellularized uterine scaffolds was effectively retained, according to Raman spectroscopy.	Arezoo et al. (2021)
	Human	Decellularized uterus	Histological immunohistochemical, CT angiographies, scanning electron microscopy	Stem cells	Sodium dodecyl sulfate was perfused into the uterine arteries using a peristaltic pump to decellularize the tissue.	The endometrium's glandular components could be seen in both native and decellularized samples.	Daryabari et al. (2022)

Murine	Decellularized matrice (uterine horns)	High hydrostatic pressure, mechanical test, transmission electron microscopy	Stroma cells Smooth muscle cells Endothelial cells	Mice's uterine horns were removed, and they were then treated with sodium dodecyl sulfate and Triton-X.	All of the layers' fibers were still present, and the ECM structure was unaffected. The glandular components of the endometrium were visible in both native and decellularized samples. Collagen fiber deposition in bioscaffolds was retained following decellularization, according to trichrome staining.	Santoso et al. (2014)
Rat	Decellularized uterine tissue	High hydrostatic histological analysis, statistical analysis, quantification analysis of biochemical molecules	Endometrial stromal cells	The uterine horns' 25 mm were removed, and the connective tissue and fat were subsequently removed. Decellularization of uterine tissue requires high hydrostatic pressure. The uterine tissue was double-packaged in a PBS solution using polyethylene bags. In this work, a cold isostatic pressurization device was used to apply high hydrostatic pressure.	After being implanted, modified tissue has the potential to be used to encourage cell growth and tissue regeneration.	Charoensombut et al. (2022)

Continued

Table 11.2 Different scaffold application.—*cont'd*

Methods	Specie	Scaffold	Evaluation/testing	Cell type	System	Result	References
Matrigel	Bovine	Matrigel	Statistical analysis, histological analysis	Bovine endometrial epithelial cells	The reproductive tract of each cow was collected and trimmed of extraneous tissue, uterine samples were rinsed with phosphate buffer saline bovine BEE three-dimensional culture in matrigel.	The bovine uterine glands were produced <i>in vitro</i> to elucidate the process of elongation of bovine embryos. In matrigel, bovine endometrial epithelial cells were cultivated and the expression of different endometrial glandular factors was investigated.	Nishino et al. (2021)
	Human	Collagen-matrigel	Morphological assessments, gene expression assessment, flow cytometric analysis, transmission electron microscopy, statistical analysis	Human endometrial cells	Endometrial CD146+ cells were purified, using magnetic activated cell sorting, and then embedded and cultured in a collagen –matrigel scaffold on top of myometrial smooth muscle cells for 10 days.	Formation of the epithelial-like cells was confirmed by morphological and ultrastructural evaluations. The results showed that the human endometrial mesenchymal CD146+ cells were able to produce endometrial glandular tube-like structures <i>in vitro</i> . Ultrastructural observation revealed some projections on the apical surfaces, appearance of basal lamina-like structures on the basal surface, and tight junctions and desmosomes on the lateral surfaces of the epithelial-like cells. The expression of studied genes at RNA level and CD9 at protein level confirmed the formation of endometrial epithelial-like cells.	Fayazi et al. (2017)

Collagen	Human	Matrigel bilayer	Pathological analysis microarray analysis	Metaplastic squamous cells	Normal cervical organoids were established, which exhibited features of squamocolumnar junction cuboidal cells and transformational zone metaplastic cells.	The obtained organoids appear to mimic cervical squamocolumnar junction cuboidal cells and, in particular, metaplastic squamous cells from the transformational zone, likely providing a novel platform in which HPV- driven cervical cancer development could be investigated.	Maru et al. (2020)
	Rat	Collagen constructs	Histological examination microscopic examination	Bone marrow- mesenchymal stem cells	Sterilized degradable collagen membranes were cut into small scaffolds which were 1.5 cm in length, 0.5 cm in width, and 0.1 cm in thickness. Collagen scaffolds were obtained from Zhenghai Biotechnology (Yantai City, Shandong Province, China).	Transplanted BM-MSCs can located to the regenerative endometrium basalis and secreted growth factors, bFGF, IGF- 1, TGF- β 1, and VEGF and induced the proliferation, migration of local endometrial cells (stromal cells, epithelial cells, and muscular cells) while promoting angiogenesis to regenerate new uterine horns.	Ding et al. (2014)
	Rat	Collagen	Functional studies, histological analysis, Statistical analysis	bFGF with CBD-bFGF [epithelium, stroma cells, vascular endothelial cells, and smooth muscle cells]	Collagen membranes (Zhenghai Biotechnology, Shandong, China) were freezing-dried collagen extraction from bovine skin.	The collagen/CBD-bFGF group notably retrieved the structure and function of uterine horns in relative to spontaneous regeneration, collagen/PBS, and collagen/NAT-bFGF groups, similar to sham operated group. The pregnancy rate of	Sun et al. (2011)

Continued

Table 11.2 Different scaffold application.—*cont'd*

Methods	Specie	Scaffold	Evaluation/testing	Cell type	System	Result	References
	Rat	Collagen	Flow cytometric analysis, histological analysis, immunofluorescence analysis, fertility test, statistical analysis	Umbilical cord-derived mesenchymal stem cells	Degradable collagen fibers (collagen type I) were obtained from the Key Laboratory of Molecular Developmental biology (Institute of Genetics and Developmental Biology, Chinese Academy of Sciences, Beijing, China).	collagen/CBD-bFGF group was also improved. Thus, this targeting bFGF delivery system shows great effectiveness. The scaffold/UC-MSCs system could facilitate collagen degradation in uterine scars via upregulation of MMP-9, which mainly derived from the transplanted UC-MSCs, and promoted regeneration of the endometrium, myometrium, and blood vessels in uterine scars. Contributing from structural reconstruction, the scaffold/UC-MSCs-treated uterine scars showed improved pregnancy outcomes.	Xu et al. (2017)
	Rat	Collagen	Histologic analysis, functional testing, statistical analysis	Vascular endothelial growth factor (VEGF)	The solubilized NAT-VEGF and CBD-VEGF was purified by nickel chelate chromatography and HiTrap heparin HP columns (GE healthcare, Chalfont St Giles, United Kingdom).	Immobilization of VEGF using collagen-binding domain (CBD-VEGF) on uterine scar tissue resulted in a high local concentration and prolonged biological effect of the growth factor. CBD-VEGF promoted remodeling of the scarred uterus and improved uterine function in the full-thickness rat uterus injury model.	Li et al. (2021)

Hydrogel	Rat	Gelatin hydrogel	Material characterization, Dynamic viscosity, Youngs modulus	Rat fetal fibroblasts	The gelatin sheets used in this study were prepared by chemical cross-linking using glutaraldehyde.	Although the gelatin sheets used in this study were prepared by chemical cross- + G2:H2 linking using GA, generation of flexible sheets by nonchemical cross-linking would be ideal for fetal tissue engineering. UV cross-linking was another option; however, it was difficult to control the degradability of the prepared materials (data not shown). Sheets prepared by dehydrothermal cross-linking shriveled and crumpled, and thus were also unsuccessful (data not shown). The generation of flexible gelatin sheets by nonchemical cross-linking or by less toxic cross-linking such as genipin will therefore be the next goal for clinical application.	Watanabe et al. (2021)
	Porcine ECM injected to rat	Endometrium extracellular matrix-derived hydrogel	Gelation kinetics, stability, ultrastructure, histology, proteomic analysis	Endometrial cells Epithelial cells Stromal cells	Porcine uterus decellularization using detergents, endometrial isolation, milling lyophilization and gelation to produce endometrial extracellular matrix hydrogel.	EndoECM is biocompatible between species supporting <i>in vitro</i> growth of different types of human endometrial cells and improving the <i>in vitro</i> proliferation rates compared to standard matrices. EndoECM contains a low amount of potentially immunoreactive molecules.	López-Martínez et al. (2021)

Continued

Table 11.2 Different scaffold application.—*cont'd*

Methods	Specie	Scaffold	Evaluation/testing	Cell type	System	Result	References
Cell sheet engineering	Rabbit	Uterus-derived ECM hydrogel	Proteomics, electron microscopy, gelling kinetics	—	Hydrogel extraction from extracellular matrix and injection.	DC protocol for whole rabbit uteri was established and endometrial ECM hydrogels were obtained. Experiment demonstrated that coatings from synchronous decellularized endometrium act as biomimetic support for rabbit embryo development and is comparable to optimal protocols, possibly because of the slow release of synchronous-specific endometrial proteins.	Campo et al. (2019)
	Mice	PIPAAm-grafted temperature-responsive cell culture dishes	Histological analysis	Mice endometrial cells	Cell sheet engineering temperature-grafted cell culture.	Endometrial cell sheets reconstructed endometrium-like tissue, which was found to form uterus-specific endometrial glands having hormonal receptor to estrogen.	Takagi et al. (2014)
PGA/PLGA	Rabbit	Subtotal uterus	Canning electron microscopy	Endometrial myometrial cells	PGA/PLGA scaffolds were tailor-made in semicircular shapes.	In vivo experimentation with full time gestation.	Magalhaes et al. (2020)
Poloxamer	Rat	Poloxamer/hydrogel	Ultrasonography measurement, histological analysis, Western blot analysis, statistical analysis	Endometrial epithelial cells	A mini-endometrial curette was used to injure the rat's endometrium to form an IUA model. 17 β -estradiol was encapsulated into the micelles of heparin-poloxamer and a thermosensitive hydrogel (E2-HP hydrogel) was formed.	Taken together, in situ administration of E2 hydrogel effectively promotes endometrial regeneration and prevent the readhesion.	Yao et al. (2020)

PLLA	Rat	PU/PLLA	Macroscopic examination, histological analysis	Endometrial cells	The PTFE graft (Fig. 1a) was commercially available (Impra Inc., Temple, AR, U.S.A.). The PU/PLLA graft (Fig. 1b) was kindly supplied by J. H. Kuit, M.Sc. (Department of Biomaterials, Technical University of Twente, Twente, the Netherlands).	Although biodegradable microporous PU/PLLA uterine horn grafts had good healing properties, mucosal folds at the graft/uterine junction impeded their ability to function. In PU/PLLA grafts, mucosal suturing is combined with the use of splints.	Jonkman et al. (1986)
PEG	Human	PEG-VS hydrogels	Statistical analysis, histological analysis, dynamic correlations	Stromal epithelial (cell lines) coculture cell	Coencapsulated stromal and epithelial cells within the functionalized PEG gels to mimic a glandular coculture. PHSRN-K-RGD, an adhesion peptide, was used to partially functionalize multiarm PEG macromers activated with vinyl sulfone (PEG-VS), and a specified peptide including substrates for both endogenous matrix metalloproteinases and exogenous SrtA was used to crosslink the materials.	Intact cells were released with little impact and SrtA mixed with the scaffold dissolved the hydrogel.	Valdez et al. (2017)

engineering and regenerative medicine. Bioprinting has the potential to overcome some of the repopulation-related limitations and to substantiate the merit of the scaffold-based uterine tissue engineering (Jung et al., 2016).

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The progress in tissue engineering of vagina

12

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12.1 Introduction

Over the last decade, a growing number and diversity of surgical implants have been established to improve the effect of genetic deficiency surgery. Everlasting synthetic implants, as well as biodegradable artificial implants were obtained from animals (xenografts) or cadavers, and all have been commercialized (allografts) (Ahmad, Alam, et al., 2008; Ahmed, Dare, et al., 2008). There's no prove of viability for these items and the rates of results such disintegrations, inconvenience, contaminations, and vaginal decay are unsatisfactorily tall, at generally 10% (Patel & Atala, 2011). Tissue and organ abnormalities or malfunction impacted by a lot of conditions are the most common causes of human health deterioration. Restoration and functional reconstruction of tissue and organ abnormalities are some of the anomalies in the medical area (Patel & Atala, 2011).

Tissue engineering (TE) has revolutionized the traditional tissue transplantation therapy concept of “repairing trauma with trauma.” Congenital vaginal anomalies usually involve wide surgical rebuilding. Surgical problems do exist owing to the minimal quantity of native tissue accessible (Ahmed, Dare, et al., 2008). The goal of this study is to objectively explore various viable approaches to addressing challenges linked to vaginal diseases and disorders, with a focus on regenerative medicine. The different regeneration therapeutic options for improving the prognosis of vaginal illnesses are explored in this article. The existing work's literature review, vaginal reconstruction is clearly the most common treatment option at this point. Additionally investigation on the progress in TE of vagina promises, advances, and challenges was carried out in this work (De Philippo et al., 2008).

12.2 Vagina anatomy

The vaginal could be a tubular structure that interfaces the cervix to the vulva and is 7–15 cm long (Fig. 12.1). Its principal functions are to enable for sexual activity and to serve as a normal birthing method. It as well plays a crucial function in the formation of the bacterial flora of infants, which contributes to the rise in caesarean section births. When we mention vagina, we are referring to the vulva, which is the exterior region of the female genitalia (Appelbaum et al., 2018).

A vaginal canal is a large tube that connects the vulva's vestibule to the uterine cervix, enabling for sexual activity as well as periodic flow and childbirth. The vaginal fibromuscular tube, which runs from the vulva to the uterus, is around 7.5 cm (2.5–3 inches) length across the front wall and approximately 8–10 cm (up to 4 inches) length across the posterior wall. The vaginal orifice is totally covered by the Hymen, the vaginal vault organ, mons, clitoris, and the entrance of the urethra are all components of the vagina (Anderson et al., 2014). The stratified squamous epithelium that borders the vaginal canal serves as a barrier against hazardous infections. It is home to a complex microbiota which is molded over a woman's lifespan by ovarian hormone fluctuations. Muscle walls are made up of a network of neurons and a highly vascularized network. Fibromuscular tissues that provide delightful sensations during sexual activity (Alvisi et al., 2018). There are several diseases/disorders associated to this critical organ that must be taken seriously; else, patients may suffer numerous issues and, in severe situations, death (Dhama et al., 2016).

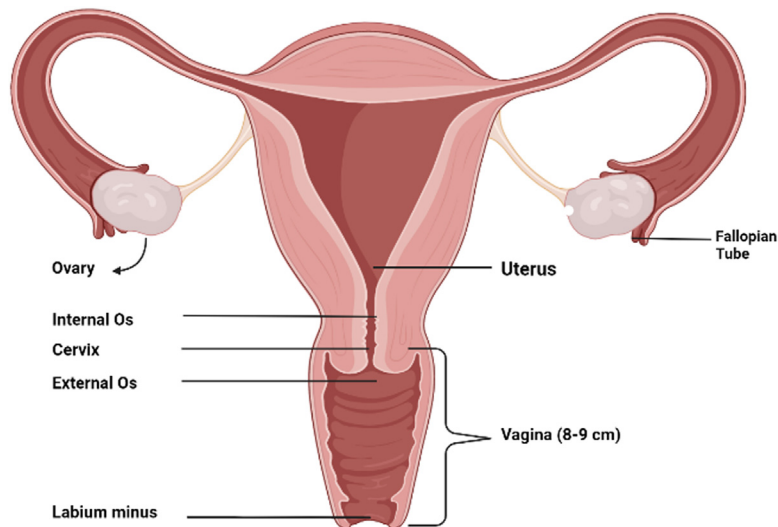


FIGURE 12.1

Anatomy of a vagina.

For women to retain a great quality of life, they need a healthy, patent vagina. In addition to congenital defects such as cloacal exstrophy, genital diseases, and the lack of the posterior two-thirds of the organ, cancer, trauma, infections, inflammatory, or iatrogenic traumas can all cause tissue damage and loss, necessitating vaginal repair or replacement (De Filippo et al., 2008). Additional problems, including as cures, contamination, maldevelopment, implant decline, and even cancer, are common when such tissue is used (De Filippo et al., 2008).

12.3 Causes of vaginal disorders

The underlisted factors are considered to be some of the causes of vaginal disorder (Ulfelder, 1980).

- Vaginal cancer
- Congenital abnormalities
- Soreness
- Contamination
- Shock
- Injuries leading to damage of tissue

12.3.1 Vaginal cancer

Vaginal cancer is a cancer that primarily affects women. Invasive vaginal cancer accounts for 0.3% of all invasive malignancies in women and 1%–2% of all gynecologic cancers in the United States (Creasman, 2005). Vaginal cancer shares many of the same risk factors as cervical cancer, including a significant relationship to human papillomavirus (HPV) infection. Anogenital cancer (especially cervical cancer), cigarette smoking, and a weakened immune system are all hazard components for the improvement of vaginal malignancies (Jemal et al., 2005).

Every year, there are 2420 new cases and 820 fatalities in the United States. Patients under the age of 50 are diagnosed in fewer than 15% of cases, while those under that age of 40 are diagnosed in 10% of cases (Shapiro et al., 1982; Smith et al., 2016). It is reported that cancer of the vagina constitutes 90% of cells determination accounting only 5% (Shapiro et al., 1982). Symptoms such as bleeding and discharge are the two most prevalent signs of vaginal cancer. Around 5% of persons get pelvic discomfort as a result of the disease spreading outside the vaginal area, whereas 5%–10% of people have no symptoms despite the disease being found during a regular pelvic examination and Pap screening. Tumors, urinary retention, pain, hematuria, and frequent urination are only some of the possible adverse effects. Vaginal cancer does not rank among the top 20 cancers in the United Kingdom, accounting for less than 1% of all new cancer cases (Ferlay et al., 2018).

12.3.2 Vaginal agenesis

Vaginal agenesis is a genetic condition that influences sexual role and productivity (Capraro & Gallego, 1976). In Rokitansky syndrome (Mayer–Rokitansky–Kuster–Hauser disorder), it is as often as possible complemented by a lost vaginal and add up to androgen lack of care disorder. After hypogonadism, Mayer–Rokitansky–Küster–Hauser (MRKH) disorder has been recognized as the most common cause of essential amenorrhea (Timmreck et al., 2003). It happens in one out of each 5000–10,000 female births, but the real cause is undetermined. Despite the appearance of typical pubertal development, the majority of women do not begin menstruation until they reach adolescence. The vaginal canal might be a few millimeters long or many centimeters long. Penetrative sexual intercourse necessitates vaginal repair (Niazi et al., 2001).

12.3.3 Posttraumatic stress disorder

Posttraumatic stress disorder (PTSD) is a typical impediment of traumatic happenings such as physical or sexual assault. Even though childbirth is generally considered as a good event, there is growing evidence that it can lead to PTSD, with some women expressing tremendous anxiety, powerlessness, and a danger to their integrity during labor (Brandes et al., 2002).

12.3.4 Vaginal candidiasis

Vaginal candidiasis (VC) is a serious condition that affects females; it affects about 75% of all females at some point in their life (Fidel, 2005). Exogenous factors that predispose women to VC include the use of antibiotics or oral contraceptives, pregnancy, hormone replacement therapy (HRT), and uncontrolled diabetic mellitus. *Candida albicans*, a gastrointestinal and reproductive system commensal organism, is the cause of VC in 85%–90% of patients. *C. albicans* interacts with epithelial cells by colonizing and multiplying, then invading, spreading, and damaging them. The most common symptoms of VC include dyspareunia, pruritus, itching, discomfort, edema, and vaginal and vulvar erythema (Johal et al., 2016).

12.3.5 Vaginal infections

Vaginal infections can be caused by bacteria, fungus, parasites, and viruses that grow in and around your vaginal and vulva (the area surrounding your vagina). Some of these organisms can be identified without producing any problems in your vaginal area. If the normal healthy conditions in your vaginal area are disturbed, however, they may spiral out of control. Douching (cleaning out your vaginal cavity), antibiotics, tampons, and other forms of contraception may all have a role (Mashburn, 2012). The symptoms include vaginal discharge that differs from normal vaginal dryness in color, odor, or thickness is a sign of vaginitis. Urination or, sexual intercourse can be painful and tissue become swollen, or damage the skin around the vagina (Gilbert et al., 2013).

12.4 Diagnosis of vagina disorder

The underlisted technologies are now used for the diagnosis of vagina disorder ([Beer-Gabel et al., 2002](#)).

- Ultrasound
- Magnetic resonance imaging
- Computer tomography
- Blood test

12.5 Current treatment

Patients may be offered nonsurgical or surgical therapies, varying depending on the how serious the condition is. The Frank/Ingram method or the Vecchiotti technique are the most common nonsurgical treatments, whereas the Abbè/McIndoe operation is the most common surgical treatment ([Caneparo et al., 2021](#)).

12.5.1 Nonsurgical treatment

According to Frank's technique ([Willemssen & Kluivers, 2015](#)), dilators are usually processed in three consecutive period for vaginal opening in the future and this. Continuously expanding in breadth and length and cleared out for approximately 20–30 min in Ingram's strategy ([Ingram, 1981](#)), the quiet embraces a bike situate to keep the vaginal dilators in staying whereas allowing the patient's weight to apply. The impediments appear to be the torment, the sum of period required for this long-standing treatment to limit the sexually develop individuals ([Santoro et al., 2015](#)). Although the method's use is limited painful procedures and the use of dilators, it is negligibly offensive and protects remaining vaginal matter. The Vecchiotti procedure ([Gambacciani et al., 2015](#)) entails the use of an acrylic olive to join the vaginal and abdominal walls. Internal traction causes dilation. A neovagina develops in a few days. The procedure is simply intrusive and supports existing vaginal tissue, although it is restricted by pain and the usage of dilators. Prolapse, long-term contractions, and urological lesions are all risks ([Gambacciani et al., 2015](#)).

12.5.2 Surgical treatment

Patients with relevant reproductive territory irregularities normally need surgical renovation. Abbè's vaginoplasty was invented. In 1898 ([Panici et al., 2007](#)), scientist used skin as an autologous transplant from the front thigh. The vaginal repair became up to date in 1938. Grafts have been chosen for vascularization locations like the abdominal rectus muscle since the 1970s. The surgery does not entail dilators, making it an excellent option for kids. Sigmoid vaginoplasty appears to give good outcomes, despite the literature's inconsistencies; however, it is accompanied

with a risk of stomach difficulties. Potential anatomical alternatives for vaginal repair have been exposed in the verbal sheath, amnionic film, peritoneum, and enhanced dermis matrices, and oxidized cellulose environments which employs skin that is an autologous transplant from the front thigh (Caneparo et al., 2021).

12.6 Tissue engineering

Vacanti authored TE in science in 1993. The concept of TE and its goals were initially conceptualized in this essay. Since then, TE has evolved significantly to provide fresh and inventive solutions to long-standing challenges (Sahakyants & Vacanti, 2020). In the context of human population aging, the emergence of continuing illnesses demanding organ auxiliary is on the rise, and sedentary behavior, even though the number of organs available for transplantation is decreasing, owing to this aging and sedentary behavior, as well as regulatory agencies' tightening of selection criteria (Scheper-Hughes, 2000). TE is the process of replacing or repairing tissues or organs in vitro, as well as creating 3D copies for study. Biomaterials, ordinary, have been examined, counting decellularized structures as scaffolds, planted or not with cells. Casting, electrospinning, and bioprinting are some of the procedures used to make the scaffold. Other techniques have been developed, such as the assembly process, which does not require any external biomaterial (Caneparo et al., 2021).

12.6.1 Synthetic biomaterials

Share a number of benefits over normal biomaterials, counting malleable quality, infection potential, accessibility, and toughness. Despite their extensive use in TE, synthetic polymers' utility as biomaterial scaffolds is partial due to insufficient external body, the potential of pro-stirring responses following replacement, and the lack of an ordinary molecular recognition site (Samavedi et al., 2014). Nonetheless, in recent decades, momentous advancement progress has been achieved in the functionalization of copied biomaterials. Hydrogels have been shown to be functionalized with a variety of proteins to mimic the microenvironment of target tissues in a range of applications, according to current study (Rice et al., 2013). These functionalized hydrogels might be employed to promote cell growth and differentiation, as well as to transport stem cells, medications, and bioactive proteins and be functionalized with diverse peptides to reproduce the microenvironment of target tissues, at least in part (Guelcher, 2008).

12.6.1.1 Poly(a-esters)

PCL that are commonly used and controlled by the Food and Drug Administration (FDA). Hydrolyzable ester connections dissolve these polymers in bulk, and their innocent degradation harvests are removed and transformed to water and carbon dioxide by metabolic pathways. The bodies are capable of dealing with its two

monomers under typical physiological settings (Samavedi et al., 2014). The rapid in vivo biodegradation, on the other hand, could result in localized concentrations which could harm cell proliferation and differentiation and trigger an inflammatory response. Multiple procedures, such as molding, extrusion, flush forming, gas foaming, and phase departure processes, are very straightforward to build three-dimensional scaffolds with satisfying qualities since they are thermoplastic. Furthermore, to produce scaffolds for TE and drug delivery systems, sponge, nanoparticles, fibers, films, and microcapsules have all been employed (Skardal et al., 2012).

12.6.1.2 Polypropylene

Polypropylene (PP) is an artificial polymer containing high carbon and hydrogen content that is nonbiodegradable. Only a few of the advantages of PP are its extended and well-controlled permeability, nontoxicity, and living lifelessness features. On the other hand, the hydrophobic surface of 1 PP makes TE difficult. To improve the PP surface, chemical processes, gamma irradiation, plasma treatment, and UV irradiation have all been tried (Sartoneva et al., 2018).

12.6.1.3 Polyethylene terephthalate

Polyethylene terephthalate (PET) is a polyester polymer with a semicrystalline structure that has a wide range of applications. Because of its inherent qualities including microporous as well as impersonal properties, and its inferior charge, PET is a popular marketable material. PET, on the other hand, has a limited range of applications. PET is crucial to change using a number of surface functionalization approaches in material building because of its surface idleness, such as damp chemistry and plasma polymer statement (Fell et al., 2019).

12.6.1.4 Acellular tissue matrices

Noncellular material media are collagen-rich resources made from allogeneic or xenogeneic tissues by decellularization techniques. Decellularization therapies attempt to eradicate all cellular components and antigenic epitopes while limiting detrimental effects on constituents, biological activity, and the extracellular matrix (ECM), which provides mechanical integrity for new tissue growth. Acellular tissue networks contain elastin, fibronectin, laminin, glycosaminoglycans (chokes), and numerous diverse sorts of collagens (Vasanthan et al., 2022). The tissue from which cellular components were extracted and the method utilized extract them on the other hand, define the makeup. A decellularized matrix's collagen fiber architecture has a significant influence on biomechanical properties (Liao et al., 2008). The useful capabilities of the source tissues from which extracellular networks are created decide the design and substance of collagen fibrils. Decellularization strategies and tissue sources can be utilized to degree the sum of development variables held in acellular platforms. Indeed, after being sanitized and put away for a long time, the bioactivity of these development variables can be protected (Hodde et al., 2005).

12.6.2 Natural biomaterials

Natural biomaterials, as opposed to synthetic materials, provide more suited settings for cell migration, proliferation, and differentiation, as well as ECM deposition, while also demonstrating great biodegradability and biocompatibility when handled properly. Natural biomaterials have a number of disadvantages, including a wide range of physicochemical qualities, the need for several purification processes, and the risk of pathogen infection, whether they come from a human being or a mammal (Caneparo et al., 2021).

12.6.2.1 Collagen is the world's most widely distributed structural protein

Biocompatible, bioresorbable, easily accessible, low allergenic, distension resistance, and the capacity to interact with neighboring tissues are just a few of the benefits of collagen. As functional supports for soft tissue regeneration, collagen-based biomaterials have a wide range of clinical and research uses. Collagen also has a great extent of elasticity and tensile strength, and its mechanical qualities can be enhanced via molecular interactions (Fig. 12.2). Collagen's cell adhesion domain sequences can help cells contact more easily while also preserving their bioactivity and phenotypic (Hubbell, 1999).

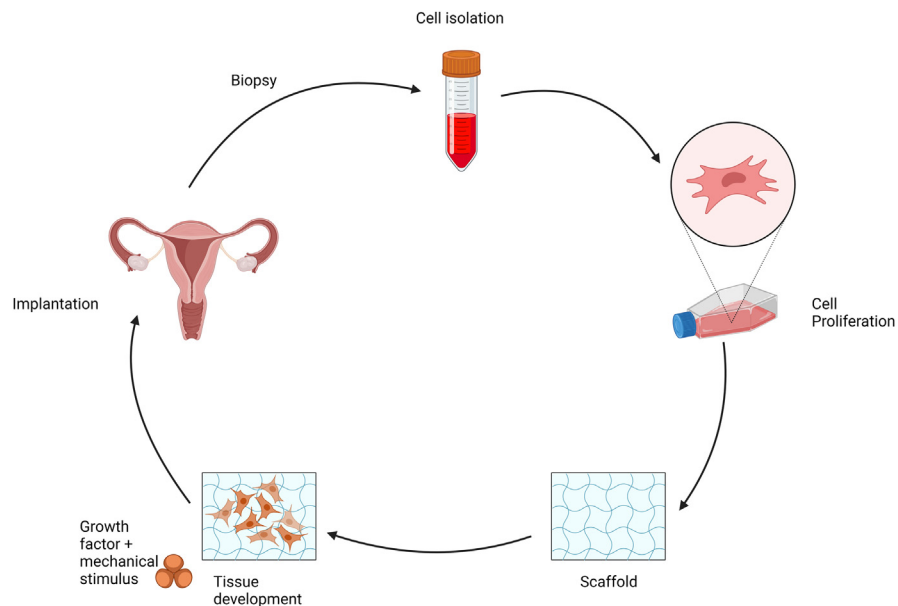


FIGURE 12.2

SEM micrograph of collagen fiber constructed in PBS (pH 7,4) (Paola et al., 2019).

12.6.2.2 Silk

Silk is a macromolecular protein polymer generated by spiders and silkworms, among other arthropods. On the silk fibroin, sericin, an adhesive-like assortment of glycoproteins with variable molecular weights, is deposited. The sericin-free silk fibroin was found to be exceedingly cytocompatibility both in vitro and in vivo., permitting cells to associate, move, multiply, and separate. Silk fibroin scaffolds include nets, electro spun fibers, absorbent sponges, films, atoms, and hydrogels are all examples of silk fibroin scaffolds. Elongation, breaking strength, and modulus are all mechanical properties of silk that contribute to its (Caneparo et al., 2021).

12.6.2.3 Alginate

Alginate is a normal polyose polymer made up of mannuronate found in two bacterial species and found in marine brown algae. *Pseudomonas* and *Pseudomonas aeruginosa* Alginate can be chemically and physically modified to tailor its features and performs to specific needs, including cellular compatibility, gelation characteristic, biodegradability, and mechanical strength. Alginate and its derivatives are used in biological and medicinal functions such as medication administration, injury treatment, TE, and cell encapsulation (Hamasaki et al., 2008).

12.6.2.4 Freeze-dried method

The polymer is liquefied in a flush and solid in fluid nitrogen though the ice evolution route is regulated in freeze-drying (Fig. 12.3). The solid solution is then

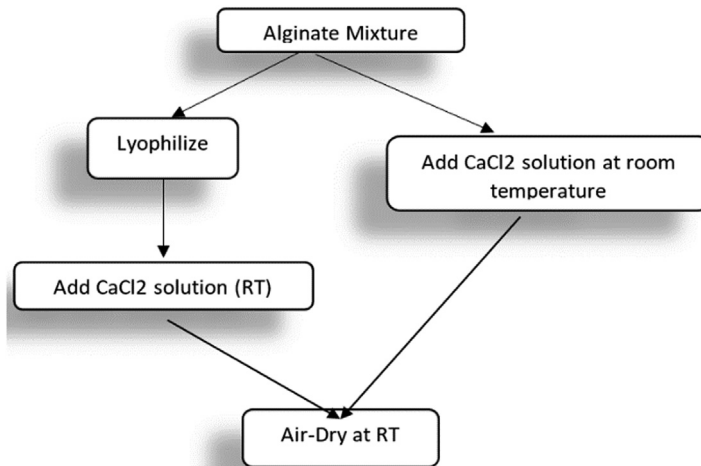
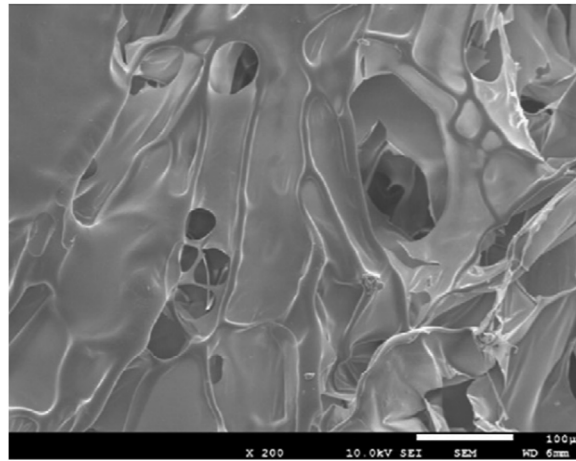


FIGURE 12.3

Illustrated freeze-dried method (Hamasaki et al., 2008).

**FIGURE 12.4**

A SEM image of the freeze-dried scaffold revealed a porous structure.

desiccated at low pressure, allowing the ice to sublime and producing macroscopically aligned pores in the freezing direction. For extra strength, the green bodies are sintered. It is possible to generate large, directed pores with porosities of more than 90%. A freeze-dried PGLA foam scaffold is seen in the SEM micrograph below (Fig. 12.4) (Hamasaki et al., 2008).

12.6.2.5 Hyaluronan

Hyaluronan (HA) could be a mucopolysaccharide made up of N-acetyl-D-glucosamine and D-glucuronic corrosive units that rehash. In an assortment of connective tissues, HA interacts with extracellular atoms to aid within the creation of the ECM. The elemental part of HA is to control ECM liquid, keep up tissue basic judgment, and associated with the viscoelastic properties of cartilage. The mucopolysaccharide HA is composed of rehashed N-acetyl-D-glucosamine and D-glucuronic acid disaccharide particles. In a number of connective tissues, HA interatomic with extracellular atoms to assist within the creation of the ECM. HA's key capacities are to control ECM liquid, protect tissue basic keenness, and connected with cartilage's viscoelastic characteristics (Aliborzi et al., 2016).

In a previous research, rats' abdomen subcutaneous area, vesicouterine space, and vagina were transplanted onto silk fibroin scaffolds. The vaginal scaffolds grew tissue at a slower rate than the pelvic and abdominal scaffolds, and there were more macrophages in the vaginal scaffolds (Zhang, 2016). An immunofluorescence experiment was used to evaluate the number of macrophages at the border of supports and host tissue, and the results showed that M2 cells were much more prevalent in all three groups (Zhang et al., 2017).

Silk fibroin scaffolds may encourage healthy tissue ingrowth rather than scar tissue ingrowth, according to macrophage categorization, that earlier was assumed to be the primary cause of vaginal contractions and pelvic pain in patients with mesh implants. [De Filippo et al. \(2003\)](#) ([Kim et al., 2019](#)) utilized vaginal epithelial and level muscle cells to assess the life expectancy of modified vaginal tissues in vivo in another examination. Female rabbit vaginal epithelial and smooth muscle cells were collected and planted at different densities on PGA frameworks. Athymic bare mice had the platforms embedded subcutaneously underneath their skin. The analysts found that cell-planted manufactured developments may create vascularized vaginal tissue in vivo that was phenotypically and practically identical to common vaginal tissue ([De Filippo et al., 2003](#)). In an ovine model, [Darzi et al. \(2021\)](#) investigated the lasting vaginal matter retort to PP lattice overlaid with noncellular matrix and PP network covered with solvable collagen. The findings revealed that both engage eventually created physical muscles that resembled normal materials via dissimilar tissue replies. M2 macrophage reactions were postponed within the acellular network coating gather, which may have contributed to the tall amounts of developed collagen. Within the collagen coating gather, there was an advancement moving forward healing reactions and biocompatibility by surging collagen maturing and vascularization interlocks on all sides ([Darzi et al., 2021](#)).

In 2014, the first human example of three-dimensional business vaginal another movement was printed. Autologous cells from four patients with congenital vaginal agenesis were seeded into commercial acellular intestinal submucosa segments. With the usage of a pattern produced individually for respectively feminine patient, the designed supports were similar to a standard vagina. The vaginotomy comes about six months after the operation uncovered that the boundary between the skin fabricating frameworks and the normal tissue was troublesome to distinguish ([Raya-Rivera et al., 2014](#)). The reconstituted tissues had physical qualities, protein expression profiles, and cellular architecture that were all identical to natural vaginal tissues. The legalized self-directed Female Sexual Purpose Key form revealed there were no difficulties following operation and that a regular series of functions was preserved during an 8-year follow-up. As a result of the discoveries, vaginal remaking locks in tissue-engineered frameworks and autologous seeding cells show up to be a reasonable choice ([Raya-Rivera et al., 2014](#)). Dissolving sutures made of bioresorbable polymers have been utilized effectively for numerous a long time. As a result, degradable polymers like poly(lactic corrosive), poly(glycolic corrosive), and poly(lactic acid-co-glycolic corrosive) (PLGA) are utilized in framework applications since they have cleared FDA necessities and can give a fast way to a commercial and clinical product ([Patel et al., 2011](#)).

Vagina from either intestine or skin tissue can now be used to produce artificial vaginas, although this can have catastrophic repercussions. The lack of lubrication provided by skin cell grafts causes pain during intercourse and can cause the vaginal membrane to shut. Intestinal cells release mucus on a regular basis, which is unclean and has an unpleasant odor.

12.7 Stem cells and vaginal reconstruction

Most known medications incorporate surgery, and the addition of synthetic/biological networks; all things considered, these strategies can cause fibrosis, destitute vascularization, and unite dismissal. The utilize of stem cells for vaginal repair has been understudied hence distant, with much of the ponder centering on muscle-derived stem cells (MDSCs). Grown-up stem cells, which shift from disciple cells in that they can create into an assortment of cell ancestries, have as of late been found in muscle as a substitute source of grown-up stem cells (Li et al., 2015). MDSCs show up to be promising helpful choices for musculoskeletal, cardiac, and urological conditions. In mouse models, MDSCs seeded on frameworks were illustrated to progress vaginal recovery by diminishing fibrosis and expanding epithelial tissue development. Moreover, MSC treatment after vaginal distension in rats can result in urinary self-control reclamation and an increment within the number of stem cells homing to the harmed range (Cruz et al., 2012). Because no human thinks about have been conducted, these medications must be considered test for the time being (Vassena et al., 2015).

The best lockups to usage are cells busy from a tissue surgery. These are removed from the swarm, separated from the matter surgery, and cultured. They have then transplanted into the same host again (2). While using autologous cells may create an early inflammatory reaction, it prevents rejection and so avoids the negative side effects of immunosuppressive medicines. As a result, most contemporary TE procedures rely on a model of autologous compartments from the host's damaged organ (Mikos et al., 2006).

12.7.1 Stem cells

Human embryonic stem cells (hESCs) have two exceptional belonging: the capability to self-repeat and the capability to recognize into nearly all the body's particular cell sorts. In vitro differences of human embryonic stem cells into cells from each of the three embryonic germ layers have been illustrated. The potential for hESCs to create teratomas in vivo, in any case, may be a limitation to their wide use in lockup recovery and tissue designing, because it is troublesome, in case not incomprehensible, to control their tumorigenicity in vivo, which raises an assortment of security concerns. Stem cells are created from social rising stem cells, lethal tissue, and grown-up stem cells (bone marrow, fat, skin (Mooney & Atala, 2006).

12.7.2 Amniotic fluid stem cells

This type of stem cell is another source of stem cells. Numerous incompletely discriminated cell types arising since the growing fetus are known to exist in

amniotic fluid. Amniotic fluid stem cells (AFSCs) were extracted from various sources, which direct emergent and adult stem cell markers (Borlongan, 2017). Without a feeder layer, the undifferentiated stem cells expand rapidly, and the people doubles every 36 h. Extensive telomeres and a normal karyotype were found in lines that had been maintained for more than 250 passage. AFS cells have a wide range of multipotency, and endodermal differentiation has been formed (Cananzi & De Coppi, 2012). In vivo, AFSCs do not form tumors, which is beneficial in scientific requests. The cells could be acquired from the developing fetus via amniocentesis or chorionic villous sampling, or from the placenta upon birth. The cells might be saved for future use, used without fear of rejection, or banked (Ilancheran et al., 2009).

12.7.3 Bone marrow-derived stem cells

BMDSCs (bone marrow-derived stem cells) have been illustrated to assist with tissue repair and recovery in an assortment of organs and tissues (Pittenger et al., 1999) BMDSCs are a source of nonhematopoietic cells all through a few endometrial cellular compartments (stroma, glandular epithelium, and luminal epithelium) in people. They essentially contribute to the creation of endometrial stromal compartment cells, as well as the glandular and luminal epithelium to a lesser sum (Vassena et al., 2015).

12.7.4 Adult stem cell and native progenitor

In spite of the reality that around organs, such as the vagina, have a solid recovering capacity in vivo, cell advancement and advancement in vitro may well be challenging. It may be conceivable to overawed the impediments that constrain cell extension in vitro by analyzing the areas of committed antecedent cells in certain organs. Urothelial (cells that line the urinary tract) cells, for case, might already be taught within the workshop, but as it were to a restricted degree. Over the final two decades, a few techniques have been established to overcome this issue (Table 12.1). A few clinical trials have utilized urothelial cells. Nonetheless, hESCs' proclivity for shaping teratomas in vitro could be a disadvantage in cell restoration and tissue building. Besides, controlling the tumorigenicity of these cells in vivo is challenging, on the off chance that not inconceivable. Also, new stem cells holder strategy provided suitable shapes, which are cell conglomerations that contain all three embryonic germ layers whereas in culture, and can create teratomas in vivo, illustrating their pluripotency (Kharazi & Badalzadeh, 2020).

Table 12.1 The major studies focused on the possibility of cell cocultured in scaffolds to treat vaginal defects in vivo investigations.

Study model	Regeneration strategy	Cells seeded on scaffolds	Benefits
Mice	Engineering vaginal tissues	Vaginal replacement and smooth muscle cells + PGA	Neovascularization
Rabbit	Vaginal replacement	Vaginal epithelial cells and smooth muscle cells + PGA	Neovascularization and appropriate physiological responses
Human	Tissue-engineered autologous vaginal organs in MRKH syndrome + SIS	Epithelial and smooth muscle cells + SIS	Organized vaginal histology
Human	MRKH syndrome	No cells; acellular dermal matrix	No complications; anatomic success 100% and normal sexual function
Human	Canal lining in patients with MRKH syndrome	Mucosal vaginal cells; no scaffold	Normal and satisfying sexual intercourse

12.8 Conclusions

Congenital defects, trauma, infection, inflammation, and cancer are all cases of pathologic disorders that can disturb normal vagina development and shape. Due to a lack of adequate native tissue, vaginal regeneration has been challenging in the past, but using created vaginal tissue in vivo will soon be viable. Because of these or other difficulties, it commonly requires reconstruction or replacement. One method of organ replacement is transplantation. A serious organ shortage already exists and this is projected to worsen as the population eternities. TE integrates cell replacement, constituents' science, and technology ideas for the generation of biological alternatives are to tolerate regular tissue role. And report investigates contemporary developments in stem cell research and regenerative medicine, as well as clinical uses of these technologies as innovative therapeutics for tissue and organ loss.

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The progress in tissue engineering of cervix

13

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13.1 Introduction

Cervical abnormalities and diseases are a major concern for the population, health, and well-being, as well as society in general, with cervical cancer at the top of the list. Cervical cancer will be the fourth most common cancer in women worldwide in 2022, according to the World Health Organization, with 604,000 new cases and 324,000 deaths in 2020, accounting for approximately 90% of new cases and deaths worldwide, attributed to low- to middle-income countries (World Health Organisation, 2022). Every year, approximately 3200 new cases of cervical cancer are diagnosed in the United Kingdom (2016–18) (Cancer Research UK, 2020). An estimated 14,100 cases of invasive cervical cancer will be diagnosed in the United States, with 690 deaths (Society, 2022).

The process of developing functional replacement tissue for the cervix, a component of the reproductive system of females that is situated at the lower end of the uterus, is known as cervical tissue engineering (TE). This field has the potential to improve treatments for conditions like cervical cancer, which frequently necessitates cervix removal, resulting in complications like infertility, incontinence, and decreased sexual function. TE techniques have produced promising results for the repair and replacement of degenerated or defective cervixes (Mabrouk et al., 2020). An effective TE (TE) process should be made up of a combination of scaffolds, growth factors (GFs), and cells. They must also be capable of replacing the damaged tissue and performing as or imitating the native tissue. The use of GFs and exogenous materials with the main purpose of speeding up and improving the body's healing processes may improve tissue conditions. Materials whose properties mimic those of the extracellular matrix (ECM) have been used for a long time and provide additional benefits in addition to providing physical structure. Biomimetic materials can enhance healing processes in all individuals as well as transport biomolecules such as GF, which promote the growth of cells. Initially, scaffolds were

thought to be only essential for the physical support of cells; however, they can now be packed with biological factors to aid the regeneration and rehabilitation of tissues (Mabrouk et al., 2020).

13.1.1 The cervix, its anatomy and physiology

The menstrual bloodshed each month must pass from the uterus and through the cervix before leaving the vagina. Sperm must pass through the cervix to reach the uterus and fallopian tubes in order to fertilize an egg. The mucus secreted by the cervix influences a woman's ability to become pregnant. This mucus is thinner and less acidic than usual around ovulation, which makes it easier for sperm to pass through to the uterus. As a result, sperm can easily reach the egg and fertilize it. During pregnancy, the cervix produces a mucus plug, which seals the entrance to a woman's uterus. This mucus plug dissolves during labor as well as the cervix softening and thinning. The cervix controls when the baby exits the uterus by dilating. Finally, objects inserted into the vagina, such as tampons or diaphragms, are prevented from slipping inside the uterus by the cervix (Cleveland Clinic., 2022).

The cervix, being a significant component of the female reproductive system, is situated in the pelvic cavity. The cervix is a tubelike structure that links the vagina and uterus. The internal cervical OS is the cervix's opening to the uterus, while the opening of the cervix into the vagina is referred to as the external OS (Koullali & House, 2019). Behind the bladder and urethra and in front of the rectum and anus are the cervix, uterus, and vagina. The transformation zone (TZ) is the area of the cervix where the endocervical canal meets the ectocervix. The TZ is the part of the cervix that shows abnormal cells as well as cervical conditions (Cleveland Clinic., 2022). The stroma, or cervix body, is made up of fibrous connective tissues rich in collagen and a minimal portion of smooth muscle (Koullali & House, 2019).

13.1.2 Disorders/diseases related to the cervix

Cervical diseases and abnormalities have a variety of causes. However, the rise in female cervical diseases may be due to pathological congenital or environmental factors, a few of which are seen in Table 13.1.

13.1.3 Risk factors of cervical conditions

Besides the pathological and congenital abnormalities that contribute to cervical conditions, there are certain factors that may increase the risk of cervical disorders and diseases. They include:

- Allergies
- Sexual history
- Virus
- Irritation
- Lifestyle (Herndon, 2022; Cancer Research UK, 2020)

Table 13.1 Brief list of disorders/diseases associated with the cervix.

Disease	Causes	Diagnosis	Treatment	References
Cervical cancer	Human Papillomavirus HPV (HPV-16, HPV-18)	Pap smear	Chemotherapy, Radical hysterectomy	Ghim et al. (2002a, 2022b)
Cervicitis	<i>Chlamydia trachomatis</i> , <i>Neisseria gonorrhoeae</i> , <i>Candida</i> spp., <i>Trichomonas vaginalis</i> , <i>Mycoplasma genitalium</i> , Herpes simplex virus, <i>Streptococcus</i> spp.	Molecular techniques and diagnostic tests	Antibiotics (Azithromycin, Moxifloxacin)	Ortiz-de la Tabla and Gutiérrez (2019)
Cervical dysplasia	HPV	Pap smear	Cryosurgery, laser therapy, loop Electrosurgical Excision procedure (LEEP), cone biopsy.	Herndon (2022)
Pelvic inflammation disease (PID)	<i>Neisseria gonorrhoeae</i> , <i>Chlamydia</i>	Molecular tests	Antibiotics (Doxycycline, metronidazole, azithromycin, Amoxycycline, ceftriaxone).	Dayan and Ooi (2003)
Cervical insufficiency	Cervical abnormalities like shorter length, abnormal shape of cervix or uterus	Physical examination, determination of pregnancy outcomes, digital examination of the cervix	Cerclage	Hassan et al. (2021)

13.1.3.1 Current treatment for cervical conditions

A variety of disorders can affect the cervix that are either congenital or acquired. Furthermore, acquired disorders like cancer could result in cervicovaginal organ loss or damage. Cervical abnormalities are very hard to treat and would normally entail a surgical procedure, with an extensive reconstruction needed to restore the

normal physiology of the internal reproductive organs. Hysterectomy was the way out of these disorders in the 1990s, resulting in the loss of all reproductive parts for patients. These procedures transitioned toward more conservative approaches, such as canalization and using autologous or allogeneic grafts for tissue regeneration. A range of autologous grafts from nonvaginal sources have now been tested for cervical reconstruction. The bowels, which have the additional advantage of being vascularized tissue, are one of the most popular approaches. Regrettably, this method causes hygienic issues. Moreover, the amount of tissue collected for transplantation may be a constraint in this method. Patients have also been tested with allogeneic tissues such as the amniotic membrane (Kajbafzadeh, 2021). However, this also suffers a drawback in terms of sterility and the possible transmission of diseases.

Various attempts have been made since then to treat, recover, and revive the cervix. Brief descriptions of some methods that have been used are mentioned below:

13.1.3.2 Surgery

A surgical procedure is the preferred method of treating cervical disorders caused by accidents, pathogens, malformations, or congenital defects. Surgical procedures are a type of treatment used to restore or remove dysfunctional reproductive tissues or organs (Fu, 2021). The procedure used is solely determined by the condition. Sometimes, these procedures may cause secondary injuries, resulting in a different medical condition for the patient.

13.1.3.3 Cervical cerclage

Cervical cerclage insertion has been demonstrated to decrease the likelihood of miscarriage or preterm birth in women at risk (Bonney et al., 2011). Cervical cerclage is a technique used for closing the cervix during gestation in order to avoid premature birth due to an incompetent cervix (Cleveland Clinic, 2022). This is done to maintain the fetus inside the uterus until delivery. One or more stitches are placed in the opening of the cervix during this procedure to keep it safely closed during pregnancy.

13.1.3.4 Loop electrosurgical excision procedure (LEEP)

This technique is used to remove abnormal cervix cells. To achieve this, the doctor utilizes a small wire loop that has been charged with an electrical current, enabling it to be used as a surgical knife. A speculum is inserted into the vagina during the procedure to spread the walls of the vaginal canal and give a better view of the cervix. A colposcope may also be used to enlarge the cervix tissue. The doctor will then use a vinegar solution to clean the cervix. This solution will turn any abnormal tissue white, making it easier to see. They could substitute iodine for vinegar. Normal cervical tissue will be stained brown by iodine, allowing abnormal cells to be identified (Fessenden, 2019).

13.1.3.5 Total hysterectomy

The womb and cervix are removed during a total hysterectomy; this is the most performed operation (NHS, 2017).

13.1.4 Medication

Treatments that can kill the pathogens that induce cervix infections. For severe infections, antibiotics may be administered orally, intravenously, or through a vein. Cryotherapy involves placing an extremely cold probe against abnormal areas of the cervix. The abnormal cells are killed by freezing, preventing cervical cancer. A high-energy laser is used in laser therapy to burn areas of abnormal cells indicated in the cervix. These abnormal cells are killed, stopping cervical cancer from developing. Chemotherapy is usually prescribed for cervical cancer that has advanced. They are typically administered via injection into a vein (Pathak, 2020).

13.1.5 Organ tissue transplantation

Congenital and pathogenic cervix problems usually lead to problems with fertility and reproduction (Babcock, 1923). Organ tissue transplantation is a technique used to replace reproductive tissues in patients undergoing therapies that are high-risk. Patients receiving cancer treatments such as pelvic radiotherapy or chemotherapy are usually required to carefully transplant nonaffected autologous organs/tissues away from the area exposed to radiation during treatment. Thrombosis, antenatal complications, infection, and graft failure have all been observed in recipients, even though graft rejection is prevalent but rarely results in graft removal (Richards et al., 2021).

13.1.5.1 Tissue engineering approaches for the cervix

Scaffolds, which are three-dimensional structures that support cell growth and organization, are one approach to cervical TE. These scaffolds can be made of a variety of materials, such as synthetic polymers, natural polymers, and ceramics (Table 13.2).

Table 13.2 Scaffolds used in tissue engineering.

Material	Scaffold	Benefits	Drawbacks	References
Natural	Alginate	Biodegradable and biocompatible encourages cell growth	There is no hydration property. It deteriorates rapidly. Mechanical support is inadequate.	Petruzzo et al. (1997)
	Chitosan	It is bio-renewable, nontoxic, bio-adhesive, and antigenic.	Organic solvent solubility is sparse. Mechanical properties are poor. Biostability is low.	Fu et al. (2018)
	Collagen hydrogel	Has the most widely available ECM and a porous structure for diffusion.	Decoupling protein concentration is challenging. Inadequate biochemical signals to generate cellular reactions.	Ellis (1979)

Continued

Table 13.2 Scaffolds used in tissue engineering.—*cont'd*

Material	Scaffold	Benefits	Drawbacks	References
Synthetic	Fibrin	Strongly biocompatible and suited for improving cellular interactions.	In vivo, the degradation process is rapid.	Lipinski and Gurewich (1976)
	Matrigel	Biocompatible and used in replicating specific ECM.	Lack mechanical integrity. Low batch-to-batch availability.	Ellis (1979)
	Silk	Biocompatible, biodegradable water-based processing.	Low quality. Fast degradation. Low level of revascularization.	Numata et al. (2012)
	PCL	Biocompatible with a slow degradation rate. No additional surgery is required.	Antimicrobial properties are lacking. Mechanical property mismatches are possible.	Franco et al. (2014)
	PEG	Appropriate for cell attachment. Cell migration and growth are stimulated.		Franco et al. (2014)
	PLA	Excellent in vivo biocompatibility, tissue development with a low rate of degradation.	Tensile strength is lost after absorption. Possibly toxic oxidative substances are released during breakdown.	Franco et al. (2014)
	PLGA	Excellent biocompatibility. Suitable for drug administration. Low rate of degradation.	Loss of mechanical stability.	Franco et al. (2014)

13.1.6 Scaffold based approach for tissue engineering

Scaffold-based TE methods entail the use of biocompatible, three-dimensional scaffolds that give structural support for cells as they grow and differentiate into functional tissues. These scaffolds can be made from a variety of materials, such as synthetic polymers, natural polymers, and bio-ceramics, and can be fabricated using a variety of techniques, such as 3D printing, electrospinning, and solvent casting.

The use of chitosan-based scaffolds for bone TE is one example of a scaffold-based TE approach. Chitosan is a natural polymer derived from chitin, which is

found in crustacean shells. Chitosan has been shown *in vitro* and *in vivo* to promote osteogenic differentiation and bone formation (Levengood & Zhang, 2014). The use of electrospun fibers for the regeneration of nerves is another example of a scaffold-based TE approach. Electrospun fibers are ultrafine fibers made from a variety of materials, such as polyethylene terephthalate and polycaprolactone. These fibers have been shown to promote nerve cell growth and have been used to develop nerve guides for the repair of peripheral nerve injuries (Xie et al., 2010). There is also the use of hydrogel scaffolds in the engineering of cartilage tissue. Hydrogels are cross-linked hydrophilic polymers that can retain large amounts of water, making them ideal for use in TE applications.

The ability to control the microenvironment in which cells are cultured, which can influence their behavior and differentiation into specific cell types, is one of the key advantages of scaffold-based approaches. Scaffolds can also be modified with different chemical and physical cues to improve cell attachment and proliferation.

13.1.7 Fabrication techniques

When fabricating scaffolds, it is critical to consider the ECM of the tissue/organ under investigation. ECM is biocompatible, capable of rapid *in-vivo* modeling to assume tissue/organ shape, and biodegradable due to its proteinoid architecture (Lu et al., 2011). As a result, researchers are turning to the development of scaffolds that can demonstrate these qualities *in vivo* and *in vitro*. Recently, various scaffold fabrication methods have been considered in recent years in order to mimic their hosts' microenvironment by simulating cell-specific responses to restore or regenerate desired tissues/organs. Some of these methods include:

Electrospinning is a technique that involves using an electric field to create ultrafine fibers from a polymer solution or melt. These fibers can be used to make 3D scaffolds with a large surface area and pore size that promote cell attachment and proliferation (Doshi & Reneker, 1995).

3D printing, also known as additive manufacturing, is the process of creating a 3D object by depositing successive layers of material using a 3D printer. This method can be used to create scaffolds with precise control over their shape, size, and porosity (Liu & Yan, 2018, p. 1789239664).

Others include phase separation, freeze-drying, material injection gas foaming, self-assembly, solvent casting, etc.

13.1.8 Scaffold selection for cervical tissue engineering

The cervix is mostly made up of fibrous connecting tissues that conform to physical changes in pregnancy and delivery. A variety of scaffold materials and designs have been proposed and evaluated in cervical TE. Some examples are listed below:

Polymeric scaffolds: For cervix TE, polymeric scaffolds made of biodegradable polymers like polylactic acid and polyglycolic acid, as well as their copolymers, have been used (Dhandayuthapani et al., 2011). These scaffolds can be made in a

variety of shapes and sizes, as well as modified with a variety of functional groups to improve biocompatibility and cell attachment (Luo et al., 2014).

Hydrogel scaffolds: Being hydrophilic polymeric networks that can retain large amounts of water, hydrogels have also been investigated for cervix TE. These scaffolds can mimic the mechanical and swelling properties of native tissue and can be functionalized with a variety of biomolecules to promote cell adhesion and proliferation (El-Sherbiny & Yacoub, 2013).

Natural-material scaffolds: Natural materials such as collagen, fibrin, and chitosan have been used as scaffolds for cervix TE (Luo et al., 2014). These materials are biocompatible and biodegradable, and they can provide an appropriate microenvironment for cell attachment and proliferation. Silk is also used as a scaffold biomaterial because techniques for controlling the morphological and functional properties of silk protein scaffolds, such as support for cell and tissue growth (House et al., 2010), are well understood in research laboratories, and the mechanical properties of scaffolds can be altered to mimic native tissue by controlling pore size (Zhang et al., 2022).

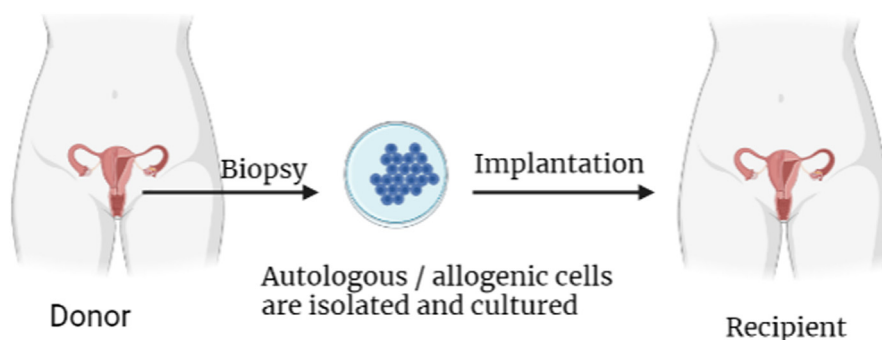
13.1.9 Cell/stem cell-based tissue engineering approach

TE is a multidisciplinary field in which cells and/or stem cells, as well as scaffolds and bioactive molecules, are used to create, regenerate, or repair damaged or diseased tissues in the body (Atala et al., 2018). Cell/stem cell TE can employ a variety of approaches, including the following:

Autologous cell transplant: Cells are isolated from the patient's own body and grown in culture before being reintroduced into the patient (Olson et al., 2011). Because the transplanted cells are derived from the same individual, this method is less immunogenic (Atala et al., 2018). Primary cells can be used to make cell-seeded scaffolds, which are structures that help cells grow and differentiate. After that, the scaffolds can be implanted into the cervix to promote tissue regeneration. However, this approach may be constrained by cell availability and the efficiency of cell expansion in culture.

Allogeneic cell transplantation: Cells are obtained from a genetically similar donor and transplanted into the patient (Fig. 13.1) in this method (Atala et al., 2018). This method may be more widely available because it does not necessitate the isolation and expansion of cells from the patient's own body (Olson et al., 2011). It may, however, be associated with a higher risk of immune rejection.

Induced pluripotent stem cells (iPSCs): Adult cells that have been conditioned to a pluripotent state are known as iPSCs (Latorre-Romero et al., 2011), which means they can separate to form specialized cells in the body (Atala, 2011; Takahashi & Yamanaka, 2006). Since this allows for the generation of an infinite supply of cells that are genetically identical to the patient, this approach has the potential to overcome the limitations of autologous and allogeneic cell transplantation (Atala et al., 2018). For the cervix, human iPSCs have been differentiated into cervical tissue-specific cell types such as cervical epithelial cells and smooth muscle cells. However, the iPSC generation efficiency of iPSC generation and differentiation into the desired cell type can be variable (Takahashi & Yamanaka, 2006).

**FIGURE 13.1**

Schematic showing biopsy, isolation, culture, and implantation of autologous/allogenic cells or stem cells from donor to recipient.

13.1.10 Possible growth factor used in tissue engineering of cervix

GFs are signaling proteins that are essential for cell growth, proliferation, and differentiation (Koria, 2012). GFs are frequently used in TE to stimulate tissue growth and repair.

TE has made use of a variety of GFs, including:

Epidermal growth factor, which stimulates cell proliferation and differentiation in the skin and other tissues, has been shown to stimulate the growth of cervical cells in culture, suggesting that it could be useful in cervix TE. Fibroblast growth factor, which is responsible for stimulating fibroblast and other cell growth. Insulin-like growth factor is a protein that promotes cell growth and survival in a variety of tissues. Platelet-derived growth factor stimulates cell proliferation and differentiation in blood vessels and connective tissue, while vascular endothelial growth factor promotes blood vessel growth (Atienza-Roca et al., 2018a, 2018b, pp. 245–269). It has been shown to stimulate blood vessel growth in the cervix and may have applications in cervix TE. GFs can be delivered to the tissue of interest via a variety of methods, including growth factor-releasing scaffolds and gene therapy.

13.2 Conclusion

Patients with various acquired or congenital cervix disorders as well as reproductive organ disorders are currently being treated using surgical, hormonal, and other invasive therapies. These procedures, notwithstanding, are not optimal. Several existing hormonal therapies, for instance, do not deliver physiologic levels of sex hormones, which can result in several problems. In addition, many surgeries, such as grafting, can fail for a variety of reasons. This has led scientists and researchers in the fields of regenerative medicine and TE to utilize cell transplantation, material science, and bioengineering principles in developing biomaterials that may inevitably enhance the life quality of these patients in several ways, ranging from delivering physiologic

hormone secretion to potentially 1 day helping to restore fertility with tissue-engineered reproductive organs (Atala, 2012).

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The progress in tissue engineering of ovarian glands

14

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14.1 Introduction

In the present time, the world is facing number of chronic diseases such as diabetes, cancer, and heart diseases. These diseases are also the major driver of health expenses around the world. Among these chronic diseases, ovarian disease is emerging as a major threat to human health. The field of biomedical engineering in collaborations with other areas of biotechnology has been battling hard to minimize the numbers that are suffering ovarian linked disorders.

This report highlights the conditions related to irregularities of ovarian gland disorders and as well as ovarian germ cell tumors. According to the statistics, ovarian epithelial cancer is the seventh most common disease among the cancers globally and eighth most common causes of deaths. Just in year 2020, 314,000 individuals were diagnosed and 207,000 died with ovarian cancer (WOCC, 2020). Some ovarian disorders are not deadly, but it has a great impact on the quality of life of an individual. According to recent published data, primary ovarian insufficiency (POI) or premature ovarian failure (POF), and polycystic ovary syndrome (PCOS) is affecting 1% of women and is caused by the hormonal imbalance (Zhang et al., 2021). The POF is one of the common endocrine diseases that leads to infertility before the age of 40 and it has been estimated that 0.1% of females are suffering from this problem are under the 20 age group and it increases up to 1.5% with age (Pennarossa et al., 2021). Likewise, PCOS is also the major problem faced by women in the reproductive age and it is affecting almost 4%–20% of females globally (Deswal et al., 2020).

Number of scientific studies have been conducted around the globe for optimal cure or treatment with the help of new techniques in ovarian related disorders. Tissue

engineering and regenerative medicine are among those techniques that can play a vital in the discovery of better treatment. The finding suggests that POI can be cured with help of stem cell therapy techniques by using stem cells and biomaterials (Zhang et al., 2021).

The tissue engineering approach uses a scaffold, combination of cells, biomaterials like extracellular matrix (ECM), collagen, alginate, synthetic polymers such as polytetrafluorethylene (PTFE) that influences the human body to make or treat the organ-like ovaries, but it needs to be biodegradable. Tissue engineering has achieved promising results in the animal model; however, experimental results show that making bioscaffold with the porcine ovary and dermal fibroblast of human beings, and it has been observed that the scaffolds that are made by decellularized ECM make suitable function ex vivo culture of ovarian cells (Pennarossa et al., 2021).

The tissue engineering of female reproductive organs is a challenging task. Over the past years, there were many strategies established to preserve the fertility of women who are suffering from the diseases like ovarian cancer. The development in the field of tissue engineering and regenerative medicine offers some hope that there would be a good result in the treatment of the ovarian gland with the help of 3D scaffolding, biodegradable materials and vascularization, and molecular and cellular spatial distribution.

This chapter highlights the ovarian gland function and structure, anatomy and physiology, ovaries-related disorders, and future research directions in the development of better treatment and discusses various studies conducted for better clinical outcome, and replace or implant of the ovaries with the help of tissue engineering strategy.

14.2 Anatomy and physiology of ovaries

The ovarian gland is a ductless reproductive gland in women that help in producing reproductive cells. The female has a pair of ovaries that have an oval shape structure, and it is located on the opposite end of the pelvic wall on both sides of the uterus. Both the ovaries are attached to the tissue called fimbria, and that tissue joins the ovaries to the fallopian tube (Sargis, 2015). There are four layers in the ovaries from the outer section to the central section, the collagenous connective tissue, large follicles, blood vessels and the medulla containing loose connective tissue, and the cortex containing preantral follicles (Gibson & Mahdy, 2019).

The ovary is one of the organized compounds of the germ cells that are also called eggs (oocytes) and some somatic cells like granulosa cells, thecal cells, and stromal cells their work to direct or give the instruction for the making of the follicles that contain oocytes, and the formation of both somatic cells and oocytes as follicles and corpus luteum, its form after the ovulation from the follicles, and its works to maintain pregnancy (Richards & Pangas, 2010). Follicles are the essential part of the ovaries that contain oocytes encircle by granulosa cells, and there is a separation between granulosa cells and stromal cells by basal lamina (Irving-Rodgers et al., 2009). The bidirectional signaling between oocytes and granulosa cells, and between

granulosa cells and theca cells can affect the development of the follicle and it can also be affected by the interaction of these cells with ECE and GFs (growth factors) (Oktem & Oktay, 2008). Folliculogenesis is the process that involves the follicle growth from the primordial stage to ovulation process or death of follicle by atresia (Galan, 2019; Orsi et al., 2014), and one of the requirements of this process is the interaction of oocyte and somatic cells that allows the external signals and maintain the oocytes during the process of maturation (Belli et al., 2013).

Ovaries produce the hormones estrogen and progesterone and some smaller hormones like relaxin and inhibin. Estrogens are used by the human body to develop the young female sex characteristics and help in the reproduction cycle (Bradford, 2017).

14.3 Diseases related to ovaries

Some serious problems are affecting the ovarian glands and some of them are very dangerous and chronic like ovarian cancer and some other diseases that are not fatal, but they need treatment for the problem like PCOS, POI, and POF, because they are affecting the lives of millions of women around the world. These problems are discussed below.

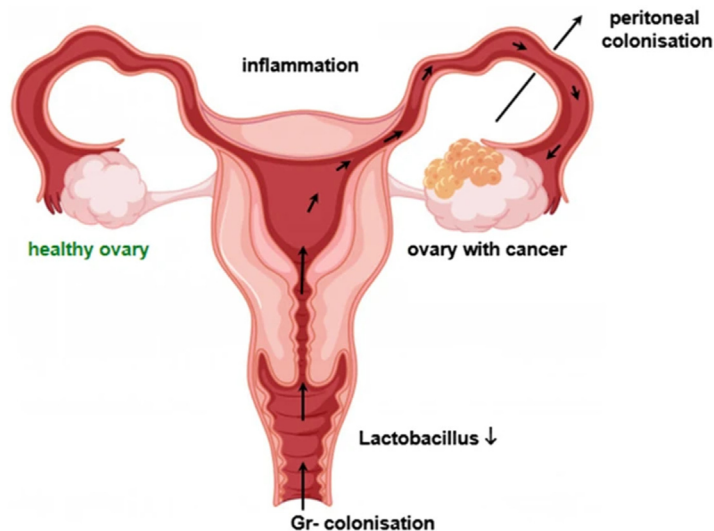
14.3.1 Ovarian cancer

Ovarian cancer is an abnormal growth of cells that forms the structure within the ovarian gland. These cells increase their number so quickly and can spread the other attached tissues (Clinic, 2018). Among all gynecological cancers, ovarian cancer is counted as the most dangerous one, and according to the reports of the patients that are diagnosed with ovarian cancer, 58% of them are in stages 3 and 4 that are the advanced stage of cancer (Fig. 14.1). It has been observed that people who are in the third stage, their survival percentage is 27% and those who are in the fourth stage, their survival is 13% that is very poor and the highest rate of survival that is 90% in those women who diagnosed or detected in the first stage of cancer. The reproductive system of women contains a pair of ovaries that produce and release different hormones by the interaction of different cells in ovaries and initially, the growth of cancer cells starts in the covering of cells that is also known as epithelial ovarian cancers and other unique types of the abnormal growth of cells are germ cell tumors, stromal tumors, and sarcomas (Cancer Research UK, 2018).

Ovarian cancer is commonly treated by surgeries, operations, and chemotherapy, but now a days, many studies are going on to cure this disease with the help of tissue engineering and regenerative medicine.

14.3.2 Polycystic ovary syndrome

PCOS is a common disorder related to ovarian glands that mostly happened during the reproductive age of females, and it can affect the cause the irregularity of the

**FIGURE 14.1**

Ovary infected by cancer (Sipos et al., 2021).

menstrual periods or prolonged periods or increase hair growth, skin issues like acne, and obesity. It may also increase the androgen level in women that is a male hormone. The ovaries can also grow various small cyst (fluid) follicles, and it can cause the problem in releasing the eggs timely (Clinic, 2017). It is also called Stein–Leventhal syndrome. It has been recorded PCOS is among those diseases that are very common in females, and it happened 8%–13% because of mental health and metabolic issues in the body. It has been recorded in one of the studies which collected the data of one region of Leicester, Leicestershire, and Rutland and they found out that for every 100 women who are affected by syndrome, 85 of them were not recorded in the health system (Mani et al., 2019). Still, the main reason that caused PCOS syndrome is unknown but if it is detected at the early stage and the patient takes proper treatment and weight control so it can reduce the long-term problems like diabetes type 2 and cardiac disease (Clinic, 2017). This disorder is the popular reason for infertility in women because PCOS prevents the process of ovulation and those patients are suffering from this syndrome has a higher number of miscarriage cases and premature delivery (Blog, 2020). Numerous studies are going on to find a suitable cure for this disease with the help of tissue engineering and regenerative medicine.

14.3.3 Primary ovarian insufficiency

POI previously known as POF is a disorder occurred before the age of 40 years, it can be caused by cancer therapy and this disorder is one of the most common causes of infertility in women; it has been estimated that 0.1% of females are suffering from

this problem is under the 20 age group and it increases up to 1%–1.5% as the age increases (Pennarossa et al., 2021). The ovaries stop working because of this problem long before the expected time (Ghahremani-Nasab et al., 2020). In this disease, the ovaries are not producing or releasing the usual amount of important hormones such as estrogens (Clinic, 2020). The reasons for POI might be radiotherapy, metabolic and autoimmune disorders. It has been observed that the patients who are suffering from POI have estrogens deficiency, and the level of gonadotropin and amenorrhea is high (Fu et al., 2021). POI can also harm the other organs or tissues of the body, and it has an impact on psychological problems, for example, cardiac disease or heart disease, depression, and osteoporosis (Pennarossa et al., 2021). POI is commonly treated by hormone therapy, in vitro activation, but now a days, many studies are going on to cure this disease with the help of tissue engineering and regenerative medicine. The recent research aims to restore the fertility of women.

14.4 Tissue engineering of ovaries

Tissue engineering is one of the emerging multidisciplinary fields in the health sciences. This engineering involves the treatment of the organ suffering from serious problems or stops working due to disease, with the help of scaffold (made with bio-materials), stem cells, and biologically active molecules like GFs for the regeneration of the organ or specific tissues. This process is carried out in the laboratory environment to prevent any kind of bacterial interference in the process of developing the cells and scaffolds. The scaffold has an effective shape for the migration, proliferation, and cell adhesion for making and changing the new tissue with the affected part or organ (Mahla, 2016).

Tissue engineering is solving and trying to solve major problems related to health sciences. There are numerous research going on in the field of regenerative medicine and tissue engineering to cure different organs like the heart, eyes, skin, breast, and ligaments, likewise, studies are going on treatment ovaries through this multidisciplinary technique that use cellular biology, stem cells biology, natural biomaterials, synthetic biomaterials to develop new therapy to reactive or replace the affected tissues or organs with the prosthetic one. By using this strategy, the ECM and matrisome that is the related factor transmit the biochemical signal in the shape of cellular adhesion within the related factor (matrisome) or paracrine and endocrine factors' orientation and availability has been controlled. It has been observed that cell morphology and the functions of cells like islets and stem cells have a good influence on it (Orsi et al., 2014; Tran et al., 2020). With the help of correlation between cellular morphology and their function, proliferation of ovarian cells has been developed (Da Silva-Buttkus et al., 2008). Through tissue engineering, researchers are trying to answer the problems related to ovaries, for example, the POI that is a common cause of infertility, the study shows some methods to cure POI with eggs donation, stem cell therapy, and tissue engineering (Burgos et al., 2017).

As is mentioned before that POF or POI can be cured by stem cells using tissue engineering approaches. The stem cells for the process can be taken from various sources though it is not easy to find different types of stem cells. It has been observed in an animal model for the treatment of POF, the stem cells transplantation helped in the regeneration of oocytes and restored the function of ovaries (Chiti et al., 2017). Tissue engineering and regenerative medicine used reproductive technology for the treatment of infertility caused by POF and it also counters limitations like the immune system and donor supply problems for the tissues transplant (Volkova et al., 2017). The stem cell-based treatment for POI and especially mesenchymal stem cells (MSCs) is very common and useful in the treatment of this disease (Sheikhan-sari et al., 2018). The MSC's ability to maintain and restore the work of the particular organ has been observed in the number of biological research, by all these studies the communication pathways, genetic modulation that is essential for the cell functioning, and paracrine messaging proteins that shape the mechanism of actions for cells. It has been recorded that the MSC treats the POI for both cellular and animal models. Furthermore, there are some tissue engineering strategies used to develop patches with the help of biomaterials like ECM, collagen, alginate to use for ovaries that can provide enough support and continued cellular delivery while the restoration of the function of ovarian glands and enhancing follicle generation or development (Huhtaniemi et al., 2018). Additionally, the signals that generate physically are essential and the studies tell that the external pressure applied physically keeps primitive follicle dormancy, to active the ovarian follicle the effect must reverse that disrupts the ECM (Hayashi et al., 2020). 3D printing is also used as a tissue engineering technique to cure ovarian problems alongside ovary-on-a-chip technology to multiply the normal tissues, which are beneficial in the making of the prosthetic ovary with help of biomaterial properties and ovarian tissue engineering (Dadashzadeh et al., 2021).

14.5 Factors involved in tissue engineering

The factors that involve in the process of tissue engineering and regenerative medicine are discussed in detail below:

14.5.1 Scaffolding

Tissue engineering has been used as an effective technique to repair damaged organs or it can be replaced by prosthetic ones. This engineering technique focuses on the regeneration of the organ that stops working because of disease or trauma, and it involves combining the living cells from the human body with the porous scaffold that can be made up with natural biomaterials or sometimes synthetic biomaterials that can act as a base ground for the regeneration and growth of particular tissue. Scaffolding is one of the main steps in the tissue engineering of ovaries and it resolves the limitations faced by the researchers in 2D cell culture. These scaffolds are highly

porous and best for cell cultivations. For in vivo structures, these scaffolds give the proper area for the attachment of cells and proliferation to mimic the desired tissues or organs. Furthermore, the survival rate of cells is very high in scaffolds because of the release of nutrients, oxygen, and biologically active agents for a long period (Xie et al., 2001). The scaffolds involve the ECM for the process of restoring the organ and the extracellular that is important for the scaffold can be extracted or decellularized from different sources and it needs decellularization protocols to carry out this (Sackett et al., 2018). Natural polymers are considered as one of the essential elements of tissue engineering techniques. Natural polymers like collagen, fibrin, and ECM are useful in the growth of the rupture tissues or the prosthetic organs (Mano et al., 2007). Some researchers carried out research based on the ECM that was collected from porcine ovaries to cure the infertility problems of ovary like POI, the cells were recombined with pig ovaries and dermal fibroblast of human beings, and it has been observed that the scaffolds that are made by decellularized ECM making suitable function ex vivo culture of the ovarian cell (Fig. 14.2) (Pennarossa et al., 2021).

These different scaffolds are very helpful in the treatment of infertility in people who are suffering from cancer. Some studies suggested that fibrin-based scaffolds

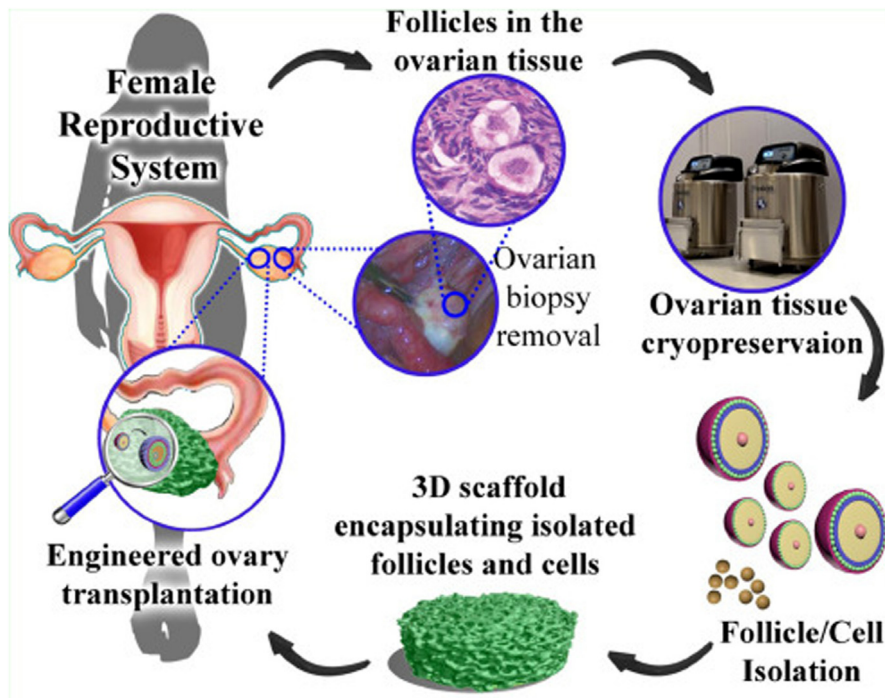


FIGURE 14.2

3D scaffold encapsulation in ovaries (Dadashzadeh et al., 2021).

are very beneficial in preserving fertility with the interaction of ECM is beneficial in improving the growth of secondary follicles that were extracted from dog's ovaries (Songsasen et al., 2011).

14.5.2 Stem cell

Stem cells have a significant role in the field of tissue engineering and regenerative medicine. The stem cells are multicellular organism that has the property of self-renewal and it can be differentiated into other cell lineages (Chen et al., 2018). This has been used as an essential tool as in vivo technique for the reconstruction of damaged tissues and organs that have stopped working because of some problems, it has the property of self-renewal, and it can differentiate between one cell or more specialized cells (McKee & Chaudhry, 2017). There are four types of stem cells present that are unipotent, multipotent, pluripotent, and totipotent stem cells, and these can be used to cure abnormalities in the tissues like old age effects, genetic problems, and cell regeneration. The main obstacle in the transplantation is the compatibility of stem cells with the patient's immune system, if it is not compatible, then the host body will reject the cells that is why the researchers are using immune suppressants like leukocyte antigens (HLAs) to make the procedures successful (Ghahremani-Nasab et al., 2020). The transmitted stem cells for the regeneration of the tissues must stay alive, proliferate, and differentiate after the transmission to the targeted somatic cell, and it should be homogenized within the circulatory system of the subject's body. It has been observed that there are a number of stem cells of different kinds used in tissue engineering, the first one is embryonic stem cells (ESCs), and the second is tissue-specific progenitor stem cells (TSPSCs), and thirdly the stem cell used is MSCs, lastly umbilical cord stem cells (UCSCs) and bone marrow MSCs (Mahla, 2016).

Many researchers are using this for the treatment of diseases related to ovaries like POI, POF, and infertility caused by cancer. Some studies indicate that the MSCs from the human body have been counted as one of the potential therapy to cure the primordial follicles quiescence that happened because of chemotherapy that is considered as the prominent cause of premature failure (Goswami & Conway, 2005). It has been observed in the previous years, the MSCs have been very successful in the treatment of the fertility problem in both young and aged women (Mohamed et al., 2018). This technique is helpful in the treatment of POI, but at the same time there are complications in this therapy, many studies are indicating that it may cause autoimmune disease, carcinomas, and there is not enough knowledge of the mechanism available. This stem cell therapy is solving many problems like it treats the ovarian cells that stop working like oocytes (Isik et al., 2017). There are various stem cells, for example, BM-MSCs, ADSCs, amniotic fluid stem cells, and human menstrual blood stem cells have significant effects on the procedures of curing POF or POI. Additionally, stem cell strategies have been proved and it is considered as one of the best therapies for the preservation of fertility in women who survived cancer (Wang et al., 2013).

14.5.3 Biomaterials

Biomaterials are counted as the essential element in the field of tissue engineering and regenerative medicine to restore or in the regeneration process of organs, biomaterials are used for clinical therapies like fabrications, but it must be biocompatible and has a good response for cellular interactions otherwise the patient's body rejects the materials (Nikolova & Chavali, 2019). Many studies show this is very helpful in the treatment of ovarian diseases, to treat the POI, and for the follicle making, ovarian patches are used that are also good in providing the mechanical support in the procedure function restoration of ovaries. The materials used for the treatment must be harmless, biologically compatible, and biodegradable so they can dissolve themselves after some time within the body, and the main focus must be on that it does not cause any inflammation during the clinical therapy of regeneration of cells or organs (Mano et al., 2007). The biomaterials used for these therapies are of two types: natural biomaterials like ECM, fibrin, collagen, alginate, and the others are synthetic biomaterials such as PTFE; comparatively, the natural biomaterials are more independent with respect to biological and mechanical properties than the synthetic biomaterials. The way biomaterials work in tissue engineering for ovaries is shown in the figure.

14.5.3.1 Extracellular matrix

This has been widely used in the field of tissue engineering and regenerative medicine to treat the problems; it provides help and support to the structure of cells to maintain homeostasis with the help of the extracellular macromolecules (Ghahremani-Nasab et al., 2020). The ECM is an essential thing in the making of scaffolds and it needs to be made sure that the three-dimensional microenvironment must be porous and should be large enough for the cell seeding process (Ermis et al., 2021). This has been used to cure many problems related to ovaries like gynecological malignant tumors and undergoing ovariectomies and POI.

14.5.3.2 Collagen

This is found in the ECM, and it is the most important structural protein. The 30% of the human body mass is made up of collagen. There are many different kinds of collagen present that are performing different functions. It has been observed that the collagen has great ability to cross-link with other molecules (Shekhter et al., 2019). It has been also noticed that collagen has a high mechanical strength, and they are good in biocompatibility with the other cells. Some researchers performed microfluidic encapsulation to develop the three-dimensional shells with the help of collagen to culture ovarian follicle (He, 2017). Collagens are helping in the treatment of ovarian disease with the help of other biomaterials.

14.5.3.3 Fibrin

Fibrin is considered an essential element in blood clotting or coagulation that provide the short-term support of fibroblast and endothelial cell invasion as a matrix during the tissue engineering process of regeneration of ruptured tissues (Bujoli et al., 2019).

Fibrin is used for scaffolding, cell carrier, and as a drug delivery system. One of the main things is that fibrin is good in biodegradability and good in biocompatibility and it has been used as the main component in the tissue engineering of ovaries, and it increases angiogenesis while the process of engraftment of ovarian tissue (Shikanov et al., 2011).

14.5.3.4 Alginates

Alginates have been used in tissue engineering and regenerative medicine as biomaterials for the treatment regeneration of the organ that stops working because of disease or trauma, ruptured tissues, clinical drug delivery, and as a cell barrier because it has a beneficial material property like good biocompatibility, nonimmunogenicity, and hydrophilicity (Tønnesen & Karlsen, 2002). Alginate has been used in the treatment of ovaries. Some researchers took 1% alginate graft isolated follicles by allografted mouse preantral follicles and ovarian cells encapsulations, and it has been observed that alginate has a very good response after the transplantation and the growth of the engineered follicles was successful (Vanacker et al., 2014). It is also used in the treatment model for the problems related to ovaries.

14.5.3.5 Synthetic biomaterials

The synthetic biomaterial is also used to treat the diseases with the help of tissue engineering techniques, mainly these materials are polymers and they have been used with other biological factors such as cells or protein to carry out the clinical therapy and increase their therapeutic efficacy. It has been observed in studies on the ovarian diseases that synthetic biomaterials like PTFE are being used to cure the ovarian problem in the mouse such as POI and how to restore the functions of ovaries again (David et al., 2017).

14.5.4 Growth factors

GFs played a vital role in the regeneration of the ruptured or death cells. They are those proteins also known as signaling proteins, and they can do proliferations in the cells. They are known as originators of signaling by the paracrine and endocrine interactions. The cytokines played an important role in the tissue engineering of the ovarian glands; these GFs can be immobilized to the ECM (Guzeloglu-Kayisli et al., 2009).

14.6 Recent trends in tissue engineering of ovaries

14.6.1 Tissue engineering ovarian follicles

Cryopreserved ovarian follicles are being transplanted as normal clinical job for quite some time. It has also shown clinical success. Similarly, preservation of ovary is currently an important trend for cancer patients who have gone through pelvic

radiotherapy and chemotherapy. It generally comprises autologous ovaries transplant to parts beyond the beams of radiation till chemotherapy completion, and the donor receives them back. The remedy uses autologous ovaries.

In utilizing such treatment, immunosuppression is not needed, also rejection issues are not there. The scholars have demonstrated first time that the fertility was recovered after complete body irradiation along with chemotherapy (Larsen et al., 2000; Rodriguez-Wallberg et al., 2015). Moreover, study conducted by Silber and colleagues reported both cryopreserved and fresh ovarian transplant while showing the successful autologous and allogeneic transplants. Furthermore, Laronda and colleagues also showed the likelihood of utilizing decellularized transplants through forming a decellularized ovarian scaffold, generate cells in it, and transplant it to start puberty in mouse (Silber et al., 2015). Nevertheless, the researchers could not demonstrate offspring, which is critical for transplanting in human if the same strategy is being used.

This part highlights and demonstrates the brilliant clinical results for the ovarian tissue transplant. However, it also shows its limitation for the ovarian cancer as the tumor might be reintroduced because ovarian tissue might contain some remains of carcinogenic material (Du et al., 2015; Silber et al., 2015). Hence, major efforts are being rendered in ovarian tissue engineering, particularly with reference to implantation.

14.6.2 Ovarian cell culture and ovarian scaffolding by tissue engineering

Scholars have advanced the understanding of oocyte maturation and isolation of cryopreserved ovarian tissues, in order to evade disease reintroduction. Pangas and colleagues conducted experiments in the mice. It showed the proliferation of granulosa cells and good growth of the oocytes with spatial arrangements.

The biomaterial such as polysaccharide is commonly utilized in tissue engineering. The researchers developed artificial ovary and used three types of follicular cells. This included oocytes, granulosa, and theca. The results demonstrated initial success as oocyte got matured. Nevertheless, the usage of polysaccharide does not have ECM protein, which is fundamental requisite for cells phenotypes. In contrast, some research works that have used biomaterials comprising fibronectin and collagen, and RGD has showed much better oocyte growth. Moreover, fibrin has also emerged as the useful biomaterial required for tissue engineering of ovary. It gels with ovary follicles and facilitates the growth of antral follicles (Pangas et al., 2003).

The level of follicle-stimulating hormone could be reduced through the fibrin graft and it also enables to reinstate cyclicity as reported in the experiment involving infertile mice. The result utilizing hydrogel supports tissues of ovary to start frozen oocytes. This development presents novel approaches in follicle biology and recognizes a major breakthrough addressing human egg bank for fertility. Furthermore, these developments could potentially form a vigorous

system, which could be utilized to mature oocyte ([Krotz et al., 2010](#)). On the other hand, problems associated with natural biomaterial like less-defined molecular composition and batch-to-batch variability are critical to tackle before the wide roll out of the aforementioned approaches.

14.6.3 Cell-based tissue engineering to reestablished ovarian function

A number of researchers have attempted to investigate tissue engineering of ovary without biomaterials. In one of the works, Lai and colleagues utilized stem cells of humans to reinstate the function of ovary. They did it through the revitalization of germline cells. They transplanted endometrial MSCs from menstrual blood and injected in the tail of mouse. This led to the weight increase and restoration of fertility in mouse ([Lai et al., 2015](#)).

Human amniotic epithelial cells (hAECs) were utilized in research to prevent cell apoptosis developed by chemotherapy and inoculate cultured hAECs into mouse suffering infertility. The findings showed that improved the growth of follicles and also enhanced the oocytes numbers in the mouse. Moreover, germline stem cells of female (FGSCs) also demonstrated to support chemoablated ovaries. The inoculation of GFP-carrying virus and the transplantation of FGSCs into ovaries of infertile mice because of chemotherapy were also to function properly, resulting in offspring ([Zhu et al., 2015](#)).

It could be summarized that scaffold-free methods based on cells are encouraging based on the recent research outcomes. Nevertheless, it is too early to establish its complete success as it needs to show effective pregnancies among the bigger animals before moving to clinical translation.

14.7 Conclusion

This review offers some of the fundamental insights into current research, technological development, and major challenges related to ovarian glands disorders leading to life-threatening disease or driving infertility in women especially young women. The report also highlights some of the recent advancements in regenerative medicine and the techniques involved in the tissue engineering of ovaries, especially strategies currently being implemented in the area of repair of the ruptured cells or regeneration of the death cells or complete transplant of the effected part. This report further summarized the factors involved in the process of tissue engineering like scaffolding, stem cells, and GFs, and it also outlined the natural and synthetic biomaterials like ECM, collagen, fibrin, alginates, and PTFE. Moreover, this comprehensive review also discussed the recent trends in tissue engineering of ovarian glands like tissue engineering of ovarian follicles, ovarian cell culturing, and cell-based established ovarian function.

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The progress in tissue engineering of endometrium

15

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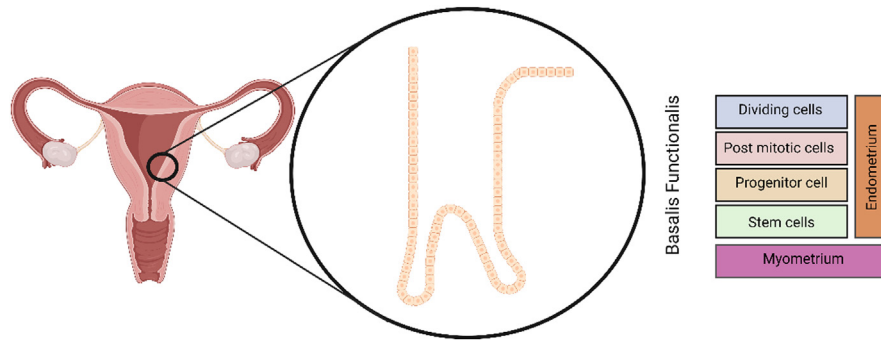
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15.1 Introduction

The human endometrium grows about 7 mm during every menstrual cycle, which is an extraordinary example of controlled tissue remodeling (McLennan & Rydell, 1965). It takes approximately 500 cycles for a woman to experience this level of angiogenesis over the course of her lifetime. An embryonic mucosal lining forms the mucosa of the endometrium during embryogenesis (Fig. 15.1). This structure consists of two main zones: (1) the functionalis, which contains glands as well as stroma supporting the surface epithelium, and (2) the basalis, which includes glands, the stroma, and lymphoid aggregates at the base. Human endometrium is composed of progenitor cells located at its base (Fig. 15.1). They later move to the functional layer (Okulicz et al., 1997) where they play an active role in rebuilding the endometrium and stimulating regeneration.

Women suffering from endometrial damage commonly have menstrual problems, infertility, recurrent pregnancy loss, and other reproductive issues. It is usually associated with repeated diagnostic curettages and intrauterine infections, all of which are referred to as Asherman syndrome (AS). Endometrial function in survi-

^a Sima Etebari and Fereshteh Asgharzadeh made equal contribution to this chapter.

**FIGURE 15.1**

The endometrium. Anatomy of the endometrium and the location of endometrial stem cells. Endometrial progenitor cells reside in the basalis at the base of glands, whereas endometrial stem cells reside near blood vessels in functionalis.

vors of cancer treatment can also be adversely affected by treatment, regardless of the impact on the ovary (Lv et al., 2021).

In modern medicine, it is common to lose tissues and to have organ dysfunctions. Our ability to care for these diverse entities has been revolutionized by the free transfer of tissue and organs over the past half century. In recent years, composite tissue allotransplants have become more common following mastectomy and solid organ transplants (Ikada, 2006). Even with these advanced surgical strategies, there are still numerous problems to be resolved. At both the autologous and allogenic levels, all three approaches require an appropriate donor source. There is a high level of morbidity associated with autologous tissue transplants, and immunosuppression is required following organ transplants for life. The effects of thrombosis can be devastating for both approaches. Moreover, some diseases remain intractable (Ikada, 2006; Ravnice et al., 2017). Therefore, engineered tissues and organs are necessary to replace virtually every surgical specialty, including reconstructive surgery and transplantation (Cui et al., 2018).

Biological replacements with mimicking properties are an interdisciplinary field that is on the rise (Al-Himdani et al., 2017; Cui et al., 2018; Nichols et al., 2018). As a multidisciplinary field, regenerative engineering combines tissue engineering with materials science, stem cell biology, and developmental biology (Laurencin & Khan, 2012). The possibility of regenerative engineering is slowly becoming a reality through the combination of cells, materials, and bioreactive molecules. A personalized treatment option could be provided by these replacement tissues based on an autologous platform. In order for ubiquitous clinical translation to occur, a number of hurdles need to be overcome. There are many advancements in both the clinical and research arenas that have been made (Bellofatto et al., 2021; Croce et al., 2019; Peloso, 2021). The following chapter will elucidate the progress in tissue engineering of endometrium.

15.2 Endometrium anatomy and physiology

Endometriums are dynamic tissue that shed periodically, repair, regenerate, and remodel when exposed to estrogen and progesterone from the ovary. For the treatment of abnormal endometrial lesions and repair of injury to the endometrium caused by repeated intrauterine operations, infections, or cancer therapy (Lv et al., 2021), it is essential to understand the cyclic physiological and biological processes of normal human endometrium.

Reproduction relies on the multilayered uterine lining. Hypothalamic—pituitary—gonadal axis regulates its makeup, which is why it undergoes reiterative cycles of proliferation, differentiation, and menstrual shedding on a monthly basis in order to serve its primordial purposes of embryonic implantation and nourishment. According to Heremans et al. (2021), the endometrium is divided into two layers: the lamina functionalis is more superficially attached to the myometrium, and the lamina basalis is the deepest and closest to it (Fig. 15.2).

Besides providing receptivity to embryo attachment, hormonal production, maintaining pregnancy, and protecting the uterus against external agents, the endometrium is also responsible for uterine receptivity for embryo attachment. There is collagen type I, III, IV, and V in the endometrium, fibronectin, laminin, heparan sulfate, elastin, and even glycoproteins such as tenascin C and HDA, which are present in nonpregnant states. As embryo implantation and trophoblast invasion during pregnancy depend on uterine extracellular matrix (ECM), it is important to note that uterine ECM plays a key role in establishing embryo receptivity. Aside from epithelial

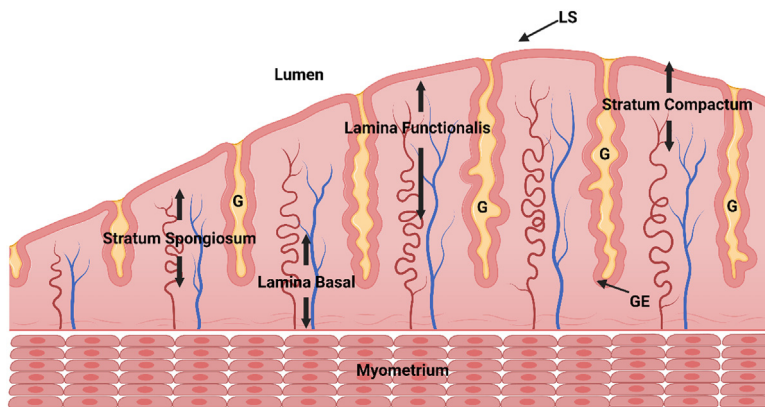


FIGURE 15.2

A schematic representation of the endometrium. Stratum compactum and spongiosum are the two superficial strata within the lamina functionalis. Lamina functionalis and uterine myometrium are separated by the persistent lamina basalis. The lamina of each layer consists of epithelial cells interspersed with stromal cells of varying densities. G, gland; GE, glandular epithelium; LE, luminal epithelium.

cells which form the endometrial epithelium, endometrial glands are composed of glandular epithelial cells surrounded by ECM in the stromal layer; endothelial cells; and mast cells, T and B cells are among the many immune cells found in the endometrium (Heremans et al., 2021).

15.2.1 Regeneration and destruction of endometrial tissue: Molecular mechanisms

The onset of menstruation is mediated by the local endometrium after withdrawal of ovarian hormones. It is interesting to note that endometrial tissues are destroyed and reepithelialized at the same time (Garry et al., 2009). As soon as menstruation begins, reepithelialization begins, and it takes approximately 48 h to be completed. Endometrial resolution during menstruation is a zonal event, as demonstrated by hysteroscopic analysis of the endometrium during the menstrual cycle, with areas adjacent to the shedding endometrium still having intact endometrium from the previous menstrual cycle and areas that have reepithelialized. A decidualized endometrial stromal cell (ESC) expresses the progesterone receptor (PR) and detects progesterone withdrawal when exposed to progesterone in the premenstrual period, indicating that DSCs may be responsible for responding to endocrine signals and transmitting paracrine signals during menstruation (Evans & Salamonsen, 2014; Garry et al., 2009). Progesterone withdrawal inhibits superoxide dismutase expression in DSCs, promoting the production of free oxygen in *in vitro* experiments. The latter upregulates nuclear factor κ B and cyclooxygenase-2, as well as the expression of inflammatory factors such as prostaglandins F2 α , and chemotaxis factors, which attract inflammatory cells into the uterine endometrium (Evans & Salamonsen, 2014; Sugino et al., 2004). Proteases and gene products linked to ECM synthesis and repair are secreted by these inflammatory cells, which contribute to the destruction of endometrial tissues while inducing endometrial repair. Those factors, including activin, vascular endothelial growth factor (VEGF), cysteine-rich secretory protein 3, and galectin-7, and development-related pathways such as the Wnt pathway and mesenchymal-epithelial transition, promote endometrial reepithelization and regeneration after menstruation but do not rely on estrogen (Evans et al., 2014, 2015; Evans & Salamonsen, 2014; Fan et al., 2012). To stimulate endometrial epithelial and stromal cell proliferation, estrogen is required after the endometrium has been completely reepithelialized.

15.3 Regeneration of the endometrium: Challenges and opportunities

Endometrial trauma results in intrauterine adhesion (IUA), also known as Asherman syndrome. The presence of IUA has been reported in the course of uterine cavity surgeries including artificial abortions, dilatation and curettages,

hemorrhage and curettages after abortions, manual placenta removals, polypectomy, myomectomy, and conization of the cervix (Abudukeyoumu et al., 2020). There is also evidence that infections can cause IUA (Sharma et al., 2008). Moreover, there are a variety of factors that can lead to the destruction of the endometrium, affecting its basal compartment as well (Cervelló et al., 2015). In the absence of a complete repair of the endometrium, fibrotic synechiae will form across the cavity, replacing the glands with inactive cubocolumnar endometrium. As fibrotic tissue forms, blood perfusion is reduced due to inactive, cystically dilated, or sparse and inactive glands, and telangiectatic vessels can be visible. The uterine cavity may also be deformed, stenosed, or scarred (Santamaria et al., 2018). There are studies that report these conditions can result in amenorrhea, dysmenorrhea, infertility, miscarriages, chronic pelvic pain, and placenta accretion (Sonan et al., 2018). However, several hypotheses have been proposed regarding the pathogenesis of IUA, including fibrosis hyperplasia, neural reflexes, abnormal stem cell differentiation, changes in the uterine cavity microenvironment and fibrosis, abnormal signal pathway regulation, and inflammation caused by adherent fibroblasts. Most studies have been conducted on the fibrosis hyperplasia theory (Cai et al., 2016; Liu, Zhu, et al., 2013; Ma et al., 2021; Sun et al., 2022). The transforming growth factor- β (TGF- β) has been identified as a critical factor in IUA pathogenesis by stimulating endometrial fibrosis on the canonical TGF- β /Smad signaling pathway.

Moreover, TGF- β is correlated with IUA pathogenesis via ER- α , CCN2, and matrix metalloprotein 9 (MMP-9) (Abudukeyoumu et al., 2020). There is a study showing the involvement of Hippo in endometrial fibrosis in IUAs as well as its cross-talk with many other signaling pathways (Liu, Ha, et al., 2013). In addition to impaired ECM homeostasis, adhesions are also caused by abnormal ECM composition. IUA has also recently been linked to impaired NF- κ B signaling. Fibrosis is often caused by such inflammatory signaling (Wang et al., 2017). IUA is also reported to trigger PP63 upregulation in endometrial stem cells, causing adhesions formation (Zhao et al., 2017).

Different studies aimed to increase our understanding about adhesion development and finding effective therapeutics and preventive methods for IUA. Besides hysteroscopy and adhesiolysis, other treatments have been investigated including hormone therapy, stem cells, TGF inhibitors, G-CSF, and KDR-miRNA interactions. IUA is characterized by endometrial fibrosis as its primary pathological feature. Investing in the specific mechanism of endometrial fibrosis can help prevent and treat IUA by inhibiting endometrial fibrosis while promoting endometrial hyperplasia by targeting fibrosis. A priority for future studies will be to improve the efficacy of IUA as well as to reduce the recurrence rate after transcervical resection of adhesion (TCRA). Those with IUA are expected to benefit from regenerative medicine in the future. Considering the evidence gathered above, stem cells or their paracrine mechanisms are potential candidates for new therapies in the treatment of IUA, which provides a theoretical method for developing potential reagents (Li et al., 2021).

15.3.1 An overview of adult stem cell research on endometrial repair and challenges

It is believed that adult stem cells are a small population of quiescent cells capable of regenerating entire tissues in which they reside. Their difference from embryonic stem cells is that they cannot produce all the types of cells found in the embryo and they do not express all the markers present in the embryo (Donovan & Gearhart, 2001). Despite their proliferating potential and the ability to persist in many tissues throughout the lifetime of the individual, adult stem cells are more committed to a particular lineage. By dividing asymmetrically, adult stem cells can maintain their own population of undifferentiated cells (self-renewal), as well as produce differentiated progenitor daughter cells that can multiply rapidly (Fuchs & Segre, 2000).

The ability to self-renew is one characteristic of adult stem cells, also known as tissue-specific stem cells. By dividing asymmetrically, they proliferate and differentiate into specific cell types. The potency of stem cells can be assessed by examining the ability of the cells to produce different types of cells. Reconstruction and repair of tissues rely heavily on adult stem cells. In 2004, Chan, Schwab, and Gargett were the first to isolate adult stem cells from the endometrium (Chan et al., 2004).

The evidence supporting the regeneration and repair of the endometrium by adult stem cells is growing (Gargett et al., 2016). The endometrial basalis layer has been damaged or dysfunctional, leading to AS/EA being treated with adult stem cells.

15.4 Biomaterials for tissue engineering

A biomaterial scaffold is used as a scaffold to seed cells growing on it from a tissue, which has been expanded in vitro. Essentially, these biomaterials serve as artificial ECM that mimics the biological and mechanical characteristics of the native ECM. Cells can adhere to biomaterials and form tissues with appropriate structure and function using the scaffolds provided by biomaterials. As a result, cells and bioactive factors can also be delivered to the desired location in the body (Kim & Mooney, 1998). It has been found that tissue engineering uses three broad classes of biomaterials: naturally derived materials (collagen and alginate), acellular tissue matrixes (small intestinal submucosa (SIS) and bladder submucosa (BSM)), and synthetic polymers like polyglycolic acid, polylactic acid, and poly(lactic-co-glycolic acid). A salt/acid extraction can be used to separate collagen from human and animal tissues, and collagen is one of the most abundant and ubiquitous structural proteins in the body. Cell adhesion domain sequences (e.g., RGD) in collagen may benefit many types of cells, including fibroblasts and chondrocytes. Tissues that are devoid of cellular components are acellular tissue matrices. Often, tissue segments are mechanically and chemically manipulated to prepare matrices (Atala, 2012). Transplanted or ingrowing host cells synthesize and secrete new ECM proteins to replace and remodel these matrices upon implantation. SIS and BSM are very common acellular tissue matrices. As well as PGA, PLA, and PLGA, tissue engineering

uses polyesters that include naturally occurring α -hydroxy acids. In 1981, Gilding (GOLDING) obtained FDA approval to use these polymers in sutures for the treatment of various ailments.

15.5 Bioengineering of endometrium

Researchers are developing tissue-engineered constructs for partial uterus repair in an effort to overcome immunogenicity and organ availability issues. These endeavors extensively utilize decellularized ECM biomaterials. Cell adhesion and migration are enhanced when native tissue proteins are present and arranged in ECMs. Accordingly, Miyazaki and Maruyama recellularized decellularized uterine matrix in order to partially regenerate and reconstruct the rat uterus. In vitro recellularization was achieved by reseeding the decellularized rat uterus, which resulted in endometrium-like tissue reconstitution after decellularization. When placed onto partially excised uterine tissues in rats (Miyazaki & Maruyama, 2014), this procedure was capable of partially regenerating structurally and functionally competent uterine tissues. Decellularized ECM may prove very useful for uterine regeneration based on these findings. It is, however, important to note that using decellularized tissue matrixes has several disadvantages, such as the possibility of residual cellular materials (e.g., DNA, cells, and antigens), which might trigger a significant immune response in the host, which was not considered in these studies (Badylak, 2004).

In another study, estrogen/progesterone stimulated endometrial cells improved the reconstructed rabbit endometrium (Wang et al., 2010). It has constructed human cervix-like tissue in vitro using collagen-coated silk scaffolds. When human cervix cells were isolated from the cervical stroma (mid-canal region) and cultured in a dynamic cell culturing system for an extended period of eight weeks, a significant increase in the concentration of ECM molecules was observed and scaffold stiffness was significantly increased (House et al., 2010). The summary of the endometrium tissue engineering studies is presented in Table 15.1.

The biomaterial part of tissue engineering is crucial since it can mimic native endometrium tissue in terms of structural support (Zhang et al., 2020). Biomedical materials may also be characterized by physical, chemical, and biological properties that may promote endometrium regeneration. Besides supporting cells, bioactive molecules are also important in tissue engineering. As well as signaling, nutrients are transported, stem cells are recruited, fertilized, proliferated, and differentiated via these two molecules. Drugs, growth factors, small molecules, and other bioactive compounds can be released in a controlled manner by biomaterials, whether they are loaded with cells or not. In recent years, research has shown that, as an alternative to traditional biomaterial-based endometrium regeneration, cells and biomaterials can be combined and modified to provide functional or structural advantages, such as cell sheets, cell-scaffold interfaces, surface-functionalized scaffolds, and decellularization of biological tissues. By inducing biomimetic changes and recreating regenerative microenvironments, injured endometrium was repaired to various extents

Table 15.1 An overview of studies on endometrium tissue engineering.

Species	Scaffold material	Cell used	Experiment	References
Human	Collagen/matrigel	Human endometrium cells	In vitro	Meng et al. (2009)
	Collagen/matrigel	Human endometrial epithelial with stromal cell	In vitro	Kim et al. (2005)
	Collagen/matrigel	Human endometrium cells	In vitro	Sengupta et al. (2008)
	Collagen/matrigel	Human endometrial stromal cell and epithelial cell	In vitro	Park et al. (2003)
	Collagen	Human endometrial carcinogenetic cell	In vitro	Benbrook et al. (2008)
	Matrigel	Human endometrial epithelial with stromal cell	In vitro	Arnold et al. (2001)
	Collagen hydrogel	Telomerase immortalized human endometrial stromal cells	In vitro	Schutte and Taylor (2012)
Rabbit	Collagen/matrigel	Rabbit endometrial cells and mouse embryos	In vitro	Wang et al. (2010)

([Liu et al., 2019](#)). This chapter evaluates the potential therapeutic effects of a variety of biomimetic endometrium scaffolds to treat female infertility and comprehensively reviews current advances in the biological interactions and applications of these scaffolds.

15.5.1 Engineering the endometrium through stem cell sheets

It is important to note that cell sheet engineering (CSE) is a promising technique that can be applied to TE. Yamato and Okano invented this technique ([Yamato & Okano, 2004](#); [Yang et al., 2005](#)). This process keeps the ECM and the entire cell together. Noninvasively harvesting of cell sheets can be achieved without disrupting cell-to-cell connections by using the temperature-responsive polymer. It involves the formation of cells in a sheet of ECM from the bottom up. The regenerative function of cell sheets will be enhanced because of a high density of cells and higher transplantation efficiency. Using temperature-responsive surfaces, CSE can collect cells easily without using any proteolytic enzymes that can cause damage to the ECM and lose cell-to-cell interaction ([Sakaguchi et al., 2015](#)). Cells of the endometrium of the uterus are called endometrial epithelial cells. For cell engineering, it makes

sense to obtain these cells (Kuramoto et al., 2018). The collection of endometrial cells in real practices, however, is not recommended since it is difficult to get adequate quantities from the uterus. Additionally, certain inflammatory or contagious diseases may interfere with the collection, apart from the harm resulting from invasive procedures. In recent years, stem cell sheet engineering (SCSE) has become an important research direction because stem cells are minimally invasive, autologous, can differentiate multiple times, proliferate widely, secrete growth factors, and transmit signals to other cells (Hsu et al., 2019; Imafuku et al., 2019; Shum et al., 2017). Kuramoto et al. applied oral mucosal epithelial cell sheets as a novel regenerative method in the treatment and prevention of IUA and also infertility. A stratified squamous epithelial cell sample was obtained from oral mucosal epithelial tissues using minimally invasive surgery (Kuramoto et al., 2015).

As a result of the high cost and time-consuming properties, the thermoresponsive approach is further modified and improved. Ascorbic acid was added to adipose-derived stem cells (ADSCs) to construct cell sheets in a novel and user-friendly method developed (Sun et al., 2018). Self-renewing, proliferative, and regenerative properties of ADSCs make them ideal for tissue engineering (Mao et al., 2019; Rogan et al., 2020). A range of ECM proteins were surrounding ADSCs after in vitro culture. VEGF and fibroblast growth factor (FGF) are among the trophic growth factors that were expressed in response to cell sheet expansion (Kang et al., 2019). Endometrial stromal cells were stimulated to proliferate and form blood vessels. Compared to this, hyperplastic fibrosis was present in the blank control group. The uterine microenvironment allowed ADSCs to differentiate into endometrial stromal cells. By providing biomimetic trophic support that was vital for cell proliferation, implantable ADSC sheets facilitated endometrial repair by attaching closely to the injured uterus. As a result, scarring associated with natural cell sheet constructs was insignificant (Sun et al., 2018). SCSE secretes growth factors and nutritional proteins that are beneficial for uterus repair. For long-term survival, however, better scaffold support is necessary for improving cell viability in vivo.

15.5.2 Regeneration of endometrium based on cell-scaffold interfaces

In order to restore the functionality and architecture of the wounded tissues, biomaterials provide structural and mechanical support. As well as mimicking native environments, these methods can induce physicochemical changes, as well as simulate changes in growth factors, signaling molecules, and extracellular vesicles (Kim et al., 2019; Liu et al., 2019). The use of scaffold biomaterial alone is insufficient for the repair of large uterine defects (Xin et al., 2019). It is important to take into account several factors, including vascularization, native cell recruitment, and scar suppression (Owusu-Akyaw et al., 2019). When cells are seeded on the scaffolds, they survive longer, proliferate, differentiate, and vascularize (Frost et al., 2019). A murine model of uterine infertility was also used to encapsulate decidualized endometrial stromal cells (dEMSCs) in hyaluronic acid (HA) hydrogel

(Kim et al., 2019). There was an increase in endometrium thickness at two weeks after injury and a decrease in fibrous tissues. Endometrial tissue from a functionally regenerated endometrium showed high levels of desmin, CD44, and platelet endothelial cell adhesion molecule expression. As a result of dEMSC-loaded HA hydrogel treatment, successfully implanted embryos developed normally and live offspring were born. As compared to BMSCs or hESC-derived endometrium-like cells, bioprocessed isotopic cells significantly shortened the recovery time. Furthermore, the presence of plenty of hyaluronidase in the endometrium seemed to indicate that HA was very suitable for the repair of endometrium. Porosity also provided the scaffold stiffness needed for endometrium regeneration due to the limiting mobility of cross-linking with porosity. Angiogenesis was restored and scar tissue formation was inhibited in selection of uterine repair cell types (Li et al., 2019). Cells in the endometrium's perivascular layer (En-PSCs) hold the same potential as stem cells. An angiogenic inducer 61 that contains cysteine-rich residues (CYR61) was found to be involved in vascular formation (Zhao et al., 2017). The scientists then developed a collagen scaffold containing CYR61-transfected En-PSCs and found it significantly accelerated neovascularization in vivo as it stimulated angiogenic factors to be released from the ECM. Since BMSCs can be isolated easily, have abundant resources, and can be regenerated (Xia et al., 2019), they were often used to regenerate endometrium and uterus in addition to endometrium-derived cells. An experimental model of wounded rat uterine horns was transplanted with BMSC-loaded collagen scaffolds (Ding et al., 2014). Regenerated endometrium basal membranes mainly recruited BMSCs. BIGF, IGF-1, VEGF, and TGF- β 1 were highly expressed in the injured tissue next to the cell scaffold composite, as well as prominent regeneration of the microvasculature. A scaffold loaded with BMSCs restored the new endometrium's receptive capacity.

A thermoresponsive gelation of Pluronic F-127 and vitamin C-encapsulated BMSCs, which enhanced membrane stability (Yang et al., 2017). Additionally, vitamin C, as an antioxidant, reduces tumor necrosis factor α (TNF- α) and interleukin 6 (IL-6) secretion, maintains redox homeostasis, and increases interleukin 10 (IL-10) levels, which facilitates pro-regenerative behavior (El Banna et al., 2019). As a result of using the BMSC/PF-127 C vitamin C hydrogel, endometrial thickness was recovered and fibrotic regions of the endometrium's stroma were decreased. Poly(glycerol sebacate) (PGS) scaffolds loaded with BMSCs might be capable of recovering different tissue deformations under dynamic conditions without external irritations (Xiao et al., 2019). A number of cell-scaffold compositions were compared, including BMSCs based on collagen, BMSCs based on poly(lactic-co-glycolic acid) (PLGA), and BMSCs based on postgrowth glycogen. As a result of PGS, BMSCs proliferated and differentiated better in the endometrium. The PGS-based scaffolds also retained BMSC in situ for a longer period of time and increased areas of vascularization in in vivo studies. To downregulate 1Np63 expression and inhibit endometrial quiescence and other pathological changes related to Asherman's syndrome, applied collagen scaffolds loaded with BMNCs

(Zhao et al., 2017). An increase in estradiol (E2) stimulation and functional endometrium growth was observed in collagen scaffolds containing BMNC-loaded proteins. There were five successful pregnancies and live births among these patients.

Embryonic stem cells derived from umbilical cords have shown excellent proliferation capabilities and have adequate sources (Lee et al., 2019). Injured uteruses of human patients were transplanted with UCMSCs on collagen scaffolds. The regenerated endometrium following the implantation of this cell scaffold mixture showed an improvement in proliferation, differentiation, and neovascularization. The most common birth defect or placental complication was not evident in most babies. UCMSCs were also examined for their ability to reduce scarring. As a result, they observed significant angiogenesis and little scarring in the injured site when UCMSCs were mixed with gelatinous degradable collagen fibers (Xu et al., 2017). Through MMP-9, FGF-2, and VEGF production, the cell-scaffold composite degrades collagen in scarred areas, leading to angiogenesis and endometrial regeneration. A paracrine pathway modulated cellular apoptosis and improved human endometrial stromal cell health in a UCMSC-loaded collagen scaffold (Xin et al., 2019). As a result of collagen remodeling in the reconstructed endometrium, the scaffold barely caused any inflammation. The collagen scaffold that was loaded with UCMSCs also resulted in a rapid reepithelization process by increasing the proliferation level and cytokeratin expression level, both vital to the subsequent repair of the endometrium. Consequently, estrogen, progesterone, and growth factors were elevated in the circulating blood. For uterine horn repair, Song and colleagues grafted hESC-derived endometrium-like cells onto collagen scaffolds (Song et al., 2015). A significant differentiation rate of over 80% was achieved by adding endometrial stromal cells, 2.2 times higher than by adding cytokines. It was demonstrated that endometrium-like cells simulating *in vivo* stem cell niches were beneficial to endometrial function and development, and they secreted growth factors that enhanced the effects of estrogen and progesterone on basal layer repair. By loading cells on the scaffolds, biomaterials are more biologically active and maintain their mechanical stability in the presence of physiochemical properties.

15.5.3 Renewal of endometrium based on a surface-functionalized scaffold

Several strategies are used to improve biocompatibility and absorption of bioactive growth factors, hormones, and extracellular vesicles for cell attachment and delivery in addition to direct cell loading in scaffolds (Li et al., 2019; Shadish et al., 2019). It is a water-adsorbable and biocompatible bacteria scaffold. Wang et al. and Rebelo et al. used it to remodel bones, vessels, and nerves. Silk fibroin supplementation and the addition of stromal cell derived factor 1a (SDF-1a) improved BC porosity (Rebelo et al., 2019; Wang et al., 2018). It also increased endometrium thickness and fetus number *in vitro* by delivering SDF-1a through SF-BC membrane carrier. Functionalized BC scaffolds facilitated glandular epithelial growth and migration, which are crucial for decidualization, embryonic implantation, and embryo

development (Rebello et al., 2019; Wang et al., 2018). A significant increase in arterial formation was also observed using this scaffold. In addition to vascularization, the SF-BC scaffolds also matured the endometrium, which is primarily responsible for these findings.

Collagen scaffolds modified with collagen-binding domains (CBDs) are designed by Li et al. (2019). As a result, bFGF delivery to the endometrium was significantly increased and random diffusion was significantly reduced. In order to maintain the effective concentration, the recombinant proteins were transported with collagen to the injured area in a location specific manner. By using this scaffold, high neovascularization, muscle fiber alignment, and thick endometrium layers were induced, which allowed for tissue repair to be very effective. The embryo rate in this study, however, was low, indicating that the endometrium has not been fully recovered. The collagen scaffold itself was loaded with CBD/VEGF to improve the angiogenesis and reepithelialization of the endometrium. CBD and native injection were compared for releasing VEGF during regeneration of rat uterine full-thickness injuries. Nutrients and oxygen were supplied to scar areas by the vascular tissue growth. Furthermore, VEGF released early in the remodeling process activated MMP and initiated ECM remodeling. In contrast to local VEGF injection alone (18.8%), CBD/VEGF collagen application showed a 31.2% improvement in pregnancy rate. A porous surface and an open porous architecture based on external/internal symmetry were developed to control bFGF release (Cai et al., 2016). A microfluidic droplet scaffold loaded with bFGF was fabricated. Precision could be achieved through the adjustable porous design. As a result, steady bFGF release was facilitated in a high-concentration setting without excessive side effects. bFGF-loaded porous scaffolds showed excellent long-term reparative effects by inhibiting adhesions, promoting vascular growth, and inducing endothelial migration. According to Xu et al., keratinocyte growth factor (KGF), part of a type of reparative factor, was incorporated into a temperature-sensitive hydrogel. Injured uteri were able to retain the drug for a long period of time due to the scaffold's controlled release and prolonged retention (Xu et al., 2017). Using KGF-modified hydrogel scaffolds, researchers found that cell autophagy was enhanced by inhibiting mTOR signaling, enhancing endometrial glandular epithelial cell migration and proliferation, as well as CD31 expression levels. Inflammation and immune responses were reduced in order to restore functional epithelial repair (Gargett et al., 2008; Zhang et al., 2017). Zhang et al. synthesized heparin-polyoxamer hydrogel loaded with 17 β -E2. They found that the hydrogel significantly lowered stress-induced apoptosis in the endoplasmic reticulum (ER) (Zhang et al., 2017). Injured tissues received more nutrients, oxygen, and hormones as a result of sustained E2 release, which reduced fibrotic tissue areas and stimulated vascularity, supported by an activated extracellular regulated protein kinase 1/2 (ERK1/2) pathway that closely participates in cellular functions, including proliferation, viability, and motility (Matsumura et al., 2017; Peng et al., 2010). Some studies have also examined extracellular vesicles released from stem cells for uterine repair, along with bioactive proteins (Azizi et al., 2018; Zhang et al., 2017). A stem cell secretome-modified

HA hydrogel was developed by Liu et al., 2019 which enhanced the release of regeneration-related growth factors, including epidermal growth factor (EGF), FGF, and IGF-1. By using cross-linked HA gel as a carrier, stem cell secretome was retained in vivo for a longer period, resulting in thicker endometrium and more glands than when gel alone was applied. Endometrium scaffolds with nano-scale functionalization mimic the natural environment, allow sustained release of bioactive molecules, and transmit signals from extracellular vesicles during uterine regeneration.

15.6 Summary

It is becoming increasingly realistic to develop uterine endometrium tissue engineering as an effective clinical treatment. It may be possible to improve human endometrium injuries by using bioactive cells and biomaterial scaffolds simultaneously. This field is also being promoted by stem cell technology. Currently, there is a lack of understanding of cell sources, structures, biological factors, and biomaterial scaffolds to create functionally engineered endometrium tissue. The integration of functional factors, the longevity of the system, and the scalability of the system should be explored further. In order to achieve functional goals, various researchers involved in this field may be able to coordinate their efforts to rebuild the uterus endometrium.

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Tissue engineering strategies to treat female infertility

16

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16.1 Introduction

The global prevalence rate of infertility has become a major concern for society, the world's economy, and most importantly the healthcare system. According to the World Health Organization (WHO), data collected on infertility suggest that between a 48 million couples and about a 186 million individuals globally, in one way or the other have infertility problems (Ngai & Loke, 2021). Though infertility may be attributed to both males and females, this review paper tends to focus on the female factor infertility. Globally, between 1990 and 2017, there has been an increase in female infertility cases from 1366.85 (1.366.85%) cases to 1571 (1.571%) cases per 100,000 females. Resulting in an increase of 206 new cases which is a 6.97% increase in the space of 27 years (Elnour et al., 2021). According to the Centers for Diseases Control and Prevention (CDC), 1 out of 10 women between the ages of 15 and 44 suffers infertility challenges in the USA. This contributes to about a 6.1 million female population in the United States (Altamimi et al., 2019). In the United Kingdom (UK), research conducted by the UK Statistics Authority for England and Wales in 2018 shows that about a 1.05 million women suffer from infertility. This brings the ratio to 1 in every 7 women of child-bearing age (Evans et al., 2019).

The use of assisted reproductive technologies (ARTs) such as in vitro fertilization (IVF), intracytoplasmic sperm injection (ICSI), gamete intrafallopian transfer (GIFT), and zygote intrafallopian transfer (ZIFT) has proven to be a key breakthrough in the issues of female infertility (Mor & Vermesh, 2005). The success rate of the first IVF cycle is about 50% for women under 35 years and 11% for women aged between 40 and 42 (Mor & Vermesh, 2005). However, ART does not provide a solution to the structural issues of infertility (Rafique & Nuzhat, 2016; Tandulwadkar, 2015). Hormonal disorders, iatrogenic causes such as chemotherapy and surgeries, and structural causes like problems with ovaries, fallopian tubes, or uterus are major contributing factors of female infertility.

Over the years, a bioengineered method of tissue and whole organ transplantations has been considered to correct the iatrogenic and dysfunctional organs of the female reproductive system (Yun et al., 2016). Such methods include the cryopreservation and transplantation of autologous organ sections or the whole organ. Fresh and cryopreserved tissues are mostly preferred due to their ability to retain the integrity of the extracellular matrix (ECM) structure and the detailed cellular composition of the natural tissue or organ (Weidenhamer et al., 2015). The cells source can either be from the same patient (autologous) or allogeneic (donor). The autologous cells are an ideal selection due to their unique ability to overcome immune rejections and the likely negative impact of immunosuppressive medications (Chisholm et al., 2001).

However, the limited bioavailability of both autologous and allogeneic tissue sources has triggered the use of decellularized tissue or organ as a replacement. The idea of using decellularized materials is to eliminate the old ECM of original cells and any possible DNA components without destroying the biochemical, mechanical, and structural features of the organ (Ma et al., 2013). The decellularized ECM is repopulated with cells from the patient to generate new tissue as the decellularization maintains the architecture of the ECM. Common decellularized tissue matrices used so far in female infertility tissue repairs include peritoneum, amniotic membrane, and porcine small intestine submucosa (SIS) (Nowocin et al., 2016).

The decellularization procedure is difficult to achieve as the real architecture is often impossible to maintain. The agents for decellularization are sometimes disruptive and almost impossible to totally get rid of the genetic materials present in the decellularized organ (McKee & Wingert, 2016). Henceforth, the production of tissue-engineered scaffolds as alternative biomaterials to replace autograft and allograft in tissue or organ repair transplantation is the current approach in curbing the reintroduction of infections by malignant cells and danger immune rejection. The tissue-engineered approach requires the use of biomaterials (scaffolds), cells, and growth factors (Zeng et al., 2010). The biomaterials used are either from a natural source (for instance, from collagen, alginate, gelatine) or synthetically produced (polylactic acid (PLA), poly(D, L-lactide-co-glycolide) (PLGA), polycaprolactone (PCL)). Natural biomaterials are mostly preferred as they retain cell adhesion, cell migration differentiation, and proliferation (Baker et al., 2009). Unfortunately, the batch-to-batch variability of the natural biomaterials is due to their characteristic compositions causing limited reproducibility (Baker et al., 2009). In addition to the nature of biomaterials, the limited availability of specific cells has resulted in scientists resorting to the use of stem cells to generate desired tissue through remodeling using tissue-specific growth factors. However, synthetic biomaterials improve biocompatibility, modifications and can be fabricated to suit the desired structural compositions of the needed tissue or organ. Despite the shortcomings of the techniques, several types of research are being conducted both in vivo and in vitro to come up with the best ways to permanently overcome female factor infertility. This review paper is focused on throwing light on some of the most recent tissue engineering approaches utilized to combat female infertility.

16.2 Female reproductive system

16.2.1 The ovary

The ovarian factor infertility contributes to about one-third of female infertility cases. It is mostly attributed to the age of the female (Levine et al., 2015). Research shows that changes occur in the hormones and the total numbers of ova produced in every female decline with age (Levine et al., 2015).

The ovaries are paired glands known as the female gonads (egg receptacles) (Yarmohammadi et al., 2011). Each ovary is connected to the upper part of the uterus by the ovarian ligaments with a length of approximately 2.5–3.5 cm long, 2 cm wide, and 1 cm thick (Sasaki & Miller, 2019). Structurally, an ovary consists of two main layers of tissues called the medulla and the cortex. The medulla is the central part of the ovary which consists of the nerves, fibrous tissues, and blood vessels. The cortex, however, is the tissue surrounding the medulla (Perven et al., 2015). It is made up of a network of connective tissues called the stroma and contains the ovarian follicles of various stages of development and is covered by an epithelial layer called the germinal epithelium (Johnson & Grier, 2018). Inside the ovarian follicle is the ovum (egg) which is inactive until the period of puberty. Primarily, the ovaries are responsible for the production of secondary oocytes and hormones such as progesterone and estrogens (Nation & Selwood, 2009).

16.2.2 The fallopian tube

According to statistics, about 20% of female infertility is attributed to faulty fallopian tubes (Carrascosa et al., 2016). This means one out of every five female infertility cases is caused by dysfunctional fallopian tubes. So far, not much research has been conducted on the tissue reconstruction of the tubes due to the less in-depth knowledge on the histological mechanisms of the structures of the tubes. The first tubal reconstruction was conducted in 1896 ended in a disastrous outcome because very little knowledge of the multiple functionalities of the fallopian tubes was available to scientists (Jiang & Li, 2021).

Fallopian tubes are paired conduits connecting to each end of the ovaries to the uterus. The tube has an average length of 11–12 cm with a lumen diameter of less than 1 cm which is anatomically composed of a finger-like projection (fimbriae), a funnel-shaped portion (infundibulum), a short, narrow, and thick-walled portion (isthmus) joining the uterus, and a lateral two-third portion (ampulla). Histologically, three major layers (mucosa, muscularis, and serosa) make up the walls of the fallopian tube (Ovrang, 2011). The inner layer (mucosa) consists of ciliated epithelium which aids in the conveying of secondary oocytes toward the uterus and smooth connective tissues (nonciliated (peg) cells) which secrete fluid that provides nutrients for the oocytes (Hagiwara et al., 1990). The middle layer (muscularis) aids in the movement of the secondary oocyte through the peristaltic contraction of its muscle coats. The outer layer (serosa), which is made of connective tissues provides

structural support to the tube. The fallopian tubes play a crucial role in pregnancy by guiding the sperm toward the fertilization site through various comprehensive mechanisms to initiate the hyperactivation of the sperm to become fertile and to transport fertilized eggs to the uterus for implantation (Govahi et al., 2021).

16.2.3 The uterus

According to statistics, slightly over 30% of female infertility may be attributed to problems with the uterus (Carrascosa et al., 2016). The uterus is mostly regarded as the root cause of infertility in most women. Unfortunately, most abnormalities of the uterus do not exhibit obvious symptoms such as pain, making detection and treatment difficult unless there are persistent issues such as miscarriages and lack of menstruations.

The uterus is a structurally muscular organ that hosts and nourishes the developing embryo. In a normal woman (no pregnancy), the average space occupied by the uterus is approximately 7 cm in length and 5 cm in width (Egbase et al., 2000). It is composed of three portions: a superior portion (fundus) which connects the uterus to the fallopian tubes, a midportion (corpus) which hosts the growing embryo, and an inferior portion (cervix opening) which serves as the pathway to the vagina (Salman et al., 2022). Histologically, the uterine wall is composed of three layers: a serous membrane (perimetrium) with epithelial tissues surrounding the entire uterus, a thick middle membrane (myometrium) containing smooth muscle fibers for uterine contractions in case of ousting menstrual blood or during childbirth, and an innermost portion (endometrium) responsible for the thickening of the uterine walls for implantation, a growing fetus, and menstrual period (Albers et al., 1995). Primarily, the uterus is responsible for hosting the embryo at implantation and throughout the period of gestation.

16.2.4 The cervix

The increased injury and mortality rate in infants are mostly attributed to the dysfunctional cervix leading to preterm births (Buck & Thiesen, 2006). It is crucial to maintain the anatomical integrity and shape of the cervix to increase the chance of normal pregnancies. Recent research proposes that the length of the cervix directly affects preterm deliveries. That is, a longer cervix increases the risk of preterm delivery (Cook & Ellwood, 2000). The cervix is located at the lower portion of the uterus which serves as the link between the uterus and the vagina. Histologically, the cervix constitutes two sections: ectocervix with stratified squamous epithelial tissue which serves as the external mucosal surface link to the vagina, and approximately 7 mm inner portion called the endocervix which serves as the canal toward the vagina (Mazza et al., 2017). The cervix primarily serves as the passageway for sperm to the fallopian tubes and a protective shield to the uterine opening during gestation.

16.3 Cause of female infertility

The incidence of female infertility may be generally attributed to endocrine problems, pathological effects, environmental conditions, and congenital abnormalities. Sometimes, the cause of infertility may not be known despite the thorough diagnosis conducted by the physician (Ngai & Loke, 2021). In some cases, the cause of infertility may be due to multiple causes such as both structural and hormonal problems. Some of these causes of infertility have been listed in Table 16.1 below.

Table 16.1 A list of physio-pathological causes of female factor infertility.

Causes	Organ(s) involved	Treatments	Impact on fertility	References
Pelvic inflammatory disease (PID). Example salpingitis	Fallopian tubes	A mixture of antibiotics to curb any bacterial type of infection	Repeated episodes of PID increase the chance of infertility from about 12% to approximately 53%.	Carson (1999)
Endometriosis	Uterine wall, fallopian tubes, ovaries	1. Hormonal treatment with birth control pills 2. Surgery	Between 30% and 50% of women experience infertility due to impaired implantation of pregnancy and altered egg quality	Swift et al. (2021)
Hydrosalpinx	Fallopian tubes	1. Tubal surgery 2. Antibiotics	May cause tubal blockage leading to infertility	Ling et al. (2022)
Ectopic pregnancy	Fallopian tubes	Surgery	Increases chance of infertility to about 35%	Khoo et al. (2014)
Ovarian torsion	Ovaries	Laparoscopic surgery	Delayed treatment may lead to the total removal of the ovary	Naffaa et al. (2018)
Uterine fibroid	Uterus	Surgery (myomectomy)	Contributes to about 10% of infertility in women with such condition	Tropeano et al. (2012)
Gynecological cancer	Ovaries, oviducts, uterus, cervix, vagina	Surgery chemotherapy radiation	The extent of infertility depends on the affected tissue/organ	Ling et al. (2022)
Polycystic ovarian syndrome	Ovary	1. Medications 2. Surgery	About 70% of women with conditions struggle to get pregnant	Salah (2013)

Continued

Table 16.1 A list of physio-pathological causes of female factor infertility.—*cont’d*

Causes	Organ(s) involved	Treatments	Impact on fertility	References
Ovarian cyst	Ovary	Surgery	Minimal chance of infertility	Tagliabue et al. (2009) Stewart and Leake (2006)
Cervicitis	Cervix	Medications (antibiotics)	Untreated cervicitis increases the chance of miscarriage and infertility	
Cervical polyps and cysts	Cervix	Surgical procedure using polyp forceps	Blockage of the cervix prevents sperm entry resulting in an increased chance of infertility	Butler (1964)

16.4 Other risk factors of female infertility

Despite the pathophysiological and congenital conditions contributing to female infertility, there are lifestyles and environmental conditions that may adversely contribute to the increase in the chances of infertility.

- Diabetes
- Eating disorders such as anorexia nervosa and bulimia nervosa
- Age (increasing age above 35 reduces the number of ova available for fertilization)
- Radiation therapy
- Smoking and excessive alcohol
- Stress
- Excessive exercising
- Exposure to environmental toxins (such as pesticides, lead, etc.)

([Dumitriu et al., 2019](#)).

16.5 Current treatments for female infertility

The idea of implementing approaches for the diagnosis and treatment of female factor infertility became a reality when the first scientist, Antoine Philips Van Leeuwenhoek first discovered spermatozoa through a microscope in 1677 ([Rakotomahe-nina et al., 2015](#)). Since then, various attempts have been made to restore the varying organ factors contributing to female fertility. Below are brief descriptions of some methods that have been deployed since it gained the attention of science ([Table 16.2](#)).

Table 16.2 A summary of existing treatment methods for treating female factor infertility and the disadvantages of each approach.

Treatment	Techniques	Organ(s) affected	Disadvantages	References	
Medication	Prescription of the problem-specific drug(s)	All organs, but mostly associated with hormonal disorders	(1) Risk of multiple pregnancies. (2) Incidence of anxiety. (3) Incidence of ovarian hyperstimulation syndrome	Mathur and Jenkins (2000)	
	Chemotherapy	All carcinogenic tissues/ organs	Risk of permanent damage to other tissues.	Elsum and Humbert (2013) , Williams et al. (2001)	
	Hormone therapy		Reintroduction of malignant cells		
Surgery	Immune therapy				
	Hysteroscopy	Uterus, oviduct	Risk of damage to tissue/organ	Mulrooney (2015) , Taylor and Gomel (2008)	
	Laparoscopy		Infection, risk of damage to internal structures		
	Myomectomy	Uterus	Injury to uterus infection	Mouttham and Comizzoli (2017) , Mulrooney (2015)	
	Uterine artery embolism	Uterus	Possible injury through microsurgical technique		
	Salpingostomy	Oviduct	(1) Risk of infections. (2) Possible injury/damage to other tissues. (3) Severe pain and discomfort. (4) Longer recovery periods.		
	Salpingectomy				
	Fimbrioplasty				
	Tubal cannulation				
Tubal anastomosis					
Organ/tissue transplantation	Autologous	All organs or tissues	Risk of infection in the case of possible malignant cells.	Elsum and Humbert (2013)	
	Allogenic		(1) Issues with immuno-rejections. (2) Bioavailability	Flate and Stalvey (2012)	
	Decellularization		(1) Possible retention of ECM structures and morphologies. (2) Loss of ECM proteins	Powell (2006)	
	Cryopreservation		Possible damages to cells due to higher intracellular concentrations	Han and Liu (2011)	

Continued

Table 16.2 A summary of existing treatment methods for treating female factor infertility and the disadvantages of each approach.—*cont'd*

Treatment	Techniques	Organ(s) affected	Disadvantages	References
Assisted reproductive technologies	Artificial insemination	Oviduct, uterus	Infections from unsterile surgical equipment	Sorger et al. (1983)
	In vitro fertilization	Ovary, oviduct, uterus	Risk of ovarian hyperstimulation, cancer, miscarriages	Mulrooney (2015)
	Tubal embryo transfer (TET)	Oviduct	Risk of pelvic infection.	Al-Hussaini et al. (1992) , Mor and Vermesh (2005)
	Gamete intrafallopian transfer (GIFT)		Puncture of delicate organs.	
	Zygote intrafallopian transfer (ZIFT)		Risks of multiple births.	

16.5.1 Medication

The use of drugs has been a preferred method for treating female infertility issues relating to conditions such as ovulation disorders, primary ovarian failures, follicle stimulation, and underlying conditions such as infections, diabetes, etc (Roness & Meiorow, 2019). In most situations, the idea of drug use is to mimic natural hormones in women with hormonal syndromes. For instance, clomiphene citrate is a popular drug recommended for inducing ovulation in women. The drug activates the pituitary gland to ensure the effective growth of ovarian follicles by releasing more luteinizing hormones (LHs) and follicle-stimulating hormones (FSHs) (Bermejo et al., 2014). Some medications are administered to control menstrual cycles, especially for women with irregular blood flow who desire to conceive. For instance, women suffering from hyperprolactinemia are treated with a combination of bromocriptine and cabergoline to suppress the over secretion of prolactin by the pituitary glands, which is the main contributing cause of low estrogen levels in the body (Jones et al., 2012).

Other drugs in the market used in the treatment of infertility relating to ovarian dysfunction include leuprolide, letrozole, chronic gonadotropin, etc., with each drug serving specific correctional functionalities (Balen et al., 2016). Despite the correctional functionalities of these medications, they mostly do not get rid of the entire conditions, but rather render managerial remedies which result in the women being subjected to close to a lifetime on medications to help remain fertile.

16.5.2 Surgical procedures

The use of the surgical procedure is the preferential approach in treating female factor infertility that may be attributed to accidents, diseases, deformities, or congenital conditions. Surgery is a form of medical procedure used for the repair or removal of dysfunctional tissues or organs in the reproductive system (Fu, 2021). The procedures relating to female infertility are currently categorized into three sections: primary standard surgical treatment, pregnancy-induced surgical treatment, and fertility preservation surgery. The type of procedure utilized is solely dependent on the treatment condition (Sato et al., 2017). For instance, neosalpingostomy is a surgical procedure for removing hydrosalpinx. The procedure has proven to improve live birth relating to IVF. Research has shown the improvement of intrauterine walls by a procedure called laparoscopy, which usually turns out to enhance instinctive pregnancy. Some of the common surgical procedures include tubal cannulation, laparoscopy, hysteroscopy, salpingectomy, myomectomy, etc. In some cases, these procedures may subject the patient to secondary injuries, creating a different medical condition for the patient (Franchini et al., 2009).

16.5.2.1 Organ/tissue transplantation

These are a series of procedures used for the replacement of reproductive tissues for patients of all ages that are undergoing high-risk therapies. Usually, patients undergoing cancer treatments such as chemotherapy or pelvic radiotherapy are required to

ensure the careful transplant of unaffected autologous organ/tissues outside the irradiation area of cancer treatment. The procedure eradicates immune rejections as well as the issues of immunosuppression drugs due to the utilization of autologous organ/tissue (Levine et al., 2015).

Since the start of this technique, there have been over 200 live babies recorded worldwide (Ozkan et al., 2013). Notwithstanding the success rate of the cryopreservation method, orthodox frozen tissues with the use of diffusing cryoprotectants of low concentrations such as dimethyl sulfoxide (DMSO) in the presence of a slow-cooling curve in a well-coordinated schematic freezing device remains the best option to preserve the fertility of prepubertal females undergoing cancer treatment (Atala, 2007). The technique cannot be recommended for patients diagnosed with tumors who are exposed to the possibility of acquiring high risks of ovarian metastasis because of a higher chance of being reinfected with possible malignant cells present in the cryopreserved ovarian tissue (Levine et al., 2015).

16.5.2.2 Ovarian follicle transplantation

Ovarian follicle transplantation has seen tremendous clinical success in recent times. Research conducted in recent times on ovarian follicle transplantation has demonstrated the total fertility recovery after chemotherapy by means of irradiation of the whole body (Roness & Meirow, 2019). From research conducted by (Silber et al., 2015), successful allogeneic and autologous transplantation of the ovary for both fresh and cryopreserved states have been achieved in a single series. Despite the breakthroughs in these clinical attempts, the technique cannot be replicated for patients treated with cancer due to the high-level threat of transferring carcinogenic cells back into the patient (Levine et al., 2015).

16.5.2.3 Uterine tissue transplantation

In the year 2000, the first whole uterus transplant surgery was performed on a 26-year-old female after being diagnosed with postpartum hemorrhage and subsequently losing her uterus. Despite a successful transplant, the patient developed acute vascular thrombosis in less than 100 days after transplantation. Further investigations conducted suggested the feasibility of an allogeneic transplant of the fresh uterus and the possibility of fertility restoration (Egbase et al., 2000). However, issues with immunogenic consequences and the availability of allogenic tissues persist.

16.5.2.4 Fallopian tube transplantation

The first case of fallopian tube transplant was recorded in 1946 when five cases of human oviducts from donors were implanted into the patient (Ritala, 1946). Tubal patency was observed in all five recipients using a postoperative hysterosalpingogram (Ritala, 1946). Similar transplantation (homograft) has been performed on four patients by Sillo Seidl in 1975 with no sign of rejection of the patients according to (Cohen, 1980). Despite none of these researchers documenting any successful pregnancy breakthrough, the viability of fallopian is reported to be dependent on

the patency of the single artery of the uterus capable of revascularizing the tube for efficient blood supply (Cohen, 1980). Technically, fallopian tube vascularization is feasible. However, immune rejection has predominantly prevailed in all attempts. The recent success of IVF has drifted the focus of further research into this area.

16.5.3 Assisted reproductive technologies

The ART comprises a series of approaches that are designed to bypass infertility in women by helping the woman conceive. The procedure is employed mostly when there is a dysfunction fallopian tube, ovary, uterus, or cervix. The techniques encompass the exploitation of egg cells or embryos to enhance the achievement of pregnancy (Mathur & Jenkins, 2000). For all the types of ART procedures available, different approaches are employed in their implementations. For instance, IVF entails the removal of eggs and nourishing them with sperms in a special lab under optimum conditions (Mathur & Jenkins, 2000).

The chances of successful IVF depend on the age of the patient. About 52% success rate is achieved for first-time patients below age 35 and patients above 40 years recording as low as 7.6% success (Al-Hussaini et al., 1992). GIFT, however, requires the fusion of egg and sperm outside the tube and finally depositing the embryo back into the fallopian tube. A frozen embryo transfer technique, however, involves thawing preserved embryos obtained from the IVF procedure and implanting them into the uterus of patients (Mor & Vermesh, 2005). Some factors such as age and health condition of the patient determine the success rate of the technology. Despite the progress made in the use of ART treatment techniques as the in recent times, over 78% of couples worldwide are battling obstinate infertility (Balen et al., 2016).

16.6 Tissue engineering-based approaches in treating female infertility

The idea of tissue engineering gained an audience of scientists in the year 1987 when a Professor at the University of California submitted a proposal to National Science Foundation (NSF) entitled “Center for the Engineering of Living Tissues” (Montaser, 2018). Today, tissue engineering may be described as a subset of regenerative medicine with the target of achieving the ability to preserve, restore, mend, and improve the functionalities of specific organs or tissues by enhancing the living cells present (Montaser, 2018). The general architecture of tissue engineering is composed of three key components including scaffolds, cells, and growth factors.

16.6.1 Scaffold-based tissue engineering approach

A biomaterial (scaffold) is any substance that has undergone modifications to interact with biological systems to repair, replace, or improve any organ or tissue in the body with no complications (Tamadon et al., 2016). Biomaterials can be

obtained by natural or synthetic means. Regardless of the source, a good biomaterial must be characterized by good biocompatibility (be able to support in vitro cell growth and the intricate in vivo regeneration of tissues), biodegradability, good biochemical and mechanical stability (withstand any physical stress caused by the complex in vivo metabolism), low toxicity, and a microporous 3D structure to enhance cell attachment (Badylak, 2002). In the case of female reproductive tissue regeneration, scaffold selected for an organ such as the uterus or the endometrial tissue should be able to work properly even during and after the menstrual cycle and the by-products formed from the scaffold during the degradation period should be easily resorbable by the metabolic cycle in the body (Yun et al., 2016). Currently, the suitable materials available for replacing reproductive tissues are biopolymers including natural and synthetic (Table 16.3).

Table 16.3 A list of biomaterials (biopolymers) used for female reproductive tissues reconstructions showing their respective advantages and disadvantages.

Material type	Material/scaffold	Advantages	Disadvantages	References
Natural	Collagen hydrogel	Most abundant ECM Porous structure to enable diffusion	Difficult to decouple protein concentration Not enough biochemical cues to induce cell response	Ellis (1979)
	Alginate	Biocompatible and biodegradable Promotes cell growth	No hydration property Rapid degradation Poor mechanical support	Petruzzo et al. (1996)
	Silk	Biocompatible Biodegradable Water-based processing	Low quality Fast degradation Low level of revascularization	Numata et al. (2012)
	Fibrin	Highly biocompatible Suitable for cellular interaction improvement	Low integrity maintenance Quick degradation process in vivo	Lipinski and Gurewich (1976)
	Chitosan	Biorenewable Nontoxic and antigenic Bioadhesive	Limited solubility in organic solvents Low mechanical properties Low biostability	Lipinski and Gurewich (1976)
	Matrigel	Biocompatible Replication of specific ECM	Lack mechanical integrity Low batch-to-batch availability	Ellis (1979)

Table 16.3 A list of biomaterials (biopolymers) used for female reproductive tissues reconstructions showing their respective advantages and disadvantages.—*cont'd*

Material type	Material/scaffold	Advantages	Disadvantages	References
Synthetic	PLA	Good in vivo biocompatibility Tissue ingrowth Low degradation rate	Loss of tensile strength after absorption Likely release of harmful oxidative substances during breakdown	Franco et al. (2014)
	PLGA	Good biocompatibility Suitable for drug delivery Low degradation rate	Loss of mechanical stability	Franco et al. (2014)
	PEG	Suitable for cells attachment Induces cell migration and proliferation		Franco et al. (2014)
	PCL	Long degradation rate No second surgery required Biocompatible	Poor antimicrobial properties Possible mismatch of mechanical properties	Franco et al. (2014)

Natural scaffolds: The natural biopolymers are obtained from the body, examples include collagen (type 1–28), chitosan, fibrin, silk, hyaluronic acid, agarose, alginate, matrigel, etc. To regenerate gynecological tissues, a composite of collagen (type I and IV) and matrigel polymers is widely used. Collagen type I is considered the main structural protein in the interstitial ECM which makes it the most widely used material for 3D in vitro simulations. Matrigel, however, corrects the limitations of collagen by supplying appropriate biochemical signals to stimulate response mimicking the in vivo models ([Tamadon et al., 2016](#)).

Synthetic scaffolds: The synthetic biomaterials are obtained from sources other than the natural. Due to the lack of mechanical integrity and bioavailability of natural polymers, most researchers now resort to synthetic biomaterials as the alternative to natural biomaterials.

Currently, the widely used synthetic biomaterials in gynecological tissue engineering are polyethylene glycol (PEG)-related biomaterials such as poly(L-lactic acid) (PLLA), PCL, PLGA, and the raw PEG ([Chen et al., 2012](#)). The key limitations of using synthetic biomaterials however are their inability to live through in vivo models without alterations to sustain the desired cell activities and tissue growth, and their bioinactive nature ([Hendow et al., 2016](#)).

16.6.1.1 Fabrication techniques

In the fabrication of biomaterials (scaffolds), it is always key to note the ECM of the tissue/organ under investigation. The proteinoid architecture of ECM makes it biocompatible, capable of swift in vivo modeling to assume the original shape of tissue/organ, and biodegradable (Lu et al., 2011). Researchers, therefore, resort to creating scaffolds that have the capacity to exhibit these attributes both in vitro and in vivo. In the past years, different fabrication methods have been considered in the production of scaffolds to mimic the microenvironment of their hosts by simulating cell-specific responses to repair or regenerate desired tissues/organs. Examples of some fabrication techniques currently in use include electrospinning, freeze-drying, gas foaming, phase separation, solvent casting, material injection, 3D printing, etc.

16.6.1.2 Scaffold selection for ovarian tissue engineering

The choice of scaffold for ovarian tissue reconstruction must consider the unique architecture of ovarian follicles (spherical shape), tissue vascularization features, and the importance of ECM recapitulation (Irving-Rodgers & Rodgers, 2005). Granulosa expansion with a unique storage modulus between 1 and 2.5 KPa during follicular growth and the stiffness levels at each stage of follicle encapsulation must also be considered in the selection of suitable biomaterial (Beg et al., 2002). With the medulla part of the ovary comprising vascular tissues and the cortex comprising the germ cells, stromal cells, and sex cells, critical attention is given to the biomaterial selection to suit both sections of the ovary and the tissues thereof. The ovarian tissue ECM consists of fibronectin, laminin, glycoprotein, and collagen protein (Beg et al., 2002). With the primordial follicle of the ovary serving as the foundation for oogenesis, it is vital to select biomaterial that supports follicular biocompatibility and engraftments, diffusion of nutrients, proliferation, and migration of follicles (Tamadon et al., 2016). Natural biomaterials used so far in ovarian tissue reconstruction are fibrin (for its physical and crosslinking properties) and alginate (for its good cell–cell attachment and differentiation) models (Chan et al., 2013). In most instances, a combination of both natural and synthetic biomaterials has shown excellent cellular microenvironment traits (Sun et al., 2012).

16.6.1.3 Scaffold selection for uterine tissue engineering

Scaffold selection for uterus reconstruct must have the capacity to mimic successful pear-shaped regenerated organs with cell-specific functions of tissues of the uterus. The ECM of the uterus is composed of collagen IV, proteoglycans, laminin, elastic fibers, fibronectin, and collagen. Based on recent studies, the natural biomaterial used in most successful uterus regeneration is the collagen or collagen-containing scaffold. With the crucial portion of the uterus for embryo implantation and successful gestation being the endometrium, most tissue reconstructs have been conducted on the endometrium. In 1894, Heinrich Fritsch has successful clinical trials on two patients

who have been treated with severe Asherman's syndrome and undergone endometrial tissue regeneration using tissue engineering technologies (Conforti et al., 2013). The procedure resulted in successful pregnancies in both patients. However, due to the poor mechanical support shown by the collagen-based scaffold, mixtures of collagen and other biomaterials have been explored and shown promising results. For instance, collagen/matrigel has been used to regenerate the full thickness of rabbit endometrium (Conforti et al., 2013). With the issues of the bioavailability and mechanical strengths of natural biomaterials, some synthetic scaffolds have also been explored in uterine tissue reconstruction. The use of PLGA, a combination of polyglycolic acid (PGA) and PLA scaffold seeded with endometrial cells, has been used in regenerating the uterine structure and function of rabbit. Out of 10 rabbits, four showed reconstructed segments of the uterus subsequently resulting in live births (Egbase et al., 2000).

16.6.1.4 Scaffold selection for fallopian tube tissue engineering

The complex architecture of the oviducts with dynamic tissues that are in continuous adjustment when responding to the fluctuating hormonal stimuli (from progesterone, estradiol) during menstrual periods plays a key role in scaffold suitable for selection (Caufriez et al., 2009).

Also, the crucial transportation mechanisms of fertilized eggs pose a huge challenge in the reconstruction of their tissues to mimic the natural cellular functions of the organ at large. The oviducts have a vascular supply and a ciliated microstructure making reconstruction very difficult to achieve. However, research suggests collagen III enhances oviduct epithelial cell invasion. Again, both collagen and alginate have been used to reconstruct the epithelial tissue of the oviduct in a 3D ex vivo model. The result showed a closed mimicking an in vivo tissue with the ability to respond to the hormones responsible for the menstrual cycle (Keator et al., 2011).

16.6.1.5 Scaffold selection for cervix tissue engineering

The cervix is composed of mainly fibrous connecting tissues which adapt to morphological changes during gestation and during the labor period. So far, the scaffold used for cervix reconstruction is silk fibroin. The scaffold was successfully proliferated with cervix cells in 3D models exhibiting almost all biochemical and ECM-like natural tissue.

16.6.2 Cell/stem cell-based tissue engineering approach

In the past few years, stem cells have been the replacement for traditional primary cell implantation. Stem cells are undifferentiated cells found in either somatic (adult) or embryonic cells. In female infertility, the main stem cells involved in tissue regenerations are the embryonic stem cells (ESCs), induced pluripotent stem cells (iPSCs), and mesenchymal stem cells (MSCs) (Dupuis & Oltra, 2021).

The ESCs are derived from the inner cell mass of blastocysts that can differentiate into other stem cells making them pluripotent in nature. The iPSCs, however, can differentiate three ESC lines (the three skin layers), with both tissues and organs developing from the endodermic portion (Pichard et al., 2020).

Cell-direct implantation: This method involves the extraction of primary cells from the healthy tissues of the patient through biopsy, culturing, and subsequently implanting at the defective site. This technique is used for both primary cell and stem cell implantations.

Decellularization, repopulation, and implantation: When the primary cells obtained are likely to be from infected tissue, the primary cells are carefully removed (decellularized) leaving the ECM framework and healthy cells are reintroduced to repopulate the matrices. This method is also employed when cells are obtained from another person (Brown & Gliedman, 1973). This is to get rid of the occurrence of immunosuppression between the donor cells and the recipient body. This technique is widely used for primary cell implantation.

Cells + scaffold implantation: Cells (primary cells or stem cells) are seeded on a scaffold and cultured in vitro before implants. This is to ensure proper cell differentiation, proliferation, and fusion of ECM to mimic the natural tissue in vivo.

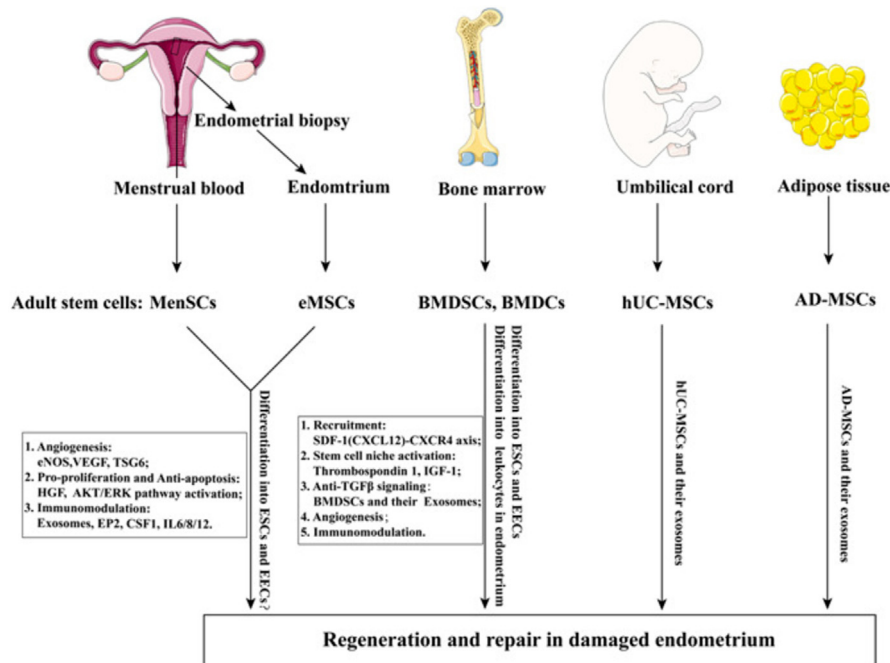
16.6.2.1 Cell-based approach for ovarian tissue engineering

Ovarian factor infertility is predictable with aging, hence the need for an amicable solution to ovarian failure. Recently, there has been a breakthrough in the discovery of ovarian stem cells (OSCs) and female germ stem cells (GSCs); two germline stem cells with the ability to maintain the functionality of the ovary through ovarian restoration. So far, the active GSCs can form oocytes when decontaminated and cultured in vitro (White et al., 2012). Human MSCs extracted from the umbilical cord have been analyzed for their regenerative function (Gadkari et al., 2014). Also, the inflammation of the ovary has been dropped and improved follicular expansion by increasing the number of oocytes using the human amniotic epithelial cells in mice that have previously been affected by excision (Somfai et al., 2012). Furthermore, the ovarian function of excised mice showed vast improvement when female germline stem cells were isolated and implanted. Stem cells of amniotic fluid can reduce follicular atresia and preserve healthy follicles when differentiated into granulosa cells (Xiao et al., 2016). Research has shown that menstrual stem cells (hMensSCs) can improve ovarian weight and the number of follicles found in mice through transplantation (Han & Liu, 2011).

The scaffold-free techniques to repair ovarian factor infertility have shown encouraging breakthroughs per the few mentioned. However, it is crucial to prove these results in larger animal versions with successful pregnancies in the next phase prior to clinical trials.

16.6.2.2 Cell-based approach for uterine tissue engineering

The construction of a stem-cell-based uterus involves natural growing cell lines as principal parts. Research in recent times suggests the use of either patient-specific

**FIGURE 16.1**

A schematic summary of different stem cell-based and scaffold-free approaches in treating uterine tissue damages (Lv et al., 2021).

primary cultures or pluripotent stem cells (PSCs) based on the purpose (Fig. 16.1). Isolated primary endometrial cell from immature mice has been cultured under low temperature and used in the regeneration of uterine endometrial tissue (Lv et al., 2021). A thicker endometrial wall was achieved after three days of reculturing. Although the cell sheets of the endometrial lining were observed to preserve uterine function in vitro, their effectiveness in encouraging the development of fertilized eggs has not been tested against other tissue engineering approaches in animal models (Lv et al., 2021). Endometrial stem cells (EndoSCs) in the endometrial tissue have been proposed to take part in endometrial regeneration and healing (Syed & Tanwar, 2020).

Tan et al. demonstrated intrauterine transplantation after in vitro growth to treat seven patients with massive uterine adhesion. Five of the patients had endometrial thickening up to 7 mm, two of the five developed pregnancies successfully when the embryo was transferred into the uterus, and one conceived naturally after the second autologous transplantation of the MenSCs (Tan et al., 2016).

16.6.3 Growth factor-based tissue engineering approach

The use of growth factors or biochemical hormones plays a critical role in tissue repair and regeneration. They can promote cell migration, proliferation, and critical in tissue morphogenesis and healing (Koria, 2012). Despite the existence of several growth hormones (GHs), the crucial component needed for tissue repair is the human GH or somatotropin. In regenerative medicine application, GHs are combined with cells and/or biomaterials for tissue growth and regeneration (Makras et al., 2004). The type of scaffold and cells used as well as the target tissue repair play critical role in the selection of suitable GHs. For instance, collagen scaffold used in endothelial tissue repair improved in the uterine thickness when collagen-based basic fibroblast growth (bFGF) was added (Ma et al., 2005). In the tissue repair of follicular oocytes, the FSH and the LHs are applied to the scaffold for implant at the damaged site.

16.7 Conclusion

Bioengineering the female reproductive system has shown great potential in resolving female infertility (Table 16.4). The most probable challenges in engineering reproductive tissues are the unique architecture, tissue vascularization, and viability of the ECM of the specific tissue/organ involved. In past decades, different approaches have been employed to safeguard the fertility of women undergoing treatments such as chemotherapy, congenital organ malformation, oviduct occlusion, hysterectomy, and immense intrauterine adhesions. Despite the success of whole organ transplants, issues such as organ availability and immune rejections from allografts remain challenging. Decellularization/repopulation of tissues with autologous cells represents an alternative remedy to overcoming the challenges of autotransplantation. Recently, advances in tissue engineering have introduced the use of biomaterials, cells, and biochemical signals to reconstruct new tissues/organs that mimic the damaged ones. The most recent strategy is the 3D printing technology with the potential of mimicking natural tissue in terms of site rating its vascularization, biomechanical structure, and cellular longitudinal redistribution. With few successful in vitro and experimentations and in vivo animal models of the causative factors of female infertility, it is empirical that more in-dept analyses are made on the larger animal trial faces for further understanding in comparison with the human anatomy before any clinical (human) trial is attempted in the future. In conclusion, regenerative medicine is a great potential to irradiate the issue of female infertility caused by either aging, congenital factors, hormonal disorder, or injuries due to previous surgeries or accidents.

Table 16.4 A table showing the various tissue engineering strategies utilized in the regeneration of female reproductive tissue/organ.

Technique	Cells used	Scaffold used	Species	Results	References
Tissue-engineered uterine patch	Uterine and mesenchymal stem cells (MSCs)	Decellularized uterine matrix	Rat	This resulted in composition and pregnancy compared to the native uterus.	Arezoo et al. (2021)
Whole organ allogeneic transplant	N/A	N/A	Human	Vascular thrombosis developed 99 days after implant forcing hysterectomy to be performed.	Pichard et al. (2020)
Whole organ autotransplant	N/A	N/A	Macaques	Transplanted uterus survived, normal menstruation, and successful pregnancy occurred.	Franco et al. (2014)
Tissue-engineered ovarian scaffold	A full oocyte	Alginate hydrogel	Mice	A live birth was achieved after successful implant.	Wang et al. (2011)
Hydrogel encapsulation	Uterine endometrium	Chitosan heparin hydrogel	Rat	Controlled release of SDF-1a.	Wenbo et al. (2020)
Decellularized ECM	Uterine and MSCs	Decellularized uterine matrix	Rat	Recellularization of the decellularized uterine matrix.	Miyazaki and Maruyama (2014)
3D printing	Ovarian follicle	3D printed microporous biomaterial	Mice	Increased interaction of cell-to-cell matrix.	Paivana et al. (2019)

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Index

Note: 'Page numbers followed by *f* indicate figures and *t* indicate tables.'

A

Absolute uterine infertility, 207
Accidents, uterus, 206
Acellular matrix
 allograft, 97
 scaffold, 59–60
Acellular tissue matrices, 241
Acute bacterial prostatitis, 124
Adipose-derived stem cells (ADSCs), 289
Adipose stem cells (ASCs), 97
Adult stem cells, 42, 148–149, 212–213, 286
Advanced age, 206
Adventitia, 90
Agarose, 34
Albuginea tunica, 114–115
Alginates, 33, 141, 243, 274
Allogeneic cell transplantation, 260
Allograft, 5
Assisted reproductive technologies, 309
Asymptomatic inflammatory prostatitis, 125
Augmentation, 64
Autologous cell transplant, 260
Autonomic nervous system, 53

B

Balanitis, 112
Balanoposthitis, 112
Benign prostatic hypertrophy (BPH), 121
Biocompatibility, 92–93
Bioengineering, cell source for
 uterus, 216–218
 embryonic stem cells (ESCs), 217
 induced pluripotent stem cells (iPSCs), 217
 scaffolds, 217–218
 uterus-specific stem cells, 216–217
Biomaterials, 6–7, 51–52, 273–274
 bladder regenerative medicine, 57–61, 58t
 definition of, 31
 natural, 31–34, 32f
 prostate, 128–129
 collagen, 129
 porous chitosan alginate, 129
 synthetic, 34–37, 35f
 testicles (testes), 137–143, 139t–140t
 decellularized testicular matrix, 141
 extracellular testicular matrix, 138–139
 natural biomaterials, 141–142

 synthetic, 142–143
 testicular prostheses, 143–144
 ureters, 95, 95t
Biomimicry, 40
Bioprinting, 64, 65f
Bioscaffolding, 6
Bladder, 18
 apex, 52
 auto-augmentation, 51
 base, 52
 body, 52
 diseases, 51, 54–56
 function, 54
 muscle structure, 52–53
 neck, 52
 nervous structure, 53–54
 physiology, 52–54
 regenerative medicine, biomaterials for, 57–61,
 58t
 cells, 60–61
 scaffold, 58–60
 tissue engineering, 61–64, 62f
Bladder cancer, 55–56
 treatment of, 55–56
Bladder smooth muscle (BSM), 77
Bowman's capsule, 28–29
Buccal mucosa, 99–100

C

Cell adhesion proteins, 4
Cell-based tissue engineering, 144, 145f
Cell culture, 41–42
 primary renal cells, 41
 progenitor cells, 42
 stem cells, 41–42
Cell-scaffold interfaces, 289–291
Cell/stem cell-based tissue engineering approach,
 260
Central nervous system (CNS), 54
Cervical abnormalities/diseases
 anatomy, 254
 disorders/diseases, 254, 255t
 medication, 257
 organ tissue transplantation, 257
 physiology, 254
 risk factors of, 254–256
 tissue engineering, 257, 257t–258t

Cervical abnormalities/diseases (*Continued*)
 treatment, 255–256
 cervical cerclage, 256
 loop electrosurgical excision procedure (LEEP), 256
 surgery, 256
 total hysterectomy, 256
 Cervical cerclage, 256
 Cervix, 302
 Chemotherapy, 124
 Chitosan, 33, 142
 Chronic bacterial prostatitis, 125
 Chronic kidney disease (CKD)
 diagnosis of, 28
 stages of, 27
 Collagen, 4, 31–32, 129, 141, 242, 242f, 273
 Collagen-binding domains (CBDs), 292–293
 Collagen scaffolds, 292–293
 Composite polymer scaffolds, 60
 Congenital anomalies, 91
 double and bifid ureters, 91
 ureteropelvic junction obstruction, 91
 Corpus cavernosum, 111, 158–159
 Corpus spongiosum, 114
 Cryptorchidism, 135
 Cysteine-rich residues (CYR61), 289–290
 Cystoplasty, 51
 Cystoscopy, 55
 Cytology, 55

D

3D bioprinting, 8, 39–40, 194
 Decellularization, 8–9, 38–39, 39f, 218
 Decellularized extracellular matrix, 193–194
 Decellularized matrices, 215–216
 Decellularized testicular matrix, 141
 Derma fibroblasts, 102
 Dialysis, 27–28
 Digital rectal examination (DRE), 122
 Direct vision internal urethrotomy, 14
 Distal convoluted tube (DCT), 29

E

Elastin, 4
 Electrospinning, 7, 37–38, 64, 65f, 147, 259
 Embryonic stem cells (ESCs), 41–42, 148, 213, 291
 Endometrium
 adult stem cell research, 286
 anatomy, 283–284
 bioengineering of, 287–293
 biomaterials, 286–287, 288t

cell-scaffold interfaces, 289–291
 challenges, 284–286
 molecular mechanisms, 284
 opportunities, 284–286
 physiology, 283–284
 regeneration of, 284–286
 stem cell sheets, 288–289
 surface-functionalized scaffold, 291–293
 Engineered corpus cavernosum, 170–173, 171f
 Engineered prosthesis, 169–170
 Epididymitis, 136
 Epispadias, 112
 Erectile dysfunction (ED), 121
 CC scaffolds
 biomaterials, 161–163, 162t
 desired properties, 163, 164f
 cells used, 164–165
 challenges, 173
 corpus cavernosum, 158–159
 diseases associated with, 159
 engineered corpus cavernosum, 170–173, 171f
 engineered prosthesis, 169–170
 mechanism of, 159
 milestones, 166t–168t
 penis
 anatomy, 158–159
 physiology, 158–159
 risk factors, 159
 tissue engineering approach, 165–169
 treatments, 160–161
 intracavernosal injection, 160
 intraurethral drugs, 160
 penile implants, 161
 penile revascularization, 161
 phosphodiesterase type-5 inhibitors (PDE5i), 160
 shockwave therapy, 161
 testosterone therapy, 160
 Extracellular matrix (ECM), 4, 6–7, 14, 273
 scaffolding technique, 147–148
 Extracellular testicular matrix, 138–139

F

Fabrication techniques, 7–8, 37–41
 freeze drying, 40–41
 gas forming, 40–41
 solvent casting, 40–41
 Fallopian tube, 301–302
 transplantation, 308–309
 Female genitalia, 10–11
 Female infertility
 assisted reproductive technologies, 309

cause of, 303, 303t–304t
 cell-based approach, 314–315
 cell/stem cell-based tissue engineering approach, 313–315
 fallopian tube transplantation, 308–309
 female reproductive system
 cervix, 302
 fallopian tube, 301–302
 ovary, 301
 uterus, 302
 growth factor-based tissue engineering approach, 316
 organ/tissue transplantation, 307–308
 ovarian follicle transplantation, 308
 risk factors of, 304
 scaffold, 309–313, 310t–311t
 cervix, 313
 fabrication techniques, 312
 fallopian tube, 313
 ovarian selection, 312
 uterine tissue engineering, 312–313
 treatments for, 304–309, 305t–306t
 medication, 307
 surgical procedures, 307–309
 uterine tissue transplantation, 308
 Female reproductive system
 cervix, 302
 fallopian tube, 301–302
 ovary, 301
 uterus, 302
 Fetal stem cells (FSCs), 42
 Fibrin, 32–33, 142, 273–274
 Fibroblast growth factor (FGF), 43, 149–150, 289
 Fibronectin, 4
 Foaming, 7
 Freeze drying, 8, 40–41, 243–244, 243f

G

Gamete donation. *See* Sperm
 Gas forming, 40–41
 Gelatine, 32
 General visceral afferent (GVA), 54
 Genetic disorders, uterus, 206
 Genitalia
 female, 10–11
 male, 11
 Genital tissues, 18–19
 Genitourinary system, 3
 anatomy and physiology, 9–11
 conditions, 11–13
 regenerative medicine, 4
 tissue engineering, 4–9, 14–20, 15t–17t

treatments and potential drawbacks, 13–14
 Genitourinary tissue engineering, 14–20
 bladder, 18
 genital tissues, 18–19
 kidney, 19–20
 review and research papers, 14, 15t–17t
 urethra, 14–18
 Glomerular filtration rate (GFR), 27–29
 Glomerulus, 28–29
 Glycosaminoglycans (GAGS), 4
 Gonadotropin-releasing hormone (GnRH), 180
 Gore-Tex, 92
 Growth factors (GFs), 9, 42–43
 urethra, 82–83

H

HA. *See* Hyaluronic acid (HA)
 Hematuria, 55
 Hemolytic-uremic syndrome, 11
 Hepatocyte growth factor (HGF), 42–43
 Hormone therapy, 124
 Hyaluronan (HA), 244–245
 Hyaluronic acid (HA), 34, 142
 Hydrocele, 136
 Hydrogels, 7
 scaffolds, 260
 Hypogonadism, 137
 Hypospadias, 12, 112

I

Immunotherapy, 124
 Induced pluripotent stem cells (iPSCs), 149, 213–214, 260
 Insulin-like growth factor 1 (IGF-1), 43
 Interstitial cystitis, 55
 Intracavernosal injection, 160
 Intraurethral drugs, 160
 Intrauterine insemination, 188
 In vitro fertilization, 188
 Isograft, 5

J

Juxtamedullary nephrons, 29–30

K

Kidney
 anatomy, 28–30, 29f
 diseases, 11–12
 function of, 10
 physiology, 28–30
 tissue engineering of, 19–20, 44
 transplants, 30–43

Kidney (*Continued*)

- biomaterials and scaffolds, 31–37
- cell culture, 41–42
- fabrication techniques, 37–41
- growth factors (GFs), 42–43
- timeline, 30–31, 30f

L

- Laminin, 4
- Living skin equivalent (LSE), 75
- Loop electrosurgical excision procedure (LEEP), 256
- Loop of Henle, 29

M

- Male genitalia, 11
- Male infertility
 - causes of, 182, 183t–186t
 - clinical and sexual anamnesis, 182
 - definition, 181–186
 - evaluation of, 182–186
 - male reproductive system
 - pennis, 180–181
 - physiology, 180–181
 - prostate, 181
 - testicles, 180
 - urethra, 181
 - vas deferens, 181
 - physical exam, 182
 - seminal analysis, 182
 - tissue engineering, 188–194
 - 3D bioprinting, 194
 - decellularized extracellular matrix, 193–194
 - scaffolds, 192–193
 - stem cells, 191–192, 191f
 - treatments
 - gamete (sperm) donation, 188
 - hormonal treatment, 187
 - solutions and methods, 189t–190t
 - surgical treatment, 186–187
- Mechanoreceptors, 54
- Medical imaging technology, 40
- Megaureter, 11
- Mesenchymal stem cells (MSCs), 61, 97, 116
- Microlithiasis, testicular, 135–136
- Milestones, 166t–168t
- Mucosa, 90
- Muscularis, 90

N

- Natural biomaterials, 31–34, 32f, 141–142
- Natural-material scaffolds, 260

- Natural/synthetic materials, 214–215
- Neotissue development, 100
- Nephrons, 28–29, 29f
- Neurogenic bladder, 11
- Nitric oxide synthase (NOS), 180

O

- Organ/tissue transplantation, 307–308
- Ovarian cancer, 267, 268f
- Ovarian failure, premature, 12
- Ovarian follicle transplantation, 308
- Ovarian glands
 - anatomy, 266–267
 - diseases, 267–269
 - ovarian cancer, 267, 268f
 - polycystic ovary syndrome, 267–268
 - primary ovarian insufficiency, 268–269
 - factors, 270–274
 - alginates, 274
 - biomaterials, 273–274
 - collagen, 273
 - extracellular matrix, 273
 - fibrin, 273–274
 - scaffolding, 270–272
 - stem cell, 272
 - synthetic biomaterials, 274
 - growth factors, 274
 - physiology, 266–267
 - tissue engineering, 269–270
- Ovary, 301

P

- Painful bladder syndrome, 55
- Paraphimosis, 113
- Parasympathetic system, 53
- Penile cancer, 112–113
- Penile corporal tissue, 111
 - albuginea tunica, 114–115
 - cell/stem cell used in, 116–117
 - disease/disorders, 111–113
 - balanitis, 112
 - epispadias, 112
 - hypospadias, 112
 - penile cancer, 112–113
 - Peyronie's disease, 113
 - phimosis and paraphimosis, 113
 - priapism, 113
 - physical bodies, 114
 - tissue engineering approaches for, 115–116
 - transplantation, materials, 113–114
- Penile implants, 161
- Penile revascularization, 161

Pennis, 180–181
 Perimetrium, 205
 Peyronie's disease, 113
 Phimosis, 113
 Phosphodiesterase type-5 inhibitors (PDE5i), 160
 Pluripotent stem cells, 61
 Poly(a-esters), 240–241
 Polycaprolactone (PCL), 36–37
 Polycystic kidney disease, 12
 Polycystic ovary syndrome, 267–268
 Polyethylene terephthalate, 241
 Polyglycolic acid (PGA), 36
 Polylactic acid (PLA), 35–36
 Poly lactic-co-glycolic acid (PLGA), 36
 Polymeric scaffolds, 259–260
 Polymer scaffold, 59
 Polypropylene (PP), 241
 Porous chitosan alginate, 129
 Posttraumatic stress disorder (PTSD), 238
 Preimplantation seeding, 97–98
 Premature ovarian failure, 12
 Priapism, 113
 Primary ovarian insufficiency, 268–269
 Primary renal cells, 41
 Progenitor cells, 42
 urethra, 73–76, 75f
 Prostate, 121–122, 181
 anatomy of, 122
 conditions of, 122–125
 histology of, 122
 tissue engineering, 125
 approach, 126–129
 Prostate cancer, 13, 121–122
 stage I, 122–123
 stage II, 123
 stage III, 123
 stage IV, 123
 treatment of, 123–124
 Prostate-specific antigen (PSA), 122–123
 Prostatitis, 124–125
 Prototyping, rapid, 7
 Proximal convoluted tubule (PCT), 29

R

Radiation, 124
 Recellularization, 218
 Regenerative medicine, 71
 biomaterials for, 57–61, 58t
 definition of, 4
 urethra, 73
 Renal corpuscle, 28–29
 Renal replacement therapy (RRT), 27–28

Renal tubule, 29
 RRT. *See* Renal replacement therapy (RRT)

S

Scaffolding, 7
 biomaterials and, 31–37
 bladder, 58–60
 acellular matrix, 59–60
 composite polymer, 60
 polymer, 59
 cervical abnormalities/diseases, 259–260
 decellularized tissue and synthetic polymers,
 98–100
 design, 8
 female infertility
 cervix, 313
 fabrication techniques, 312
 fallopian tube, 313
 ovarian selection, 312
 uterine tissue engineering, 312–313
 male infertility, 192–193
 ovarian glands, 270–272
 testis cell culture, 146–148
 decellularization, 148
 3D printing, 146
 3D testicular prostheses, 146, 147f
 electrospinning technique, 147
 extracellular matrix (ECM), 147–148
 uterus, 220t–227t
 Scrotum, 134–135
 Shockwave therapy, 161
 Silk, 142, 243
 Small intestinal submucosa (SIS), 79
 Solvent casting, 40–41
 Sperm, 188
 Spermatogonia, 134–135
 Spermatogonia stem cells (SSC), 149
 Stem cells, 5–6, 41–42, 246–247
 adult stem cell and native progenitor, 247
 adult stem cells, 42
 amniotic fluid stem cells, 246–247
 bone marrow-derived stem cells, 247
 definition, 246
 embryonic stem cells (ESCs), 41–42
 female infertility, 313–315
 fetal stem cells (FSCs), 42
 male infertility, 191–192, 191f
 ovarian glands, 272
 sheets, 288–289
 testes, 148–149
 urethra, 76
 Surface-functionalized scaffold, 291–293

Sympathetic system, 53
 Synthetic biomaterials, 34–37, 35f, 240–241, 274
 testicular implant, 142–143

T

Testicles (testes), 133–134, 180
 anatomy of, 134–135
 biomaterials, 137–144, 139t–140t
 pathophysiology of, 135–137
 cryptorchidism, 135
 epididymitis, 136
 hydrocele, 136
 hypogonadism, 137
 spermatocele, 136–137
 testicular cancer, 137, 138t
 testicular microlithiasis, 135–136
 testicular torsion, 137
 varicocele, 136
 physiology of, 134–135
 tissue engineering, 144–150
 Testicular cancer, 137, 138t
 Testicular microlithiasis, 135–136
 Testicular stem cells, 149
 Testicular torsion, 12, 137
 Testosterone, 134–135
 therapy, 160
 Tissue-engineered tracheal graft (TETG), 99
 Tissue engineering, 28, 51–52, 71
 bladder, 61–64, 62f
 bioprinting and electrospinning, 64, 65f
 neurogenic bladder, 63–64
 stress incontinence, 63
 cervical abnormalities/diseases, 257, 257t–258t
 cell/stem cell-based tissue engineering
 approach, 260
 fabrication techniques, 259
 possible growth factor, 261
 scaffold, 259–260
 endometrium, 286–287, 288t
 female infertility, 309–313, 310t–311t
 genitourinary, 14–20, 15t–17t
 kidney, 44
 male infertility, 188–194
 ovarian glands, 269–270
 cell-based tissue engineering, 276
 ovarian cell culture, 275–276
 ovarian follicles, 274–275
 ovarian scaffolding, 275–276
 overview of, 5f
 penile corporal tissue, 115–116

 principles of, 4–9
 biomaterials, 6–7
 cells, 5–6
 decellularization, 9
 fabrication, 7–8
 growth factors (GFs), 9
 material properties, 8
 scaffold design, 8
 structural properties, 8
 prostate, 125
 biomaterials, 128–129
 injection stem cell therapy, 126, 127f
 tissue generation, 127–128, 128f
 testicular, 137–150
 cell-based, 144, 145f
 fibroblast growth factors, 149–150
 scaffold-based, 145–148
 stem cells, 148–149
 ureter, 92–102, 93f, 93t–94t
 urethra, 72–83
 uterus, 211
 vagina, 240–245
 acellular tissue matrices, 241
 poly(a-esters), 240–241
 polyethylene terephthalate, 241
 polypropylene (PP), 241
 synthetic biomaterials, 240–241
 Tissue engineering triad, 72–73
 Total hysterectomy, 256
 Transformation zone (TZ), 254
 Transurethral resection of the prostate (TURP), 123
 Traumas, uterus, 206
 Tunica albuginea, 134–135

U

Ureteropelvic junction obstruction, 91
 Ureters, 89–90
 anatomy, 90
 general, 90
 microscopic, 90
 animal models, 101–102
 biomaterials, 95, 95t
 congenital anomalies, 91
 decellularized tissue and synthetic polymers, 98–100
 diseases of, 91–92
 dysfunction, repair of, 94–95
 full-circumference ureteral defects, 100
 physiology, 90–91

- preimplantation seeding, 97–98
 - regenerative cells, 96–97
 - stones, 92
 - tissue engineering, 92–102
 - tumors and tumor-like lesions, 91
 - Urethra, 71–72
 - anatomy, 72
 - biomaterial scaffolds for, 76–82, 78t
 - decellularized matrixes, 77–79, 79f
 - natural material, 79–80, 80f
 - synthetic material, 80–82, 81f
 - cell sources, 73–76, 74t
 - progenitor cells, 73–76, 75f
 - stem cells, 76
 - clinical issues and discussion, 84–85
 - etiology, 72
 - growth factors (GFs), 82–83
 - male reproductive system, 181
 - tissue engineering of, 72–83
 - Urethral dilation (UD), 14
 - Urethral restoration, 14
 - Urethral strictures, 12
 - Urinary incontinence, 12, 51, 54–55
 - Urinary tract infections, 11
 - Urothelium, 52–53
 - Uterine histoarchitecture, 207–208
 - Uterine tissue engineering, 216–218
 - Uterine tissue transplantation, 308
 - Uterus, 302
 - absolute uterine infertility, 207
 - accidents and traumas, 206
 - adult stem cells, 212–213
 - advanced age, 206
 - anatomy, 204
 - bioengineering, cell source for, 216–218
 - embryonic stem cells (ESCs), 217
 - induced pluripotent stem cells (iPSCs), 217
 - scaffolds, 217–218
 - uterus-specific stem cells, 216–217
 - cell source, 212–214
 - clinical research, 219
 - decellularization, 218
 - decellularized matrices, 215–216
 - diseases and illnesses, 206
 - 2D models, 211–212
 - 3D models, 211–212
 - embryonic stem cells, 213
 - genetic disorders, 206
 - induced pluripotent stem cells, 213–214
 - labor, 205
 - menstruation, 205
 - natural and synthetic materials, 214–215
 - physiology, 204–205
 - pregnancy, 205
 - recellularization, 218
 - scaffold, 209t, 214
 - application, 220t–227t
 - tissue engineering, 211
 - uterine histoarchitecture, 207–208
 - uterine tissue engineering, 216–218
 - UTx animal research, 208–210
 - UTx human research, 210–211
- ## V
- Vagina
 - anatomy, 236–237, 236f
 - diagnosis of, 239
 - disorders, causes of, 237–238
 - posttraumatic stress disorder (PTSD), 238
 - vaginal agenesis, 238
 - vaginal cancer, 237
 - vaginal candidiasis (VC), 238
 - vaginal infections, 238
 - natural biomaterials, 242–245
 - alginate, 243
 - collagen, 242, 242f
 - freeze-dried method, 243–244, 243f
 - hyaluronan (HA), 244–245
 - silk, 243
 - stem cells, 246–247
 - adult stem cell and native progenitor, 247
 - amniotic fluid stem cells, 246–247
 - bone marrow-derived stem cells, 247
 - definition, 246
 - tissue engineering, 240–245
 - acellular tissue matrices, 241
 - poly(a-esters), 240–241
 - polyethylene terephthalate, 241
 - polypropylene (PP), 241
 - synthetic biomaterials, 240–241
 - treatment, 239–240
 - nonsurgical treatment, 239
 - surgical treatment, 239–240
 - vaginal reconstruction, 246–247
 - Vaginal agenesis, 238
 - Vaginal cancer, 237
 - Vaginal candidiasis (VC), 238

Vaginal infections, 238

Varicocele, 136

Vascular endothelial growth factor (VEGF),
82–83

Vascularization, 20

Vas deferens, 181

X

Xenograft, 5

Y

Youngs Modulus, 7